

Fundamental Theorem of Matrix Product States

A Formalization Blueprint

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Contents

1	Introduction	6
2	Matrix Product Vectors	8
2.1	Basic definitions	8
2.2	Site-periodic tensor families	10
2.3	Gauge equivalence and same MPV	10
2.4	Injectivity and normality	13
2.5	Canonical form	14
2.6	Blocking	15
2.7	MPV overlap	16
3	The Single-Block Fundamental Theorem	18
3.1	The multiplicative linear extension	18
3.2	Inner automorphism and the single-block theorem	19
4	Quantum Channels and Positive Maps	21
4.1	Positive and completely positive maps	21
4.2	Kraus representations and complete positivity	22
4.3	Quantum channels	22
4.4	Rectangular positive maps, trace bounds, and density matrices	22
4.5	Trace adjoints and rectangular Kraus maps	23
4.6	The Choi–Jamiołkowski representation of maps between matrix algebras	24
4.7	The Kraus representation theorem for rectangular matrix algebras	27
4.8	Irreducibility	27
4.9	Transfer maps	29
4.10	Peripheral spectrum and primitivity	30
4.11	Fixed-point projection	32
4.12	Peripheral eigenvalues and powering	33
	4.12.1 Periodicity removal by powering	33
	4.12.2 Peripheral closure via adjoint fixed point	34
4.13	Primitivity and the complementary transfer-map gap	35
4.14	Anti-linear conjugation of the spectrum	36
5	Schwarz Inequalities and Multiplicative Domains	37
5.1	Kadison–Schwarz inequality	37
5.2	Multiplicative domain	39
5.3	Trace adjoints and positive retractions	40
5.4	Douglas-type factorization	41

5.5	Peripheral Schwarz equality with a faithful fixed point	41
5.6	Douglas Factorization	42
5.7	Block Schur Complements	42
6	Perron–Frobenius Theory for Channels and Transfer Maps	46
6.1	Cesàro fixed points for quantum channels	46
6.2	The phase-weighted Cesàro formula for the peripheral projection	51
6.3	Positive definiteness	52
6.4	Uniqueness	55
6.5	Existence and the Perron–Frobenius theorem	57
6.6	Right- and left-canonical gauges	57
6.7	Similarity preserves irreducibility	58
6.8	Perron–Frobenius eigenvector existence	59
6.9	Spectral radius at the Perron eigenvalue	60
6.10	General Brouwer and stationary states	61
6.11	Fixed-point decomposition and span	61
7	Peripheral Channel Structure and Transfer-Operator Gaps	63
7.1	Peripheral eigenvalue group structure	64
7.2	Mixed transfer operator	64
7.3	MPV overlap as transfer trace	65
7.4	Eigenvalue bound and transfer-operator gap	65
7.5	Transfer-operator gap — rectangular case	67
7.6	MPV overlap decay	67
7.7	Transfer-operator gap under irreducible-TP hypotheses	68
7.8	Overlap rigidity for normal blocks	69
7.9	Peripheral spectral refinements	70
7.9.1	Cyclic decomposition of irreducible finite Kraus maps	71
7.10	Self-overlap and the period-one criterion	72
7.11	Primitive overlap convergence (complementary transfer-map gap form)	72
8	Wielandt Bound	74
8.1	Cumulative span	74
8.1.1	Paper primitivity and indices	75
8.2	Nonzero trace product	80
8.3	Eigenvector spreading	82
8.4	Quantum Wielandt bounds	83
8.5	Fixed-length matrix spanning	85
8.6	Primitive MPS tensors	87
8.7	Primitivity and normality	87
8.7.1	From primitivity to normality	88
8.7.2	Primitive implies irreducible	88
8.7.3	From irreducibility to full spanning via Burnside’s theorem	88
8.7.4	From primitivity to strong irreducibility and normality	89
8.8	One-step padding and exact word spans	90
8.9	Block injectivity from TP/primitive/irreducible blocks	90

9	Canonical Form Reduction	91
9.1	Invariant-subspace decomposition and irreducible blocks	92
9.1.1	Iterated reduction to irreducible blocks	93
9.2	Nonzero blocks and the trace-preserving gauge	94
9.2.1	Trace-preserving gauge for arbitrary tensors	94
9.3	Blocking, period removal, and primitive blocks	95
9.3.1	Period removal by cyclic sectors	95
9.4	Cyclic-sector irreducibility	97
9.5	Translation-invariant canonical form	97
9.6	Normal tensors and canonical forms	100
9.6.1	Normalization conventions	106
9.7	Projector criterion for canonical decomposition	106
9.8	Fixed points and canonical gauges of irreducible blocks	107
9.9	Normal canonical form	110
9.9.1	Separated one-copy normal canonical forms	111
9.9.2	Canonical form from primitivity	112
9.10	Left-canonical and dual diagonalization of irreducible blocks	112
9.11	Cyclic-sector isometries and primitive reduction	113
9.11.1	Reduction to primitive blocks	114
10	Basis of Normal Tensors	116
10.1	Bases of normal tensors	116
10.2	Normal blocks and overlap rigidity	119
10.3	Block-injective consequences for BNT families	122
10.4	Permutation rigidity for bases of normal tensors	125
10.5	Newton–Girard identities and power-sum recovery	127
10.6	The BNT canonical form on a sector decomposition	127
10.6.1	Sector decompositions and phase matching	128
10.6.2	Single-copy sector decompositions	135
10.6.3	Prepared-block construction of BNT canonical forms	135
10.7	Coefficient comparison and copy-weight recovery	142
10.7.1	Matched-sector weight multiset equality	142
10.8	Strong existential matching by exact linear independence	143
10.8.1	Bijective matching by symmetry	144
11	Proof of the Fundamental Theorem	145
11.1	Coefficient identities and copy weights	145
11.2	The equal-MPV global gauge	148
11.3	The proportional and equal cases	150
11.3.1	Unitary gauges on normalized sectors	151
11.3.2	The equal case	152
12	Symmetries, Physical String Order, and Virtual-Boundary Nondecay	154
12.1	Virtual Local Covariance and Projective Bond Actions	154
12.2	Cohomology Classes of Virtual Symmetry Cocycles	157
12.2.1	Non-triviality via the commutator phase	159
12.3	Physical Endpoint String Order and Virtual-Boundary Nondecay	159
12.4	SPT Phase Labels and Virtual-Boundary Nondecay	166

13 Auxiliary Quantum-Channel Theory	169
13.1 Bipartite systems and the Schmidt decomposition	169
13.2 Quantum steering	170
13.3 Maximal weight in convex decomposition	171
13.4 Complete positivity from positivity	172
13.5 Partial trace over subsystems	173
13.6 Extending completely positive maps from operator systems	174
13.7 Extending completely positive maps from operator systems in a direct sum of matrix algebras	179
13.8 Set spectra and spectral multiplicities	181
13.9 Pure states, projective rays, and Wigner’s theorem	183
13.10 Maximally entangled states and Choi matrices	187
13.11 No information without disturbance	189
13.12 Additional results on multiplicative domains and order	190
13.12.1 Abstract Schwarz maps and their multiplicative domains	190
13.12.2 Kraus specialization and full algebraic structure	192
13.12.3 Positive maps preserve order and spectral intervals	193
13.13 Positive Schwarz maps outside complete positivity	193
13.13.1 A positive Schwarz map that is not completely positive	193
13.13.2 Two-variable operator Schwarz inequality	194
13.14 Exponential positivity for irreducible CP maps	196
13.15 Ergodicity of irreducible channels	197
13.16 A CP spectral characterization of irreducibility	197
 I Appendices	 199
A Schwarz Inequalities and Multiplicative Domains: Supporting Results	200
A.1 Trace duality and elementary order preservation	200
A.2 Positive functionals and rank-one retractions	201
A.3 Faithful weighted traces and peripheral Schwarz equality	203
 B Perron–Frobenius Theory for Channels and Transfer Maps: Supporting Results	 204
B.1 Density matrices, Brouwer’s theorem, and Cesàro limits	204
B.2 Canonical-gauge algebra	205
B.3 Similarity bookkeeping	206
B.4 Auxiliary Perron reductions	207
B.5 Exponential truncation and scalar reformulations	208
 C Peripheral Channel Structure and Transfer-Operator Gaps: Supporting Results	 209
C.1 Mixed-transfer powers and overlap traces	209
C.2 Auxiliary trace positivity	210
C.3 Rectangular spectral-radius formulation	210
C.4 Peripheral intertwiners and gauge rigidity	210
C.5 Spectral-radius decay and overlap limits	211
C.6 Rank-one Perron projection	212
C.7 Periodicity removal	213
C.7.1 Cyclic corners and restricted primitivity	214

C.7.2	Group structure and divisibility	216
D	Wielandt Bound: Supporting Results	217
D.1	Exact-word and block-injectivity support	217
D.2	Cumulative-span and spectral linear algebra	218
D.3	One-step augmentation	220
D.4	Blocking and fixed-length spanning	221
D.5	Complementary-gap consequences	222
D.6	Identity in the one-step span	223
E	Canonical Form Reduction: Supporting Results	225
E.1	Similarities, strict splitting, and zero blocks	225
E.2	Projector closure and isometric corner decompositions	226
E.3	Perron gauges and preservation under conjugation	228
E.4	Blocking identities and primitive stability	229
E.5	Support compression and cyclic-sector families	233
E.6	Prepared normal-form consequences	235
F	Basis of Normal Tensors: Supporting Results	236
F.1	Elementary normal-tensor and Gram-matrix support	236
F.2	Change of basis, power sums, and sector bookkeeping	238
F.2.1	Proportional matching and finite bijection extraction	240
F.3	Direct-sum separation and prepared-family bookkeeping	241
G	Proof of the Fundamental Theorem: Supporting Results	246
G.1	Matched phases and coefficient extraction	246
G.2	Coefficient identities from gauge-phase equalities	247
G.3	The scalar-threaded proportional identity	247
G.4	Direct-sum conjugation	248
G.5	Additional proportional consequences	248
G.6	Matched flattened coordinates	249
G.7	Permutation gauges and literal coordinates	250
G.8	Unitary witness refinements	251
H	Symmetries and Virtual-Boundary Nondecay: Supporting Results	252
H.1	Permutation reindexing and identity bookkeeping	252
H.1.1	Linear physical-index rotations	253
H.2	Gauge ratios and scalar uniqueness	254
H.3	Cocycle equivalence and coboundaries	255
H.4	Stationary-boundary twist support	255
H.5	Kraus mixing, transfer expansions, and a common TP gauge	256
H.6	Boundary and limit support	257
H.7	Routine universality corollary	257

Chapter 1

Introduction

The central result is the Fundamental Theorem for translation-invariant matrix product vectors. The basic object is a family of matrices $A = \{A^i\}_{i=0}^{d-1}$, which assigns to each chain length N the vector $|V^{(N)}(A)\rangle$ with coefficients indexed by $\sigma = (i_1, \dots, i_N)$,

$$V^{(N)}(A)_\sigma = \text{tr}(A^{i_1} \dots A^{i_N}). \quad (1.1)$$

The main argument, for translation-invariant tensors, follows the canonical-form and basis-of-normal-tensors argument of [CPGSV16, CPGSV21], with the single-block injective case anchored in the earlier representation theorem of [PGVWC07].

Notation

$\mathbb{C}, \mathbb{N}, \mathbb{Z}$	complex numbers, natural numbers, integers
$M_D(\mathbb{C}), \text{GL}_D(\mathbb{C})$	complex matrices and invertible complex matrices
$\text{tr}(X), \text{span}_{\mathbb{C}}(S)$	trace and complex linear span
$A = \{A^i\}_{i=0}^{d-1}$	a translation-invariant MPS tensor
$w = (i_1, \dots, i_L), A^w$	a word and the product $A^{i_1} \dots A^{i_L}$
$ V^{(N)}(A)\rangle$	the length- N matrix product vector
$V^{(N)}(A)_\sigma$	the coefficient of $ V^{(N)}(A)\rangle$ at σ
$O_{AB}(N)$	the bilinear overlap $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}$
$\mathcal{E}_A(X)$	the transfer map $\sum_i A^i X (A^i)^\dagger$
$A^{[L]}$	the length- L blocked tensor

In canonical form one writes

$$A^i = \bigoplus_{j=1}^g \mu_j A_j^i, \quad (1.2)$$

$$B^i = \bigoplus_{k=1}^h \nu_k B_k^i. \quad (1.3)$$

The blocks in equations 1.2 and 1.3 are normal tensors. If $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every positive chain length N , then the bases of normal tensors have the same size. After relabelling the blocks of B , each matched pair satisfies

$$B_j^i = \zeta_j X_j A_j^i X_j^{-1}, \quad (1.4)$$

for some invertible matrix X_j and nonzero scalar ζ_j , and the equal case compares the coefficient power sums attached to the relabelled bases. The block gauges in equation 1.4 and the weight comparison then combine into a single gauge conjugacy of the two weighted block-diagonal tensors in equations 1.2 and 1.3. For a single injective block this reduces to the familiar conclusion $B^i = X A^i X^{-1}$ for all physical indices i , the single-block specialization of equation 1.4.

Chapters 2–11 carry out this argument. Chapter 2 fixes the matrix product vector notation, word evaluations, transfer maps, gauge equivalence, injectivity, blocking, and site-periodic tensor families. Chapter 3 proves the single-block theorem by trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether. Chapters 4, 5, 6, and 7 develop the positive-map, multiplicative-domain, Perron–Frobenius, and overlap-decay tools used to control normal blocks. Chapters 8 and 9 supply the blocking, primitivity, irreducibility, and canonical-form reductions. Chapter 9 includes the invariant-projection splitting. Chapter 10 supplies bases of normal tensors and the power-sum comparison.

Chapter 12 uses the Fundamental Theorem to obtain virtual symmetry gauges, projective bond representations, and string-order criteria; its Kraus-mixing step appeals only to the later channel-representation chapter, not to any additional input for the Fundamental Theorem.

Beyond Chapter 12, the later chapters develop parent-Hamiltonian consequences, correlation estimates, examples, channel representations, normal forms, quantum dynamical semigroups, operator convexity, and entropy. They also include the standard examples of positive maps that fail to be completely positive. The final three chapters introduce matrix product operators and density operators, develop the renormalization fixed points of pure matrix product states, and then treat the zero-correlation-length and fixed-point conditions for mixed states.

Chapter 2

Matrix Product Vectors

Fix a physical dimension d and a bond dimension D . This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors here are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1.1 (MPS tensor). [Lean](#) [✓ Stmt](#) A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices $\{A^i\}_{i=0}^{d-1}$, where $A^i \in M_D(\mathbb{C})$, indexed by a physical index $i \in \{0, \dots, d-1\}$. Such a tensor defines an MPV family. Diagrammatically,

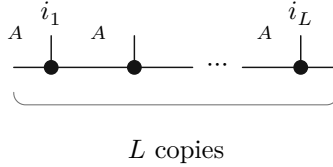


The black node denotes the tensor A , the horizontal legs are virtual, and the upper leg is the physical index i .

Definition 2.1.2 (Word evaluation). [Lean](#) [✓ Stmt](#) Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \dots A^{i_L} \in M_D(\mathbb{C}). \quad (2.1)$$

The empty word evaluates to the identity, $A^\emptyset = \mathbb{1}_D$. In tensor-network notation,



in which each black node denotes the same local tensor A , the virtual legs remain open, and the physical legs are labelled by the word $(i_1, \dots, i_L)$.

Lemma 2.1.3 (Word evaluation along a listed configuration). [Lean](#) [✓ Stmt](#) For a configuration $\sigma = (\sigma_0, \dots, \sigma_{N-1})$, let $w_\sigma = [\sigma_0, \dots, \sigma_{N-1}]$. Then $A^{w_\sigma} = A^{\sigma_0} A^{\sigma_1} \dots A^{\sigma_{N-1}}$.

Proof. ✓ Proof This follows by induction on N from the recursive definition of word evaluation. $\square$

Lemma 2.1.4 (Multiplicativity of word evaluation). Lean ✓ Stmt For any words w_1, w_2 , $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$.

Proof. ✓ Proof The claim follows from the definition of A^w . $\square$

Lemma 2.1.5 (Centre of $M_D(\mathbb{C})$). Lean ✓ Stmt If a matrix $Z \in M_D(\mathbb{C})$ commutes with every element of $M_D(\mathbb{C})$, then there exists $\lambda \in \mathbb{C}$ such that $Z = \lambda \mathbb{1}_D$.

Proof. ✓ Proof Since Z commutes with every matrix unit, all off-diagonal entries of Z vanish and all diagonal entries are equal. $\square$

Lemma 2.1.6 (Rectangular amplified commutant). Lean ✓ Stmt Let $(A_i)_{i \in I}$ span $M_D(\mathbb{C})$, and let $C : \mathbb{C}^{m_2} \otimes \mathbb{C}^D \rightarrow \mathbb{C}^{m_1} \otimes \mathbb{C}^D$ satisfy, for every $i \in I$,

$$(\mathbb{1}_{m_1} \otimes A_i)C = C(\mathbb{1}_{m_2} \otimes A_i).$$

Then there is a matrix $F : \mathbb{C}^{m_2} \rightarrow \mathbb{C}^{m_1}$ such that $C = F \otimes \mathbb{1}_D$.

Proof. ✓ Proof Each (a, b) multiplicity block of C commutes with every A_i . Since the A_i span the full matrix algebra, Lemma 2.1.5 makes this block a scalar multiple of the identity. These scalars are the entries of F . $\square$

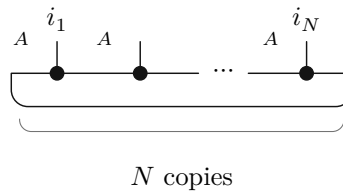
Definition 2.1.7 (Matrix product vector). Lean ✓ Stmt The *matrix product vector* (MPV) of a tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1, \dots, i_N=0}^{d-1} \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}. \quad (2.2)$$

Equivalently, for a configuration $\sigma = (i_1, \dots, i_N) \in \{0, \dots, d-1\}^N$, we write

$$V^{(N)}(A)_\sigma := \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}). \quad (2.3)$$

The coefficient function $\sigma \mapsto V^{(N)}(A)_\sigma$ gives the components of the vector in (2.2); the displayed ket is the corresponding vector in $(\mathbb{C}^d)^{\otimes N}$. For a general word w , we also write $c_w(A) := \text{tr}(A^w)$. The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$. The coefficient $V^{(N)}(A)_\sigma$ is the periodic contraction

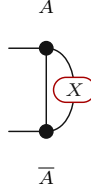


of N copies of the local tensor A , with the outer virtual legs closed by the trace.

Definition 2.1.8 (Transfer map). Lean ✓ Stmt The *transfer map* associated to a tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger. \quad (2.4)$$

Diagrammatically, the transfer map is the double-layer contraction



in which the upper node denotes A , the lower node denotes $A^\dagger$, and the physical index is summed over between them.

Remark 2.1.9. ✓ Stmt The transfer map is automatically positive: if $X \geq 0$ then $\mathcal{E}_A(X) \geq 0$, because each summand $A^i X (A^i)^\dagger$ is positive semidefinite. The abstract positive-map framework appears in Chapter 4.

2.2 Site-periodic tensor families

Definition 2.2.1 (Site-periodic tensor family). Lean ✓ Stmt For a fixed period m , a *site-periodic tensor family* is a family of local tensors indexed by $\{0, \dots, m-1\}$, with local physical dimension d and bond dimension D .

Definition 2.2.2 (Periodic same-state and gauge-equivalence relations). Lean Lean ✓ Stmt For site-periodic tensors A, B at fixed period m :

- equality of the induced site-periodic states;
- cyclic gauge equivalence.

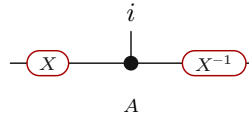
Lemma 2.2.3 (Equivalence structures for periodic relations). Lean Lean ✓ Stmt Both periodic relations are equivalence relations (reflexive, symmetric, transitive).

2.3 Gauge equivalence and same MPV

Definition 2.3.1 (Gauge equivalence). Lean ✓ Stmt Two tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that, for every $i \in \{0, \dots, d-1\}$,

$$B^i = X A^i X^{-1}. \quad (2.5)$$

In tensor-network notation, (2.5) is represented by



where the black node denotes the tensor A , the upper leg is the physical index i , and the two red side nodes denote the gauge matrices acting on the virtual legs.

Definition 2.3.2 (Same MPV). Lean ✓ Stmt Two tensors A and B (of the same bond dimension) generate the same MPV family if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size N .

Definition 2.3.3 (Same MPV - different bond dimensions). [Lean](#) [✓ Stmt](#) For tensors A and B of possibly different bond dimensions D_1 and D_2 , we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every word length N , including the empty word. This algebraic relation is used for exact direct-sum identities.

Definition 2.3.4 (Positive-length same MPV - different bond dimensions). [Lean](#) [✓ Stmt](#) For tensors A and B of possibly different bond dimensions, we write $\mathcal{V}^+(A) = \mathcal{V}^+(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every $N > 0$.

Remark 2.3.5. [✓ Stmt](#) Definition 2.3.2 is the same-bond-dimension specialization of Definition 2.3.3. We keep both because later results use both the same-dimension and different-dimension forms.

Definition 2.3.6 (Proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors A and B (of possibly different bond dimensions) generate weakly proportional MPV families if for every $N > 0$ there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$.

Definition 2.3.7 (Nonzero proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors A and B (of possibly different bond dimensions) generate nonzero proportional MPV families if for every $N > 0$ there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$. This is the projective reading of the proportionality hypothesis in [CPGSV16, Theorem 2.10].

Definition 2.3.8 (Eventually nonzero proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors A and B are eventually nonzero proportional if, for all sufficiently large N , there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$. This auxiliary form is used with the eventual linear-independence statement Lemma F.1.10.

Lemma 2.3.9 (Elementary nonzero proportionality rules). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Nonzero proportionality implies weak proportionality and eventual nonzero proportionality, is symmetric, and equality of MPV families is the special case with scalar 1, at every positive length.*

Proof. [✓ Proof](#) Forgetting the nonvanishing condition gives weak proportionality. The eventual form follows by viewing an all-length statement as an eventual one. Symmetry follows by inverting the scalar c_N at each length, and equality is the case $c_N = 1$. $\square$

Lemma 2.3.10 (Coefficient isolation after complementary matching). [Lean](#) [✓ Stmt](#) *Let R be a ring, let V be an R -module, and let I and K be finite index sets. Let $T \subseteq I$, and let $\{u_i\}_{i \in I}$ and $\{v_k\}_{k \in K}$ be vectors in V . Suppose that, for every $i \notin T$, there are a scalar α_i and an index $\phi(i) \in K$ such that $u_i = \alpha_i v_{\phi(i)}$. If*

$$\sum_{i \in I} a_i u_i = \sum_{k \in K} b_k v_k$$

and the coproduct-indexed family $w: T \amalg K \rightarrow V$, defined by $w(\text{inl}(i)) = u_i$ and $w(\text{inr}(k)) = v_k$, is linearly independent, then $a_i = 0$ for every $i \in T$.

Proof. [✓ Proof](#) Substitute $u_i = \alpha_i v_{\phi(i)}$ for $i \notin T$ and collect the resulting coefficients over the fibres of ϕ . The given equality is then a linear relation in the combined family. Linear independence makes every coefficient vanish, in particular each a_i with $i \in T$. $\square$

Lemma 2.3.11 (Scaling of word evaluation). [Lean](#) [✓ Stmt](#) *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then, for any word w , $(\zeta A)^w = \zeta^{|w|} A^w$, where $|w|$ is the length of w .*

Proof. ✓ **Proof** Induction on w : the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ . □

Lemma 2.3.12 (Scaling of MPV components). Lean ✓ **Stmt** Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then, for every N and every configuration $\sigma = (i_1, \dots, i_N)$, $V^{(N)}(\zeta A)_\sigma = \zeta^N V^{(N)}(A)_\sigma$.

Proof. ✓ **Proof** Apply Lemma 2.3.11 to the word $w = (i_1, \dots, i_N)$ of length N , then take the trace; the scalar ζ^N pulls out by linearity. □

Definition 2.3.13 (Gauge-phase equivalence). Lean Lean ✓ **Stmt** Two tensors A and B are *gauge-phase equivalent* if there exist $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that, for every physical index i ,

$$B^i = \zeta X A^i X^{-1}. \quad (2.6)$$

Proof. ✓ **Proof** Gauge equivalence is the special case $\zeta = 1$. □

Definition 2.3.14 (Unit gauge-phase equivalence). Lean ✓ **Stmt** Two tensors are *unit gauge-phase equivalent* when the gauge-phase equation holds with $|\zeta| = 1$.

Lemma 2.3.15 (Unit gauge phase implies gauge phase). Lean ✓ **Stmt** Unit gauge-phase equivalence implies gauge-phase equivalence.

Proof. ✓ **Proof** A unit-modulus scalar is nonzero. □

Lemma 2.3.16 (Gauge-phase equivalence across equal bond dimensions). Lean Lean ✓ **Stmt** Gauge-phase equivalence is symmetric after identifying equal bond dimensions. Moreover, suppose that A , B , and C have identified bond dimensions, that X_1 and X_2 are invertible, that $\zeta_1, \zeta_2 \neq 0$, and that, for every physical index i ,

$$\begin{aligned} B^i &= \zeta_1 X_1 A^i X_1^{-1}, \\ C^i &= \zeta_2 X_2 A^i X_2^{-1}. \end{aligned}$$

Then

$$C^i = (\zeta_2 \zeta_1^{-1}) (X_2 X_1^{-1}) B^i (X_1 X_2^{-1}), \quad (2.7)$$

so B and C are gauge-phase equivalent after the corresponding identification of their bond dimensions.

Proof. ✓ **Proof** Symmetry follows by replacing (X, ζ) with (X^{-1}, ζ^{-1}) . Formula (2.7) follows by substituting the first gauge formula into the second. □

Lemma 2.3.17 (Gauge covariance of word evaluation). Lean ✓ **Stmt** If $B^i = X A^i X^{-1}$ for all i , then $B^w = X A^w X^{-1}$ for every word w .

Proof. ✓ **Proof** Induction on the word length, using $XX^{-1} = \mathbb{1}$. □

Theorem 2.3.18 (Gauge invariance of MPVs). Lean ✓ **Stmt** If A and B are gauge equivalent, then they generate the same MPV family.

Proof. ✓ **Proof** Let $B^i = X A^i X^{-1}$. By Lemma 2.3.17, $B^w = X A^w X^{-1}$ for every word w . Then $\text{tr}(B^w) = \text{tr}(X A^w X^{-1}) = \text{tr}(A^w)$ by cyclicity of the trace. □

Theorem 2.3.19 (Gauge-phase scaling of MPVs). Lean ✓ **Stmt** If $B^i = \zeta X A^i X^{-1}$ for every physical index i , then for every system size N and configuration σ one has $V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma$.

Proof. ✓ **Proof** Combine gauge covariance of word evaluation with scalar scaling, then take traces. □

2.4 Injectivity and normality

Definition 2.4.1 (Injectivity). [Lean](#) [✓ Stmt](#) A tensor A is *injective* if its matrices span the full matrix algebra: $\text{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

The inverse-tensor characterization of injectivity says that A is injective exactly when there is a tensor A^{-1} such that

$$\sum_i (A^i)_{\alpha,\beta} (A^{-1,i})_{\alpha',\beta'} = \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}. \quad (2.8)$$

Thus (2.8) identifies the two pairs of virtual indices when A^{-1} is contracted with A through their common physical index. In shorthand, $A^{-1}A = \mathbb{1}$. This is the identity used in [CPGSV16, Section 2.3, lines 324–331].

Definition 2.4.2 (L -block injectivity). [Lean](#) [✓ Stmt](#) A tensor A is *L -block injective* if

$$\text{span}_{\mathbb{C}}\{A^{i_1} \dots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C}). \quad (2.9)$$

An injective tensor is 1-block injective.

Lemma 2.4.3 (One-site injectivity gives one-block injectivity). [Lean](#) [✓ Stmt](#) If the one-site matrices $\{A^i\}_{i=0}^{d-1}$ span $M_D(\mathbb{C})$, then the length-one word products span $M_D(\mathbb{C})$, so A is 1-block injective.

Proof. [✓ Proof](#) The length-one words are exactly the letters $i \in \{0, \dots, d-1\}$, and evaluating a length-one word gives the matrix A^i . Thus the length-one word span is the same span as in one-site injectivity. $\square$

Definition 2.4.4 (Normal tensor). [Lean](#) [✓ Stmt](#) A tensor A is *normal* [CPGSV21] if there exists a positive integer L_0 such that A is L_0 -block injective.

Lemma 2.4.5 (Gauge invariance of block injectivity). [Lean](#) [✓ Stmt](#) If A and B are gauge equivalent and A is L -block injective, then B is L -block injective.

Proof. [✓ Proof](#) Conjugation by an invertible matrix maps the span of the length- L words of A onto the span of the length- L words of B . Since conjugation is an automorphism of the full matrix algebra, fullness of the first span implies fullness of the second. $\square$

Theorem 2.4.6 (Gauge transport of normality). [Lean](#) [✓ Stmt](#) If A and B are gauge equivalent and A is normal, then B is normal.

Proof. [✓ Proof](#) Apply Lemma 2.4.5 at a blocking length for which A is block-injective. $\square$

Remark 2.4.7. [✓ Stmt](#) The algebraic characterization of normality in Definition 2.4.4 — eventual full matrix span, equivalently eventual block injectivity — is equivalent to the spectral characterization used in [CPGSV21, Definition IV.1]: after TP normalization, the transfer map $\mathcal{E}_A$ is a primitive channel, equivalently irreducible with trivial peripheral spectrum. This equivalence is proved in [SPGWC10, Proposition 3]. In what follows, the directions of the equivalence are supplied by Lemma 8.7.2.1, Theorem 8.7.4.2 for primitive normalized tensors with positive-definite fixed point, the Burnside route through Theorem 8.8.2 for the aperiodic algebraic criterion, and Theorem 9.9.2.1. In particular, every subsequent use of “normal” here refers to this single notion; Definition 9.9.2 employs the spectral formulation, but it describes the same class of tensors.

2.5 Canonical form

Definition 2.5.1 (Block-injective canonical form). [Lean](#) [✓ Stmt](#) Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,
4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i. \quad (2.10)$$

This block-injective formulation of the canonical form is used in [\[PGVWC07\]](#); see also [\[CPGSV21\]](#) for the normal canonical form.

Remark 2.5.2. [✓ Stmt](#) Here we use the block-injective canonical form: each block is injective in the sense of Definition 2.4.1. This is stronger than the NT-block canonical form of [\[CPGSV21\]](#), where the blocks are only normal in the spectral sense. Chapter 9 treats the normal canonical form needed for the blocking-based multi-block theory, while the periodic irreducible-form extension is kept separate. This section isolates the stronger block-injective form used in the injective and block-injective arguments.

Definition 2.5.3 (Block projection). [Lean](#) [✓ Stmt](#) For a finite direct sum $\bigoplus_k \mathbb{C}^{D_k}$, the block projection P_j is the matrix that is the identity on the j -th summand and zero on all other summands.

Definition 2.5.4 (Block-diagonal matrix). [Lean](#) [✓ Stmt](#) A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if it is a direct sum of square matrices, one on each summand.

Lemma 2.5.5 (Off-block characterization). [Lean](#) [✓ Stmt](#) A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if and only if every entry from one summand to a distinct summand is zero.

Proof. [✓ Proof](#) The forward direction follows from the definition of a direct sum of matrices. Conversely, if all off-block entries vanish, the diagonal blocks reconstruct the original matrix. $\square$

Lemma 2.5.6 (Commutation with block projections). [Lean](#) [✓ Stmt](#) If a matrix on $\bigoplus_k \mathbb{C}^{D_k}$ commutes with every block projection P_j , then it is block diagonal.

Proof. [✓ Proof](#) Compare the (i, j) block of the identity $XP_j = P_jX$ for $i \neq j$. The right multiplication by P_j keeps that block, while the left multiplication by P_j kills it. Hence every off-block entry is zero, and Lemma 2.5.5 applies. $\square$

Definition 2.5.7 (Block-diagonal tensor from blocks). [Lean](#) [✓ Stmt](#) Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the associated *block-diagonal tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

Lemma 2.5.8 (Word evaluation of a block-diagonal tensor). [Lean](#) [✓ Stmt](#) For a word w , the word evaluation of a block-diagonal tensor remains block diagonal:

$$A^w = \bigoplus_k \mu_k^{|w|} A_k^w. \quad (2.11)$$

Proof. [✓ Proof](#) Induct on the word and multiply block-diagonal matrices block by block; every letter contributes one scalar factor μ_k on the k -th block. $\square$

Theorem 2.5.9 (MPV decomposition). [Lean](#) [✓ Stmt](#) The MPV of a block-diagonal tensor decomposes as

$$|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(A_k)\rangle. \quad (2.12)$$

Equivalently, for each configuration σ one has

$$V^{(N)}(A)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(A_k)_\sigma.$$

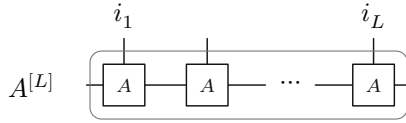
Proof. [✓ Proof](#) By Lemma 2.5.8, the full word evaluation is block diagonal with blocks $\mu_k^N A_k^w$ for a word w of length N . Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\text{tr}(A^w) = \sum_k \mu_k^N \text{tr}(A_k^w)$. $\square$

2.6 Blocking

Definition 2.6.1 (Blocked tensor). [Lean](#) [✓ Stmt](#) Given a tensor A and a blocking length L , the L -blocked tensor is the tensor with physical index set $\{0, \dots, d-1\}^L$ (hence physical dimension d^L) and the same bond dimension D , whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1, \dots, i_L)} = A^{i_1} A^{i_2} \dots A^{i_L}. \quad (2.13)$$

Blocking coarse-grains L neighbouring sites into one tensor: the inner virtual bonds are contracted, the L physical legs merge into a single composite index, and the two outer bonds remain as the bond of $A^{[L]}$.



Lemma 2.6.2 (Blocked word evaluation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $w = (J_1, \dots, J_m)$ is a word in the blocked alphabet, where each J_k is an L -tuple in $\{0, \dots, d-1\}^L$, then concatenating the block words gives a word $\tilde{w}$ of length mL in the original alphabet and $(A^{[L]})^w = A^{\tilde{w}}$.

Proof. [✓ Proof](#) Induct on the blocked word w and split off the first block word. Lemma 2.1.4 turns concatenation of blocks into multiplication of the corresponding evaluations. $\square$

Lemma 2.6.3 (Blocking preserves MPV coefficients). [Lean](#) [✓ Stmt](#) For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system, where each σ_k is an L -tuple in $\{0, \dots, d-1\}^L$, there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that $V^{(N)}(A^{[L]})_\sigma = V^{(NL)}(A)_{\tilde{\sigma}}$.

Proof. ✓ Proof Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.6.2). $\square$

Theorem 2.6.4 (Blocking preserves same MPV). Lean ✓ Stmt *If A and B generate the same MPV family, then for any blocking length L , the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.*

Proof. ✓ Proof By Lemma 2.6.3, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the corresponding concatenated configuration. The same holds for $B^{[L]}$ (with the same concatenated configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. $\square$

2.7 MPV overlap

Definition 2.7.1 (MPV as Hilbert-space vector). Lean ✓ Stmt We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \dots, d-1\}^N$, with σ -component $V^{(N)}(A)_\sigma$ as in Definition 2.1.7. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.7.2 (MPV inner product). Lean ✓ Stmt The *MPV inner product* between two tensors A and B at system size N is

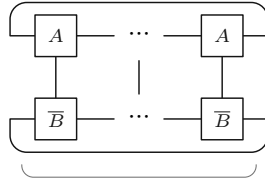
$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle := \sum_{\sigma \in \{0, \dots, d-1\}^N} \overline{V^{(N)}(A)_\sigma} V^{(N)}(B)_\sigma, \quad (2.14)$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.7.3 (MPV overlap). Lean ✓ Stmt The *MPV overlap* between two tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) := \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}. \quad (2.15)$$

Diagrammatically, the overlap contracts the MPV ring of A against the conjugate ring of B along their shared physical indices, both virtual rings closed by the trace:



N sites

This pairing is linear in the first MPV coefficient and conjugate-linear in the second. By Lemma 2.7.4, it is the complex conjugate of the Hilbert-space inner product from Definition 2.7.2.

Lemma 2.7.4 (Overlap equals conjugate inner product). Lean ✓ Stmt $O_{AB}(N) = \overline{\langle V^{(N)}(A) | V^{(N)}(B) \rangle}$.

Proof. ✓ Proof The overlap sums $V_\sigma \overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma} W_\sigma$. Complex conjugation swaps these. $\square$

Lemma 2.7.5 (Overlap determined by positive-length MPV). Lean ✓ Stmt If $V^{(N)}(A)_\sigma = V^{(N)}(A')_\sigma$ and $V^{(N)}(B)_\sigma = V^{(N)}(B')_\sigma$ for all $N > 0$ and all σ , then $O_{AB}(N) = O_{A'B'}(N)$ for all positive N .

Proof. ✓ Proof Direct pointwise rewriting: if the MPV coefficients agree for all positive chain lengths, the overlap sums are identical. □

Chapter 3

The Single-Block Fundamental Theorem

An injective MPS tensor that generates the same matrix-product vectors as a second tensor of equal bond dimension is conjugate to it [CPGSV21, Corollary IV.5]: there is an invertible X with $B^i = XA^iX^{-1}$ for every physical index i . The argument is purely algebraic. Equality of the matrix-product vectors gives $\text{tr}(A^w) = \text{tr}(B^w)$ for every word w ; the length-two and length-three traces produce a linear map T with $T(A^i) = B^i$ that is multiplicative and nonzero; a nonzero multiplicative endomorphism of the simple algebra $M_D(\mathbb{C})$ is a unital automorphism; and by Skolem–Noether every automorphism of $M_D(\mathbb{C})$ is conjugation by an invertible X . Evaluating at A^i closes the proof. The same conjugation, one normal block at a time, is the central step of the multi-block theorem in Chapter 11.

3.1 The multiplicative linear extension

Nondegeneracy of the trace pairing on $M_D(\mathbb{C})$ is a generic algebra fact, proved as Theorem A.1.3 in Appendix A.

Lemma 3.1.1 (Same matrix-product vectors give trace agreement). Lean ✓ Stmt *If A and B generate the same matrix-product vectors, then $\text{tr}(A^w) = \text{tr}(B^w)$ for every word w .*

Proof. ✓ Proof By (2.3), the configuration $\sigma = w$ gives $V^{(|w|)}(A)_w = \text{tr}(A^w)$. Equality of the matrix-product vectors gives $V^{(N)}(A) = V^{(N)}(B)$ for all N , so $\text{tr}(A^w) = \text{tr}(B^w)$. $\square$

The trace conditions constrain every product of the matrices A^i . To bring the algebra structure of $M_D(\mathbb{C})$ to bear, record how a matrix pairs against the family $\{A^i\}$ under the trace.

Definition 3.1.2 (Trace pairing map). Lean ✓ Stmt For a tensor A , the trace pairing map $\Phi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ is defined by

$$\Phi_A(M)_i = \text{tr}(MA^i). \quad (3.1)$$

Theorem 3.1.3 (The trace pairing map is injective for injective tensors). Lean Lean ✓ Stmt *If A is injective, then $\ker \Phi_A = \{0\}$.*

Proof. ✓ Proof If $\Phi_A(M) = 0$, then (3.1) gives $\text{tr}(MA^i) = 0$ for every i . The $\{A^i\}$ span $M_D(\mathbb{C})$, so $\text{tr}(MN) = 0$ for all N , and $M = 0$ by Theorem A.1.3. $\square$

Lemma 3.1.4 (Injectivity transfers across a range inclusion). Lean ✓ Stmt Let $\Phi, \Psi : V \rightarrow W$ be $\mathbb{C}$ -linear maps with V finite-dimensional. If $\ker \Phi = \{0\}$ and $\text{range } \Phi \subseteq \text{range } \Psi$, then $\ker \Psi = \{0\}$.

Proof. ✓ Proof By rank-nullity, $\ker \Phi = \{0\}$ gives $\dim \text{range } \Phi = \dim V$, and the range inclusion gives

$$\dim V = \dim \text{range } \Phi \leq \dim \text{range } \Psi \leq \dim V.$$

so $\dim \text{range } \Psi = \dim V$ and $\ker \Psi = \{0\}$. □

Theorem 3.1.5 (The linear extension exists and is unique). Lean ✓ Stmt If A is injective and A, B generate the same matrix-product vectors, then there is a unique linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ with $T(A^i) = B^i$ for all i .

Proof. ✓ Proof Length-two trace agreement gives $\Phi_A(A^i) = \Phi_B(B^i)$ for every i . The $\{A^i\}$ span $M_D(\mathbb{C})$, so $\text{range } \Phi_A \subseteq \text{range } \Phi_B$. With $\ker \Phi_A = \{0\}$ (Theorem 3.1.3), Lemma 3.1.4 gives $\ker \Phi_B = \{0\}$. Let g be a left inverse of Φ_B and set $T = g \circ \Phi_A$; then

$$T(A^i) = g(\Phi_A(A^i)) = g(\Phi_B(B^i)) = B^i.$$

Uniqueness holds because the $\{A^i\}$ span $M_D(\mathbb{C})$. □

Theorem 3.1.6 (The linear extension is multiplicative). Lean ✓ Stmt Under the hypotheses of Theorem 3.1.5, the map T satisfies $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.

Proof. ✓ Proof Length-two trace agreement gives $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$; since the $\{A^i\}$ span $M_D(\mathbb{C})$, this extends to $\Phi_B \circ T = \Phi_A$, so $\text{range } \Phi_A = \text{range}(\Phi_B \circ T) \subseteq \text{range } \Phi_B$. With $\ker \Phi_A = \{0\}$ (Theorem 3.1.3), Lemma 3.1.4 makes Φ_B injective. Length-three trace agreement gives, for all i, j , and k ,

$$\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k.$$

Hence injectivity of Φ_B yields $T(A^i A^j) = B^i B^j$. Extend in two stages: for fixed j , the maps $M \mapsto T(MA^j)$ and $M \mapsto T(M)B^j$ agree on the spanning $\{A^i\}$, hence $T(MA^j) = T(M)B^j$ for all M ; then for fixed M , the maps $N \mapsto T(MN)$ and $N \mapsto T(M)T(N)$ agree on the spanning $\{A^j\}$, hence $T(MN) = T(M)T(N)$ for all N . □

3.2 Inner automorphism and the single-block theorem

The rigidity of the full matrix algebra now finishes the proof. A nonzero multiplicative map on $M_D(\mathbb{C})$ is forced to be bijective (simplicity), unit-preserving (surjectivity), and conjugation by a single invertible matrix (Skolem–Noether). The next four results establish these steps and the nonvanishing of T , which the final theorem combines into the gauge X .

Theorem 3.2.1 (A nonzero multiplicative endomorphism of $M_D(\mathbb{C})$ is bijective). Lean ✓ Stmt A nonzero multiplicative $\mathbb{C}$ -linear endomorphism of $M_D(\mathbb{C})$ is bijective.

Proof. ✓ Proof The kernel of a multiplicative linear map is a two-sided ideal. The algebra $M_D(\mathbb{C})$ is simple, so the kernel is $\{0\}$ or all of $M_D(\mathbb{C})$; the latter forces the map to be zero. Hence the map is injective, and surjective by finite-dimensionality. □

Lemma 3.2.2 (A surjective multiplicative map preserves the unit). [Lean](#) [✓ Stmt](#) If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a surjective multiplicative $\mathbb{C}$ -linear map, then $T(\mathbb{1}) = \mathbb{1}$, so T is a $\mathbb{C}$ -algebra homomorphism.

Proof. [✓ Proof](#) Choose X with $T(X) = \mathbb{1}$. Then

$$T(\mathbb{1}) = T(X)T(\mathbb{1}) = T(X\mathbb{1}) = T(X) = \mathbb{1}.$$

□

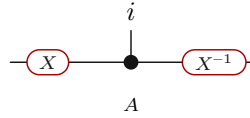
Lemma 3.2.3 (An injective tensor forces a nonzero partner). [Lean](#) [✓ Stmt](#) Let $D \geq 1$. If A is injective and A, B generate the same matrix-product vectors, then the B^i are not all zero.

Proof. [✓ Proof](#) If $B^i = 0$ for every i , then Lemma 3.1.1 on length-one words gives $\text{tr}(A^i) = 0$ for all i . The $\{A^i\}$ span $M_D(\mathbb{C})$, so the trace functional vanishes identically, contradicting $\text{tr}(\mathbb{1}) = D \neq 0$. □

Theorem 3.2.4 (Skolem–Noether). [Lean](#) [✓ Stmt](#) Every $\mathbb{C}$ -algebra automorphism f of $M_D(\mathbb{C})$ is inner: there is $X \in \text{GL}_D(\mathbb{C})$ with $f(M) = XMX^{-1}$ for all M .

Proof. [✓ Proof](#) Under the identification $M_D(\mathbb{C}) \cong \text{End}_{\mathbb{C}}(\mathbb{C}^D)$, every algebra automorphism of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$ is conjugation by an invertible linear map. Translating back gives the invertible X . □

Theorem 3.2.5 (Single-block Fundamental Theorem). [Lean](#) [✓ Stmt](#) Let A be an injective MPS tensor and B a tensor of the same bond dimension. If A and B generate the same matrix-product vectors, then there is $X \in \text{GL}_D(\mathbb{C})$ with $B^i = XA^iX^{-1}$ for all i . In tensor-network notation the gauge conjugates the local tensor on its virtual legs, with the physical index i on the upper leg:



Proof. [✓ Proof](#) Theorem 3.1.5 gives the unique linear T with $T(A^i) = B^i$, and Theorem 3.1.6 makes it multiplicative. It is nonzero: otherwise $B^i = 0$ for all i , against Lemma 3.2.3. By Theorem 3.2.1, T is bijective, and by Lemma 3.2.2 it fixes $\mathbb{1}$, so T is a $\mathbb{C}$ -algebra automorphism. Theorem 3.2.4 gives $X \in \text{GL}_D(\mathbb{C})$ with $T(M) = XMX^{-1}$; evaluating at $M = A^i$ gives $B^i = XA^iX^{-1}$. □

Chapter 4

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

4.1 Positive and completely positive maps

Definition 4.1.1 (Positive map). Lean ✓ Stmt A linear map $E : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite.

Definition 4.1.2 (Trace-preserving map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *trace-preserving* if $\text{tr}(E(X)) = \text{tr}(X)$ for all X .

Remark 4.1.3. ✓ Stmt We use the Loewner order on matrices: $X \leq Y$ means $Y - X \geq 0$.

Theorem 4.1.4 (Finite frame criterion). Lean ✓ Stmt Let $(v_i)_{i \in I}$ be a finite family in $\mathbb{C}^n$. Then

$$\sum_{i \in I} |v_i\rangle\langle v_i| > 0 \iff \text{span}\{v_i : i \in I\} = \mathbb{C}^n.$$

Proof. ✓ Proof Let $B : \mathbb{C}^I \rightarrow \mathbb{C}^n$ be the synthesis map with columns v_i . The frame operator is $BB^\dagger$, while the vectors span exactly when B is surjective. A right inverse of B makes $B^\dagger$ injective, which gives positive definiteness of $BB^\dagger$. Conversely, if $BB^\dagger$ is positive definite, then $B^\dagger(BB^\dagger)^{-1}$ is a right inverse of B . $\square$

Definition 4.1.5 (Completely positive map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that, for every $X \in M_D(\mathbb{C})$,

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger. \tag{4.1}$$

The Kraus representation also gives entrywise positivity on every positive block matrix by Theorem 4.2.1, and hence the associated completely positive map between matrix C^* -algebras in Theorem 4.2.2.

4.2 Kraus representations and complete positivity

This section proves the two complete-positivity consequences stated after Definition 4.1.5: entrywise positivity on block matrices and the associated completely positive map between matrix C^* -algebras.

Theorem 4.2.1 (Entrywise complete positivity from a Kraus representation). [Lean](#) [Stmnt](#) *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be completely positive in the Kraus sense. For every k and every positive block matrix $M \in M_k(M_D(\mathbb{C}))$, the entrywise image $(E(M_{ab}))_{a,b}$ is again positive in $M_k(M_D(\mathbb{C}))$.*

Proof. [Proof](#) Write E in the Kraus form (4.1). For each i , let d_i be the block-diagonal matrix with K_i on every diagonal block. Then

$$(E(M_{ab}))_{a,b} = \sum_i d_i M d_i^\dagger.$$

Each term on the right is positive because conjugation preserves positivity, and a finite sum of positive matrices is positive. $\square$

Theorem 4.2.2 (Kraus maps as abstract completely positive maps). [Lean](#) [Stmnt](#) *Every Kraus-represented completely positive map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ determines a completely positive map between the matrix C^* -algebras in the entrywise positivity sense.*

Proof. [Proof](#) The defining entrywise positivity condition is exactly Theorem 4.2.1. $\square$

Thus the Kraus formulation of Definition 4.1.5 has the two consequences cited there: entrywise complete positivity and the associated abstract completely positive map.

Theorem 4.2.3 (CP maps are positive). [Lean](#) [Stmnt](#) *Every completely positive map is positive.*

Proof. [Proof](#) By the Kraus representation (4.1), if $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ because conjugation preserves positive semidefiniteness. Hence their sum is positive semidefinite. $\square$

4.3 Quantum channels

Definition 4.3.1 (Quantum channel). [Lean](#) [Stmnt](#) A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

Theorem 4.3.2 (Channels are positive). [Lean](#) [Stmnt](#) *Every quantum channel is positive.*

Proof. [Proof](#) A quantum channel is completely positive by definition, so Theorem 4.2.3 applies. $\square$

4.4 Rectangular positive maps, trace bounds, and density matrices

Definition 4.4.1 (Associated positive linear map). [Lean](#) [Stmnt](#) A positive matrix map $E : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ determines the same linear map regarded as a positive linear map: if $A, B \in M_n(\mathbb{C})$ and $A \leq B$, then $E(A) \leq E(B)$ in $M_m(\mathbb{C})$.

Definition 4.4.2 (Positive map between matrix algebras). A linear map $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ between matrix algebras of possibly different dimensions is *positive* if $T(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite. For $D' = D$ this is the notion of Definition 4.1.1.

Definition 4.4.3 (Trace-preserving map between matrix algebras). A linear map $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ between matrix algebras of possibly different dimensions is *trace-preserving* if $\text{tr}(T(X)) = \text{tr}(X)$ for all X . For $D' = D$ this is the notion of Definition 4.1.2.

Theorem 4.4.4 (The trace bounds a positive matrix). Lean ✓ Stmt Let $D \geq 1$ and let $P \in M_D(\mathbb{C})$ be positive semidefinite. Then

$$P \preceq \text{tr}(P)\mathbb{1}. \quad (4.2)$$

Proof. ✓ Proof Diagonalize P with non-negative eigenvalues $\lambda_1, \dots, \lambda_D$. Each is at most their sum $\text{tr}(P)$, so every eigenvalue of $\text{tr}(P)\mathbb{1} - P$ is non-negative. $\square$

Theorem 4.4.5 (Positive maps preserve Hermiticity). Lean ✓ Stmt If $E : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ is positive and $X \in M_n(\mathbb{C})$ satisfies $X = X^\dagger$, then $E(X) = E(X)^\dagger$ in $M_m(\mathbb{C})$.

Proof. ✓ Proof Regard E as the associated positive linear map (Definition 4.4.1). For positive linear maps between matrix C^* -algebras, $E(Y^*) = E(Y)^*$. Since $X = X^\dagger$ means $X = X^*$, one obtains $E(X)^\dagger = E(X)^* = E(X^*) = E(X)$. $\square$

Definition 4.4.6 (Density matrices). Lean Lean ✓ Stmt The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \text{tr}(\rho) = 1\}. \quad (4.3)$$

4.5 Trace adjoints and rectangular Kraus maps

Definition 4.5.1 (Trace-pairing adjoint). Lean ✓ Stmt The trace-pairing adjoint $E^* : M_{D'}(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ of a linear map $E : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ between matrix algebras of possibly different dimensions is the adjoint for the bilinear pairing $(A, B) \mapsto \text{tr}(AB)$.

Definition 4.5.2 (Completely positive map in rectangular Kraus form). Lean ✓ Stmt A linear map $\mathcal{S}$ between matrix algebras is *completely positive in rectangular Kraus form* if there are finitely many operators $A_i : H \rightarrow K$ such that

$$\mathcal{S}(X) = \sum_i A_i X A_i^\dagger.$$

No trace-preservation normalization is imposed.

Lemma 4.5.3 (Agreement with square-map complete positivity). Lean ✓ Stmt When the input and output matrix algebras agree, rectangular Kraus complete positivity is equivalent to the square-map notion of complete positivity.

Lemma 4.5.4 (Basic properties of rectangular Kraus complete positivity). Lean Lean ✓ Stmt A map in rectangular Kraus form sends positive semidefinite matrices to positive semidefinite matrices. Every trace-preserving completely positive Kraus map is a completely positive Kraus map.

Lemma 4.5.5 (Sum of rectangular Kraus maps). [Lean](#) [✓ Stmt](#) *The sum of two completely positive maps in rectangular Kraus form is completely positive in rectangular Kraus form.*

Proof. [✓ Proof](#) Concatenating Kraus families for the two summands gives a Kraus family for their sum. $\square$

Definition 4.5.6 (Trace-preserving completely positive map). [Lean](#) [✓ Stmt](#) A linear map $\mathcal{S}$ between matrix algebras is *trace-preserving completely positive* if it has a Kraus form $\mathcal{S}(X) = \sum_i A_i X A_i^\dagger$ with $\sum_i A_i^\dagger A_i = I$. The Kraus operators may be rectangular, so the input and output dimensions need not agree.

Theorem 4.5.7 (Trace preservation completes a Kraus channel). [Lean](#) [✓ Stmt](#) *A completely positive map in rectangular Kraus form that preserves the matrix trace is trace-preserving completely positive.*

Proof. [✓ Proof](#) If $(A_i)_i$ is a Kraus family, trace preservation and cyclicity give $\text{tr}((\sum_i A_i^\dagger A_i)X) = \text{tr}(X)$ for every X . Nondegeneracy of the trace pairing implies $\sum_i A_i^\dagger A_i = \mathbb{1}$. $\square$

Theorem 4.5.8 (Trace preservation from a Kraus resolution). [Lean](#) [✓ Stmt](#) *Every trace-preserving completely positive map in rectangular Kraus form preserves the matrix trace.*

Proof. [✓ Proof](#) If $\mathcal{S}(X) = \sum_a A_a X A_a^\dagger$ and $\sum_a A_a^\dagger A_a = I$, cyclicity gives

$$\text{tr } \mathcal{S}(X) = \sum_a \text{tr}(A_a X A_a^\dagger) = \text{tr} \left(\left(\sum_a A_a^\dagger A_a \right) X \right) = \text{tr } X.$$

$\square$

Theorem 4.5.9 (The trace adjoint of a channel is unital). [Lean](#) [✓ Stmt](#) *If $\mathcal{S} : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ is trace-preserving and completely positive, then its trace-pairing adjoint satisfies*

$$\mathcal{S}^*(\mathbb{1}_{D'}) = \mathbb{1}_D.$$

This is the Schrödinger–Heisenberg duality of [Wol12, Section 1.2].

Proof. [✓ Proof](#) For every $X \in M_D(\mathbb{C})$, trace duality and trace preservation give

$$\text{tr}(\mathcal{S}^*(\mathbb{1}_{D'})X) = \text{tr}(\mathcal{S}(X)) = \text{tr}(X) = \text{tr}(\mathbb{1}_D X).$$

Nondegeneracy of the trace pairing proves the identity. $\square$

Lemma 4.5.10 (Positivity preservation for Kraus maps). [Lean](#) [✓ Stmt](#) *If $\mathcal{S}$ is trace-preserving completely positive and $X \geq 0$, then $\mathcal{S}(X) \geq 0$.*

4.6 The Choi–Jamiołkowski representation of maps between matrix algebras

Fix $d, d' \geq 1$. For a linear map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ between matrix algebras of possibly different dimensions, write Ω_d for the maximally entangled vector on the *input* factor and τ for the associated operator on $\mathbb{C}^{d'} \otimes \mathbb{C}^d$, with the output factor first. The partial trace over the output factor is written tr_A and the partial trace over the input factor is written tr_B .

Definition 4.6.1 (Choi matrix of a map between matrix algebras). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The Choi matrix of a linear map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is

$$\tau = (T \otimes \text{id}_d)(|\Omega_d\rangle\langle\Omega_d|),$$

an operator on $\mathbb{C}^{d'} \otimes \mathbb{C}^d$. This is the correspondence of [Wol12, Proposition 2.1].

Lemma 4.6.2 (Linearity of the Choi assignment). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Write τ_T for the Choi matrix of T . For all linear maps $T, S : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$, all $c \in \mathbb{C}$, all finite index sets s and all families $(T_i)_{i \in s}$,

$$\tau_{T+S} = \tau_T + \tau_S, \quad (4.4)$$

$$\tau_{cT} = c \tau_T, \quad (4.5)$$

$$\tau_{\sum_{i \in s} T_i} = \sum_{i \in s} \tau_{T_i}. \quad (4.6)$$

Proof. [✓ Proof](#) Each entry $\tau_{(i_1, i_2), (j_1, j_2)}$ is the value of T at a fixed matrix, read off at a fixed pair of indices, so (4.4) and (4.5) hold entrywise. They make $T \mapsto \tau_T$ a complex linear map, and (4.6) is the image of a finite sum under it. $\square$

Theorem 4.6.3 (Mutual inverses of the Choi correspondence). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The assignments $T \mapsto \tau$ and $\tau \mapsto T$, the latter defined entrywise by

$$T(B)_{i_1 j_1} = d \sum_{i_2, j_2=0}^{d-1} \tau_{(i_1, i_2), (j_1, j_2)} B_{i_2 j_2},$$

are mutual inverses: every linear map is recovered from its Choi matrix, and every operator on $\mathbb{C}^{d'} \otimes \mathbb{C}^d$ is the Choi matrix of the map it defines. This is [Wol12, Proposition 2.1, Equation (2.4)].

Proof. [✓ Proof](#) The (i_2, j_2) -slice of $|\Omega_d\rangle\langle\Omega_d|$ is the matrix unit $E_{i_2 j_2}$ scaled by $1/d$; substituting it into the defining entries and using linearity gives both compositions. $\square$

Theorem 4.6.4 (Trace-pairing formula). [Lean](#) [✓ Stmt](#) For every $A \in M_{d'}(\mathbb{C})$ and $B \in M_d(\mathbb{C})$,

$$\text{tr}(AT(B)) = d \text{tr}(\tau(A \otimes B^T)).$$

This is the second line of [Wol12, Proposition 2.1, Equation (2.1)].

Proof. [✓ Proof](#) Expand $\text{tr}(AT(B))$ entrywise, insert the inverse assignment of Theorem 4.6.3, and identify the resulting four-fold sum with $d \text{tr}(\tau(A \otimes B^T))$. $\square$

Theorem 4.6.5 (Hermiticity preservation). [Lean](#) [✓ Stmt](#) The Choi matrix is Hermitian, $\tau = \tau^\dagger$, if and only if $T(B^\dagger) = T(B)^\dagger$ for every $B \in M_d(\mathbb{C})$. This is the Hermiticity clause of [Wol12, Proposition 2.1].

Proof. [✓ Proof](#) The entries of τ along the diagonal slices are the entries of $T(E_{i_2 j_2})$ scaled by $1/d$, so Hermiticity of τ is the relation $T(E_{j_2 i_2}) = T(E_{i_2 j_2})^\dagger$ on matrix units; both directions then follow by linearity. $\square$

Theorem 4.6.6 (Complete positivity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The map T is completely positive, in the sense that it admits a Kraus representation $T(X) = \sum_j K_j X K_j^\dagger$ with $K_j \in M_{d' \times d}(\mathbb{C})$, if and only if $\tau \geq 0$. This is the complete-positivity clause of [Wol12, Proposition 2.1].

Proof. ✓ Proof If $T(X) = \sum_j K_j X K_j^\dagger$, then $\tau = \sum_j |\psi_j\rangle\langle\psi_j|$ with $(\psi_j)_{(i_1, i_2)} = (K_j)_{i_1 i_2} / \sqrt{d}$, so $\tau \geq 0$. Conversely, a positive semidefinite τ decomposes into rank-one outer products $\sum_j |\psi_j\rangle\langle\psi_j|$, and rescaling the vectors ψ_j by $\sqrt{d}$ gives Kraus operators via the inverse assignment. $\square$

Theorem 4.6.7 (Unitality). Lean Lean ✓ Stmt *The input-factor partial trace of the Choi matrix is $\text{tr}_B(\tau) = T(\mathbb{1}_d)/d$; hence $T(\mathbb{1}_d) = \mathbb{1}_{d'}$ if and only if $\text{tr}_B(\tau) = \mathbb{1}_{d'}/d$. This is the unitality clause of [Wol12, Proposition 2.1].*

Proof. ✓ Proof The diagonal slices of $|\Omega_d\rangle\langle\Omega_d|$ sum to $\mathbb{1}_d/d$, so linearity gives $\text{tr}_B(\tau) = T(\mathbb{1}_d)/d$; cancelling the factor $1/d$ gives the equivalence. $\square$

Theorem 4.6.8 (Preservation of the trace). Lean Lean Lean ✓ Stmt *The output-factor partial trace of the Choi matrix is $\text{tr}_A(\tau) = (T^*(\mathbb{1}_{d'}))^T/d$, where T^* is the trace-pairing adjoint; hence $T^*(\mathbb{1}_{d'}) = \mathbb{1}_d$, equivalently T preserves the trace, if and only if $\text{tr}_A(\tau) = \mathbb{1}_d/d$. This is the trace-preservation clause of [Wol12, Proposition 2.1].*

Proof. ✓ Proof The output-factor partial trace has entries $\text{tr}(T(E_{i_2 j_2}))/d$, which are the entries of $(T^*(\mathbb{1}_{d'}))^T/d$ by the defining trace pairing of the adjoint. The equivalence $\text{tr}(T(X)) = \text{tr}(X) \Leftrightarrow T^*(\mathbb{1}_{d'}) = \mathbb{1}_d$ is the nondegeneracy of the trace pairing, and transposition and the factor $1/d$ cancel. $\square$

Theorem 4.6.9 (Trace normalization). Lean ✓ Stmt *The full trace of the Choi matrix is $\text{tr}(\tau) = \text{tr}(T^*(\mathbb{1}_{d'}))/d$. This is the normalization clause of [Wol12, Proposition 2.1].*

Proof. ✓ Proof Take the trace of the identity $\text{tr}_A(\tau) = (T^*(\mathbb{1}_{d'}))^T/d$ of Theorem 4.6.8. $\square$

Theorem 4.6.10 (Doubly-stochastic maps). Lean ✓ Stmt *The conditions $T(\mathbb{1}_d) \propto \mathbb{1}_{d'}$ and $T^*(\mathbb{1}_{d'}) \propto \mathbb{1}_d$ hold if and only if $\text{tr}_B(\tau) \propto \mathbb{1}_{d'}$ and $\text{tr}_A(\tau) \propto \mathbb{1}_d$. This is the doubly-stochastic clause of [Wol12, Proposition 2.1].*

Proof. ✓ Proof Immediate from the partial-trace identities $\text{tr}_B(\tau) = T(\mathbb{1}_d)/d$ and $\text{tr}_A(\tau) = (T^*(\mathbb{1}_{d'}))^T/d$ of Theorems 4.6.7 and 4.6.8. $\square$

Theorem 4.6.11 (Decomposition into completely positive maps). Lean Lean Lean Lean ✓ Stmt *Every linear map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is a complex linear combination of four completely positive maps. If T is Hermitian, that is $T(B^\dagger) = T(B)^\dagger$ for every $B \in M_d(\mathbb{C})$, then T is a real linear combination of two of them. This is [Wol12, Proposition 2.2].*

Proof. ✓ Proof The assignment $T \leftrightarrow \tau$ is linear by Lemma 4.6.2 and one-to-one by Theorem 4.6.3, so it suffices to decompose τ . The identity

$$\tau = \frac{\tau + \tau^\dagger}{2} + i \frac{i\tau^\dagger - i\tau}{2} \quad (4.7)$$

presents τ as a complex linear combination of two Hermitian operators, and the spectral decomposition writes each of them as the difference of its positive and negative parts, both positive semidefinite. Every positive semidefinite operator on $\mathbb{C}^{d'} \otimes \mathbb{C}^d$ is the Choi matrix of a completely positive map, so carrying the four positive semidefinite pieces back through the correspondence produces the four completely positive maps, with coefficients $1, -1, i, -i$. If T is Hermitian then τ is Hermitian by Theorem 4.6.5, so the second summand of (4.7) vanishes and the positive and negative parts of τ alone give two completely positive maps, with coefficients 1 and -1 . $\square$

4.7 The Kraus representation theorem for rectangular matrix algebras

Theorem 4.7.1 (Kraus normalization conditions). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $T(X) = \sum_{j=1}^r K_j X K_j^\dagger$ with $K_j \in M_{d' \times d}(\mathbb{C})$. Then T is trace preserving if and only if $\sum_j K_j^\dagger K_j = \mathbb{1}_d$, and unital if and only if $\sum_j K_j K_j^\dagger = \mathbb{1}_{d'}$. This is item 1 of [Wol12, Theorem 2.1].

Proof. [✓ Proof](#) Cyclicity of the trace gives $\text{tr}((\sum_j K_j^\dagger K_j)X) = \text{tr}(T(X))$ for every X , so the trace-preserving equivalence follows from nondegeneracy of the trace pairing; the unital equivalence is the evaluation at $X = \mathbb{1}_d$. $\square$

Definition 4.7.2 (Kraus cardinality and Kraus rank). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ has *Kraus cardinality* r if it has an exact r -operator Kraus representation. Its *Kraus rank* (Choi rank) is the rank of its Choi matrix, $r = \text{rank}(\tau)$; following [Wol12, Theorem 2.1, footnote], this is distinguished from the rank of T as a linear map.

Theorem 4.7.3 (Minimality of the Kraus rank). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The minimal number of Kraus operators of a completely positive map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ is $r = \text{rank}(\tau) \leq dd'$. This is item 2 of [Wol12, Theorem 2.1].

Proof. [✓ Proof](#) Every r -operator Kraus family gives $\tau = \sum_{j=1}^r |\psi_j\rangle\langle\psi_j|$, so $\text{rank}(\tau) \leq r$. Conversely, the spectral decomposition of $\tau \geq 0$ into $\text{rank}(\tau)$ rank-one outer products yields a Kraus family of exactly $\text{rank}(\tau)$ operators by Theorem 4.6.6, and $\text{rank}(\tau) \leq dd'$ since τ acts on a dd' -dimensional space. $\square$

Theorem 4.7.4 (Orthogonal minimal Kraus family). [Lean](#) [✓ Stmt](#) Every completely positive map $T : M_d(\mathbb{C}) \rightarrow M_{d'}(\mathbb{C})$ admits a Kraus representation with $r = \text{rank}(\tau)$ Hilbert–Schmidt orthogonal Kraus operators, $\text{tr}[K_i^\dagger K_j] \propto \delta_{ij}$; the off-diagonal traces vanish and the diagonal traces are nonzero. This is item 3 of [Wol12, Theorem 2.1].

Proof. [✓ Proof](#) Choose the ψ_j in the spectral decomposition of Theorem 4.7.3 to be orthogonal eigenvectors; the corresponding Kraus operators then satisfy $\text{tr}[K_i^\dagger K_j] = d \lambda_i \delta_{ij}$, where λ_i is the i -th nonzero eigenvalue of τ . $\square$

Theorem 4.7.5 (Unitary freedom of Kraus representations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Two sets of Kraus operators $\{K_j\}$ and $\{\tilde{K}_\ell\}$ represent the same completely positive map if and only if there is a unitary U with $K_j = \sum_\ell U_{j\ell} \tilde{K}_\ell$, where the smaller set is padded with zero operators; equivalently, after padding, the families are related by an isometry V with $V^\dagger V = \mathbb{1}$. This is item 4 of [Wol12, Theorem 2.1].

Proof. [✓ Proof](#) An isometric combination preserves the map by the orthogonality relations of the mixing matrix. Conversely, equal maps have equal trace-pairing adjoints, hence equal Gram matrices for the vectorized Kraus operators $(K_j)_{ab}$; equal Gram matrices are related by a unitary, whose rectangular submatrix is the claimed isometry after zero-padding. $\square$

4.8 Irreducibility

Definition 4.8.1 (Orthogonal projection). [Lean](#) [✓ Stmt](#) A matrix $P \in M_D(\mathbb{C})$ is an *orthogonal projection* if $P = P^\dagger$ and $P^2 = P$.

Definition 4.8.2 (Irreducible map). [Lean](#) [✓ Stmt](#) A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Theorem 6.2(1)]. The definition applies to any linear map; complete positivity is not required.

Theorem 4.8.3 (Orthogonal projections and involutive-algebra projections). [Lean](#) [Lean](#) [✓ Stmt](#) For a complex square matrix, the two descriptions $P = P^\dagger$, $P^2 = P$ and $P^* = P$, $P^2 = P$ are equivalent.

Proof. [✓ Proof](#) Since the involution on $M_D(\mathbb{C})$ is conjugate transpose, $P^* = P^\dagger$. Therefore $P^\dagger = P$, $P^2 = P$ if and only if $P^* = P$, $P^2 = P$. $\square$

Theorem 4.8.4 (Coisometric transport of a star projection). [Lean](#) [✓ Stmt](#) Let Q be a self-adjoint idempotent on a finite-dimensional space, and let U be a coisometry onto that space, so that $UU^\dagger = \mathbb{1}$. Then $U^\dagger QU$ is self-adjoint and idempotent.

Proof. [✓ Proof](#) Self-adjointness follows from $(U^\dagger QU)^\dagger = U^\dagger Q^\dagger U = U^\dagger QU$. Idempotence follows from

$$(U^\dagger QU)^2 = U^\dagger Q(UU^\dagger)QU = U^\dagger Q^2 U = U^\dagger QU.$$

$\square$

Theorem 4.8.5 (Positive-definite transport by a coisometry). [Lean](#) [✓ Stmt](#) Let $A > 0$ be a square complex matrix and let V be a possibly rectangular matrix satisfying $VV^\dagger = \mathbb{1}$. Then

$$VAV^\dagger > 0. \tag{4.8}$$

Proof. [✓ Proof](#) For a row vector x , one has

$$x(VAV^\dagger)x^\dagger = (xV)A(xV)^\dagger. \tag{4.9}$$

Moreover, $xV = 0$ implies

$$x = x(VV^\dagger) = 0. \tag{4.10}$$

Thus every nonzero x gives $xV \neq 0$, and positive definiteness of A makes the displayed quadratic form strictly positive. $\square$

Theorem 4.8.6 (Vectorized identity realizes the trace pairing). [Lean](#) [✓ Stmt](#) For every square complex matrix X ,

$$\text{vec}(\mathbb{1}) \cdot \text{vec}(X) = \text{tr}(X). \tag{4.11}$$

Proof. [✓ Proof](#) Expanding column stacking gives

$$\sum_{i,j} (\mathbb{1})_{ij} X_{ij} = \sum_{i,j} \delta_{ij} X_{ij} = \sum_i X_{ii} = \text{tr}(X). \tag{4.12}$$

$\square$

Lemma 4.8.7 (Pairwise orthogonality from a projection resolution). [Lean](#) [✓ Stmt](#) Let $(Q_k)_k$ be a finite family of orthogonal projections satisfying $\sum_k Q_k = \mathbb{1}$. If $k \neq \ell$, then $Q_k Q_\ell = 0$.

Proof. ✓ Proof Multiplying the resolution of the identity on both sides by Q_k gives $\sum_j Q_k Q_j Q_k = Q_k$. Each summand is positive semidefinite because $Q_k Q_j Q_k = (Q_j Q_k)^*(Q_j Q_k)$, while the $j = k$ summand is already Q_k . Subtracting that term and taking traces gives

$$0 = \sum_{j \neq k} \text{tr}((Q_j Q_k)^*(Q_j Q_k)) = \sum_{j \neq k} \|Q_j Q_k\|_F^2.$$

Hence $Q_j Q_k = 0$ for $j \neq k$, and taking adjoints gives $Q_k Q_j = 0$. □

Theorem 4.8.8 (Complement of an orthogonal projection). Lean ✓ Stmt *If $P \in M_D(\mathbb{C})$ is an orthogonal projection, then $\mathbb{1} - P$ is an orthogonal projection.*

Proof. ✓ Proof In the involutive algebra $M_D(\mathbb{C})$, P satisfies $P^\dagger = P$ and $P^2 = P$. Hence $(\mathbb{1} - P)^\dagger = \mathbb{1} - P$ and $(\mathbb{1} - P)^2 = \mathbb{1} - 2P + P^2 = \mathbb{1} - P$. Thus $\mathbb{1} - P$ is an orthogonal projection. □

Theorem 4.8.9 (Injectivity implies irreducibility). Lean ✓ Stmt *If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.*

Proof. ✓ Proof If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^i P = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M , so $P = 0$ or $P = \mathbb{1}$. □

4.9 Transfer maps

Theorem 4.9.1 (Transfer map is CP). Lean ✓ Stmt *The transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is completely positive.*

Proof. ✓ Proof It is already written in the Kraus form (4.1), with Kraus operators $\{A^i\}$. □

Theorem 4.9.2 (Trace adjointness of a Kraus map). Lean ✓ Stmt *For matrices ρ and X and a finite family $(K_i)_i$,*

$$\text{tr} \left(\rho \sum_i K_i X K_i^\dagger \right) = \text{tr} \left(\left(\sum_i K_i^\dagger \rho K_i \right) X \right). \quad (4.13)$$

Proof. ✓ Proof Apply Lemma A.3.1 to the Kraus map determined by $(K_i)_i$. □

Lemma 4.9.3 (Trace adjointness of a transfer map). Lean ✓ Stmt *The trace-adjoint identity of Theorem 4.9.2 holds when the Kraus family is the family of matrices of an MPS tensor and the two Kraus maps are written as transfer maps.*

Proof. ✓ Proof Identify the two Kraus maps with the corresponding transfer maps. □

Lemma 4.9.4 (Transfer map under scalar multiplication). Lean ✓ Stmt *Let A be an MPS tensor and let $w \in \mathbb{C}$. Then, for every bond matrix X ,*

$$\mathcal{E}_{wA}(X) = w\bar{w} \mathcal{E}_A(X).$$

Proof. ✓ Proof Expand the transfer map and pull the scalar w and its conjugate out of each summand. □

Lemma 4.9.5 (Left-canonical transfer maps preserve trace). [Lean](#) [✓ Stmt](#) If A is left canonical, then, for every bond matrix X ,

$$\mathrm{tr}(\mathcal{E}_A(X)) = \mathrm{tr}(X).$$

Proof. [✓ Proof](#) Cyclicity of trace gives

$$\mathrm{tr}(\mathcal{E}_A(X)) = \mathrm{tr}\left(\left(\sum_i (A^i)^\dagger A^i\right) X\right) = \mathrm{tr}(X),$$

since left canonicity makes the sum in parentheses equal to the identity. $\square$

Theorem 4.9.6 (Transfer map is a channel). [Lean](#) [✓ Stmt](#) If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a quantum channel (CPTP).

Proof. [✓ Proof](#) Complete positivity is Theorem 4.9.1. Trace preservation follows by cyclicity and the normalization hypothesis:

$$\mathrm{tr}(\mathcal{E}_A(X)) = \sum_i \mathrm{tr}(A^i X (A^i)^\dagger) = \mathrm{tr}\left(\left(\sum_i (A^i)^\dagger A^i\right) X\right) = \mathrm{tr}(X).$$

$\square$

4.10 Peripheral spectrum and primitivity

Definition 4.10.1 (Peripheral spectrum). [Lean](#) [✓ Stmt](#) The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $|\mu|$ equal to the spectral radius of T .

Definition 4.10.2 (Peripheral eigenvalues). [Lean](#) [✓ Stmt](#) The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are the eigenvalues λ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Proposition 6.1], so these coincide with the eigenvalues of maximal modulus.

The condition $|\lambda| = 1$ is appropriate for channels, since the spectral radius of a channel is 1; for a general linear map with spectral radius $r \neq 1$ the condition would generalise to $|\lambda| = r$.

Lemma 4.10.3 (Unit spectral radius forces positive matrix size). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a bounded linear map. If its spectral radius is one, then $D \geq 1$.

Proof. [✓ Proof](#) If $D = 0$, then the matrix space and its endomorphism algebra contain only zero, so every endomorphism has spectral radius zero. $\square$

Remark 4.10.4 (Peripheral conventions for channels). [✓ Stmt](#) The spectral-radius convention of Definition 4.10.1 is the usual bounded-operator notion. For channels the spectral radius is 1, so the channel arguments below use the unit-circle set of Definition 4.10.2.

Definition 4.10.5 (Primitive map). [Lean](#) [✓ Stmt](#) A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\mathrm{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Theorem 6.7]. This definition applies to any linear endomorphism, not only to channels.

Theorem 4.10.6 (Eigenvalue bound for a trace-preserving Kraus map). [Lean](#) [✓ Stmt](#) Let $\{K_i\}_{i=0}^{d-1}$ be matrices in $M_D(\mathbb{C})$ with $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$, and let

$$\mathcal{K}_K(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$$

be the associated Kraus map. Every eigenvalue μ of $\mathcal{K}_K$ satisfies $|\mu| \leq 1$. This is the trace-preserving specialization of [Wol12, Proposition 6.1].

Proof. [✓ Proof](#) Trace preservation of $\mathcal{K}_K$ is the cyclicity computation

$$\mathrm{tr}(\mathcal{K}_K(X)) = \sum_{i=0}^{d-1} \mathrm{tr}(K_i X K_i^\dagger) = \mathrm{tr}\left(\left(\sum_{i=0}^{d-1} K_i^\dagger K_i\right)X\right) = \mathrm{tr}(X),$$

and $\mathcal{K}_K$ is completely positive by Definition 5.1.1. For $D \geq 1$, Theorem ?? therefore gives $|\mu| \leq 1$. For $D = 0$ the algebra $M_0(\mathbb{C})$ is trivial, so $\mathcal{K}_K$ has no eigenvector and the bound is vacuous. $\square$

Theorem 4.10.7 (Spectral-radius bound for a trace-preserving Kraus map). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\{K_i\}_{i=0}^{d-1}$ be a trace-preserving Kraus family on $M_D(\mathbb{C})$. The spectral radius of its Kraus map $\mathcal{K}_K$ is at most 1.

Proof. [✓ Proof](#) The spectral radius is the supremum of the moduli of the spectral values,

$$\varrho(\mathcal{K}_K) = \sup_{\mu \in \mathrm{spec}(\mathcal{K}_K)} |\mu|,$$

and on the finite-dimensional space $M_D(\mathbb{C})$ the spectral values are exactly the eigenvalues. Theorem 4.10.6 bounds each of them by 1. $\square$

Definition 4.10.8 (Spectral radius of a rectangular mixed Kraus map). [Lean](#) [✓ Stmt](#) Let $\{A_i\}_{i=0}^{d-1} \subseteq M_{D_1}(\mathbb{C})$ and $\{B_i\}_{i=0}^{d-1} \subseteq M_{D_2}(\mathbb{C})$. Their rectangular mixed Kraus map is the endomorphism of $M_{D_1 \times D_2}(\mathbb{C})$ given by

$$\mathcal{M}_{A,B}(X) = \sum_{i=0}^{d-1} A_i X B_i^\dagger.$$

Its spectral radius is denoted by $\varrho(\mathcal{M}_{A,B})$.

Theorem 4.10.9 (Eigenvalue bound for a trace-preserving mixed Kraus map). [Lean](#) [✓ Stmt](#) Suppose that

$$\begin{aligned} \sum_{i=0}^{d-1} A_i^\dagger A_i &= \mathbb{1}_{D_1}, \\ \sum_{i=0}^{d-1} B_i^\dagger B_i &= \mathbb{1}_{D_2}. \end{aligned}$$

Every eigenvalue μ of $\mathcal{M}_{A,B}$ satisfies $|\mu| \leq 1$.

Proof. ✓ **Proof** For each i , set $C_i = \begin{pmatrix} A_i & 0 \\ 0 & B_i \end{pmatrix}$ and embed a rectangular matrix X as $J(X) = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}$. Then

$$\sum_{i=0}^{d-1} C_i^\dagger C_i = \begin{pmatrix} \mathbb{1}_{D_1} & 0 \\ 0 & \mathbb{1}_{D_2} \end{pmatrix},$$

$$\mathcal{K}_C(J(X)) = J(\mathcal{M}_{A,B}(X)).$$

Hence a nonzero eigenvector X of $\mathcal{M}_{A,B}$ with eigenvalue μ gives the nonzero eigenvector $J(X)$ of the trace-preserving Kraus map $\mathcal{K}_C$ with the same eigenvalue. Theorem 4.10.6 gives $|\mu| \leq 1$. $\square$

Theorem 4.10.10 (Spectral-radius bound for a trace-preserving mixed Kraus map). Lean

✓ **Stmnt** Suppose that

$$\sum_{i=0}^{d-1} A_i^\dagger A_i = \mathbb{1}_{D_1},$$

$$\sum_{i=0}^{d-1} B_i^\dagger B_i = \mathbb{1}_{D_2}.$$

Then $\varrho(\mathcal{M}_{A,B}) \leq 1$.

Proof. ✓ **Proof** On the finite-dimensional space $M_{D_1 \times D_2}(\mathbb{C})$, every spectral value is an eigenvalue. Theorem 4.10.9 bounds the modulus of each spectral value by 1, so the spectral radius is at most 1. $\square$

4.11 Fixed-point projection

Definition 4.11.1 (Fixed-point projection). Lean ✓ **Stmnt** Given a linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho. \tag{4.14}$$

This projects onto the span of ρ along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

Theorem 4.11.2 (Power decomposition). Lean ✓ **Stmnt** If E is trace-preserving and $E(\rho) = \rho$, then, for $n \geq 1$,

$$E^n = P + (E - P)^n, \tag{4.15}$$

where P is the fixed-point projection.

Proof. ✓ **Proof** By (4.14), P is idempotent. The fixed-point relation $E(\rho) = \rho$ gives $E \circ P = P$, and trace preservation gives $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, yielding (4.15). $\square$

4.12 Peripheral eigenvalues and powering

Theorem 4.12.1 (1 is a peripheral eigenvalue). [Lean](#) [✓ Stmt](#) *If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then 1 is a peripheral eigenvalue of E .*

Proof. [✓ Proof](#) ρ is an eigenvector with eigenvalue 1, and $|1| = 1$. □

Theorem 4.12.2 (Finiteness of peripheral eigenvalues). [Lean](#) [✓ Stmt](#) *On a finite-dimensional space, the set of peripheral eigenvalues is finite.*

Proof. [✓ Proof](#) Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. □

Theorem 4.12.3 (Power-stable peripheral eigenvalues are roots of unity). [Lean](#) [✓ Stmt](#) *Let E be a linear endomorphism on a finite-dimensional space. If μ is a peripheral eigenvalue of E and μ^n is an eigenvalue of E for every $n \geq 1$, then μ is a root of unity.*

Proof. [✓ Proof](#) Since μ^n is an eigenvalue for every positive n , the pigeonhole principle on the finite eigenvalue set gives $\mu^a = \mu^b$ for some $a \neq b$, hence $\mu^{|a-b|} = 1$. □

Theorem 4.12.4 (Peripheral powers imply root of unity). [Lean](#) [✓ Stmt](#) *If the set of peripheral eigenvalues of a linear endomorphism E on a finite-dimensional space is closed under powers ($\mu \in \text{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \text{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.*

Proof. [✓ Proof](#) The closure hypothesis provides $\mu^n \in \text{peripheral}(E)$ for all positive n , which in particular means μ^n is an eigenvalue. Theorem 4.12.3 then gives $\mu^p = 1$ for some $p > 0$. □

4.12.1 Periodicity removal by powering

Remark 4.12.1.1. [✓ Stmt](#) The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its p -blocked tensor replaces $\mathcal{E}_A$ by $\mathcal{E}_A^p$, so this powering step is the operator-theoretic form of blocking.

Lemma 4.12.1.2 (Common exponent for a finite set of roots of unity). [Lean](#) [✓ Stmt](#) *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

Proof. [✓ Proof](#) Take p to be the least common multiple of the exponents p_μ . □

Lemma 4.12.1.3 (Powering collapses peripheral eigenvalues). [Lean](#) [✓ Stmt](#) *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a linear map, and assume that E has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue μ of E satisfies $\mu^p = 1$, then $\text{peripheral}(E^p) = \{1\}$.*

Proof. [✓ Proof](#) The fixed point ρ gives $1 \in \text{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \text{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \text{spec}(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \text{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. □

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

Remark 4.12.1.4 (Bi-canonical special case). ✓ Stmt When a Kraus map happens to be both unital and trace-preserving (the “bi-canonical” case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 5.2.3) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 4.12.4. It requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

4.12.2 Peripheral closure via adjoint fixed point

The Kraus map, its adjoint, and the unital normalization are those of Definitions 5.1.1, 5.1.4, and 5.1.5. The following results combine the canonical Kadison–Schwarz and multiplicative-domain theorems of Chapter 5 with a positive definite fixed point of the adjoint map, replacing trace preservation by the weighted-trace argument in [CPGSV16, Appendix A].

Lemma 4.12.2.1 (Peripheral closure via adjoint fixed point). Lean ✓ Stmt Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital ($\sum_i K_i K_i^\dagger = \mathbb{1}$), and assume:

1. the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point $\rho > 0$, and
2. E is irreducible.

Then $\text{peripheral}(E)$ is closed under powers: if $\mu \in \text{peripheral}(E)$, then $\mu^n \in \text{peripheral}(E)$ for every $n \in \mathbb{N}$.

Proof. ✓ Proof Let $E(X) = \mu X$ with $|\mu| = 1$, and set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By (5.3), $G \geq 0$. Since $E^*(\rho) = \rho$,

$$\begin{aligned} \text{tr}(\rho G) &= \text{tr}(\rho E(X^\dagger X)) - \text{tr}(\rho E(X)^\dagger E(X)) \\ &= \text{tr}(E^*(\rho) X^\dagger X) - \text{tr}(\rho E(X)^\dagger E(X)) \\ &= \text{tr}(\rho X^\dagger X) - |\mu|^2 \text{tr}(\rho X^\dagger X) = 0. \end{aligned} \tag{4.16}$$

Because $\rho > 0$ and $G \geq 0$, this forces $G = 0$, so $E(X^\dagger X) = E(X)^\dagger E(X) = X^\dagger X$. Hence $X^\dagger X$ is a positive semidefinite fixed point, which is positive definite by irreducibility (Theorem 6.3.11). Thus X is invertible. Theorem 5.2.2 gives the corresponding intertwining relation with each Kraus operator. Iterating the multiplicative identity of Theorem 5.2.5 gives $E(X^n) = \mu^n X^n$, so μ^n is peripheral. $\square$

Theorem 4.12.2.2 (Roots of unity via adjoint fixed point). Lean ✓ Stmt Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital, and assume that the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point and that E is irreducible. Then every peripheral eigenvalue of E is a root of unity.

Proof. ✓ Proof Combine the closure-under-powers result (Lemma 4.12.2.1) with Theorem 4.12.4. $\square$

Theorem 4.12.2.3 (Unitality of the conjugate-transposed Kraus family). Lean ✓ Stmt If a Kraus family $(K_i)_i$ satisfies $\sum_i K_i^\dagger K_i = \mathbb{1}$, then the family $(K_i^\dagger)_i$ is unital.

Theorem 4.12.2.4 (Adjoint map of the conjugate-transposed family). Lean ✓ Stmt The adjoint Kraus map associated with $(K_i^\dagger)_i$ is the Kraus map associated with $(K_i)_i$.

Theorem 4.12.2.5 (Irreducibility under conjugate transposition). [Lean](#) [✓ Stmt](#) *If the Kraus map associated with $(K_i)_i$ is irreducible, then so is the Kraus map associated with $(K_i^\dagger)_i$.*

Theorem 4.12.2.6 (Peripheral spectrum under conjugate transposition). [Lean](#) [✓ Stmt](#) *The peripheral spectrum of the Kraus map associated with $(K_i^\dagger)_i$ is the complex conjugate of the peripheral spectrum associated with $(K_i)_i$.*

Theorem 4.12.2.7 (Roots of unity for irreducible channels). [Lean](#) [✓ Stmt](#) *Let $D \geq 1$, and let $E: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be an irreducible quantum channel. Then every peripheral eigenvalue of E is a root of unity.*

Proof. [✓ Proof](#) Choose a normalized Kraus representation $E(X) = \sum_i K_i X K_i^\dagger$ using Theorem ??, and set $L_i = K_i^\dagger$. The normalization $\sum_i K_i^\dagger K_i = 1$ makes the Kraus map associated with $(L_i)_i$ unital. Theorem 6.1.19 gives a nonzero positive-semidefinite fixed point ρ of E , and irreducibility makes ρ positive definite by Theorem 6.3.11.

The adjoint of the Kraus map associated with $(L_i)_i$ is E , so ρ is a positive-definite fixed point of that adjoint map. Irreducibility is preserved on passing from $(K_i)_i$ to $(K_i^\dagger)_i$, while a peripheral eigenvalue μ of E becomes the peripheral eigenvalue $\bar{\mu}$ of the latter Kraus map. Theorem 4.12.2.2 gives $\bar{\mu}^p = 1$ for some $p > 0$. Taking complex conjugates yields $\mu^p = 1$. $\square$

Remark 4.12.2.8. [✓ Stmt](#) Lemma 4.12.2.1 and Theorem 4.12.2.2 cover the bi-canonical (unital + TP) case as well: trace preservation gives $E^*(1) = 1$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV16, Appendix A], which uses a faithful invariant state rather than bi-canonicity. Theorem 4.12.2.2 establishes the root-of-unity conclusion in this adjoint-fixed-point formulation. In the same adjoint-fixed-point formulation, Chapter 7 proves the corresponding cyclic structure: the cyclic group of peripheral eigenvalues is Theorem 7.1.1, and the cyclic projection decomposition is Theorem 7.9.1.1.

Definition 4.12.2.9 (Channel period). [Lean](#) [✓ Stmt](#) *The period of a channel E is the number of peripheral eigenvalues, counted without multiplicity: $p(E) := |\text{peripheral}(E)|$.*

4.13 Primitivity and the complementary transfer-map gap

Theorem 4.13.1 (Primitivity equals period one). [Lean](#) [✓ Stmt](#) *A linear map with a nonzero fixed point is primitive if and only if its period is 1.*

Proof. [✓ Proof](#) If $\text{peripheral}(E) = \{1\}$, then $|\text{peripheral}(E)| = 1$. Conversely, if $|\text{peripheral}(E)| = 1$, then $1 \in \text{peripheral}(E)$ by Theorem 4.12.1, so the only element is 1. $\square$

Remark 4.13.2 (Period-one characterization). [✓ Stmt](#) The period $p(E)$ counts peripheral eigenvalues without multiplicity. Theorem 4.13.1 is therefore the period-one characterization of primitivity. The spectral estimates below are stated directly with Definition 4.10.5, since they use the absence of peripheral eigenvalues other than 1, not a separate counting hypothesis.

Theorem 4.13.3 (Complementary transfer-map gap for primitive maps). [Lean](#) [✓ Stmt](#) *Let $E: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be trace-preserving, and let $\rho \neq 0$ satisfy $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$. Let P be the fixed-point projection associated to ρ , as defined in (4.14). Assume that:*

1. E is primitive;
2. every eigenvalue μ of E satisfies $|\mu| \leq 1$;

3. whenever $E(X) = X$ and $\text{tr}(X) = 0$, one has $X = 0$.

Then every eigenvalue ν of $E - P$ satisfies $|\nu| < 1$.

Proof. ✓ Proof Let $(E - P)(X) = \nu X$ with $X \neq 0$. Then $E(X) = \nu X + P(X)$. If $\text{tr}(X) = 0$, then $P(X) = 0$, so $E(X) = \nu X$ and hence ν is an eigenvalue of E with $|\nu| \leq 1$. If $|\nu| = 1$, primitivity forces $\nu = 1$, so X is a trace-zero fixed point of E ; by assumption this implies $X = 0$, a contradiction. Thus $|\nu| < 1$.

If instead $\text{tr}(X) \neq 0$, trace preservation and $\text{tr}(P(X)) = \text{tr}(X)$, which follows from (4.14), give

$$\text{tr}(X) = \text{tr}(E(X)) = \nu \text{tr}(X) + \text{tr}(P(X)) = \nu \text{tr}(X) + \text{tr}(X).$$

Hence $\nu \text{tr}(X) = 0$ and therefore $\nu = 0$. In either case, $|\nu| < 1$. □

4.14 Anti-linear conjugation of the spectrum

Entrywise complex conjugation is an anti-linear involution of the matrix space. Conjugating a linear map on both sides by this involution yields a linear map again, and all spectral data conjugate accordingly. These facts transport the spectral clauses of the normal-tensor definition of [CPGSV16] across entrywise conjugation of a tensor.

Definition 4.14.1 (Entrywise-conjugation transport). Lean Lean ✓ Stmt Let $\sigma : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ denote entrywise complex conjugation, $\sigma(X) = \overline{X}$. The *transport along entrywise conjugation* of a linear map $\Phi : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is

$$\Phi \mapsto \sigma \circ \Phi \circ \sigma. \tag{4.17}$$

Both factors σ conjugate the scalars, so the transported map is again linear. The transport is an involution, respects sums and composition, and sends the scalar μ to $\bar{\mu}$; it is therefore a ring automorphism of the endomorphism ring that conjugates the scalars.

Theorem 4.14.2 (Spectrum under anti-linear conjugation). Lean Lean ✓ Stmt For every linear map $\Phi : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$,

$$\text{spec}(\sigma \circ \Phi \circ \sigma) = \overline{\text{spec}(\Phi)}, \tag{4.18}$$

and μ is an eigenvalue of $\sigma \circ \Phi \circ \sigma$ if and only if $\bar{\mu}$ is an eigenvalue of Φ .

Proof. ✓ Proof The transport sends $\bar{\mu} \mathbf{1} - \Phi$ to $\mu \mathbf{1} - \sigma \circ \Phi \circ \sigma$, and a ring automorphism preserves invertibility, which gives (4.18). For the eigenvalue statement, if $(\sigma \circ \Phi \circ \sigma)(V) = \mu V$ with $V \neq 0$, then applying σ to both sides gives $\Phi(\bar{V}) = \bar{\mu} \bar{V}$ with $\bar{V} \neq 0$; the converse follows because the transport is an involution. □

Theorem 4.14.3 (Peripheral data under anti-linear conjugation). Lean Lean Lean ✓ Stmt For every linear map $\Phi : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the peripheral eigenvalues of $\sigma \circ \Phi \circ \sigma$ are the complex conjugates of the peripheral eigenvalues of Φ , the spectral radii agree,

$$r(\sigma \circ \Phi \circ \sigma) = r(\Phi), \tag{4.19}$$

and $\sigma \circ \Phi \circ \sigma$ is primitive if and only if Φ is primitive.

Proof. ✓ Proof Complex conjugation preserves the modulus. Conjugating the spectrum and the eigenvalues by Theorem 4.14.2 therefore preserves the supremum of the moduli of the spectral values, which gives (4.19), and maps the unit-circle eigenvalues of Φ onto those of $\sigma \circ \Phi \circ \sigma$. Since $\{1\}$ is fixed by conjugation, the peripheral eigenvalues equal $\{1\}$ after transport if and only if they do before. □

Chapter 5

Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory of [Wol12, Chapter 5]: the Kadison–Schwarz inequality for Kraus maps and the multiplicative domain as a $*$ -subalgebra on which the map restricts to a $*$ -homomorphism. Trace adjoints, positive retractions, and the faithful-fixed-point form of peripheral Schwarz equality supply the operator-theoretic results used by the Fundamental Theorem of Chapter 11. The k -positive and Schmidt-rank theory appears later in the full blueprint.

5.1 Kadison–Schwarz inequality

A linear map is completely positive (Definition 4.1.5) if and only if it has a Kraus form. The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map defined in (2.4) is a Kraus map (Theorem 4.9.1), so these results apply directly to transfer maps.

Definition 5.1.1 (Kraus map). Lean ✓ Stmt Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is

$$E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger. \quad (5.1)$$

A linear map is completely positive (Definition 4.1.5) if and only if it can be written in this form, as in (4.1).

Theorem 5.1.2 (Kraus-map form of the transfer map). Lean ✓ Stmt For a finite matrix family A , write $\mathcal{K}_A$ for its Kraus map. Then

$$\mathcal{K}_A = \mathcal{E}_A.$$

Proof. ✓ Proof For every $X \in M_D(\mathbb{C})$,

$$\mathcal{K}_A(X) = \sum_i A^i X (A^i)^\dagger = \mathcal{E}_A(X).$$

□

Theorem 5.1.3 (Irreducibility in Kraus-map notation). [Lean](#) [✓ Stmt](#) If the transfer map $\mathcal{E}_A$ of a finite matrix family is irreducible, then its Kraus map $\mathcal{K}_A$ is irreducible.

Proof. [✓ Proof](#) The two maps are equal by Theorem 5.1.2. $\square$

Definition 5.1.4 (Adjoint Kraus map). [Lean](#) [✓ Stmt](#) The *adjoint Kraus map* is

$$E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i. \quad (5.2)$$

When the $\{K_i\}$ are the matrices of an MPS tensor A , this is the transfer map of the conjugate-transposed family $i \mapsto (A^i)^\dagger$.

Definition 5.1.5 (Unital Kraus map). [Lean](#) [✓ Stmt](#) A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$. Equivalently, $E(\mathbb{1}) = \mathbb{1}$.

Definition 5.1.6 (Trace-preserving Kraus map). [Lean](#) [✓ Stmt](#) A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

Theorem 5.1.7 (Kadison–Schwarz inequality). [Lean](#) [Lean](#) [✓ Stmt](#) Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map, so that $\sum_i K_i K_i^\dagger = \mathbb{1}$. Then, for every $X \in M_D(\mathbb{C})$,

$$E(X^\dagger X) \geq E(X)^\dagger E(X). \quad (5.3)$$

This is [Wol12, Equation (5.2)].

Proof. [✓ Proof](#) Set

$$v = \begin{bmatrix} X^\dagger \\ \mathbb{1} \end{bmatrix}, \quad vv^\dagger = \begin{bmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{bmatrix} \geq 0.$$

Applying the unital completely positive map blockwise preserves positive semidefiniteness; unitality ensures that the $(2, 2)$ -block maps to $\mathbb{1}$. The Schur complement of this block gives (5.3). $\square$

Theorem 5.1.8 (Operator-norm contraction of a unital Kraus map). [Lean](#) [✓ Stmt](#) Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a Kraus map satisfying $E(\mathbb{1}) = \mathbb{1}$. Then, for every $X \in M_D(\mathbb{C})$,

$$\|E(X)\|_\infty \leq \|X\|_\infty. \quad (5.4)$$

Proof. [✓ Proof](#) The operator inequality $X^\dagger X \leq \|X\|_\infty^2 \mathbb{1}$, positivity of E , and unitality give $E(X^\dagger X) \leq \|X\|_\infty^2 \mathbb{1}$. Combining this with (5.3) yields

$$E(X)^\dagger E(X) \leq E(X^\dagger X) \leq \|X\|_\infty^2 \mathbb{1}.$$

The C^* -identity now gives $\|E(X)\|_\infty^2 \leq \|X\|_\infty^2$, which is (5.4). $\square$

Theorem 5.1.9 (Eigenvalue bound for a unital Kraus map). [Lean](#) [✓ Stmt](#) Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a Kraus map satisfying $E(\mathbb{1}) = \mathbb{1}$. Every eigenvalue μ of E satisfies

$$|\mu| \leq 1. \quad (5.5)$$

This is the unital Kraus specialization of [Wol12, Proposition 6.1].

Proof. ✓ Proof Choose a nonzero matrix X such that $E(X) = \mu X$. Then Theorem 5.1.8 gives $|\mu| \|X\|_\infty = \|E(X)\|_\infty \leq \|X\|_\infty$. Since $\|X\|_\infty > 0$, division by this norm proves (5.5). $\square$

Theorem 5.1.10 (Kadison–Schwarz for the adjoint channel). Lean ✓ Stmt *If $\sum_i K_i^\dagger K_i = \mathbb{1}$, then the adjoint Kraus map satisfies, for every $X \in M_D(\mathbb{C})$,*

$$E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X). \quad (5.6)$$

Proof. ✓ Proof The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 5.1.7 to $\{K_i^\dagger\}$. $\square$

Theorem 5.1.11 (Hilbert–Schmidt contraction). Lean ✓ Stmt *If a Kraus map is both unital and trace-preserving, then $\text{tr}(E(X)^\dagger E(X)) \leq \text{tr}(X^\dagger X)$ for all $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof By (5.3), $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\text{tr}(E(X^\dagger X)) \geq \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. $\square$

5.2 Multiplicative domain

Theorem 5.2.1 (KS gap decomposition). Lean ✓ Stmt *For a unital Kraus map E , the Kadison–Schwarz gap decomposes as*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i, \quad (5.7)$$

$$R_i := XK_i^\dagger - K_i^\dagger E(X). \quad (5.8)$$

Proof. ✓ Proof Expand the right-hand side of (5.7) using (5.8), distribute the products, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to recover the Kadison–Schwarz gap. $\square$

Theorem 5.2.2 (KS equality implies Kraus commutation). Lean ✓ Stmt *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$ for some X , then, for every i ,*

$$XK_i^\dagger = K_i^\dagger E(X). \quad (5.9)$$

Proof. ✓ Proof By (5.7), the vanishing Kadison–Schwarz gap is $\sum_i R_i^\dagger R_i = 0$, with R_i given by (5.8). Each summand is positive semidefinite, so each R_i vanishes. This gives (5.9). $\square$

Theorem 5.2.3 (KS equality for peripheral eigenvectors). Lean ✓ Stmt *Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then the Kadison–Schwarz gap vanishes:*

$$E(X^\dagger X) = E(X)^\dagger E(X). \quad (5.10)$$

Proof. ✓ Proof Set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By (5.3), $G \geq 0$. Trace preservation, $E(X) = \mu X$, and $|\mu| = 1$ give

$$\begin{aligned} \text{tr}(G) &= \text{tr}(E(X^\dagger X)) - \text{tr}(E(X)^\dagger E(X)) \\ &= \text{tr}(X^\dagger X) - |\mu|^2 \text{tr}(X^\dagger X) = 0. \end{aligned}$$

A positive semidefinite matrix with zero trace vanishes, which proves (5.10). $\square$

Theorem 5.2.4 (Left multiplicative identity from KS equality). [Lean](#) [✓ Stmt](#) Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then, for every $Y \in M_D(\mathbb{C})$,

$$E(X^\dagger Y) = E(X)^\dagger E(Y). \quad (5.11)$$

Proof. [✓ Proof](#) By (5.9), $XK_i^\dagger = K_i^\dagger E(X)$ for every i . Taking conjugate transposes gives $K_i X^\dagger = E(X)^\dagger K_i$. Therefore,

$$E(X^\dagger Y) = \sum_i K_i X^\dagger Y K_i^\dagger = \sum_i E(X)^\dagger K_i Y K_i^\dagger = E(X)^\dagger E(Y).$$

□

Theorem 5.2.5 (Right multiplicative identity from KS equality). [Lean](#) [✓ Stmt](#) Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then, for every $Y \in M_D(\mathbb{C})$,

$$E(YX) = E(Y)E(X). \quad (5.12)$$

Proof. [✓ Proof](#) By (5.9), $XK_i^\dagger = K_i^\dagger E(X)$ for every i . Hence

$$E(YX) = \sum_i K_i Y X K_i^\dagger = \sum_i K_i Y K_i^\dagger E(X) = E(Y)E(X).$$

□

5.3 Trace adjoints and positive retractions

The trace-duality identities and positivity results behind the applications below are proved in Section A.1. The density and rank-one support arguments for the retraction theorem are collected in Section A.2.

Theorem 5.3.1 (Positive retractions onto one matrix factor). [Lean](#) [✓ Stmt](#) Let $m, d > 0$, and let $E : M_{md}(\mathbb{C}) \rightarrow M_{md}(\mathbb{C})$ be a positive complex-linear map. Suppose that the image of E is contained in $\mathbb{1}_m \otimes M_d(\mathbb{C})$ and that $E(\mathbb{1}_m \otimes X) = \mathbb{1}_m \otimes X$ for every $X \in M_d(\mathbb{C})$. Then there is a density matrix $\rho \in M_m(\mathbb{C})$ such that, for every $A \in M_{md}(\mathbb{C})$,

$$E(A) = \mathbb{1}_m \otimes \text{tr}_m((\rho \otimes \mathbb{1}_d)A). \quad (5.13)$$

This is the one-factor case of [Wol12, Proposition 1.5 and Equation (1.40)].

Proof. [✓ Proof](#) Let $R(A)$ be the right tensor factor of $E(A)$, so that $E(A) = \mathbb{1}_m \otimes R(A)$. For $Y \geq 0$ and $P_\psi = |\psi\rangle\langle\psi|$, the inequality $Y \leq \text{tr}(Y)\mathbb{1}_m$ and positivity give

$$0 \leq R(Y \otimes P_\psi) \leq \text{tr}(Y)P_\psi.$$

By Lemma A.2.2, for each $v \in \mathbb{C}^d$ there is a scalar c_v such that $R(Y \otimes P_v) = c_v P_v$. By Theorem A.2.3, applied to the map $X \mapsto R(Y \otimes X)$, there is a scalar c_Y such that $R(Y \otimes X) = c_Y X$ for every $X \in M_d(\mathbb{C})$.

Fix a coordinate projection P_0 and define $f(Y) = R(Y \otimes P_0)_{00}$. Rank-one polarization gives $R(Y \otimes X) = f(Y)X$ for all $Y \in M_m(\mathbb{C})$ and $X \in M_d(\mathbb{C})$. Positivity of R gives $f(Y) \geq 0$ when $Y \geq 0$, and the retraction property gives $f(\mathbb{1}_m) = 1$. Theorem A.2.1 therefore supplies a density matrix ρ satisfying $f(Y) = \text{tr}(\rho Y)$.

Finally, on simple tensors,

$$\text{tr}_m((\rho \otimes \mathbb{1}_d)(Y \otimes X)) = \text{tr}(\rho Y)X = R(Y \otimes X).$$

Since matrix units are simple tensors, linearity proves (5.13) for every $A \in M_{md}(\mathbb{C})$. □

Theorem 5.3.2 (Positive retractions onto finite-dimensional star-algebras). [Lean](#) [Stmt](#) Let $S \subseteq M_D(\mathbb{C})$ be a unital $*$ -subalgebra, and let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive complex-linear map whose image lies in S and whose restriction to S is the identity. There are positive integers d_k, m_k , a unitary U , and density matrices $\rho_k \in M_{m_k}(\mathbb{C})$ such that

$$U^* S U = \bigoplus_k \mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C}), \quad (5.14)$$

and, for every $A \in M_D(\mathbb{C})$,

$$E(A) = U \left(\bigoplus_k \mathbb{1}_{m_k} \otimes \text{tr}_{m_k}((\rho_k \otimes \mathbb{1}_{d_k})(U^* A U)_{kk}) \right) U^*. \quad (5.15)$$

Equivalently, cyclicity of the partial trace permits the factor $\rho_k \otimes \mathbb{1}_{d_k}$ to be placed on the right of $(U^* A U)_{kk}$. This is [Wol12, Proposition 1.5 and Equation (1.40)].

Proof. [Proof](#) The standard finite-dimensional Wedderburn theorem in [Wol12, Proposition 1.5] supplies U and the block form (5.14). Conjugate by U to identify the ambient space with $\bigoplus_k \mathbb{C}^{m_k} \otimes \mathbb{C}^{d_k}$. Let P_k be the central projection onto the k -th summand and let T_k be the k -th diagonal block of the conjugated map. Since $T_k(\mathbb{1} - P_k) = 0$, positivity implies $T_k((\mathbb{1} - P_k)A) = T_k(A(\mathbb{1} - P_k)) = 0$. Hence $T_k(A) = T_k(P_k A P_k)$. The restriction to the k -th diagonal summand is a positive retraction onto $\mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C})$, so the one-factor theorem gives the density matrix ρ_k . Taking the direct sum and conjugating back proves (5.15). Entrywise expansion of the partial trace gives

$$\text{tr}_{m_k}((\rho_k \otimes \mathbb{1}) A_{kk}) = \text{tr}_{m_k}(A_{kk}(\rho_k \otimes \mathbb{1})).$$

□

5.4 Douglas-type factorization

Theorem 5.4.1 (Range inclusion implies right factorization). [Lean](#) [Stmt](#) If $\text{ran}(A) \subseteq \text{ran}(B)$, then there exists C such that $A = BC$.

Proof. In finite dimensions, $\text{ran}(A) \subseteq \text{ran}(B)$ implies $A = B(B^\dagger B)^\dagger B^\dagger A$, where $(B^\dagger B)^\dagger$ denotes the Moore–Penrose pseudoinverse. Set $C = (B^\dagger B)^\dagger B^\dagger A$. □

Theorem 5.4.2 (Vectorwise range criterion implies factorization). [Lean](#) [Stmt](#) If $Av \in \text{ran}(B)$ for every vector v , then there exists C with $A = BC$.

Proof. [Proof](#) The pointwise hypothesis is exactly $\text{ran}(A) \subseteq \text{ran}(B)$. Apply the preceding range-inclusion factorization theorem. □

5.5 Peripheral Schwarz equality with a faithful fixed point

Theorem 5.5.1 (Fixed-point peripheral eigenvectors satisfy KS equality). [Lean](#) [Stmt](#) Let E be unital and let E^* be trace-preserving with a positive definite fixed point $\rho > 0$. If $E(X) = \mu X$ with $|\mu| = 1$, then $E(X^\dagger X) = E(X)^\dagger E(X)$.

Proof. ✓ Proof Define the Kadison–Schwarz gap $G = E(X^\dagger X) - E(X)^\dagger E(X)$. By Theorem 5.1.7, $G \succeq 0$. The adjointness identity of Lemma A.3.1, the fixed-point equation $E^*(\rho) = \rho$, the eigenvalue equation $E(X) = \mu X$, and $|\mu| = 1$ give

$$\begin{aligned} \operatorname{tr}(\rho G) &= \operatorname{tr}(\rho E(X^\dagger X)) - \operatorname{tr}(\rho E(X)^\dagger E(X)) \\ &= \operatorname{tr}(E^*(\rho) X^\dagger X) - \operatorname{tr}(\rho (\mu X)^\dagger (\mu X)) \\ &= \operatorname{tr}(\rho X^\dagger X) - |\mu|^2 \operatorname{tr}(\rho X^\dagger X) = 0. \end{aligned}$$

Since $\rho > 0$ and $G \succeq 0$, Lemma A.3.2 yields $G = 0$, which is the asserted equality. $\square$

5.6 Douglas Factorization

Definition 5.6.1 (Support projection of a Hermitian matrix). Lean Lean Lean ✓ Stmt Let A be Hermitian, with spectral decomposition $A = U \operatorname{diag}(\lambda_i) U^\dagger$. Its support projection is

$$P_A = U \operatorname{diag}(\mathbf{1}_{\lambda_i \neq 0}) U^\dagger, \quad (5.16)$$

the orthogonal projection onto the range of A .

Theorem 5.6.2 (Douglas factorisation). Lean ✓ Stmt Let $A, B \in M_D(\mathbb{C})$. The following are equivalent:

1. $\operatorname{range}(A) \subseteq \operatorname{range}(B)$,
2. $AA^\dagger \leq \mu BB^\dagger$ for some $\mu \geq 0$,
3. $A = BC$ for some $C \in M_D(\mathbb{C})$.

Definition 5.6.3 (Pseudoinverse). Lean ✓ Stmt The Moore–Penrose pseudoinverse B^+ is defined via the support-generalised inverse of $BB^\dagger$: $B^+ := B^\dagger \cdot (BB^\dagger)_{\operatorname{supp}}^{-1}$.

Lemma 5.6.4 (Pseudoinverse-factorization identity). Lean ✓ Stmt $B \cdot B^+ = P_{\operatorname{supp}}(BB^\dagger)$, where $P_{\operatorname{supp}}(X)$ denotes the support projection of X (the orthogonal projection onto the range of X).

Theorem 5.6.5 (Explicit factorization via pseudoinverse). Lean ✓ Stmt If $\operatorname{range}(A) \subseteq \operatorname{range}(B)$, then $A = B \cdot (B^+ A)$.

Theorem 5.6.6 (Uniqueness of the pseudoinverse factorization). Lean ✓ Stmt If $A = BC_1 = BC_2$ and $\operatorname{range}(C_1), \operatorname{range}(C_2) \subseteq \operatorname{range}(B^\dagger)$, then $C_1 = C_2$. In particular, $C = B^+ A$ is the unique such factor.

Theorem 5.6.7 (Norm-minimal factorization). Lean ✓ Stmt For every factorisation $A = BC$, the pseudoinverse witness attains the minimal operator norm: $\|B^+ A\| \leq \|C\|$.

5.7 Block Schur Complements

Theorem 5.7.1 (Wolf’s printed block Schur complement statement). Not ready Let $P \in M_{D_1}(\mathbb{C})$, $R \in M_{D_2}(\mathbb{C})$ be positive semi-definite and let $Q \in M_{D_1, D_2}(\mathbb{C})$. Wolf’s Theorem 5.2 states that the following are equivalent:

1. The block matrix $\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix}$ is PSD,

2. $\ker(R) \subseteq \ker(Q)$ and $P \geq QR^+Q^\dagger$,
3. $\ker(R) \subseteq \ker(Q)$ and $\|P^{-1/2}QR^{-1/2}\|_\infty \leq 1$.

Here R^+ denotes the pseudoinverse (inverse on the support) and the inverse square roots are extended by zero on the respective kernels.

Local fix (Wolf Theorem 5.2, condition (3)): The third condition is false as printed when P is singular. It becomes equivalent to the first two after adjoining $\ker(P) \subseteq \ker(Q^\dagger)$. The counterexample and corrected theorem are recorded below and in `docs/paper-gaps/schur_complement_tfae.tex`.

Definition 5.7.2 (Block matrix). [Lean](#) [✓ Stmt](#) A 2×2 block matrix $\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix}$ indexed by $\{0, \dots, D_1-1\} \oplus \{0, \dots, D_2-1\}$.

Lemma 5.7.3 (Hermiticity of the block matrix). [Lean](#) [✓ Stmt](#) If P and R are Hermitian, then the block matrix $\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix}$ is Hermitian.

Theorem 5.7.4 (Quadratic form of the block matrix). [Lean](#) [✓ Stmt](#) For vectors $x \in \mathbb{C}^{D_1}$ and $y \in \mathbb{C}^{D_2}$,

$$\begin{pmatrix} x \\ y \end{pmatrix}^\dagger \begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = x^\dagger Px + x^\dagger Qy + y^\dagger Q^\dagger x + y^\dagger Ry. \quad (5.17)$$

Proof. [✓ Proof](#) Multiplication by the block matrix sends (x, y) to $(Px + Qy, Q^\dagger x + Ry)$. Taking the inner product with (x, y) and distributing over both sums gives the four displayed terms. $\square$

Definition 5.7.5 (Schur complement). [Lean](#) [✓ Stmt](#) The pseudoinverse Schur complement $P - QR^+Q^\dagger$.

Theorem 5.7.6 (Support absorption under kernel inclusion). [Lean](#) [✓ Stmt](#) Let R be Hermitian and let Q be a matrix with the same column index such that $\ker(R) \subseteq \ker(Q)$. Then $Q \operatorname{supp}(R) = Q$.

Proof. [✓ Proof](#) Since $R(1 - \operatorname{supp}(R)) = 0$, the kernel inclusion gives $Q(1 - \operatorname{supp}(R)) = 0$. Expanding this equality yields $Q - Q \operatorname{supp}(R) = 0$, and hence $Q \operatorname{supp}(R) = Q$. $\square$

Theorem 5.7.7 (Pseudoinverse support identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If R is positive semi-definite, then $RR^+ = R^+R = \operatorname{supp}(R)$, $\operatorname{supp}(R)R^+ = R^+$, and R^+ is Hermitian.

Proof. [✓ Proof](#) The pseudoinverse-factorization identity, together with equality of the supports of R^2 and R , gives $RR^+ = \operatorname{supp}(R)$. The spectral calculus shows that the support inverse of R^2 commutes with R ; the definition of R^+ then shows that R^+ is Hermitian. Taking adjoints in the first identity gives $R^+R = \operatorname{supp}(R)$. Finally, write $R = \sqrt{R}\sqrt{R}$. The support projection absorbs $\sqrt{R}$, and the defining factorization of R^+ therefore gives $\operatorname{supp}(R)R^+ = R^+$. $\square$

Theorem 5.7.8 (Kernel inclusion from block positivity). [Lean](#) [✓ Stmt](#) If $\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix}$ is positive semi-definite, then $\ker(R) \subseteq \ker(Q)$.

Proof. [✓ Proof](#) A vector in $\ker(R)$ has zero quadratic form in the lower-right principal subspace. Positivity therefore places the corresponding block vector in the kernel of the full matrix, whose upper component is Qy . $\square$

Theorem 5.7.9 (Pseudoinverse completion of the square). [Lean](#) [✓ Stmt](#) If R is positive semi-definite and $\ker(R) \subseteq \ker(Q)$, then

$$\begin{pmatrix} x \\ y \end{pmatrix}^\dagger \begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (R^+ Q^\dagger x + y)^\dagger R (R^+ Q^\dagger x + y) + x^\dagger (P - Q R^+ Q^\dagger) x. \quad (5.18)$$

Proof. [✓ Proof](#) The kernel inclusion gives $Q \operatorname{supp}(R) = Q$. Expanding the right-hand side and using $RR^+ = R^+R = \operatorname{supp}(R)$ and $(R^+)^\dagger = R^+$ yields the four terms of the block quadratic form. $\square$

Theorem 5.7.10 (Schur-complement positivity from block positivity). [Lean](#) [✓ Stmt](#) If P and R are positive semi-definite and $\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix}$ is positive semi-definite, then $P - Q R^+ Q^\dagger$ is positive semi-definite.

Proof. [✓ Proof](#) Since P and R^+ are Hermitian, so are $Q R^+ Q^\dagger$ and $P - Q R^+ Q^\dagger$. By Theorem 5.7.8, $\ker(R) \subseteq \ker(Q)$. In Theorem 5.7.9, take $y = -R^+ Q^\dagger x$. The completed-square term vanishes, and positivity of the block matrix gives $x^\dagger (P - Q R^+ Q^\dagger) x \geq 0$ for every x . $\square$

Theorem 5.7.11 (Block positivity from Schur-complement positivity). [Lean](#) [✓ Stmt](#) Suppose that P and R are positive semi-definite, $\ker(R) \subseteq \ker(Q)$, and $P - Q R^+ Q^\dagger$ is positive semi-definite. Then $\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix}$ is positive semi-definite.

Proof. [✓ Proof](#) Theorem 5.7.9 writes every quadratic form of the block matrix as the sum of quadratic forms of R and of $P - Q R^+ Q^\dagger$. Both are nonnegative. $\square$

Theorem 5.7.12 (Block positivity and the pseudoinverse Schur complement). [Lean](#) [✓ Stmt](#) Let P and R be positive semi-definite. Then

$$\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix} \geq 0 \iff \ker(R) \subseteq \ker(Q) \text{ and } P - Q R^+ Q^\dagger \geq 0. \quad (5.19)$$

Scope restriction (Wolf Theorem 5.2, conditions (1) and (2)): This proves exactly the first two conditions of the source theorem. The printed contraction condition (3) is false in the singular case, as recorded in `docs/paper-gaps/schur_complement_tfae.tex`.

Proof. [✓ Proof](#) The forward implication combines kernel inclusion with positivity of the Schur complement. The reverse implication is Theorem 5.7.11. $\square$

Theorem 5.7.13 (Pseudoinverse as inverse on the support). [Lean](#) [✓ Stmt](#) If R is positive semi-definite, then its Moore–Penrose pseudoinverse agrees with the inverse obtained by inverting R on its support and setting it to zero on the kernel.

Definition 5.7.14 (Support-normalized off-diagonal block). [Lean](#) [✓ Stmt](#) For positive semi-definite P and R , define

$$K = P^{-1/2} Q R^{-1/2}, \quad (5.20)$$

where both inverse square roots vanish on the respective kernels.

Theorem 5.7.15 (Normalized Schur term). [Lean](#) [✓ Stmt](#) The support-normalized off-diagonal block satisfies

$$K K^\dagger = P^{-1/2} (Q R^+ Q^\dagger) P^{-1/2}. \quad (5.21)$$

Theorem 5.7.16 (Positivity of the Schur term). [Lean](#) [✓ Stmt](#) If R is positive semi-definite, then $QR^+Q^\dagger$ is positive semi-definite.

Theorem 5.7.17 (Congruence criterion for domination). [Lean](#) [✓ Stmt](#) Let A and B be positive semi-definite, let $c > 0$, and suppose that $\ker(A) \subseteq \ker(B)$. Then

$$A - cB \geq 0 \iff c\|A^{-1/2}BA^{-1/2}\|_\infty \leq 1, \quad (5.22)$$

where the inverse square root vanishes on $\ker(A)$.

Theorem 5.7.18 (Left support from Schur-complement positivity). [Lean](#) [✓ Stmt](#) Suppose that P and R are positive semi-definite, $\ker(R) \subseteq \ker(Q)$, and $P - QR^+Q^\dagger$ is positive semi-definite. Then $\ker(P) \subseteq \ker(Q^\dagger)$.

Theorem 5.7.19 (Corrected Schur-complement contraction criterion). [Lean](#) [✓ Stmt](#) Let P and R be positive semi-definite and suppose that $\ker(R) \subseteq \ker(Q)$. Then

$$P - QR^+Q^\dagger \geq 0 \iff \ker(P) \subseteq \ker(Q^\dagger) \text{ and } \|P^{-1/2}QR^{-1/2}\|_\infty \leq 1. \quad (5.23)$$

Local fix (Wolf Theorem 5.2, condition (3)): The left-support condition is necessary and is absent from the printed statement. See `docs/paper-gaps/schur_complement_tf.ae.tex`.

Proof. [✓ Proof](#) Put $B = QR^+Q^\dagger$ and $K = P^{-1/2}QR^{-1/2}$. The two support conditions allow congruence by $P^{-1/2}$ to identify $P - B \geq 0$ with $\|P^{-1/2}BP^{-1/2}\|_\infty \leq 1$. By Theorem 5.7.15, the matrix inside this norm is $KK^\dagger$, whose norm is $\|K\|_\infty^2$. $\square$

Theorem 5.7.20 (Counterexample to Wolf's printed condition (3)). [Lean](#) [✓ Stmt](#) For the scalar blocks $P = 0$ and $Q = R = 1$, the printed kernel condition and contraction bound hold, but $\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix}$ is not positive semi-definite.

Proof. [✓ Proof](#) The normalized off-diagonal block vanishes because $P^{-1/2} = 0$. However, the quadratic form of $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ at $(1, -1)$ equals -1 . $\square$

Theorem 5.7.21 (Corrected block contraction criterion). [Lean](#) [✓ Stmt](#) Let P and R be positive semi-definite. Then

$$\begin{pmatrix} P & Q \\ Q^\dagger & R \end{pmatrix} \geq 0 \iff \ker(R) \subseteq \ker(Q), \quad (5.24)$$

$$\ker(P) \subseteq \ker(Q^\dagger), \quad (5.25)$$

$$\|P^{-1/2}QR^{-1/2}\|_\infty \leq 1. \quad (5.26)$$

Local fix (Wolf Theorem 5.2, condition (3)): This is the source statement with its missing left-support condition restored. See `docs/paper-gaps/schur_complement_tf.ae.tex`.

Proof. [✓ Proof](#) Combine the pseudoinverse Schur-complement criterion with the corrected contraction criterion. $\square$

Chapter 6

Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes elementary fixed-point existence for quantum channels and the Perron–Frobenius theory: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, and the spectral-radius identification of Perron eigenvalues. The ergodicity of irreducible channels and the CP-map irreducibility criterion modeled on Wolf’s spectral characterization appear in the auxiliary channel-theory chapter following the Fundamental Theorem sequence. The conceptual source is Wolf’s treatment of irreducible positive maps [Wol12, Chapter 6, especially Theorems 6.2–6.5 and Corollary 6.3] and [EHK78].

6.1 Cesàro fixed points for quantum channels

Definition 6.1.1 (Cesàro mean). Lean ✓ Stmt The *Cesàro mean* of a linear map E at order N , applied to a starting matrix ρ_0 , is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0). \quad (6.1)$$

Theorem 6.1.2 (Cesàro telescope). Lean ✓ Stmt For $N \geq 1$, the Cesàro mean satisfies

$$E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0). \quad (6.2)$$

Proof. ✓ Proof Applying E to (6.1) gives $E(\sigma_N) = \frac{1}{N} \sum_{n=1}^N E^n(\rho_0)$. Subtracting σ_N cancels all but the first and last terms, yielding (6.2). $\square$

Lemma 6.1.3 (Dirichlet’s simultaneous approximation). Lean ✓ Stmt Let $x_1, \dots, x_m$ be m real numbers and $q > 1$ any integer. Then there exist integers $n, p_1, \dots, p_m$ such that $1 \leq n \leq q^m$ and $|x_k n - p_k| \leq 1/q$ for all k . This is [Wol12, Lemma 6.1].

Proof. ✓ Proof Pigeonhole principle on the integer lattice of side q^m : the $q^m + 1$ points $(\{nx_1\}, \dots, \{nx_m\})$ for $n = 0, \dots, q^m$ lie in q^m cells; two share a cell, and their difference yields n . $\square$

Corollary 6.1.4 (Recurrent powers of finitely many phases). [Lean](#) [Lean](#) [✓ Stmt](#) For finitely many complex numbers θ_j satisfying $|\theta_j| = 1$, there is a strictly increasing sequence of positive integers n_i such that $\theta_j^{n_i} \rightarrow 1$ simultaneously for every j .

Proof. [✓ Proof](#) Write each phase as $\theta_j = e^{2\pi i x_j}$. Apply simultaneous approximation with successively larger denominators and choose the approximating exponents recursively beyond the preceding one. $\square$

Definition 6.1.5 (Peripheral spectral splitting). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a finite-dimensional complex vector space. Its peripheral subspace is the sum of the maximal generalized eigenspaces for eigenvalues μ with $|\mu| = 1$; its non-peripheral subspace is the corresponding sum for the remaining eigenvalues. This is the splitting induced by the full spectral decomposition of [Wol12, Equation (6.5)], with the peripheral projection itself defined in [Wol12, Equation (6.12)].

Theorem 6.1.6 (Complementarity of the peripheral splitting). [Lean](#) [✓ Stmt](#) The peripheral and non-peripheral spectral subspaces are complementary.

Lemma 6.1.7 (Powers vanish on the non-peripheral subspace). [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a finite-dimensional complex vector space whose every eigenvalue has modulus at most one. Then $T^n(Y) \rightarrow 0$ for every Y in the non-peripheral subspace:

$$T^n(Y) \rightarrow 0. \quad (6.3)$$

Proof. [✓ Proof](#) Every non-peripheral eigenvalue μ has modulus strictly below one. On the corresponding maximal generalized eigenspace write $T = \mu\mathbb{1} + N$ with $N^\ell = 0$; the binomial expansion gives

$$(\mu\mathbb{1} + N)^n = \sum_{m=0}^{\ell-1} \binom{n}{m} \mu^{n-m} N^m \rightarrow 0, \quad (6.4)$$

since each summand carries the factor $\mu^n \rightarrow 0$ against a polynomial in n . The non-peripheral subspace is the sum of these eigenspaces, and convergence to zero is preserved by finite sums. $\square$

Theorem 6.1.8 (Powers on an eigenspace). [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a vector space over a field. If $T(x) = \mu x$, then

$$T^n(x) = \mu^n x$$

for every $n \geq 0$.

Proof. [✓ Proof](#) Induct on n . The induction step follows from linearity: $T(\mu^n x) = \mu^n T(x) = \mu^{n+1} x$. $\square$

Theorem 6.1.9 (Recurrence on a finite sum of eigenspaces). [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a complex normed space, let $S \subseteq \mathbb{C}$ be finite, and let $(n_k)_{k \geq 0}$ be a sequence of nonnegative integers such that $\mu^{n_k} \rightarrow 1$ for every $\mu \in S$. If x belongs to the sum of the μ -eigenspaces for $\mu \in S$, then $T^{n_k}(x) \rightarrow x$.

Proof. [✓ Proof](#) Write $x = \sum_{\mu \in S} x_\mu$ with $T(x_\mu) = \mu x_\mu$. By Theorem 6.1.8, $T^{n_k}(x_\mu) = \mu^{n_k} x_\mu \rightarrow x_\mu$. The conclusion follows by summing over the finite set S . $\square$

Lemma 6.1.15 (Closedness of the completely positive maps). [Lean](#) [Lean](#) [✓ Stmt](#) Let $D > 0$. The completely positive maps form a closed subset of the endomorphisms of $M_D(\mathbb{C})$ in the operator norm. In particular, if every $S_N : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is completely positive and $S_N \rightarrow P$ in the operator norm, then P is completely positive.

Proof. [✓ Proof](#) By Theorem 13.10.10 a map is completely positive precisely when its Choi matrix is positive semidefinite. The Choi matrix depends linearly, hence continuously, on the map, and the positive semidefinite matrices are closed. The set of completely positive maps is therefore the preimage of a closed set under a continuous map. $\square$

Corollary 6.1.16 (Channel structure of the mean-ergodic projection). [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be completely positive with bounded orbits. Then the mean-ergodic projection T_∞ of [Wol12, Equation (6.14)] is completely positive. If T is completely positive and trace-preserving, the bounded orbits are automatic and T_∞ is completely positive and trace-preserving, hence a quantum channel. Together with Corollary 6.1.13 this completes the preservation assertion of [Wol12, Proposition 6.3] for all three maps T_∞ , T_φ , and T'_φ .

Proof. [✓ Proof](#) For $D = 0$ the algebra is trivial and every linear map is completely positive, so assume $D > 0$. Write

$$S_N = \frac{1}{N} \sum_{n=0}^{N-1} T^n \quad (6.8)$$

for the N -th Cesàro average. Powers of a completely positive map are completely positive, and so are finite sums and nonnegative multiples of completely positive maps, so every S_N is completely positive. For every $X \in M_D(\mathbb{C})$, (??) gives

$$S_N(X) \longrightarrow T_\infty(X), \quad (6.9)$$

and Lemma 6.1.14 upgrades this to

$$S_N \longrightarrow T_\infty \quad (6.10)$$

in the operator norm. Lemma 6.1.15 then makes T_∞ completely positive. Trace preservation of T_∞ is Theorem ??, and the bounded orbits it requires come from Theorem ??. $\square$

Theorem 6.1.17 (Peripheral absorption of the mean-ergodic projection). [Lean](#) [✓ Stmt](#) If T_∞ denotes the mean-ergodic projection of a positive trace-preserving map, then

$$T_\varphi T_\infty = T_\infty. \quad (6.11)$$

This is the absorption identity adjacent to [Wol12, Equation (6.14)].

Proof. [✓ Proof](#) The range of T_∞ consists of fixed points of T , hence every power of T fixes this range. Passing to the recurrent-subsequence limit shows that T_φ also fixes it pointwise. $\square$

Theorem 6.1.18 (Asymptotic image). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving, with $D > 0$, and let

$$\mathcal{X}_T = \text{span}\{X \in M_D(\mathbb{C}) \mid \exists \varphi \in \mathbb{R} : T(X) = e^{i\varphi} X\} \quad (6.12)$$

be the span of its peripheral eigenvectors. Then

1. $T_\varphi(M_D(\mathbb{C})) = \mathcal{X}_T$,

2. there are positive semidefinite matrices ρ_i with $\mathcal{X}_T = \text{span}\{\rho_i\}$,
3. $T(\mathcal{X}_T) = \mathcal{X}_T$.

This is [Wol12, Proposition 6.12 (Asymptotic image)].

Proof. ✓ Proof Write $V_\mu = \ker(T - \mu)$ for the eigenspace of a peripheral eigenvalue μ , so that $\mathcal{X}_T = \sum_{|\mu|=1} V_\mu$ by the definition of $\mathcal{X}_T$. The image of the projection T_φ is its set of fixed points, and the fixed points of T_φ are the linear combinations of peripheral eigenvectors:

$$T_\varphi(M_D(\mathbb{C})) = \{X \in M_D(\mathbb{C}) \mid T_\varphi(X) = X\} \quad (6.13)$$

$$= \sum_{|\mu|=1} V_\mu \quad (6.14)$$

$$= \mathcal{X}_T. \quad (6.15)$$

The equality (6.13) holds because T_φ is a projection; a matrix fixed by T_φ lies in the peripheral subspace, whose Jordan blocks are trivial for a positive trace-preserving map (Theorem 6.8.7), which gives (6.14); and (6.15) is the definition of $\mathcal{X}_T$. This is the first claim.

The projection T_φ is itself positive and trace-preserving, so its fixed-point space is spanned by its stationary density matrices:

$$\mathcal{X}_T = \{X \in M_D(\mathbb{C}) \mid T_\varphi(X) = X\} = \text{span}\{\rho \mid \rho \text{ a stationary density matrix of } T_\varphi\}. \quad (6.16)$$

Every such ρ is positive semidefinite, which is the second claim.

On V_μ the map T acts as multiplication by μ , and $|\mu| = 1$ forces $\mu \neq 0$, so multiplication by μ carries V_μ onto itself and $T(V_\mu) = \mu V_\mu = V_\mu$. Summing over the peripheral eigenvalues,

$$T(\mathcal{X}_T) = T\left(\sum_{|\mu|=1} V_\mu\right) = \sum_{|\mu|=1} T(V_\mu) = \sum_{|\mu|=1} V_\mu = \mathcal{X}_T, \quad (6.17)$$

which is the third claim. □

Theorem 6.1.19 (Cesàro fixed-point theorem). Lean ✓ Stmt Every quantum channel $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, with $D \geq 1$, has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.

Proof. ✓ Proof Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem B.1.3). Lemma B.1.6 keeps every σ_N , $N \geq 1$, in the compact set $\mathcal{D}_D$, so a subsequence converges to some $\rho \in \mathcal{D}_D$. The telescope identity (6.2) gives

$$\|E(\sigma_N) - \sigma_N\| = \frac{1}{N} \|E^N(\rho_0) - \rho_0\| \longrightarrow 0,$$

because the iterates $E^N(\rho_0)$ lie in the compact set $\mathcal{D}_D$ and are therefore uniformly bounded. Along the convergent subsequence, continuity of E now gives $E(\rho) = \rho$. In [Wol12, Theorem 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead. □

6.2 The phase-weighted Cesàro formula for the peripheral projection

Lemma 6.2.1 (Orthogonality of unit-modulus phases). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let s be a finite set of complex numbers of modulus one and let $\lambda \in s$. Then

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \sum_{\mu \in s} (\bar{\mu} \lambda)^n = 1. \quad (6.18)$$

Proof. [✓ Proof](#) Every ratio $\bar{\mu} \lambda$ again has modulus one, and $\bar{\mu} \lambda = 1$ forces $\lambda = \mu$, since $\mu \bar{\mu} = 1$. The term $\mu = \lambda$ therefore equals 1 for every n and contributes 1 to the mean. For $\mu \neq \lambda$ set $\zeta = \bar{\mu} \lambda \neq 1$; the geometric sum

$$\sum_{n=1}^N \zeta^n = \frac{\zeta - \zeta^{N+1}}{1 - \zeta} \quad (6.19)$$

is bounded by $2/|1 - \zeta|$ uniformly in N , so dividing by N sends it to zero. $\square$

Lemma 6.2.2 (Cesàro means of a vanishing sequence). [Lean](#) [✓ Stmt](#) Let $u_1, u_2, \dots$ be a sequence in a complex normed space with $u_n \rightarrow 0$. Then

$$\frac{1}{N} \sum_{n=1}^N u_n \rightarrow 0. \quad (6.20)$$

Proof. [✓ Proof](#) The Cesàro mean of a convergent sequence converges to the same limit. $\square$

Definition 6.2.3 (Phase-weighted Cesàro mean). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a complex vector space and let s be a finite set of complex numbers. The *phase-weighted Cesàro mean* of T over s is

$$C_N(T, s) := \frac{1}{N} \sum_{n=1}^N \sum_{\mu \in s} (\bar{\mu} T)^n. \quad (6.21)$$

Taking for s the eigenvalues of T of modulus one gives the averages of [Wol12, Equation (6.15)].

Lemma 6.2.4 (Action of the phase-weighted means on the spectral splitting). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be an endomorphism of a complex normed space and let s be a finite set of complex numbers of modulus one.

- (i) On the span of the eigenspaces of T belonging to the elements of s , the means $C_N(T, s)$ converge to the identity.
- (ii) At a vector X with $T^n(X) \rightarrow 0$, the means $C_N(T, s)(X)$ converge to zero.

Proof. [✓ Proof](#) For (i) it suffices to treat a λ -eigenvector X with $\lambda \in s$, because the span is the sum of the eigenspaces. There $(\bar{\mu} T)^n(X) = (\bar{\mu} \lambda)^n X$, so $C_N(T, s)(X)$ is the scalar $\frac{1}{N} \sum_{n=1}^N \sum_{\mu \in s} (\bar{\mu} \lambda)^n$ times X , which tends to X by (6.18). For (ii), multiplication by a unit-modulus scalar does not change norms, so $\|(\bar{\mu} T)^n(X)\| = \|T^n(X)\| \rightarrow 0$ for every $\mu \in s$; the inner sum is finite, hence also tends to zero, and its Cesàro means tend to zero. $\square$

Theorem 6.2.5 (Phase-weighted Cesàro formula for T_φ). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive and trace-preserving, with $D > 0$, and let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T of modulus one. Then

$$T_\varphi = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \sum_{k=1}^m (\bar{\lambda}_k T)^n, \quad (6.22)$$

both pointwise and in operator norm. This is [Wol12, Equation (6.15)].

Proof. [Proof](#) Split $X = T_\varphi(X) + (X - T_\varphi(X))$ along the complementary spectral subspaces. Bounded forward orbits force every eigenvalue to have modulus at most one, so every eigenvalue outside the peripheral spectrum has modulus strictly below one and $T^n(Y) \rightarrow 0$ for every Y in the non-peripheral subspace; part (ii) of Lemma 6.2.4 sends the second summand to zero. Triviality of the peripheral Jordan blocks identifies the peripheral subspace with the span of the peripheral eigenspaces, on which part (i) of the same lemma acts as the identity, so the first summand converges to $T_\varphi(X)$. Pointwise convergence of endomorphisms of a finite-dimensional space is convergence in operator norm. $\square$

6.3 Positive definiteness

Theorem 6.3.1 (Positive-definite growth for irreducible CP maps). [Lean](#) [Stmnt](#) Let E be an irreducible completely positive map on $M_D(\mathbb{C})$ and let $A \geq 0$ be nonzero. Then

$$(\mathbb{1} + E)^{D-1}(A) > 0. \quad (6.23)$$

This is the completely positive specialization of the implication (1) $\Rightarrow$ (2) in [Wol12, Theorem 6.2].

Proof. [Proof](#) Choose a Kraus decomposition

$$E(X) = \sum_i K_i X K_i^\dagger$$

and set $T(B) = B + E(B)$ for nonzero $B \geq 0$. Always $\ker T(B) \subseteq \ker B$: if $v \in \ker T(B)$, then

$$0 = v^\dagger B v + v^\dagger E(B) v,$$

and both terms are non-negative, so $v^\dagger B v = 0$ and hence $B v = 0$. If B is not positive definite, irreducibility forces

$$\dim \ker T(B) < \dim \ker B.$$

Indeed, equality of the kernel dimensions would turn the inclusion above into $\ker B \subseteq \ker T(B)$. Thus, for $v \in \ker B$,

$$0 = v^\dagger E(B) v = \sum_i (K_i^\dagger v)^\dagger B (K_i^\dagger v).$$

Every summand is non-negative, so equality would imply, for every i ,

$$K_i^\dagger(\ker B) \subseteq \ker B.$$

Equivalently, $\text{supp } B$ would be invariant under every K_i , making the support projection of B a nontrivial invariant projection.

We prove by induction on n that every nonzero $B \geq 0$ with $\dim \ker B \leq n$ satisfies $T^n(B) > 0$. For $n = 0$, the kernel is trivial, so $B > 0$. Suppose the claim holds for n and let $\dim \ker B \leq n+1$. If $B > 0$, then $T(B) = B + E(B) > 0$, and every subsequent iterate remains positive definite. Otherwise the strict kernel decrease gives $\dim \ker T(B) \leq n$; the induction hypothesis applied to $T(B)$ yields $T^n(T(B)) = T^{n+1}(B) > 0$. Since $A \neq 0$, rank-nullity gives $\dim \ker A \leq D-1$, which proves (6.23). $\square$

Theorem 6.3.2 (Kernel inclusion forces positive definiteness). [Lean](#) [✓ Stmt](#) *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. If $\rho \geq 0$, $\rho \neq 0$, and*

$$\ker \rho \subseteq \ker E(\rho),$$

then ρ is positive definite.

Proof. [✓ Proof](#) Write $E(X) = \sum_i K_i X K_i^\dagger$ and let Q be the support projection of ρ . For $v \in \ker \rho$, the kernel inclusion gives

$$0 = v^\dagger E(\rho) v = \sum_i (K_i^\dagger v)^\dagger \rho (K_i^\dagger v).$$

Every summand is non-negative, so $K_i^\dagger(\ker \rho) \subseteq \ker \rho$ for every i . Equivalently, for every X ,

$$QE(QXQ)Q = E(QXQ).$$

Irreducibility forces $Q \in \{0, I\}$, while $\rho \neq 0$ rules out $Q = 0$. Thus $Q = I$, so $\ker \rho = \{0\}$ and $\rho > 0$. $\square$

Theorem 6.3.3 (PSD eigenvectors are positive definite). [Lean](#) [✓ Stmt](#) *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. If $\rho \geq 0$, $\rho \neq 0$, $r > 0$, and $E(\rho) = r\rho$, then ρ is positive definite.*

Proof. [✓ Proof](#) The eigenvector equation gives

$$\ker \rho = \ker E(\rho).$$

Hence $\ker \rho \subseteq \ker E(\rho)$. Theorem 6.3.2 then forces ρ to be positive definite. $\square$

Definition 6.3.4 (Support projection). [Lean](#) [✓ Stmt](#) For a positive semidefinite matrix $\rho \geq 0$, the *support projection* P is the orthogonal projection onto the range of ρ . Via the spectral decomposition $\rho = U \operatorname{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$, it is

$$P = U \operatorname{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger, \tag{6.24}$$

where $\mathbf{1}_{\lambda_j > 0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

Theorem 6.3.5 (Vanishing complementary Kraus corners). [Lean](#) [✓ Stmt](#) *Let $\rho \geq 0$ be fixed by the Kraus map of a finite matrix family K , and let Q be the support projection of ρ . Then Q is an orthogonal projection and*

$$(\mathbb{1} - Q)K_i Q = 0$$

for every i .

Proof. ✓ Proof The matrices Q and ρ have the same kernel. For a vector in the orthogonal complement of the range of Q , the fixed-point identity writes its zero quadratic form against ρ as a sum of nonnegative quadratic forms. Each summand vanishes, so $QK_i^\dagger(\mathbb{1} - Q) = 0$. Taking adjoints gives $(\mathbb{1} - Q)K_iQ = 0$. $\square$

Theorem 6.3.6 (Vanishing lower corners imply invariant compression). Lean ✓ Stmt *Let K be a finite matrix family with Kraus map E , and let Q be an orthogonal projection. If $(\mathbb{1} - Q)K_iQ = 0$ for every i , then*

$$QE(QXQ)Q = E(QXQ)$$

for every $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof The lower-corner identity gives $K_iQ = QK_iQ$; taking adjoints gives $QK_i^\dagger = QK_i^\dagger Q$. Substituting these two equalities in every summand of the Kraus representation of $E(QXQ)$ proves the compression identity. $\square$

Theorem 6.3.7 (Invariant compression implies vanishing lower corners). Lean ✓ Stmt *Let K be a finite matrix family with Kraus map E , and let Q be an orthogonal projection. If*

$$QE(QXQ)Q = E(QXQ)$$

for every $X \in M_D(\mathbb{C})$, then $(\mathbb{1} - Q)K_iQ = 0$ for every i .

Proof. ✓ Proof Set $M_i = (\mathbb{1} - Q)K_iQ$. Multiplying the compression identity by $\mathbb{1} - Q$ on the left gives the first equality below. Taking $X = \mathbb{1}$, multiplying on the right by $\mathbb{1} - Q$, and expanding the Kraus sum gives the second:

$$\begin{aligned} (\mathbb{1} - Q)E(QXQ) &= 0, \\ (\mathbb{1} - Q)E(Q)(\mathbb{1} - Q) &= \sum_i M_i M_i^\dagger = 0. \end{aligned}$$

Each matrix $M_i M_i^\dagger$ is positive semidefinite. Hence every term in the vanishing sum is zero, and therefore $M_i = 0$ for every i . $\square$

Theorem 6.3.8 (Invariant support compression for a CP fixed point). Lean ✓ Stmt *Let E be completely positive, let $\rho \geq 0$ satisfy $E(\rho) = \rho$, and let Q be the support projection of ρ . Then Q is an orthogonal projection and*

$$QE(QXQ)Q = E(QXQ). \tag{6.25}$$

for every $X \in M_D(\mathbb{C})$.

Proof. ✓ Proof The fixed-point equation makes the complementary Kraus corners vanish, $(\mathbb{1} - Q)K_iQ = 0$. Expanding the Kraus sum then gives Equation (6.25). $\square$

Theorem 6.3.9 (Nonzero matrix has nonzero support). Lean ✓ Stmt *If $\rho \geq 0$ and $\rho \neq 0$, then its support projection is nonzero.*

Proof. ✓ Proof A zero support projection would give $\rho = Q\rho = 0$. $\square$

Theorem 6.3.10 (Full support implies positive definiteness). Lean ✓ Stmt *If $\rho \geq 0$ and its support projection is the identity, then ρ is positive definite.*

Proof. ✓ **Proof** The support is the orthogonal complement of the kernel. Thus full support makes the kernel trivial, so ρ is invertible and positive definite. $\square$

Theorem 6.3.11 (PSD fixed point is positive definite — irreducible version). Lean ✓ **Stmnt** *If E is completely positive and irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point of E , then ρ is positive definite.*

Proof. ✓ **Proof** Let Q be the support projection of ρ . Theorem 6.3.8 makes Q an invariant orthogonal projection, so irreducibility gives $Q = 0$ or $Q = 1$. Theorem 6.3.9 excludes $Q = 0$, and Theorem 6.3.10 turns $Q = 1$ into positive definiteness. $\square$

Theorem 6.3.12 (Transfer-map PSD fixed point — irreducible version). Lean ✓ **Stmnt** *Let A be an MPS tensor whose transfer map $\mathcal{E}_A$ is irreducible. If $\rho \geq 0$, $\rho \neq 0$, and $\mathcal{E}_A(\rho) = \rho$, then ρ is positive definite.*

Proof. ✓ **Proof** The transfer map is completely positive, so Theorem 6.3.11 applies. $\square$

Theorem 6.3.13 (PSD fixed point is positive definite). Lean ✓ **Stmnt** *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of $\mathcal{E}_A$. Then ρ is positive definite.*

Proof. ✓ **Proof** Injectivity implies irreducibility of the transfer map. Apply Theorem 6.3.12. $\square$

Theorem 6.3.14 (Orthogonal trace condition). Lean ✓ **Stmnt** *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, and let $A, B \geq 0$ be nonzero positive semidefinite matrices with $\text{tr}(BA) = 0$. Then there exists t with $1 \leq t \leq D - 1$ such that $\text{tr}(BE^t(A)) > 0$.*

Proof. ✓ **Proof** Theorem 6.3.1 gives $(1 + E)^{D-1}(A) > 0$. Since $B \geq 0$ is nonzero, $\text{tr}(B(1 + E)^{D-1}(A)) > 0$. Expanding by the binomial theorem gives

$$\text{tr}\left(B \sum_{k=0}^{D-1} \binom{D-1}{k} E^k(A)\right) > 0.$$

Each term $\text{tr}(BE^k(A))$ is non-negative since both factors are positive semidefinite. The $k = 0$ term vanishes by the hypothesis $\text{tr}(BA) = 0$. Hence at least one $k \in \{1, \dots, D - 1\}$ contributes a strictly positive term. This proves the implication (2) $\Rightarrow$ (4) in [Wol12, Theorem 6.2(4)] by a binomial expansion and positivity. $\square$

6.4 Uniqueness

Lemma 6.4.1 (Critical scalar for positive-definite matrices). Lean ✓ **Stmnt** *Let $D \geq 1$, and let $\rho, \sigma \in M_D(\mathbb{C})$ be positive definite. There is a real number $c > 0$ such that $\sigma - c\rho$ is positive semidefinite but not positive definite.*

Proof. ✓ **Proof** Write $\rho = SS^\dagger$ with S invertible and set $H = S^{-1}\sigma(S^\dagger)^{-1}$. Take c to be the smallest eigenvalue of H . Then $H - c1$ is positive semidefinite and singular. Congruence by S gives the corresponding assertions for $\sigma - c\rho$. $\square$

Theorem 6.4.2 (Fixed-point proportionality from boundary definiteness). Lean ✓ **Stmnt** *Let $E: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be linear, and let ρ and σ be positive-definite fixed points of E . Suppose every nonzero positive-semidefinite fixed point of E is positive definite. Then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. ✓ Proof Apply Lemma 6.4.1 to choose $c > 0$ for which $\sigma - c\rho$ is positive semidefinite but not positive definite. This difference is fixed by E . If it were nonzero, the hypothesis would make it positive definite, a contradiction. Hence $\sigma - c\rho = 0$. $\square$

Theorem 6.4.3 (Irreducible fixed-point uniqueness — CP version). Lean ✓ Stmt *Let E be a completely positive irreducible map on $M_D(\mathbb{C})$. If $\rho, \sigma \geq 0$ are fixed points of E and $\rho \neq 0$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. ✓ Proof If $\sigma = 0$, take $c = 0$. Otherwise, Theorem 6.3.11 makes both ρ and σ positive definite. The same theorem makes every nonzero positive-semidefinite fixed point positive definite. Apply Theorem 6.4.2. $\square$

Theorem 6.4.4 (Irreducible fixed-point uniqueness). Lean ✓ Stmt *Let A be an MPS tensor whose transfer map is irreducible. If $\rho, \sigma \geq 0$ are fixed points of $\mathcal{E}_A$ and $\rho \neq 0$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. ✓ Proof The transfer map is completely positive, so Theorem 6.4.3 applies. $\square$

Theorem 6.4.5 (Uniqueness of PSD fixed point). Lean ✓ Stmt *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. ✓ Proof Injectivity implies irreducibility of the transfer map. Apply Theorem 6.4.4. $\square$

Theorem 6.4.6 (Uniqueness of positive eigenvalue for irreducible CP maps). Lean ✓ Stmt *Let $D \geq 1$ and let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho, \sigma \geq 0$ are nonzero positive semidefinite matrices satisfying $E(\rho) = r_1\rho$ and $E(\sigma) = r_2\sigma$ for real scalars $r_1, r_2 > 0$. Then $r_1 = r_2$.*

Proof. ✓ Proof Extract Kraus operators K for E . Irreducibility of E passes to the adjoint transfer map $E^\dagger(X) = \sum_i K_i^\dagger X K_i$, which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 6.8.2) together with the irreducible positive-definiteness upgrade (Theorem 6.3.3) gives a positive definite eigenvector $\tau > 0$ with eigenvalue $t > 0$: $E^\dagger(\tau) = t\tau$. For any nonzero positive semidefinite X with $E(X) = sX$, the trace-pairing identity (A.1) yields

$$s \operatorname{tr}(\tau X) = \operatorname{tr}(\tau E(X)) = \operatorname{tr}(E^\dagger(\tau)X) = t \operatorname{tr}(\tau X).$$

Since $\tau > 0$ and $X \geq 0$, $X \neq 0$, the scalar $\operatorname{tr}(\tau X) > 0$, so $s = t$. Applying this to both (ρ, r_1) and (σ, r_2) gives $r_1 = t = r_2$. This is the completely positive specialization of [Wol12, Theorem 6.3(3)]: uniqueness of a positive eigenvalue admitting a nonzero positive semidefinite eigenvector. The proof follows Wolf’s dual-map trace argument. $\square$

Theorem 6.4.7 (PSD eigenvector uniqueness for irreducible CP maps). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho \neq 0$ and $\rho, \sigma \geq 0$ satisfy $E(\rho) = r\rho$ and $E(\sigma) = r\sigma$ for a real number $r > 0$. Then $\sigma = c\rho$ for some $c \in \mathbb{C}$. This is the completely positive specialization of the positive-eigenvector uniqueness in [Wol12, Theorem 6.3(2–3)].*

Proof. ✓ Proof Choose Kraus operators for E and rescale each by $r^{-1/2}$. Their transfer map is $r^{-1}E$, so both ρ and σ are fixed points. The rescaled map remains irreducible and completely positive, and Theorem 6.4.3 gives the claimed proportionality. $\square$

6.5 Existence and the Perron–Frobenius theorem

Theorem 6.5.1 (Existence of PSD fixed point). [Lean](#) [✓ Stmt](#) Assume $D \geq 1$. Let A be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so that $\mathcal{E}_A$ is trace-preserving. Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.

Proof. [✓ Proof](#) Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 4.9.6). The Cesàro fixed-point theorem (Theorem 6.1.19) then provides the fixed point; injectivity is not needed for this step. $\square$

Theorem 6.5.2 (Quantum Perron–Frobenius). [Lean](#) [✓ Stmt](#) Assume $D \geq 1$. Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique positive semidefinite fixed point ρ up to scaling, and ρ is positive definite.

Proof. [✓ Proof](#) Combine Theorems 6.5.1, 6.3.13, and 6.4.5. $\square$

Theorem 6.5.3 (Quantum Perron–Frobenius for irreducible transfer maps). [Lean](#) [✓ Stmt](#) Assume $D \geq 1$. Let A be an MPS tensor such that its transfer map is irreducible and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so that $\mathcal{E}_A$ is trace-preserving. Then $\mathcal{E}_A$ has a unique positive definite fixed point, up to scalar multiple.

Proof. [✓ Proof](#) Existence is the channel fixed-point theorem (Theorem 6.5.1). Irreducibility upgrades the fixed point to positive definite, and Theorem 6.4.4 gives uniqueness up to scalar multiple. $\square$

6.6 Right- and left-canonical gauges

Theorem 6.6.1 (Right-canonical (unital) gauge). [Lean](#) [✓ Stmt](#) Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A'^i = S^{-1}A^iS$. Then $\sum_i A'^i(A'^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital. This is the right-canonical normalization.

Proof. [✓ Proof](#) Conjugating each summand gives $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i\rho(A^i)^\dagger(S^\dagger)^{-1}$. The substitution $\mathcal{E}_A(\rho) = \rho$ leaves $S^{-1}\rho(S^\dagger)^{-1}$, and $\rho = SS^\dagger$ cancels the outer factors. The term-by-term calculation is in Section B.2. $\square$

Definition 6.6.2 (Left-canonical normalization). [Lean](#) [✓ Stmt](#) An MPS tensor $A = (A^i)_i$ is left-canonical when

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Equivalently, its Kraus map is trace-preserving.

Theorem 6.6.3 (Left-canonical (trace-preserving) gauge). [Lean](#) [Lean](#) [✓ Stmt](#) Let σ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger \sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization.

Proof. [✓ Proof](#) Conjugating each summand gives $(SA^iS^{-1})^\dagger(SA^iS^{-1}) = (S^\dagger)^{-1}(A^i)^\dagger\sigma A^iS^{-1}$. The adjoint fixed-point substitution leaves $(S^\dagger)^{-1}\sigma S^{-1}$, and $\sigma = S^\dagger S$ cancels the outer factors. See Section B.2 for the full calculation. $\square$

Theorem 6.6.4 (Square-root trace-preserving gauge from an adjoint fixed point). [Lean](#) [✓ Stmt](#)
Let σ be positive definite and satisfy $\sum_i (A^i)^\dagger \sigma A^i = \sigma$. Define

$$B^i = \sigma^{1/2} A^i \sigma^{-1/2}. \quad (6.26)$$

Then B is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

Proof. [✓ Proof](#) This is the specialization $K_i = A^i$ of Theorem B.2.1. Each conjugated summand is $(B^i)^\dagger B^i = \sigma^{-1/2} (A^i)^\dagger \sigma A^i \sigma^{-1/2}$. Substituting $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ and cancelling the outer inverse square roots gives $\mathbb{1}$. The complete calculation appears in Section B.2. $\square$

Theorem 6.6.5 (Square-root trace-preserving gauge from an adjoint eigenvector). [Lean](#) [✓ Stmt](#)
Let σ be positive definite and satisfy $\sum_i (A^i)^\dagger \sigma A^i = r \sigma$ for $r > 0$. Define

$$B^i = \sigma^{1/2} (r^{-1/2} A^i) \sigma^{-1/2}. \quad (6.27)$$

Then B is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

Proof. [✓ Proof](#) Rescale A by $r^{-1/2}$. The adjoint transfer map is quadratic in the tensor, so the eigenvector equation becomes a fixed-point equation. Theorem 6.6.4 then gives the result. $\square$

Remark 6.6.6. [✓ Stmt](#) The right-canonical gauge of Theorem 6.6.1 and the left-canonical gauge of Theorem 6.6.3 are generally different similarity transforms: one is built from a fixed point of $\mathcal{E}_A$, the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

6.7 Similarity preserves irreducibility

Lemma 6.7.1 (Similarity transform preserves irreducibility). [Lean](#) [✓ Stmt](#) Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $C \in M_D(\mathbb{C})$ be invertible, with $\det C \neq 0$, and let $c > 0$. Define the similarity-transformed map by

$$E'(X) = cC^{-1}E(CXC^\dagger)(C^\dagger)^{-1}.$$

Then E' is also irreducible.

Proof. [✓ Proof](#) Suppose $Q \neq 0, \mathbb{1}$ is an invariant projection for E' . Thus $(\mathbb{1} - Q)E'(QXQ)Q = 0$ for every X . Setting $R = CQC^\dagger$, the support projection P of R is an invariant projection for E . Irreducibility of E forces $P = 0$ or $P = \mathbb{1}$. The invertibility of C then forces $Q = 0$ or $Q = \mathbb{1}$, a contradiction. This is the completely positive specialization of the similarity result in [Wol12, Proposition 6.6]; the scalar case $C = \mathbb{1}$ gives the scaling result stated next. $\square$

Theorem 6.7.2 (Scaling preserves irreducibility). [Lean](#) [✓ Stmt](#) If E is an irreducible map and $c \neq 0$, then cE is also irreducible. For positive real c , this is the scalar completely positive case of [Wol12, Proposition 6.6]; the abstract irreducibility statement above also allows any nonzero complex c .

Proof. [✓ Proof](#) An invariant projection for cE is an invariant projection for E , since the nonzero constant may be removed from the invariance relation. $\square$

6.8 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Theorem 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Theorem 6.3]. The TP-gauge application follows [CPGSV16, Appendix A].

Theorem 6.8.1 (PSD eigenvector existence for a positive map). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $D > 0$. Then there are a nonzero positive semidefinite matrix ρ and a real number $r \geq 0$ such that*

$$E(\rho) = r\rho. \quad (6.28)$$

This is the eigenvector-existence part of [Wol12, Theorem 6.5]. It does not by itself identify r with the spectral radius.

Proof. ✓ Proof If E annihilates a nonzero positive semidefinite matrix ρ , take $r = 0$. Otherwise $E(\sigma) \neq 0$ for every nonzero $\sigma \geq 0$, and Theorem 6.8.2 gives (6.28) with $r > 0$. $\square$

Theorem 6.8.2 (Positive-eigenvalue specialization). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $D > 0$, and assume that $E(\sigma) \neq 0$ for every nonzero positive semidefinite matrix σ . Then there are a nonzero positive semidefinite matrix ρ and a real number $r > 0$ such that $E(\rho) = r\rho$.*

Proof. ✓ Proof Normalize E on the density matrices by

$$\rho \mapsto \frac{E(\rho)}{\text{tr}(E(\rho))}.$$

Positivity and the nonvanishing hypothesis make this a continuous self-map. Theorem B.1.5 gives a fixed point ρ . Clearing the positive denominator yields $E(\rho) = r\rho$ with $r = \text{tr}(E(\rho)) > 0$. Thus this theorem is the positive-eigenvalue specialization of Theorem 6.8.1. $\square$

Theorem 6.8.3 (Irreducibility of the transfer map of an irreducible tensor). Lean ✓ Stmt *If an MPS tensor A is irreducible, then its transfer map $\mathcal{E}_A$ is irreducible.*

Proof. ✓ Proof If P is a nontrivial invariant projection for $\mathcal{E}_A$, then

$$(\mathbb{1} - P)A^iP = 0$$

for every i , which is exactly a nontrivial invariant projection for the family $\{A^i\}$, contradicting irreducibility of A . $\square$

Theorem 6.8.4 (Positive-definite adjoint eigenvector for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that $\mathcal{E}_A^\dagger(\sigma) = r\sigma$.*

This combines the eigenvector-existence part of [Wol12, Theorem 6.5] with the irreducible upgrade to positive definiteness ([Wol12, Theorem 6.3(2)–(3)]); the application to the adjoint transfer map follows [CPGSV16, Appendix A].

Proof. ✓ Proof By Theorem 6.8.3, $\mathcal{E}_A$ is irreducible, and since $\mathcal{K}_A = \mathcal{E}_A$ (Theorem 5.1.2), its Kraus map $\mathcal{K}_A$ is irreducible too (Theorem 5.1.3). Applying Theorem B.4.4 to $K_i = A^i$ gives a positive definite σ and a positive real $r > 0$ with $\sum_i (A^i)^\dagger \sigma A^i = r\sigma$, i.e. $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. $\square$

Theorem 6.8.5 (TP-gauge existence for irreducible tensors). [Lean](#) [✓ Stmt](#) *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r , a positive definite matrix σ , and a tensor B gauge-equivalent to $r^{-1/2}A$ such that*

$$B^i = \sigma^{1/2}(r^{-1/2}A^i)\sigma^{-1/2}, \quad (6.29)$$

and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

This is the “spectral rescaling + TP gauge” step of [CPGSV16, Appendix A]. The similarity is generally non-unitary.

Proof. [✓ Proof](#) By Theorem 6.8.4, obtain σ positive definite and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. Set $c = r^{-1/2}$ and $A^i = cA^i$. Then $\mathcal{E}_{A^i}^\dagger(\sigma) = \sigma$. By the square-root trace-preserving eigenvector gauge construction (Theorem 6.6.5), the tensor defined by (6.29) satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. Lemma B.2.2 gives the required gauge equivalence to $r^{-1/2}A$. $\square$

Theorem 6.8.6 (Unital-gauge existence for irreducible tensors). [Lean](#) [✓ Stmt](#) *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r , a positive definite matrix ρ , and a tensor B gauge-equivalent to $r^{-1/2}A$ such that*

$$B^i = r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}, \quad (6.30)$$

and $\sum_i B^i(B^i)^\dagger = \mathbb{1}$.

This is the Perron–Frobenius unital-gauge orientation used in [PGVWC07, Theorem 4, lines 765–770], stated for one irreducible nonzero block.

Proof. [✓ Proof](#) Apply Theorem 6.8.4 to the conjugate-transposed Kraus family. Irreducibility transfers to that family, and the nonzero Kraus hypothesis is preserved by taking adjoints. Translating the resulting adjoint eigenvector equation back gives $\mathcal{E}_A(\rho) = r\rho$ with ρ positive definite and $r > 0$. The unital gauge theorem then gives the representative (6.30), and the same square-root similarity gives gauge equivalence to $r^{-1/2}A$. $\square$

Theorem 6.8.7 (Trivial Jordan blocks for peripheral eigenvalues). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive trace-preserving or unital linear map with $D > 0$. If λ is an eigenvalue of T with $|\lambda| = 1$, then $\ker(T - \lambda)^k = \ker(T - \lambda)$ for all $k \geq 1$. This is [Wol12, Proposition 6.2].*

Proof. [✓ Proof](#) If a rank-2 generalized eigenvector existed, the binomial expansion $T^n X = \lambda^n X + n\lambda^{n-1}(T - \lambda)X$ (when $(T - \lambda)^2 X = 0$) would make $\|T^n X\|$ grow linearly with n . This contradicts the bounded-orbit theorem, which holds under either hypothesis: trace preservation or unitality each give $\text{tr}[1T(1)] = D$, and positivity turns this into a uniform bound on $\text{tr}[AT^n(B)]$ over all powers of T . The rank- k case reduces to rank 2 by induction. $\square$

6.9 Spectral radius at the Perron eigenvalue

Theorem 6.9.1 (Spectral radius is the Perron eigenvalue). [Lean](#) [✓ Stmt](#) *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho > 0$ and $E(\rho) = r\rho$ with $r > 0$. Then the spectral radius of E is r . This is the completely positive specialization of [Wol12, Theorem 6.3(4)].*

Proof. [✓ Proof](#) Choose Kraus operators $\{K_i\}$ for E ; their Kraus map is E , so it is irreducible. Theorem B.4.4 therefore gives a positive definite matrix $\sigma > 0$ and $t > 0$ such that $E^\dagger(\sigma) = t\sigma$. The weighted trace pairing with ρ gives

$$r \text{tr}(\sigma\rho) = \text{tr}(\sigma E(\rho)) = \text{tr}(E^\dagger(\sigma)\rho) = t \text{tr}(\sigma\rho).$$

Since $\text{tr}(\sigma\rho) > 0$, one has $t = r$. Rescale by r^{-1} and conjugate by $\sigma^{1/2}$ as in Theorem B.2.1; this produces a trace-preserving Kraus map F similar to $r^{-1}E$. The trace-preserving spectral-radius bound (Theorem 4.10.7) gives $\rho_{\text{spec}}(F) \leq 1$. The conjugate of ρ is a nonzero fixed point of F , so 1 is an eigenvalue and $\rho_{\text{spec}}(F) \geq 1$. Hence $\rho_{\text{spec}}(F) = 1$. Similarity preserves the spectrum, and undoing the scalar rescaling gives $\rho_{\text{spec}}(E) = r$. $\square$

Theorem 6.9.2 (Eigenvalue 1 for positive trace-preserving maps). [Lean](#) [✓ Stmt](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive trace-preserving linear map with $D > 0$. Then there exists a nonzero positive semidefinite matrix ρ such that $T(\rho) = \rho$. This is the eigenvalue-1 assertion of [Wol12, Proposition 6.1].*

Proof. [✓ Proof](#) The Perron–Frobenius theorem (Theorem 6.8.2) gives a PSD eigenvector $T(\rho) = r\rho$ with $r > 0$. Trace preservation forces $r = 1$ because $\text{tr}(T(\rho)) = \text{tr}(\rho)$. $\square$

6.10 General Brouwer and stationary states

Theorem 6.10.1 (Brouwer’s fixed point theorem (simplex)). *Let $f : \Delta^n \rightarrow \Delta^n$ be a continuous map from the standard n -simplex to itself. Then there exists $x \in \Delta^n$ such that $f(x) = x$.*

Theorem 6.10.2 (Stationary states — nonlinear). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $D \geq 1$ and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a continuous, trace-preserving, positive (not necessarily linear) map. Then T has at least one stationary state: a density matrix ρ such that $T(\rho) = \rho$.*

This matches Wolf Theorem 6.11 exactly: the hypotheses are continuity, positivity ($X \geq 0 \Rightarrow T(X) \geq 0$), and trace preservation ($\text{tr}(T(X)) = \text{tr}(X)$). Linearity is not assumed.

The proof is Wolf’s argument: positivity and trace preservation imply that T restricts to a continuous self-map of the compact convex set of density matrices; the density-matrix version of Brouwer (Theorem B.1.5) then yields a fixed point.

Proof. [✓ Proof](#) Positivity and trace preservation give $T(\rho) \in \mathcal{D}_D$ for every $\rho \in \mathcal{D}_D$; continuity of T gives continuity on $\mathcal{D}_D$. Theorem B.1.5 applied to the restriction $T|_{\mathcal{D}_D}$ yields $\rho \in \mathcal{D}_D$ with $T(\rho) = \rho$. $\square$

6.11 Fixed-point decomposition and span

Theorem 6.11.1 (Positive fixed-point decomposition). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive, trace-preserving linear map and $X = E(X)$ a fixed point. Decompose X into Hermitian and anti-Hermitian parts and then each into orthogonal positive and negative parts, yielding four positive semidefinite operators $P_1, \dots, P_4$. Then $E(P_j) = P_j$ for $j = 1, \dots, 4$; in other words, every fixed point is a $\mathbb{C}$ -linear combination of four positive-semidefinite fixed points. In particular, if E is a channel and $H = H^\dagger$ is fixed by E , then there are positive-semidefinite fixed points Q_1, Q_2 such that $H = Q_1 - Q_2$.*

This is Wolf Proposition 6.8.

Proof. [✓ Proof](#) The proof follows Wolf: first handle Hermitian fixed points by the orthogonality argument using the support projection (equation (6.46)–(6.47) in the source). This gives the two-part decomposition for a Hermitian channel fixed point. Decompose an arbitrary fixed point into Hermitian and skew-Hermitian parts and apply the Hermitian case to each. $\square$

Corollary 6.11.2 (Linearly independent stationary states). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive, trace-preserving linear map. Let $\mathcal{F}_E = \{X \mid$*

$E(X) = X\}$ be the fixed-point subspace and $r = \dim_{\mathbb{C}} \mathcal{F}_E$. Then there exist r linearly independent stationary density matrices $\rho_1, \dots, \rho_r$ (positive semidefinite, trace 1, fixed by E) whose $\mathbb{C}$ -linear span equals $\mathcal{F}_E$.

This is Wolf Corollary 6.5 (Linearly independent stationary states).

Proof. ✓ Proof By Theorem 6.11.1, every fixed point is a $\mathbb{C}$ -linear combination of four positive-semidefinite fixed points. Each nonzero positive-semidefinite fixed point can be scaled to a stationary density matrix, so the stationary density matrices span $\mathcal{F}_E$. Since $\mathcal{F}_E$ is finite-dimensional, the spanning set contains a basis; extracting one yields r linearly independent stationary density matrices. $\square$

Chapter 7

Peripheral Channel Structure and Transfer-Operator Gaps

This chapter studies two spectral phenomena for finite-dimensional quantum channels and MPS transfer maps, following [PGVWC07] and [Wol12].

The first is *peripheral* structure: for an irreducible unital Schwarz map with a faithful adjoint fixed point, the eigenvalues on the unit circle form a finite cyclic group (Theorem 7.1.1), and under trace preservation the order of this group divides the bond dimension D (Theorem 7.9.2). Removing this period by unitary conjugation and cyclic corner projections produces the periodicity-free cyclic-sector decomposition (Theorem 7.9.1.1) used by the canonical-form reduction of Chapter 9.

The second is the *mixed transfer operator* F_{AB} of two MPS tensors A and B , and its spectral radius $\rho_{\text{spec}}(F_{AB})$. We show that $\rho_{\text{spec}}(F_{AB}) \leq 1$ under trace-preserving normalization (Theorem 7.4.1), that equality forces A and B to be gauge-phase equivalent (Theorem 7.4.4), and that mismatched bond dimensions always force a strict gap (Theorem 7.5.2). Since the MPV overlap $O_{AB}(N)$ equals the operator trace $\text{Tr}(F_{AB}^N)$ (Theorem 7.3.1), these gaps translate directly into the overlap-decay theorems (Theorem 7.6.5 and its rectangular and irreducible-trace-preserving variants) used throughout the fundamental theorem. The complementary case — a single tensor whose transfer map has a trivial peripheral spectrum away from the fixed-point projection — gives the rank-one Perron limit $\text{Tr}(\mathcal{E}_A^n) \rightarrow 1$ (Theorem 7.11.1) and hence primitive self-overlap convergence $O_{AA}(N) \rightarrow 1$ (Theorem 7.11.2).

Throughout, $\rho_{\text{spec}}(\cdot)$ denotes the spectral radius of a linear endomorphism of a finite-dimensional matrix space: $\rho_{\text{spec}}(F_{AB}) < 1$ for a mixed transfer operator and $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ for a transfer map after removing its fixed-point projection P . The unqualified phrase *spectral gap* is reserved for the energy gap of a many-body Hamiltonian, discussed later in the full blueprint.

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}. \quad (7.1)$$

Right-canonical (unital) and left-canonical (trace-preserving) gauges are generally different similarity transforms of the same tensor, recorded in Remark 7.4.5; both are used below, depending on which fixed point — of the transfer map or of its adjoint — drives the gauge.

The proofs of the technical mixed-transfer identities, Frobenius-norm estimates, gauge-rigidity

intertwiners, spectral-radius decay estimates, the rank-one Perron projection, and the cyclic-decomposition machinery removing peripheral periodicity are collected in Appendix C.

7.1 Peripheral eigenvalue group structure

This section proves a finite-Kraus specialization of the cyclicity conclusion in [Wol12, Theorem 6.6(1)]: the peripheral eigenvalues form a finite cyclic group. It does not establish the bound on the group order in item (1), nor the conclusions in items (2)–(4). Wolf’s theorem assumes an irreducible positive unital Schwarz map, whereas the statement below assumes a Kraus representation and hence complete positivity. The remaining abstract Schwarz-map case is not included. The trace-preserving refinement that the order divides the bond dimension is proved in Theorem 7.9.2. The invertibility of peripheral eigenvectors and the product closure of the peripheral eigenvalues used below are proved in Section C.7.

Theorem 7.1.1 (Peripheral eigenvalues: cyclic structure for Kraus maps). Lean ✓ Stmt *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point (trace-preservation is not required). Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. ✓ Proof Let S be the set of peripheral eigenvalues and let $m = |S|$. Finiteness of S and $1 \in S$ give $m > 0$. Every element of S is a root of unity, while S is closed under products and powers. Hence μ^{-1} belongs to S whenever $\mu \in S$. Multiplication by such a μ therefore permutes S . Since no element of S vanishes,

$$\prod_{\nu \in S} \nu = \prod_{\nu \in S} (\mu \nu) = \mu^m \prod_{\nu \in S} \nu.$$

Cancelling the nonzero product gives $\mu^m = 1$.

Let $\gamma = \exp(2\pi i/m)$, a primitive m -th root of unity. The set S is contained in the set of all m -th roots of unity, and both sets have cardinality m . Thus they are equal. Consequently every element of S is γ^j for some $0 \leq j < m$, and power closure gives the reverse inclusion. $\square$

7.2 Mixed transfer operator

Definition 7.2.1 (Mixed transfer operator). Lean ✓ Stmt The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors A and B of the same bond dimension D is the linear map $F_{AB} : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$F_{AB}(X) := \sum_{i=0}^{d-1} A^i X (B^i)^\dagger. \quad (7.2)$$

Definition 7.2.2 (Rectangular mixed transfer operator). Lean ✓ Stmt For MPS tensors A of bond dimension D_1 and B of bond dimension D_2 , possibly with $D_1 \neq D_2$, the *rectangular mixed transfer operator* is the map $F_{AB}^{\text{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \rightarrow M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F_{AB}^{\text{rect}}(X) := \sum_{i=0}^{d-1} A^i X (B^i)^\dagger. \quad (7.3)$$

The self-transfer identity $F_{AA} = \mathcal{E}_A$, the iterated word expansion of F_{AB} , and the trace-expansion lemma over matrix units are proved in Section C.1.

7.3 MPV overlap as transfer trace

Recall from (2.15) that $O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}$. This section rewrites that scalar as the operator trace of the mixed transfer power.

Lemma 7.3.1 (Overlap equals transfer trace). [Lean](#) [✓ Stmt](#) *For tensors A and B of bond dimension $D \geq 1$,*

$$O_{AB}(N) = \text{Tr}(F_{AB}^N). \quad (7.4)$$

Proof. [✓ Proof](#) Section C.1 expands the operator trace over matrix units (Lemma C.1.3) and the transfer power by words (Lemma C.1.2); reassembling the sums over matrix-unit indices gives

$$\text{Tr}(F_{AB}^N) = \sum_{\sigma} \text{tr}(A^\sigma) \overline{\text{tr}(B^\sigma)} = \sum_{\sigma} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N).$$

The rectangular case, Lemma 7.3.2, is the same argument over the rectangular matrix-unit basis. $\square$

Lemma 7.3.2 (Rectangular overlap as transfer trace). [Lean](#) [✓ Stmt](#) *For tensors A of bond dimension $D_1 \geq 1$ and B of bond dimension $D_2 \geq 1$,*

$$O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N). \quad (7.5)$$

Proof. [✓ Proof](#) The detailed word-index calculation is the same as the proof of Theorem 7.3.1, over the rectangular matrix-unit basis; see Section C.1. $\square$

7.4 Eigenvalue bound and transfer-operator gap

The rectangular eigenvalue bound is the MPS formulation of the generic bound for trace-preserving mixed Kraus maps. Under the identification of the rectangular mixed transfer with the mixed Kraus map, the two tensor normalizations are precisely the trace-preserving conditions. For the rigidity theorem, one gauges A and B separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument ([Wol12, Theorem 5.3]) to a block embedding of a mixed-transfer eigenvector.

Theorem 7.4.1 (Eigenvalue bound). [Lean](#) [✓ Stmt](#) *Let A and B be normalized MPS tensors. Every eigenvalue μ of F_{AB} satisfies $|\mu| \leq 1$.*

Proof. [✓ Proof](#) This is the equal-bond-dimension case $D_1 = D_2 = D$ of Theorem 7.4.2. The argument uses only the trace-preserving normalization (7.1) and the general eigenvalue bound for trace-preserving Kraus maps; it does not appeal to Perron–Frobenius theory. $\square$

Theorem 7.4.2 (Rectangular eigenvalue bound). [Lean](#) [✓ Stmt](#) *Let A and B be normalized MPS tensors of bond dimensions $D_1 \geq 1$ and $D_2 \geq 1$. Every eigenvalue μ of F_{AB}^{rect} satisfies $|\mu| \leq 1$.*

Proof. [✓ Proof](#) This is the MPS formulation of Theorem 4.10.9: the map F_{AB}^{rect} is the mixed Kraus map of the two matrix families, and the normalizations of A and B are exactly its two trace-preserving hypotheses. $\square$

Lemma 7.4.3 (Spectral radius bound). [Lean](#) [✓ Stmt](#) *For normalized tensors, $\rho_{\text{spec}}(F_{AB}) \leq 1$.*

Proof. ✓ Proof The spectral radius is the maximum of $|\mu|$ over the eigenvalues, and each satisfies $|\mu| \leq 1$ by Theorem 7.4.1. $\square$

Theorem 7.4.4 (Spectral radius ≥ 1 implies gauge-phase equivalence). Lean ✓ Stmt *Let A and B be injective normalized MPS tensors. If $\rho_{\text{spec}}(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof Since $\rho_{\text{spec}}(F_{AB}) \leq 1$ by Lemma 7.4.3, the hypothesis gives an eigenvalue μ with $|\mu| = 1$ and a nonzero eigenvector X satisfying $F_{AB}(X) = \mu X$. The proof proceeds in five steps, elaborated in Section C.4.

Step 1 (Separate gauging). Choose positive definite fixed points ρ_A and ρ_B of $\mathcal{E}_A$ and $\mathcal{E}_B$ by Theorem 6.5.2. Gauge the tensors separately to unital Kraus families by Theorem 6.6.1: $A^i = \rho_A^{-1/2} A^i \rho_A^{1/2}$ and similarly for B . Set $X' = \rho_A^{-1/2} X \rho_B^{-1/2}$.

Step 2 (Block Kraus family and off-diagonal embedding). Form

$$C^i := \begin{pmatrix} A'^i & 0 \\ 0 & B'^i \end{pmatrix}, \quad Y := \begin{pmatrix} 0 & X' \\ 0 & 0 \end{pmatrix}.$$

The matrices $\{C^i\}$ form a unital Kraus family on $M_{2D}(\mathbb{C})$, and $E_C(Y) = \mu Y$.

Step 3 (Kadison–Schwarz equality and Kraus relations). The positive definite fixed points of the adjoint maps give a faithful weighted trace for the block map. Applying the Kadison–Schwarz inequality to Y and pairing its positive semidefinite gap with this fixed point shows that the gap vanishes. The Kraus relation of Theorem 5.2.2 then gives $Y(C^i)^\dagger = \mu(C^i)^\dagger Y$. Applying the same argument to the complementary product gives $C^i Y = \mu Y C^i$.

Step 4 (Intertwining identity and invertibility). Expanding the block relations from Step 3 gives

$$X'(B'^i)^\dagger = \mu(A'^i)^\dagger X', \quad A'^i X' = \mu X' B'^i.$$

The corresponding multiplicative identities make $X'^\dagger X'$ and $X' X'^\dagger$ nonzero positive semidefinite fixed points of the appropriate transfer maps. Injectivity upgrades them to positive definite matrices, so X' is invertible.

Step 5 (Skolem–Noether conclusion). The intertwining identity and invertibility of X' give an algebra isomorphism between the full matrix algebras generated by the two families. By Theorem 3.2.4, this isomorphism is inner. Undoing the separate gauges gives $B^i = \bar{\mu} X A^i X^{-1}$ for every i , which is the gauge-phase relation (2.6).

The gauging is applied separately to $\mathcal{E}_A$ and $\mathcal{E}_B$; the mixed transfer map F_{AB} need not be simultaneously unital and trace-preserving. $\square$

Remark 7.4.5 (Two gauge conventions). ✓ Stmt The right-canonical gauge used here, $A'^i = \rho^{-1/2} A^i \rho^{1/2}$, makes $\{A'^i\}$ a unital Kraus family: $\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$. The convention $\tilde{A}^i = \rho^{1/2} A^i \rho^{-1/2}$ makes the family trace-preserving: $\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$. Both conventions appear in practice: the unital form is used in the transfer-operator gap proof above, while the trace-preserving form is used in the canonical-form reduction of Chapter 9.

Theorem 7.4.6 (Strict transfer-operator gap). Lean ✓ Stmt *Let A and B be injective normalized MPS tensors that are not gauge-phase equivalent. Then $\rho_{\text{spec}}(F_{AB}) < 1$.*

Proof. ✓ Proof Lemma 7.4.3 gives $\rho_{\text{spec}}(F_{AB}) \leq 1$. Equality would imply gauge-phase equivalence by Theorem 7.4.4, contradicting the hypothesis. $\square$

7.5 Transfer-operator gap — rectangular case

Lemma 7.5.1 (Rectangular intertwining forces equal bond dimension). [Lean](#) [✓ Stmt](#) *Let A and B be injective tensors of bond dimensions D_1 and D_2 . If there are $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ and $\mu \in \mathbb{C}$ such that, for every physical index i ,*

$$X(B^i)^\dagger = \mu(A^i)^\dagger X, \quad A^i X = \mu X B^i, \quad (7.6)$$

then $D_1 = D_2$.

Proof. [✓ Proof](#) The first relation in (7.6) shows that $\ker(X)$ is invariant under every $(B^i)^\dagger$: if $Xv = 0$, then $X(B^i)^\dagger v = \mu(A^i)^\dagger Xv = 0$. Since the $\{B^i\}$ span the full matrix algebra, so do the $\{(B^i)^\dagger\}$ (conjugate transposition is a conjugate-linear bijection and therefore preserves complex linear spans), so $\ker(X)$ is invariant under every matrix; injectivity of the resulting map forces $\ker(X) = 0$, giving $D_2 \leq D_1$. Taking adjoints of the second relation gives $X^\dagger(A^i)^\dagger = \bar{\mu}(B^i)^\dagger X^\dagger$, so the same argument applied to $X^\dagger$ shows $\ker(X^\dagger)$ is invariant under every $(A^i)^\dagger$ and hence trivial, giving $D_1 \leq D_2$. $\square$

Theorem 7.5.2 (Rectangular transfer-operator gap). [Lean](#) [✓ Stmt](#) *Let A be an injective normalized tensor of bond dimension D_1 and B an injective normalized tensor of bond dimension D_2 , with $D_1 \neq D_2$. Then $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$.*

Proof. [✓ Proof](#) Theorem 7.4.2 gives $|\mu| \leq 1$ for every eigenvalue of F_{AB}^{rect} . If its spectral radius were 1, there would be a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$.

Gauge A and B to unital families A' and B' and transport X to a nonzero rectangular intertwiner X' satisfying (C.3). Lemma 7.5.1 would then give $D_1 = D_2$, contradicting the hypothesis. $\square$

Theorem 7.5.3 (Rectangular transfer powers below unit spectral radius). [Lean](#) [✓ Stmt](#) *Let A and B have bond dimensions D_1 and D_2 , respectively. If $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$, then, for every $X \in M_{D_1 \times D_2}(\mathbb{C})$,*

$$(F_{AB}^{\text{rect}})^n(X) \longrightarrow 0.$$

This is the decay mechanism for the off-diagonal transfer operators in [CPGSV16, Appendix B, equations (B.3)–(B.4) and the paragraph following equation (B.6)].

Proof. [✓ Proof](#) The Gelfand formula implies convergence of $(F_{AB}^{\text{rect}})^n$ to zero in operator norm. Evaluation at a fixed rectangular matrix is continuous, and therefore preserves this limit. $\square$

Theorem 7.5.4 (Rectangular overlap decay). [Lean](#) [✓ Stmt](#) *If $D_1 \neq D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. [✓ Proof](#) The gap $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$ from Theorem 7.5.2 and Lemma C.5.1 imply operator-norm convergence $(F_{AB}^{\text{rect}})^N \rightarrow 0$, so its operator trace tends to zero. The identity (7.5) therefore gives $O_{AB}(N) \rightarrow 0$. $\square$

7.6 MPV overlap decay

We first record two spectral-radius identities for complex normed algebras. They will also control the scalar weights of normal fixed-point blocks in Chapter ??.

Theorem 7.6.1 (Spectral radius under nonzero scaling). [Lean](#) [✓ Stmt](#) *Let a belong to a complex Banach algebra and let $c \in \mathbb{C} \setminus \{0\}$. Then*

$$\rho_{\text{spec}}(ca) = |c| \rho_{\text{spec}}(a).$$

Proof. [✓ Proof](#) Multiplication by c sends the spectrum of a onto the spectrum of ca . Taking the supremum of the moduli of the spectral values gives the formula. $\square$

Theorem 7.6.2 (Spectral radius of a nonzero idempotent). [Lean](#) [✓ Stmt](#) *Let a be a nonzero element of a complex normed algebra. If $a^2 = a$, then*

$$\rho_{\text{spec}}(a) = 1.$$

Proof. [✓ Proof](#) The identity $a^2 = a$ gives $a(a - 1) = 0$. Since $a \neq 0$, the element $a - 1$ is not invertible, so 1 belongs to the spectrum of a . Every spectral value λ satisfies $\lambda^2 = \lambda$, hence the spectrum is contained in $\{0, 1\}$. $\square$

The Banach-algebra fact that powers below unit spectral radius vanish is proved in Section C.5.

Lemma 7.6.3 (An idempotent below unit spectral radius vanishes). [Lean](#) [✓ Stmt](#) *Let a be an element of a complex Banach algebra with $a^2 = a$ and $\rho_{\text{spec}}(a) < 1$. Then $a = 0$.*

Proof. [✓ Proof](#) Lemma C.5.1 gives $a^n \rightarrow 0$. Idempotence gives $a^{n+1} = a$ for every n , so uniqueness of limits yields $a = 0$. $\square$

Theorem 7.6.4 (Transfer powers converge to zero). [Lean](#) [✓ Stmt](#) *Let A and B be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \rightarrow 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. [✓ Proof](#) The gap $\rho_{\text{spec}}(F_{AB}) < 1$ from Theorem 7.4.6 and the equal-bond-dimension case of Theorem 7.5.3 give $F_{AB}^n(X) \rightarrow 0$ for every X . $\square$

Theorem 7.6.5 (Overlap decay). [Lean](#) [✓ Stmt](#) *Let A and B be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. [✓ Proof](#) By (7.4) and (C.2),

$$O_{AB}(N) = \text{Tr}(F_{AB}^N) = \sum_{p,q} (F_{AB}^N(E_{pq}))_{pq}.$$

Theorem 7.6.4 makes every summand tend to zero. Since the sum is finite, $O_{AB}(N) \rightarrow 0$. $\square$

The block-separation consequences of this gap — cross-correlation decay between distinct blocks and the persistence of self-correlation at a fixed point — used in the canonical-form separation of Chapter 9, are recorded in Section C.5.

7.7 Transfer-operator gap under irreducible-TP hypotheses

The preceding transfer-operator gap results require *injectivity* of both tensors. The following variants weaken this to *irreducibility* together with the trace-preserving normalization (7.1), following Cirac et al. [CPGSV16] and the irreducibility theory of [Wol12, Section 6.2]. The argument replaces the Kraus-commutation step with a Cauchy–Schwarz rigidity argument for the Perron–Frobenius fixed point.

Theorem 7.7.1 (Gauge-equivalence from irreducible-TP spectral radius). [Lean](#) [✓ Stmt](#) *Let A and B be irreducible trace-preserving normalized MPS tensors of bond dimension D . If $\rho_{\text{spec}}(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. [✓ Proof](#) Choose a modulus-one eigenvector as in Theorem 7.4.4. The proof follows its five steps, with one change. Separate gauging, the block Kraus family, and the Kadison–Schwarz equality proceed as before. The equality makes the relevant positive products of the transported intertwiner nonzero positive semidefinite fixed points. Under irreducibility, Theorem 6.3.12 makes these fixed points positive definite, so the transported intertwiner is invertible. Lemma C.4.1 supplies (C.3), and Lemma C.4.2 then gives gauge-phase equivalence. $\square$

Theorem 7.7.2 (Strict transfer-operator gap (irreducible-TP)). [Lean](#) [✓ Stmt](#) *Let A and B be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then $\rho_{\text{spec}}(F_{AB}) < 1$.*

Proof. [✓ Proof](#) Lemma 7.4.3 gives $\rho_{\text{spec}}(F_{AB}) \leq 1$, while the contrapositive of Theorem 7.7.1 excludes equality. $\square$

Theorem 7.7.3 (Overlap decay (irreducible-TP)). [Lean](#) [✓ Stmt](#) *Let A and B be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. [✓ Proof](#) Theorem 7.7.2 gives $\rho_{\text{spec}}(F_{AB}) < 1$, so $F_{AB}^N \rightarrow 0$. The overlap–trace identity (7.4) gives $O_{AB}(N) \rightarrow 0$. $\square$

Theorem 7.7.4 (Rectangular transfer-operator gap (irreducible-TP)). [Lean](#) [✓ Stmt](#) *Let A and B be irreducible trace-preserving normalized tensors of bond dimensions D_1 and D_2 , respectively, with $D_1 \neq D_2$. Then $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$.*

Proof. [✓ Proof](#) Suppose instead that $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) = 1$, and choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$. Gauge A and B separately to unital families A' and B' and transport X to $X' \neq 0$. The matrix $X'X'^{\dagger}$ is a nonzero positive semidefinite fixed point of the appropriate transfer map on $M_{D_1}(\mathbb{C})$, so Theorem 6.3.12 makes it positive definite. Thus X' has full row rank and $D_1 \leq D_2$. Similarly, $X'^{\dagger}X'$ is positive definite, so X' has full column rank and $D_2 \leq D_1$. Hence $D_1 = D_2$, a contradiction. $\square$

Theorem 7.7.5 (Rectangular overlap decay (irreducible-TP)). [Lean](#) [✓ Stmt](#) *If $D_1 \neq D_2$ and both tensors are irreducible and trace-preserving normalized, then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. [✓ Proof](#) Theorem 7.7.4 gives $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$, so $(F_{AB}^{\text{rect}})^N \rightarrow 0$. The rectangular overlap–trace identity (7.5) gives $O_{AB}(N) \rightarrow 0$. $\square$

7.8 Overlap rigidity for normal blocks

The BNT matching argument ultimately turns a non-decaying mixed overlap into a gauge-phase match. The following two results isolate that spectral step.

Lemma 7.8.1 (Mixed-transfer spectral radius from unit-modulus overlap). [Lean](#) [✓ Stmt](#) *Let A and B have the same bond dimension. If $\|\langle V^{(N)}(A) | V^{(N)}(B) \rangle\| \rightarrow 1$, then the mixed transfer map has spectral radius at least 1.*

Proof. [✓ Proof](#) If the mixed-transfer spectral radius were less than 1, the iterated mixed transfer map would tend to zero in operator norm, hence so would its trace. By (7.4), the overlap would tend to zero, contradicting the hypothesis. $\square$

Theorem 7.8.2 (Gauge-phase equivalence from unit-modulus overlap). [Lean](#) [✓ Stmt](#) *Let A and B be irreducible MPS tensors of the same bond dimension with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If $\|V^{(N)}(A)|V^{(N)}(B)\| \rightarrow 1$, then A and B are gauge-phase equivalent.*

Proof. [✓ Proof](#) By Lemma 7.8.1, the mixed-transfer spectral radius is at least 1, and Theorem 7.7.1 then gives gauge-phase equivalence. $\square$

7.9 Peripheral spectral refinements

Every result in this section concerning peripheral eigenvalues, cyclic projections, or peripheral eigenspaces is stated for a finite Kraus map and therefore assumes complete positivity. These results do not supply the abstract Schwarz-map case of Wolf Theorem 6.6.

Theorem 7.9.1 (Transfer-map peripheral eigenvalues: cyclic structure). [Lean](#) [✓ Stmt](#) *Let $\mathcal{E}_K$ be the Kraus map of a finite matrix family K . Assume that $\mathcal{E}_K$ is irreducible and unital and has a positive-definite adjoint fixed point. Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that the peripheral eigenvalues of $\mathcal{E}_K$ are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. [✓ Proof](#) The peripheral eigenvalues form a finite subgroup of $\mathbb{C}^\times$: they contain 1, are nonzero, and are closed under multiplication and inversion. Every finite subgroup of $\mathbb{C}^\times$ is cyclic. Choosing a generator γ gives the asserted primitive root and enumeration. $\square$

Theorem 7.9.2 (Transfer-map peripheral eigenvalues form a cyclic group). [Lean](#) [✓ Stmt](#) *Let K be a finite matrix family on $M_D(\mathbb{C})$ with Kraus map $\mathcal{E}_K$. Assume that $\mathcal{E}_K$ is irreducible, that K is unital and trace-preserving, and that the adjoint Kraus map has a positive-definite fixed point. Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that $m \mid D$ and the peripheral eigenvalues of $\mathcal{E}_K$ are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. [✓ Proof](#) Apply Theorem 7.9.1 for the cyclic structure. Divisibility $m \mid D$ follows from the cyclic decomposition: m mutually orthogonal projections summing to $\mathbb{1}$ each have equal trace by trace preservation, so $m \operatorname{tr}(P_0) = D$ with $\operatorname{tr}(P_0) \in \mathbb{N}$. $\square$

Theorem 7.9.3 (Transfer-map period divides bond dimension). [Lean](#) [✓ Stmt](#) *Let K be a finite matrix family on $M_D(\mathbb{C})$ with Kraus map $\mathcal{E}_K$. Assume that $\mathcal{E}_K$ is irreducible, that K is unital and trace-preserving, and that the adjoint Kraus map fixes a positive-definite matrix. Let γ be a primitive m -th root of unity such that the peripheral eigenvalues are exactly $\{\gamma^k : k \in \{0, \dots, m-1\}\}$. Then $m \mid D$.*

Proof. [✓ Proof](#) The cyclic decomposition gives orthogonal projections $P_0, \dots, P_{m-1}$ with $\sum_k P_k = \mathbb{1}$ and $E(P_{k+1}) = P_k$. Trace preservation gives $\operatorname{tr}(P_{k+1}) = \operatorname{tr}(P_k)$ for every k , hence

$$D = \operatorname{tr}(\mathbb{1}) = \sum_{k=0}^{m-1} \operatorname{tr}(P_k) = m \operatorname{tr}(P_0).$$

Since the trace of an orthogonal projection is an integer, $m \mid D$. $\square$

7.9.1 Cyclic decomposition of irreducible finite Kraus maps

The construction — normalizing a peripheral eigenvalue to a unitary eigenvector, forming its discrete Fourier (cyclic) spectral projections, and the supporting corner-algebra apparatus (restriction to an invariant corner, corner rank, and the compression isometry $PM_D(\mathbb{C})P \cong M_n(\mathbb{C})$) — is given in Section C.7.

Theorem 7.9.1.1 (Cyclic decomposition for irreducible finite Kraus maps). [Lean](#) [✓ Stmt](#) *Let K be a finite unital matrix family whose Kraus map is irreducible, and suppose the adjoint Kraus map fixes a positive-definite matrix. If the peripheral spectrum is generated by a primitive m -th root γ , then there are a unitary U and orthogonal projections $P_0, \dots, P_{m-1}$ summing to $\mathbb{1}$ such that $E_K(U^k) = \gamma^k U^k$, $U^m = \mathbb{1}$, $U = \sum_{k=0}^{m-1} \gamma^k P_k$, and $E_K(P_{k+1}) = P_k$.*

Proof. [✓ Proof](#) Normalize an eigenvector for γ to a unitary U . Its powers remain eigenvectors, and the Fourier projections of U give the stated cyclic decomposition. $\square$

Lemma 7.9.1.2 (Cyclic shift of the Kraus operators). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $P_0, \dots, P_{m-1}$ be orthogonal projections summing to the identity and cyclically permuted by a Kraus map, $\sum_v K_v P_{k+1} K_v^\dagger = P_k$. Then the projections are mutually orthogonal, none of them vanishes, and each Kraus operator intertwines consecutive projections:*

$$K_v P_{k+1} = P_k K_v. \quad (7.7)$$

Proof. [✓ Proof](#) Compressing $\sum_v K_v P_{k+1} K_v^\dagger = P_k$ by $\mathbb{1} - P_k$ exhibits a vanishing sum of positive semidefinite matrices, so each summand vanishes: $(\mathbb{1} - P_k) K_v P_{k+1} = 0$. Mutual orthogonality follows by compressing the resolution of the identity by a fixed P_k , another vanishing sum of positive semidefinite matrices. Expanding $K_v P_{k+1}$ and $P_k K_v$ through the resolution of the identity, the off-diagonal compressions vanish by orthogonality, leaving (7.7). If some projection vanished, the cyclic action would propagate the vanishing around the cycle, contradicting the resolution of the identity. $\square$

Theorem 7.9.1.3 (Cyclic shift along a Kraus word). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Lemma 7.9.1.2, a product of ℓ Kraus operators moves each cyclic projection back by ℓ steps. In particular, every product of m Kraus operators commutes with every cyclic projection.*

Proof. [✓ Proof](#) Induct on the length of the word using (7.7). $\square$

The primitivity and irreducibility of the restricted sector dynamics, the invariance of each cyclic corner under the period, and the closure of the peripheral eigenvalues under products and inverses are proved in Section C.7.

Theorem 7.9.1.4 (Peripheral eigenspaces are one-dimensional). [Lean](#) [✓ Stmt](#) *Let K be a finite unital matrix family whose Kraus map is irreducible, and suppose that the adjoint Kraus map fixes a positive-definite matrix. For every peripheral eigenvalue γ of the transfer map, the γ -eigenspace is one-dimensional.*

Proof. [✓ Proof](#) Given a peripheral unitary eigenvector U for γ , the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ -eigenvector X . By the scalar fixed-point lemma for irreducible unital maps, $XU^\dagger = c\mathbb{1}$, hence $X = cU$ and the eigenspace is one-dimensional. $\square$

7.10 Self-overlap and the period-one criterion

Theorem 7.10.1 (Self-overlap along peripheral-period multiples). [Lean](#) [✓ Stmt](#) *Let $D > 0$ and $m > 0$. Let A be a left-canonical irreducible MPS tensor with bond dimension D , and suppose that the peripheral eigenvalues of its transfer map are exactly $\{z \in \mathbb{C} : z^m = 1\}$. Then $O_{AA}(mn) \rightarrow m$.*

Proof. [✓ Proof](#) Apply Theorem 9.11.1 to the period- m blocking $A^{[m]}$. It gives m nonzero cyclic-sector tensors $C_0, \dots, C_{m-1}$, each left-canonical, whose unit-weight direct sum has the same MPV family as $A^{[m]}$. The cyclic-sector relations make every C_u primitive and irreducible. Distinct sectors are not gauge-phase equivalent. Hence the primitive self-overlap limit and the equal- and unequal-dimension overlap-decay theorems give

$$O_{C_u C_v}(n) \rightarrow \begin{cases} 1, & u = v, \\ 0, & u \neq v. \end{cases}$$

Expanding the overlap of the unit-weight direct sum and taking the finite sum of these limits yields

$$O_{A^{[m]} A^{[m]}}(n) = \sum_{u,v \in \mathbb{Z}_m} O_{C_u C_v}(n) \rightarrow m.$$

Finally, blocking identifies $O_{A^{[m]} A^{[m]}}(n) = O_{AA}(mn)$, which proves the claim. $\square$

Theorem 7.10.2 (Primitivity and normality from normalized self-overlap). [Lean](#) [✓ Stmt](#) *Let A be an irreducible left-canonical tensor of positive bond dimension. If $O_{AA}(N) \rightarrow 1$, then the transfer map of A is primitive and A is normal.*

Proof. [✓ Proof](#) Let m be the peripheral period. Along its multiples, $O_{AA}(mn) \rightarrow m$. The assumed full limit gives $O_{AA}(mn) \rightarrow 1$, hence $m = 1$. The transfer map is therefore primitive, and irreducibility and left-canonicity give normality. $\square$

Theorem 7.10.3 (Normality from normalized self-overlap). [Lean](#) [✓ Stmt](#) *Let A be an irreducible left-canonical tensor of positive bond dimension. If $O_{AA}(N) \rightarrow 1$, then A is normal.*

Proof. [✓ Proof](#) This is the normality conclusion of Theorem 7.10.2. $\square$

7.11 Primitive overlap convergence (complementary transfer-map gap form)

The theorems in this section are stated directly in terms of the complementary map $E - P$. This is the analytic form needed for the later overlap argument, corresponding to the primitive-case specialization of [Wol12, Theorem 6.7] where T_ϕ is a rank-one projection. For primitive normalized transfer maps, Chapter 4 supplies this complementary transfer-map input under explicit hypotheses; the rank-one Perron projection $P = P_\rho$, its trace, and the concrete irreducible instantiation of the complementary spectral-radius gap are given in Section C.6.

Theorem 7.11.1 (Complementary transfer-map gap implies trace convergence to 1). [Lean](#) [✓ Stmt](#) *Let E be trace-preserving with fixed point ρ , where $\text{tr}(\rho) \neq 0$, and let P be the fixed-point projection of Definition 4.11.1. If $\rho_{\text{spec}}(E - P) < 1$, then $\text{Tr}(E^n) \rightarrow 1$ as $n \rightarrow \infty$.*

Proof. ✓ **Proof** By the power decomposition (4.15), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho_{\text{spec}}(E - P) < 1$, the Gelfand formula gives $(E - P)^n \rightarrow 0$, so $\text{Tr}(E^n) \rightarrow \text{Tr}(P)$. Lemma C.6.1 (Section C.6) gives $\text{Tr}(P) = 1$. $\square$

Theorem 7.11.2 (Self-overlap convergence from a complementary transfer-map gap). Lean
✓ **Stmt** Let A be a normalized MPS tensor with a nonzero PSD fixed point ρ of the transfer map. If $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$, where P is the fixed-point projection, then $O_{AA}(N) \rightarrow 1$ as $N \rightarrow \infty$.

Proof. ✓ **Proof** By the overlap–trace identity (7.4) with $A = B$, one has $O_{AA}(N) = \text{Tr}(\mathcal{E}_A^N)$. By Theorem 7.11.1, $\text{Tr}(\mathcal{E}_A^N) \rightarrow 1$. $\square$

Chapter 8

Wielandt Bound

This chapter proves the quantum Wielandt bounds of [SPGWC10]. For a primitive normalized tensor, it compares the uniform vector-spreading index $q(\mathcal{E}_A)$ with the first exact word length $\iota(A)$ satisfying $S_{\iota(A)}(A) = M_D(\mathbb{C})$ and proves $\iota(A) \leq (D^2 - \text{kr}(A) + 1)D^2$. The cumulative bound $T_{D^2}(A) = M_D(\mathbb{C})$ is an intermediate step in the proof of Lemma 1.

8.1 Cumulative span

The exact-word identities used in this section are proved in Section D.1. The stabilization and spectral linear algebra behind the cumulative estimates are collected in Section D.2.

Definition 8.1.1 (Word evaluation for a finite matrix family). [Lean](#) [✓ Stmt](#) Let $K = \{K^i\}_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices. For a word $w = (i_1, \dots, i_n)$, its *word evaluation* is

$$K^w = K^{i_1} \dots K^{i_n},$$

with the empty word evaluated as the identity matrix.

Definition 8.1.2 (Word span). [Lean](#) [✓ Stmt](#) The *word span* at length n is the subspace

$$S_n(A) = \text{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, Section II].

Definition 8.1.3 (Cumulative span). [Lean](#) [✓ Stmt](#) The *cumulative span* at level n is $T_n(A) = S_0(A) + S_1(A) + \dots + S_n(A)$.

Definition 8.1.4 (Fixed-length vector span). [Lean](#) [✓ Stmt](#) The *fixed-length vector span* at length n is

$$H_n(K, \varphi) = \text{span}_{\mathbb{C}}\{K^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(K)|\varphi\rangle$ in the notation of [SPGWC10].

Definition 8.1.5 (Fixed-length vector span of an MPS tensor). [Lean](#) [✓ Stmt](#) For an MPS tensor A , its fixed-length vector span is the fixed-length vector span of the underlying finite matrix family.

8.1.1 Paper primitivity and indices

For the quantitative bounds, write $d' = \dim S_1(A)$ for the number of linearly independent Kraus operators. The source definitions and Proposition 3 of [SPGWC10] distinguish uniform vector spreading, eventual exact-word spanning, and spectral strong irreducibility.

Definition 8.1.1.1 (Kraus rank). [Lean](#) [✓ Stmt](#) The *Kraus rank* is $\text{kr}(A) = \dim S_1(A)$.

Definition 8.1.1.2 (Eventually full Kraus rank). [Lean](#) [✓ Stmt](#) The Kraus family has *eventually full Kraus rank* if there is a positive integer i such that $S_i(A) = M_D(\mathbb{C})$.

Definition 8.1.1.3 (Paper primitivity). [Lean](#) [✓ Stmt](#) The transfer map is *primitive in the sense of [SPGWC10]* if there is a positive integer q such that

$$H_q(A, \varphi) = \mathbb{C}^D \quad (8.1)$$

for every nonzero $\varphi \in \mathbb{C}^D$. The same q works for all nonzero vectors.

Definition 8.1.1.4 (Full-Kraus-rank index). [Lean](#) [✓ Stmt](#) The *full-Kraus-rank index* is

$$\iota(A) = \min\{n \geq 1 : S_n(A) = M_D(\mathbb{C})\}.$$

If the set is empty, its value is defined to be zero.

Definition 8.1.1.5 (Primitivity index). [Lean](#) [✓ Stmt](#) The *primitivity index* is

$$q(\mathcal{E}_A) = \min\{q \geq 1 : H_q(A, \varphi) = \mathbb{C}^D \text{ for every } \varphi \neq 0\}.$$

If the set is empty, its value is defined to be zero.

Definition 8.1.1.6 (Strengthened strong irreducibility). [Lean](#) [✓ Stmt](#) We call the tensor *strongly irreducible* here if there is a matrix $\rho > 0$ such that

$$\begin{aligned} \mathcal{E}_A(\rho) &= \rho, \\ \text{spec}_{\text{per}}(\mathcal{E}_A) &= \{1\}, \end{aligned}$$

and $\mathcal{E}_A$ is irreducible. The condition in [SPGWC10, Proposition 3(c)] requires that 1 be the only peripheral eigenvalue and that its eigenspace be one-dimensional, generated by a positive-definite fixed point. The definition above strengthens this condition by also requiring irreducibility explicitly.

The next seven results use the following notation. For nonnegative integers d and D , let $K = \{K^i\}_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices, and let $\mathcal{E}_K$ be its Kraus map.

Theorem 8.1.1.7 (Iterated Kraus map on a rank-one matrix). [Lean](#) [✓ Stmt](#) For every choice of d , D , and K as above, every $q \geq 0$, and every $\varphi \in \mathbb{C}^D$,

$$\mathcal{E}_K^q(|\varphi\rangle\langle\varphi|) = \sum_{|w|=q} |K^w\varphi\rangle\langle K^w\varphi|.$$

Proof. [✓ Proof](#) Expand the Kraus representation of the q -fold iterate. Each summand is $K^w|\varphi\rangle\langle\varphi|(K^w)^\dagger = |K^w\varphi\rangle\langle K^w\varphi|$. □

Theorem 8.1.1.8 (Positive rank-one image from vector spreading). [Lean](#) [✓ Stmt](#) For every choice of d , D , and K as above and every $q \geq 0$, suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero $\varphi \in \mathbb{C}^D$. Then $\mathcal{E}_K^q(|\varphi\rangle\langle\varphi|) > 0$ for every $\varphi \neq 0$.

Proof. ✓ Proof The preceding theorem expresses the image as the frame operator of the spanning family $(K^w \varphi)_{|w|=q}$. A finite frame operator is positive definite exactly when its vectors span the ambient space. $\square$

Theorem 8.1.1.9 (Positive-semidefinite preservation by Kraus iterates). Lean ✓ Stmt *For every choice of d , D , and K as above and every $n \geq 0$, the iterate $\mathcal{E}_K^n$ maps positive-semidefinite matrices to positive-semidefinite matrices.*

Proof. ✓ Proof Decompose a positive-semidefinite matrix into a finite sum of rank-one positive matrices and apply Theorem 8.1.1.7 to every summand. $\square$

Theorem 8.1.1.10 (Positivity improvement from vector spreading). Lean ✓ Stmt *For every choice of d , D , and K as above and every $q \geq 0$, suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ . Then $\mathcal{E}_K^q(\rho) > 0$ for every nonzero positive-semidefinite matrix ρ .*

Proof. ✓ Proof Write ρ as a sum of rank-one positive matrices. At least one vector in this decomposition is nonzero, so its image is positive definite by Theorem 8.1.1.8; all remaining images are positive semidefinite by Theorem 8.1.1.9. $\square$

Theorem 8.1.1.11 (Definiteness of fixed points from vector spreading). Lean ✓ Stmt *For every choice of d , D , and K as above and every $q \geq 0$, suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ . Then every nonzero positive-semidefinite fixed point of $\mathcal{E}_K$ is positive definite.*

Proof. ✓ Proof A fixed point of $\mathcal{E}_K$ is fixed by $\mathcal{E}_K^q$, whose action on every nonzero positive-semidefinite matrix is positive definite. $\square$

Theorem 8.1.1.12 (Definiteness of fixed points of positive powers). Lean ✓ Stmt *For every choice of d , D , and K as above and every $q \geq 0$, suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ . For every $p > 0$, each nonzero positive-semidefinite fixed point of $\mathcal{E}_K^p$ is positive definite.*

Proof. ✓ Proof If $\mathcal{E}_K^p(\rho) = \rho$, then $\mathcal{E}_K^{pq}(\rho) = \rho$. Factor this iterate as $\mathcal{E}_K^q \mathcal{E}_K^{(p-1)q}$. The inner image is nonzero and positive semidefinite, so the outer application is positive definite. $\square$

Theorem 8.1.1.13 (Irreducibility from vector spreading). Lean ✓ Stmt *For every choice of d , D , and K as above and every $q \geq 0$, if $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ , then the Kraus map $\mathcal{E}_K$ is irreducible.*

Proof. ✓ Proof Let P be an invariant orthogonal projection. Every word K^w preserves $\text{ran } P$. If $P \neq 0$, choose a nonzero vector in this range. Its length- q word orbit spans $\mathbb{C}^D$ by hypothesis and remains in $\text{ran } P$, hence $P = \mathbb{1}$. $\square$

Theorem 8.1.1.14 (Conjugate eigenmatrix of a positive map). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive. If $E(X) = \mu X$, then $E(X^\dagger) = \bar{\mu} X^\dagger$.*

Proof. ✓ Proof Positivity implies that E preserves adjoints. Taking the adjoint of $E(X) = \mu X$ gives the result. $\square$

Theorem 8.1.1.15 (Powers of a transfer-map eigenmatrix). Lean ✓ Stmt *If $\mathcal{E}_A(X) = \mu X$, then $\mathcal{E}_A^n(X) = \mu^n X$ for every $n \geq 0$.*

Proof. ✓ Proof This is Theorem 6.1.8 for the eigenspace of μ . $\square$

Theorem 8.1.1.16 (Finite-order eigenmatrix is power-fixed). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be linear. If $E(X) = \mu X$ and $\mu^p = 1$, then $E^p(X) = X$.*

Proof. ✓ **Proof** Theorem 6.1.8 gives $E^p(X) = \mu^p X = X$. □

Theorem 8.1.1.17 (Conjugate finite-order eigenmatrix is power-fixed). Lean ✓ **Stmt** Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive. If $E(X) = \mu X$ and $\mu^p = 1$, then $E^p(X^\dagger) = X^\dagger$.

Proof. ✓ **Proof** The conjugate eigenvalue is $\bar{\mu}$, and $(\bar{\mu})^p = \overline{\mu^p} = 1$. □

Theorem 8.1.1.18 (Trace of a non-fixed eigenmatrix). Lean ✓ **Stmt** Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ preserve the trace. If $E(X) = \mu X$ and $\mu \neq 1$, then $\text{tr}(X) = 0$.

Proof. ✓ **Proof** Trace preservation gives $\mu \text{tr}(X) = \text{tr}(E(X)) = \text{tr}(X)$. Since $\mu - 1 \neq 0$, the trace vanishes. □

Theorem 8.1.1.19 (Hermitian power-fixed point from a finite-order eigenmatrix). Lean ✓ **Stmt** Let E be a channel, and suppose $E(X) = \mu X$ with $X \neq 0$, $\mu \neq 1$, and $\mu^p = 1$. Then there is a nonzero Hermitian matrix H such that $\text{tr}(H) = 0$ and $E^p(H) = H$.

Proof. ✓ **Proof** At least one of $X + X^\dagger$ and $i(X^\dagger - X)$ is nonzero. Both are Hermitian, both have trace zero, and each is fixed by E^p . □

Theorem 8.1.1.20 (Uniqueness of power-fixed positive matrices for a Kraus map). Lean ✓ **Stmt** Suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ . If $p > 0$ and ρ, σ are nonzero positive-semidefinite fixed points of $\mathcal{E}_K^p$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.

Proof. ✓ **Proof** Fixed-length spreading makes every nonzero positive-semidefinite fixed point of $\mathcal{E}_K^p$ positive definite. The critical-scalar argument then gives proportionality. □

Theorem 8.1.1.21 (Powers of completely positive maps). Lean ✓ **Stmt** If E is completely positive, then E^p is completely positive for every $p \geq 0$.

Proof. ✓ **Proof** Induct on p , using complete positivity of the identity map at $p = 0$ and closure of completely positive maps under composition for the induction step. □

Theorem 8.1.1.22 (Powers of channels). Lean ✓ **Stmt** If E is a channel, then E^p is a channel for every $p \geq 0$.

Proof. ✓ **Proof** Complete positivity and trace preservation are both closed under composition and hold for the identity map. □

Theorem 8.1.1.23 (Vanishing of Hermitian power-fixed points). Lean ✓ **Stmt** Let K be trace-preserving and suppose $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero φ . If $p > 0$ and $H = H^\dagger$ satisfies $\text{tr}(H) = 0$ and $\mathcal{E}_K^p(H) = H$, then $H = 0$.

Proof. ✓ **Proof** Decompose H as a difference of positive-semidefinite fixed points of the channel $\mathcal{E}_K^p$. Both are proportional to one nonzero fixed point. Their traces and $\text{tr}(H) = 0$ force equal coefficients, hence $H = 0$. □

Theorem 8.1.1.24 (Hermitian power-fixed point from a peripheral eigenvector). Lean ✓ **Stmt** Let A be a normalized MPS tensor, and suppose $\mathcal{E}_A(X) = \mu X$ with $X \neq 0$, $\mu \neq 1$, and $\mu^p = 1$. Then there is a nonzero Hermitian matrix H such that

$$\begin{aligned} \text{tr}(H) &= 0, \\ \mathcal{E}_A^p(H) &= H. \end{aligned}$$

Moreover, H is not positive semidefinite.

Proof. ✓ Proof Apply Theorem 8.1.1.19 to the transfer channel. The resulting matrix is nonzero, Hermitian, trace zero, and fixed by $\mathcal{E}_A^p$. A nonzero positive-semidefinite matrix has positive trace, so this matrix is not positive semidefinite. $\square$

Theorem 8.1.1.25 (Uniqueness of power-fixed positive matrices). Lean ✓ Stmt Suppose there is a length $q \geq 0$ such that $H_q(A, \varphi) = \mathbb{C}^D$ for every nonzero $\varphi \in \mathbb{C}^D$. Let $p > 0$. If ρ and σ are nonzero positive-semidefinite fixed points of $\mathcal{E}_A^p$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.

Proof. ✓ Proof This is Theorem 8.1.1.20 for the Kraus family formed by the matrices of the tensor. $\square$

Theorem 8.1.1.26 (Positive-definite fixed point from vector spreading). Lean ✓ Stmt Let $D > 0$, and let $K = \{K^i\}_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices satisfying

$$\sum_{i=0}^{d-1} (K^i)^\dagger K^i = \mathbb{1}.$$

Suppose there is a common length $q \geq 0$ such that $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero $\varphi \in \mathbb{C}^D$. Then there is a positive-definite matrix ρ such that $\mathcal{E}_K(\rho) = \rho$.

Proof. ✓ Proof The normalization makes $\mathcal{E}_K$ a channel. The Cesàro fixed-point theorem, Theorem 6.1.19, gives a nonzero positive-semidefinite fixed point. The common fixed-length spreading hypothesis makes this fixed point positive definite by Theorem 8.1.1.11. $\square$

Theorem 8.1.1.27 (Primitivity from vector spreading). Lean ✓ Stmt Let $D > 0$, and let $K = \{K^i\}_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices satisfying

$$\sum_{i=0}^{d-1} (K^i)^\dagger K^i = \mathbb{1}.$$

Suppose there is a common length $q \geq 0$ such that $H_q(K, \varphi) = \mathbb{C}^D$ for every nonzero $\varphi \in \mathbb{C}^D$. Then the Kraus map $\mathcal{E}_K$ is primitive: it has a nonzero fixed point and 1 is its only peripheral eigenvalue.

Proof. ✓ Proof The normalization makes $\mathcal{E}_K$ a channel, so Theorem 6.1.19 gives a nonzero positive-semidefinite fixed point. This supplies the peripheral eigenvalue 1. Theorem 8.1.1.13 gives irreducibility. Thus $\mathcal{E}_K$ is an irreducible channel. By Theorem 4.12.2.7, every peripheral eigenvalue μ is a root of unity; choose $p > 0$ with $\mu^p = 1$.

Suppose $\mu \neq 1$. Theorem 8.1.1.19 gives a nonzero Hermitian matrix H with $\text{tr}(H) = 0$ and $\mathcal{E}_K^p(H) = H$. Theorem 8.1.1.23 gives $H = 0$, a contradiction. Thus $\mu = 1$. $\square$

Theorem 8.1.1.28 (Strengthened form of the primitivity equivalence). Lean Lean Lean ✓ Stmt Let $D > 0$ and suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. The following are equivalent:

1. there is a positive q for which $H_q(A, \varphi) = \mathbb{C}^D$ for every $\varphi \neq 0$;
2. there is a positive i for which $S_i(A) = M_D(\mathbb{C})$;
3. $\mathcal{E}_A$ is strongly irreducible.

Proposition 3 of [SPGWC10] states the same equivalence with item (3) replaced by the literal source condition: 1 is the only peripheral eigenvalue, and its one-dimensional eigenspace is generated by a positive-definite fixed point. Here item (3) includes the additional irreducibility condition.

Proof. ✓ Proof If $S_i(A) = M_D(\mathbb{C})$ and $\varphi \neq 0$, then $S_i(A)\varphi = \mathbb{C}^D$, proving (2) $\Rightarrow$ (1). For (1) $\Rightarrow$ (3), regard the matrices of A as a finite Kraus family. Theorems 8.1.1.26 and 8.1.1.27 give a positive-definite fixed point and show that 1 is the only peripheral eigenvalue. Theorem 8.1.1.13 gives irreducibility. Together these facts say that $\mathcal{E}_A$ is strongly irreducible.

For (3) $\Rightarrow$ (2), write $\mathcal{E}_A^m = P_\rho + (\mathcal{E}_A - P_\rho)^m$. The complementary powers tend to zero. If $S_m(A) \neq M_D(\mathbb{C})$ for every m , choose a nonzero matrix B orthogonal to $S_m(A)$. The associated trace pairing vanishes on the left, while positive definiteness gives a lower bound $c\|B\|^2$ for the P_ρ term and the complementary term is $o(\|B\|^2)$. For large m this is a contradiction. $\square$

Theorem 8.1.1.29 (Primitive-channel specialization of the index bound). Lean ✓ Stmt *Under the hypotheses of Theorem 8.1.1.28, paper primitivity implies*

$$q(\mathcal{E}_A) \leq \iota(A).$$

Proposition 1 of [SPGWC10] states this inequality for every quantum channel; the present theorem records its specialization under normalization and paper primitivity.

Proof. ✓ Proof By Theorem 8.1.1.28, $\iota(A)$ is a positive length with $S_{\iota(A)}(A) = M_D(\mathbb{C})$. Therefore $H_{\iota(A)}(A, \varphi) = \mathbb{C}^D$ for every $\varphi \neq 0$, so the defining minimum for $q(\mathcal{E}_A)$ is at most $\iota(A)$. $\square$

Theorem 8.1.1.30 (Unique fixed points of full rank). Lean ✓ Stmt *Let $D > 0$ and suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. If there is a positive n for which $S_n(A) = M_D(\mathbb{C})$, then there is a unique positive definite density matrix $\rho \in M_D(\mathbb{C})$ such that every fixed point of $\mathcal{E}_A$ is a scalar multiple of ρ .*

This is [Wol12, Theorem 6.15].

Proof. ✓ Proof By Theorem 8.1.1.28, the span hypothesis makes $\mathcal{E}_A$ strongly irreducible, giving a positive-definite fixed point ρ_0 together with peripheral primitivity (unique peripheral eigenvalue 1) and irreducibility of $\mathcal{E}_A$. Peripheral primitivity and irreducibility of $\mathcal{E}_A$ together give the complementary transfer-map gap at ρ_0 (Lemma D.5.5): the restriction of $\mathcal{E}_A$ off the fixed-point projection at ρ_0 has spectral radius strictly less than one. This gap forces every fixed point of $\mathcal{E}_A$ to be a scalar multiple of ρ_0 ; rescaling ρ_0 by the inverse of its (positive) trace gives the unique density-matrix representative ρ .

The source proves this instead by passing to $\mathcal{E}_A^n$, showing every $\mathcal{E}_A^n(|\psi\rangle\langle\psi|)$ has full rank from the span hypothesis, and ruling out a fixed-point space of dimension larger than one via Corollary 6.5 of [Wol12]. Both routes use only Wolf’s own Chapter 6 machinery; the present proof reuses the already-formalized Proposition 3 equivalence. $\square$

Lemma 8.1.1.31 (Monotonicity of cumulative span). Lean ✓ Stmt *For every n , one has $T_n(A) \subseteq T_{n+1}(A)$.*

Proof. ✓ Proof The span T_{n+1} includes all words of length at most $n+1$ and therefore contains all words of length at most n . $\square$

Lemma 8.1.1.32 (Dimension bound). Lean ✓ Stmt *For every n , one has $\dim T_n(A) \leq D^2$.*

Proof. ✓ Proof Since $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$, the result follows. $\square$

Theorem 8.1.1.33 (Cumulative span stabilisation). Lean ✓ Stmt *If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for every $m \geq n$.*

Proof. ✓ Proof If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of S_{n+1} by A^i gives an element of S_{n+2} , so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for every $m > n$, hence $T_m = T_n$. $\square$

Theorem 8.1.1.34 (Dimension growth). [Lean](#) [✓ Stmt](#) If $T_n(A) \neq T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.

Proof. [✓ Proof](#) The inclusion $T_n \leq T_{n+1}$ from Lemma 8.1.1.31, together with $T_n \neq T_{n+1}$, gives strict subspace inclusion and hence strict dimension growth. $\square$

Lemma 8.1.1.35 (Word span characterises block injectivity). [Lean](#) [✓ Stmt](#) One has $S_L(A) = M_D(\mathbb{C})$ if and only if A is L -block injective.

Proof. [✓ Proof](#) Both conditions assert that the span of all products of length L equals $M_D(\mathbb{C})$. $\square$

Theorem 8.1.1.36 (Positive-length propagation of block injectivity). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every n ,

$$S_{n+1}(A) = \text{span}_{\mathbb{C}}\{A^i\}_{i=0}^{d-1} S_n(A).$$

Consequently, if $L > 0$ and A is L -block injective, then A is m -block injective for every $m \geq L$. This auxiliary length-shift statement combines the block-injectivity discussion in [CPGSV16, Section II] with [CPGSV16, Appendix C.3, Lemma L]; it is not stated there as a separate lemma.

Proof. [✓ Proof](#) Every word of length $n + 1$ factors into its first letter and a word of length n , which proves the displayed equality by bilinearity. Write $L = n + 1$. Since $S_L(A) = M_D(\mathbb{C})$, one has $S_n(A) \leq S_L(A)$. Hence

$$M_D(\mathbb{C}) = S_L(A) = \text{span}_{\mathbb{C}}\{A^i\}_{i=0}^{d-1} S_n(A) \leq \text{span}_{\mathbb{C}}\{A^i\}_{i=0}^{d-1} S_L(A) = S_{L+1}(A).$$

Iterating this implication gives $S_m(A) = M_D(\mathbb{C})$ for every $m \geq L$. $\square$

8.2 Nonzero trace product

Theorem 8.2.1 (Cumulative span reaches $M_D(\mathbb{C})$). [Lean](#) [✓ Stmt](#) Let $D > 0$. If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$. This cumulative estimate supports the proof of [SPGWC10, Lemma 1]; it is not a conclusion of source Theorem 1.

Proof. [✓ Proof](#) This is Theorem D.2.7. $\square$

Theorem 8.2.2 (Cumulative span reaches $M_D(\mathbb{C})$ — sharp bound). [Lean](#) [✓ Stmt](#) Let $D > 0$. If A is normal, then $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$, where $d' = \dim S_1(A) = \text{kr}(A)$. This is the cumulative-span estimate used to prove the sharp trace conclusion in [SPGWC10, Lemma 1].

Proof. [✓ Proof](#) If $T_n(A) \neq T_{n+1}(A)$, then the dimension strictly increases. Since $S_1(A) \subseteq T_1(A)$ and $\dim S_1(A) = d'$, one has $\dim T_1(A) \geq d'$. After at most $D^2 - d'$ additional strict-growth steps, the dimension reaches $D^2 = \dim M_D(\mathbb{C})$. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality. Hence $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$. $\square$

Theorem 8.2.3 (Eigenvector extraction from a full cumulative span). [Lean](#) [✓ Stmt](#) Let $T_N(K)$ be the span of the products of at most N matrices from a finite family K . If $D > 0$ and $T_N(K) = M_D(\mathbb{C})$, then there are a word w , a nonzero scalar μ , and a nonzero vector $\varphi \in \mathbb{C}^D$ such that

$$\begin{aligned} |w| &\leq N, \\ K^w \varphi &= \mu \varphi. \end{aligned}$$

Proof. ✓ Proof Since the identity lies in $T_N(K)$ and has nonzero trace, some word of length at most N has nonzero trace. Such a matrix has a nonzero eigenvalue and a corresponding nonzero eigenvector by Theorem D.2.9. $\square$

Theorem 8.2.4 (Eigenvector extraction for an MPS tensor). Lean ✓ Stmt *The conclusion of Theorem 8.2.3 holds for the matrix family of an MPS tensor whenever its cumulative span at level N is the full matrix algebra.*

Proof. ✓ Proof Apply the finite-family theorem to the matrices of the tensor. $\square$

Theorem 8.2.5 (Sharp nonzero trace product for a primitive channel). Lean ✓ Stmt *Let $D > 0$, suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume that A is primitive in the sense of Definition 8.1.1.3. Then there is a word w with*

$$1 \leq |w| \leq D^2 - \text{kr}(A) + 1, \\ \text{tr}(A^w) \neq 0.$$

This is the sharp trace conclusion of [SPGWC10, Lemma 1].

Proof. ✓ Proof Theorem 8.1.1.28 gives eventual full Kraus rank, hence normality. Put $b = D^2 - \text{kr}(A) + 1$ and let V_n be the span of A^w with $1 \leq |w| \leq n$. Then $S_1(A) \subseteq V_1$, so $\dim V_1 \geq \text{kr}(A)$. If $V_n = V_{n+1}$ for some $n \geq 1$, then V_n is closed under left multiplication by every A^i . It therefore contains every positive-length word product; normality forces $V_n = M_D(\mathbb{C})$. Thus, until the full matrix algebra is reached, the dimensions increase strictly, and the bound on $\dim M_D(\mathbb{C})$ gives $V_b = M_D(\mathbb{C})$. Express $\mathbb{1}$ as a linear combination of the positive-length words defining V_b . Since $\text{tr}(\mathbb{1}) = D \neq 0$, at least one has nonzero trace. $\square$

Theorem 8.2.6 (Sharp nonzero trace product). Lean ✓ Stmt *Let $D > 0$. If A is normal, then there exists a word w of length at most $D^2 - \text{kr}(A) + 1$ such that $\text{tr}(A^w) \neq 0$. This normality variant has the same sharp length bound as [SPGWC10, Lemma 1]; the source hypotheses are stated in Theorem 8.2.5.*

Proof. ✓ Proof By Theorem 8.2.2, the identity $\mathbb{1}$ lies in $T_{D^2-d'+1}(A)$ and is therefore a linear combination of word products of length at most $D^2 - d' + 1$. At least one such product has nonzero trace. $\square$

Theorem 8.2.7 (Sharp nonzero trace product — positive length). Lean ✓ Stmt *If A is normal and $D \geq 1$, then there exists a word w of positive length $1 \leq |w| \leq D^2 - \text{kr}(A) + 1$ such that $\text{tr}(A^w) \neq 0$. This strengthens Theorem 8.2.6 by requiring $|w| \geq 1$, a feature needed for the blocking argument in the general Wielandt bound below.*

Proof. ✓ Proof Define the *positive-level* cumulative span $V_n = \text{span}_{\mathbb{C}}\{A^w : 1 \leq |w| \leq n\}$. The same stabilization-or-strict-growth dichotomy shows $V_{D^2-d'+1} = M_D(\mathbb{C})$: if V_n stabilises and is not $M_D(\mathbb{C})$, then V_n is closed under left multiplication by each A^i , so $S_N(A) \subseteq V_n \neq M_D(\mathbb{C})$ for every $N \geq 1$, contradicting normality. Since $V_{D^2-d'+1} = M_D(\mathbb{C})$, the identity $\mathbb{1}$ lies in $V_{D^2-d'+1}$ and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace because $\text{tr}(\mathbb{1}) = D \neq 0$ for $D \geq 1$. $\square$

8.3 Eigenvector spreading

Definition 8.3.1 (Cumulative vector span). Lean ✓ Stmt Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w \varphi : |w| \leq n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

Theorem 8.3.2 (Eigenvector spreading). Lean ✓ Stmt Let A be a normal MPS tensor and let $A^{i_0} \varphi = \mu \varphi$, where $\mu \neq 0$ and $\varphi \neq 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This cumulative form supplies the dimension-growth step in [SPGWC10, Lemma 2(a)].

Proof. ✓ Proof The vector span K_n is monotone and $\dim K_n \leq D$. If $K_n = K_{n+1}$ for some $n < D - 1$, the span stabilises by Theorem D.2.11. But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 8.2.1, so Lemma D.2.12 gives $K_{D^2}(A, \varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by at least 1 at each step, reaching D by step $D - 1$. $\square$

Theorem 8.3.3 (Cumulative eigenvector spreading for finite families). Lean ✓ Stmt Let $K = \{K^i\}_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices, where $D > 0$, and let $\varphi \neq 0$. If, for some N , words in K of length at most N span $M_D(\mathbb{C})$, then the vectors $K^w \varphi$ with $|w| \leq D - 1$ span $\mathbb{C}^D$.

Proof. ✓ Proof The cumulative vector spaces are monotone and have dimension at most D . Once two consecutive spaces agree, the common space is invariant under every K^i . Full cumulative matrix span then forces it to contain $M_D(\mathbb{C})\varphi = \mathbb{C}^D$. Since the initial space is nonzero, it reaches dimension D by step $D - 1$. $\square$

Lemma 8.3.4 (Eigenvector padding for finite families). Lean ✓ Stmt Suppose $K^{i_0} \varphi = \mu \varphi$ with $\mu \neq 0$. For every n , the span of $K^w \varphi$ over words with $|w| \leq n$ equals $H_n(K, \varphi)$.

Proof. ✓ Proof Repeating the letter i_0 pads a word of length $m \leq n$ to length n . Its action on φ gains the nonzero factor μ^{n-m} , so every shorter-word vector belongs to the fixed-length span. The reverse inclusion is immediate. $\square$

Theorem 8.3.5 (Fixed-length spreading from eventual word-span fullness). Lean ✓ Stmt Let $K = (K^i)_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices, where $D > 0$, and suppose that $S_n(K) = M_D(\mathbb{C})$ for every sufficiently large n . If $\varphi \neq 0$, $\mu \neq 0$, and $K^{i_0} \varphi = \mu \varphi$, then $H_{D-1}(K, \varphi) = \mathbb{C}^D$.

Proof. ✓ Proof Eventual fullness gives a level at which the cumulative matrix span is all of $M_D(\mathbb{C})$. The cumulative vector spaces therefore reach $\mathbb{C}^D$ by level $D - 1$. The eigenvector relation then pads shorter words to length $D - 1$ by nonzero powers of μ . $\square$

Theorem 8.3.6 (Fixed-length eigenvector spreading for a primitive channel). Lean ✓ Stmt Let $D > 0$, suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume that A is primitive in the sense of Definition 8.1.1.3. If $\varphi \neq 0$, $\mu \neq 0$, and $A^{i_0} \varphi = \mu \varphi$, then $H_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].

Proof. ✓ Proof Proposition 3, in the form of Theorem 8.1.1.28, gives eventual full Kraus rank and hence normality. The cumulative spaces $K_n(A, \varphi)$ cannot stabilise below $\mathbb{C}^D$, because every matrix eventually lies in an exact word span. Their dimensions therefore reach D by $n = D - 1$, giving $K_{D-1}(A, \varphi) = \mathbb{C}^D$. Finally, the eigenvector relation pads every shorter word to length $D - 1$ by a nonzero power of μ , so $K_{D-1}(A, \varphi) = H_{D-1}(A, \varphi)$. $\square$

8.4 Quantum Wielandt bounds

Section D.3 proves the one-step augmentation facts used in cases (2) and (3). The blocking identities used in the general case are proved in Section D.4.

Remark 8.4.1 (Explicit blocking-length estimates). ✓ Stmt Theorem 8.2.1 gives the D^2 cumulative-span bound. The full-rank index bound $\iota(A) \leq (D^2 - d' + 1)D^2$ ([SPGWC10, Theorem 1, case (1)]) is proved in Theorem 8.4.8 below. The left-canonical normal blocked-injectivity estimate D^4 is proved below in Theorem 8.5.5. A sharper $3D^5$ canonical-form block-separation estimate, developed in the full blueprint, combines this D^4 estimate with the direct-sum argument of [PGVWC07].

Theorem 8.4.2 (Wielandt case (2), one-step invertible element). Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive in the sense of Definition 8.1.1.3. If $S_1(A)$ contains an invertible matrix X , then*

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}). \quad (8.2)$$

In the terminology of [SPGWC10], this is the case where the first application space $S_1(A)$ contains an invertible operator. This is the exact word-span form of [SPGWC10, Theorem 1, case (2)].

Proof. ✓ Proof Adjoin X as an extra first Kraus operator, obtaining $\tilde{A}$. Theorem 8.1.1.28 gives normality of A , hence normality of $\tilde{A}$; the Kraus rank and all exact word spans are unchanged. The first Kraus operator of $\tilde{A}$ is invertible, so Theorem D.3.5 gives $S_{D^2 - \text{kr}(\tilde{A}) + 1}(\tilde{A}) = M_D(\mathbb{C})$. Transfer this equality back to A . □

Corollary 8.4.3 (Wielandt case (2), index bound). Lean ✓ Stmt *Under the hypotheses of Theorem 8.4.2, one has $\iota(A) \leq D^2 - \text{kr}(A) + 1$, where $\iota(A) = \min\{n \geq 1 : S_n(A) = M_D(\mathbb{C})\}$.*

Proof. ✓ Proof The word-span equality (8.2) gives an admissible value in the defining minimum for $\iota(A)$. □

Lemma 8.4.4 (Rank-one extraction from a non-invertible eigenvector). Lean ✓ Stmt *Let A be a normal MPS tensor with bond dimension $D \geq 1$. Suppose that a Kraus operator A^{i_0} is non-invertible and that there exist a nonzero vector $\varphi \in \mathbb{C}^D$ and a nonzero scalar μ such that $A^{i_0}\varphi = \mu\varphi$. Then, for every $\psi \in \mathbb{C}^D$, $|\varphi\rangle\langle\psi| \in S_{D^2 - D + 1}(A)$.*

Proof. ✓ Proof Let $\mathcal{L}(M) = A^{i_0}M$, and choose r so that $\mathcal{L}^r$ annihilates the generalized zero-eigenspace of $\mathcal{L}$; equivalently, $\text{range}(\mathcal{L}^r)$ is its stabilized nonzero-spectral part. The dimension-growth argument gives a level n_0 for which

$$(A^{i_0})^r S_{n_0}(A) = \{(A^{i_0})^r M : M \in M_D(\mathbb{C})\}.$$

Since $A^{i_0}\varphi = \mu\varphi$ and $\mu \neq 0$, iterating gives $(A^{i_0})^r\varphi = \mu^r\varphi$ and hence $\varphi = \mu^{-r}(A^{i_0})^r\varphi \in \text{range}((A^{i_0})^r)$. Therefore, for every $\psi \in \mathbb{C}^D$, $|\varphi\rangle\langle\psi| \in S_{r+n_0}(A)$. The corresponding Fitting-index estimate gives $r + n_0 \leq D^2 - D + 1$. Since

$$A^{i_0}|\varphi\rangle\langle\psi| = |A^{i_0}\varphi\rangle\langle\psi| = \mu|\varphi\rangle\langle\psi|,$$

one has $|\varphi\rangle\langle\psi| = \mu^{-1}A^{i_0}|\varphi\rangle\langle\psi|$. Thus $|\varphi\rangle\langle\psi| \in S_k(A)$ implies $|\varphi\rangle\langle\psi| \in S_{k+1}(A)$, because left multiplication by A^{i_0} maps $S_k(A)$ into $S_{k+1}(A)$. Iterating from $k = r + n_0$ and using $r + n_0 \leq D^2 - D + 1$ gives $|\varphi\rangle\langle\psi| \in S_{D^2 - D + 1}(A)$. □

Theorem 8.4.5 (Sharp rank-one extraction for a primitive channel). [Lean](#) [✓ Stmt](#) Let $D > 0$, suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume that A is primitive in the sense of Definition 8.1.1.3. Suppose A^{i_0} is not invertible and $A^{i_0}\varphi = \mu\varphi$ with $\varphi \neq 0$ and $\mu \neq 0$. Then, for every $\psi \in \mathbb{C}^D$,

$$|\varphi\rangle\langle\psi| \in S_{D^2-D+1}(A).$$

This is the quantitative rank-one conclusion of [SPGWC10, Lemma 2(b)].

Proof. [✓ Proof](#) Theorem 8.1.1.28 first gives normality. Apply the Fitting decomposition to left multiplication $\mathcal{L}(M) = A^{i_0}M$. Choose r so that $\mathcal{L}^r$ kills its generalized zero-eigenspace. Dimension growth then gives n_0 with

$$(A^{i_0})^r S_{n_0}(A) = \{(A^{i_0})^r M : M \in M_D(\mathbb{C})\}, \\ r + n_0 \leq D^2 - D + 1.$$

Since $(A^{i_0})^r \varphi = \mu^r \varphi$ and $\mu \neq 0$, the vector φ lies in the range of $(A^{i_0})^r$. Hence $|\varphi\rangle\langle\psi| \in S_{r+n_0}(A)$ for every ψ . The identity $|\varphi\rangle\langle\psi| = \mu^{-1} A^{i_0} |\varphi\rangle\langle\psi|$ pads this element to the exact length $D^2 - D + 1$. $\square$

Theorem 8.4.6 (Wielandt case (3), one-step non-invertible element). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive in the sense of Definition 8.1.1.3. If $S_1(A)$ contains a non-invertible matrix X with a nonzero eigenvalue, then $S_{D^2}(A) = M_D(\mathbb{C})$. More explicitly, it is enough to have $X\varphi = \mu\varphi$ with $\varphi \neq 0$ and $\mu \neq 0$. In the terminology of [SPGWC10], this is the case where the first application space $S_1(A)$ contains a non-invertible operator with a nonzero eigenvalue. This is the exact word-span form of [SPGWC10, Theorem 1, case (3)].

Proof. [✓ Proof](#) Theorem 8.1.1.28 gives normality. Adjoin X as an extra first Kraus operator. The augmented tensor remains normal and has the same exact word spans as A . The eigenvector relation for X is now a one-generator eigenvector relation. The eigenvector-spreading argument gives full fixed-length vector span by length $D-1$, and Lemma 8.4.4 gives the rank-one operators $|\varphi\rangle\langle\psi|$ at length $D^2 - D + 1$. Multiplying these two fixed-length spans gives S_{D^2} for the augmented tensor, hence for A . $\square$

Corollary 8.4.7 (Wielandt case (3), index bound). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 8.4.6, one has $\iota(A) \leq D^2$.

Proof. [✓ Proof](#) The equality $S_{D^2}(A) = M_D(\mathbb{C})$ gives an admissible value in the defining minimum for $\iota(A)$. $\square$

Theorem 8.4.8 (Wielandt's inequality — general bound). [Lean](#) [✓ Stmt](#) Let A be a normalized MPS tensor with $D > 0$, $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, that is primitive in the sense of Definition 8.1.1.3. Then

$$\iota(A) \leq (D^2 - \text{kr}(A) + 1)D^2,$$

where $\iota(A) = \min\{n \geq 1 : S_n(A) = M_D(\mathbb{C})\}$ is the full-Kraus-rank index. This is [SPGWC10, Theorem 1, case (1)].

Proof. [✓ Proof](#) Theorem 8.1.1.28 gives normality. By Theorem 8.2.5, there exists a positive-length word w with $|w| \leq D^2 - d' + 1$ and $\text{tr}(A^w) \neq 0$, where $d' = \text{kr}(A)$. The matrix A^w has nonzero trace, hence a nonzero eigenvalue μ and eigenvector φ by Theorem D.2.9. Set $n = |w|$ and let $B = A^{[n]}$ be the blocked tensor. Then B is normal by Theorem D.4.1, and the encoding of w as a single block index gives $B^{i_0} = A^w$.

1. If A^w is invertible, then $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2n}(A) = M_D(\mathbb{C})$.
2. If A^w is non-invertible, then the eigenvector φ spans a proper subspace, and the rank-one extraction argument gives $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2n}(A) = M_D(\mathbb{C})$.

In the invertible case, the invertible Kraus operator B^{i_0} allows direct dimension growth, giving $S_{D^2-k_B+1}(B) = M_D(\mathbb{C})$, where $k_B = \text{kr}(B)$. In the non-invertible case, the bound $S_{D^2}(B) = M_D(\mathbb{C})$ follows from tracking indices through the rank-one extraction argument of [SPGWC10, Lemma 2(b)]; the separate blocked support construction in Theorem D.4.6 does not supply this explicit bound. In both cases, $\iota(A) \leq D^2n \leq D^2(D^2 - d' + 1)$. $\square$

Remark 8.4.9 (Quantitative and qualitative Wielandt bounds). ✓ Stmt The quantitative bound $\iota(A) \leq (D^2 - d' + 1)D^2$ is cited from [SPGWC10, Theorem 1, case (1)]. The later Fundamental Theorem needs only the qualitative conclusion $S_N(A) = M_D(\mathbb{C})$ for some N . The blocked construction in Theorem D.4.6 supplies this conclusion directly from normality; the explicit index bound is not needed there.

8.5 Fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$. The next stage is to pass to a span at one fixed length, namely to find n_0 such that $S_{n_0}(A) = M_D(\mathbb{C})$ (i.e. word products of *exactly* one length span all matrices). The product-span facts used here are proved in Section D.1; the blocked fixed-length construction is in Section D.4.

Lemma 8.5.1 (Eigenvector padding). Lean ✓ Stmt Suppose $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$. Then $K_n(A, \varphi) = H_n(A, \varphi)$ for every n . In particular, if $K_n(A, \varphi) = \mathbb{C}^D$, then $H_n(A, \varphi) = \mathbb{C}^D$.

Proof. ✓ Proof Any word w with $|w| \leq n$ can be padded to exact length n by appending $k = n - |w|$ copies of the index i_0 . The eigenvector relation gives $A^{w \cdot i_0^k}\varphi = \mu^k A^w\varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of K_n lies in H_n . The reverse inclusion $H_n \leq K_n$ is immediate. $\square$

Theorem 8.5.2 (Fixed-length vector spanning implies matrix spanning). Lean ✓ Stmt Assume:

1. $H_n(A, \varphi) = \mathbb{C}^D$ (length- n word products applied to φ span all of $\mathbb{C}^D$), and
2. for each standard basis vector e_j , the rank-one operator $|\varphi\rangle\langle e_j|$ lies in $S_m(A)$.

Then $S_{n+m}(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof Fix a matrix unit E_{ij} . By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i\varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i|\varphi\rangle\langle e_j| = |M_i\varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij}.$$

Lemma D.1.1 therefore gives $E_{ij} \in S_{n+m}(A)$. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. $\square$

Lemma 8.5.3 (Block injectivity and blocked-tensor injectivity). Lean ✓ Stmt A tensor A is L -block injective if and only if its L -blocked tensor $A^{[L]}$ is injective.

Proof. ✓ Proof The blocked physical alphabet is a reindexing of words of length L . Decoding blocked indices therefore identifies the two spanning families, and hence the two injectivity conditions are equivalent. $\square$

Lemma 8.5.4 (Blocking transfer). Lean ✓ Stmt *If the blocked tensor $A^{[L]}$ satisfies $S_n(A^{[L]}) = M_D(\mathbb{C})$, then $S_{nL}(A) = M_D(\mathbb{C})$. In particular, A is L -block injective if and only if $A^{[L]}$ is one-site injective.*

Proof. ✓ Proof By the blocked-tensor definition (2.13), each length- n word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length- nL word in the original alphabet $\{0, \dots, d - 1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. For $n = 1$, the d^L blocked letters are indexed by the length- L words of A , giving $S_1(A^{[L]}) = S_L(A)$, which proves the stated equivalence. $\square$

Theorem 8.5.5 (Bounded injective blocking for left-canonical normal tensors). Lean ✓ Stmt *Let A be a left-canonical normal MPS tensor with bond dimension $D > 0$. Then there is a positive blocking length $L \leq D^4$ such that the blocked tensor $A^{[L]}$ is one-site injective.*

This is the blocked-injectivity form of the quantum Wielandt input used in the Fundamental Theorem for translation-invariant tensors. It corresponds to the statement in [CPGSV16, Section II] that every normal tensor becomes injective after at most D^4 blockings, using the index estimate of [SPGWC10, Theorem 1]. The left-canonical/trace-preserving hypothesis is included because this is the canonical-form context in which the blocked-injectivity input is used.

Proof. ✓ Proof Normality is eventual full Kraus rank, so Theorem 8.1.1.28 gives paper primitivity. Theorem 8.4.8 then yields

$$\iota(A) \leq (D^2 - \text{kr}(A) + 1)D^2.$$

Trace preservation gives a nonzero Kraus operator, hence $\text{kr}(A) \geq 1$ and the right-hand side is at most D^4 . Set $L = \iota(A)$. By definition, $L > 0$ and $S_L(A) = M_D(\mathbb{C})$. Lemma 8.5.4 identifies the latter equality with one-site injectivity of $A^{[L]}$. $\square$

Theorem 8.5.6 (Exact quantum-Wielandt blocking length). Lean ✓ Stmt *Let A be a left-canonical normal MPS tensor with bond dimension $D > 0$. Then $S_{D^4}(A) = M_D(\mathbb{C})$; equivalently, A is D^4 -block injective.*

Proof. ✓ Proof Choose L with $0 < L \leq D^4$ and $S_L(A) = M_D(\mathbb{C})$, and write $D^4 = L + k$. Induction on k , using positive-length propagation at each step, gives $S_{L+k}(A) = M_D(\mathbb{C})$. $\square$

Theorem 8.5.7 (Common exact block-injectivity length). Lean ✓ Stmt *Let $(A_j)_{j=1}^g$ be a finite family of positive-bond-dimension, left-canonical normal tensors. There is a positive integer L such that every A_j is L -block injective.*

Proof. ✓ Proof Apply Theorem 8.5.5 to each block. The product of the finitely many positive lengths is a common positive multiple, and exact block injectivity persists under positive multiples. $\square$

Corollary 8.5.8 (Uniform injective blocking for separated normal-canonical families). Lean ✓ Stmt *Let $(A_x)_{x \in X}$ be a finite separated normal-canonical family whose blocks have positive virtual bond dimension. Then there is a positive integer L such that every blocked tensor $A_x^{[L]}$ is one-site injective. The same conclusion holds simultaneously for two finite separated normal-canonical families whose blocks have positive virtual bond dimension.*

Here the family is a basis of normal-canonical representatives in the sense of Definition 9.9.1.2; the blocks are distinct and the weight moduli are non-increasing but need not be strictly ordered.

The representative family is assumed to have been selected; this statement does not reconstruct the full CPSV sector decomposition with repeated copies inside one sector. This scope restriction is recorded in `docs/paper-gaps/ft_one_copy_scope_restriction.tex`.

Proof. ✓ Proof Apply Theorem 8.5.5 to each block. Taking the product of the finitely many resulting positive blocking lengths gives a common positive multiple, and injectivity persists under positive further blocking. $\square$

Remark 8.5.9 (Quantitative and blocked rank-one arguments). ✓ Stmt The quantitative rank-one statement of [SPGWC10, Lemma 2(b)] is Theorem 8.4.5. Theorem 8.5.2 turns that rank-one conclusion and fixed-length vector spreading into matrix spanning. Section D.4 also proves a separate blocked construction, Theorem D.4.6, whose fixed-length conclusion follows from its normality hypothesis.

Remark 8.5.10 (Use of the quantum Wielandt theorem in the Fundamental Theorem). The cumulative estimate $T_{D^2}(A) = M_D(\mathbb{C})$ in Theorem 8.2.1 supports the proof of Lemma 1; it is not a conclusion of source Theorem 1. The source theorem instead bounds the first positive N for which $S_N(A) = M_D(\mathbb{C})$. This exact-length statement matches injectivity of Γ_N and passes through blocking by Lemma 8.5.4.

The blocked normality and rank-one constructions used below are proved in Section D.4.

Remark 8.5.11 (Blocking argument for fixed-length spanning). ✓ Stmt Theorem D.4.6 first extracts a rank-one operator in a blocked tensor and then applies the fixed-length matrix-spanning argument. It is a separate consequence of normality, not the quantitative statement of source Lemma 2(b).

8.6 Primitive MPS tensors

Definition 8.6.1 (Primitive MPS tensor). Lean ✓ Stmt Assume $D > 0$. A tensor A together with a matrix $\rho \in M_D(\mathbb{C})$ is *primitive* if

$$\begin{aligned} \sum_i (A^i)^\dagger A^i &= \mathbb{1}, \\ \rho &\geq 0, \quad \rho \neq 0, \quad \mathcal{E}_A(\rho) = \rho, \\ \rho_{\text{spec}}(\mathcal{E}_A - P) &< 1, \end{aligned}$$

where P is the fixed-point projection associated to ρ . When the choice of ρ is irrelevant, we simply say that A is a primitive MPS tensor. This condition combines a complementary spectral gap with a nonzero positive semidefinite fixed point. It is not the paper definition in Definition 8.1.1.3, which is the uniform spreading condition (8.1). With the additional hypothesis $\rho > 0$, the complementary-gap condition implies strong irreducibility and hence paper primitivity by Theorem 8.1.1.28.

8.7 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 8.6.1: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius < 1 .

8.7.1 From primitivity to normality

The primitivity condition of Definition 8.6.1 combines TP normalization, a nonzero PSD fixed point, and a complementary transfer-map gap. It is connected to normality through the following results.

Remark 8.7.1.1 (Complementary-gap and paper primitivity). ✓ Stmt Definition 8.6.1 assumes TP normalization, a nonzero positive semidefinite fixed point ρ , and $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$. Without $\rho > 0$, these hypotheses do not imply irreducibility or normality; for example, a fixed point supported on $\text{diag}(1, 0)$ need not see the complementary invariant subspace.

If $\rho > 0$, the transfer map is irreducible and the complementary gap gives peripheral spectrum $\{1\}$. Thus the tensor is strongly irreducible in the sense of Definition 8.1.1.6. Proposition 3 then gives uniform spreading and eventual full Kraus rank. This implication compares the two notions; it does not identify Definition 8.6.1 with the paper definition.

8.7.2 Primitive implies irreducible

Lemma 8.7.2.1 (Primitive tensors with positive-definite fixed point are irreducible). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ and ρ is positive definite, then A is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV16, Section 2.3]).*

Proof. ✓ Proof By contrapositive. Assume A has an invariant projection P ($P \neq 0$, $P \neq 1$, $(1 - P)A^i P = 0$ for all i). Set $\sigma = P\rho P$.

1. *Invariance of σ under $\mathcal{E}_A^n$:* the projection relation $(1 - P)A^i P = 0$ implies $(1 - P)\mathcal{E}_A^n(\sigma) = 0$ for all n (by induction on n).
2. *Convergence:* By Lemma D.5.6, $\mathcal{E}_A^n(\sigma) \rightarrow \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$. Hence $(1 - P)\frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho = 0$.
3. *Nonzero scalar:* Since $P \neq 0$ and ρ is positive definite, $\text{tr}(P\rho P) > 0$, so $\frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \neq 0$. Therefore $(1 - P)\rho = 0$.
4. *Contradiction:* Since ρ is positive definite, it is a unit in $M_D(\mathbb{C})$. From $(1 - P)\rho = 0$ we get $1 - P = 0$, i.e. $P = 1$, contradicting $P \neq 1$.

□

8.7.3 From irreducibility to full spanning via Burnside's theorem

The irreducibility condition (Definition 9.1.1.1) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of $\mathbb{C}^D$ under a finite matrix family.

Definition 8.7.3.1 (Algebra span). Lean ✓ Stmt For a finite matrix family A , let $\text{alg}(A)$ be the unital $\mathbb{C}$ -subalgebra of $M_D(\mathbb{C})$ generated by the matrices $\{A^i\}_{i=0}^{d-1}$: $\text{alg}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d-1 \rangle \subseteq M_D(\mathbb{C})$.

Definition 8.7.3.2 (Invariant submodule). Lean ✓ Stmt A subspace $W \leq \mathbb{C}^D$ is *A-invariant* if $A^i W \subseteq W$ for every i .

Definition 8.7.3.3 (Irreducible action). Lean ✓ Stmt The matrices in A act *irreducibly* on $\mathbb{C}^D$ if every A -invariant subspace is trivial, i.e. equal to 0 or to $\mathbb{C}^D$.

Theorem 8.7.3.4 (Irreducible tensor implies irreducible action). Lean ✓ Stmt *If A is irreducible as a tensor (Definition 9.1.1.1), then the Kraus operators act irreducibly on $\mathbb{C}^D$ (Definition 8.7.3.3).*

Proof. ✓ Proof Let $W \leq \mathbb{C}^D$ be an A -invariant subspace. In finite dimension the orthogonal projection P onto W exists. Invariance of W implies $(\mathbb{1} - P)A^i P = 0$ for all i . Thus P is a nontrivial invariant projection unless $W = 0$ or $W = \mathbb{C}^D$, contradicting irreducibility of the tensor. $\square$

Theorem 8.7.3.5 (Burnside’s theorem for matrix algebras). Lean ✓ Stmt *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then the generated unital subalgebra is the full matrix algebra: $\text{alg}(A) = M_D(\mathbb{C})$. This is the complex finite-dimensional case of Burnside’s theorem (equivalently, Jacobson’s density theorem); see [Jac09].*

Proof. ✓ Proof Identify $\text{alg}(A)$ with its image under the algebra equivalence $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \text{End}_{\mathbb{C}}(\mathbb{C}^D)$. The irreducible-action hypothesis makes $\mathbb{C}^D$ a simple module for this algebra. Over the algebraically closed field $\mathbb{C}$, Schur’s lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$. $\square$

Remark 8.7.3.6. ✓ Stmt This Burnside/Jacobson-density step is a substantial external algebraic ingredient in the chapter. We use it here as the finite-dimensional complex Burnside theorem, namely the statement that an irreducible matrix algebra action already generates the full endomorphism algebra.

8.7.4 From primitivity to strong irreducibility and normality

The results in this subsection remove the one-step padding assumption from the normality conclusions. They connect primitivity in Definition 8.6.1, formulated via a complementary transfer-map gap, directly to normality (eventually full Kraus rank), using the convergence of the transfer-map iterates to the projection onto the fixed-point subspace.

Lemma 8.7.4.1 (Primitive tensors with positive-definite fixed point are strongly irreducible). Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ , then it satisfies the strengthened condition in Definition 8.1.1.6: $\mathcal{E}_A$ is irreducible, the fixed point ρ is positive definite, and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$. The source clause is [SPGWC10, Proposition 3(c)]; the definition used here records irreducibility separately.*

Proof. ✓ Proof Lemma D.5.7 gives irreducibility of $\mathcal{E}_A$. The complementary-gap hypothesis gives peripheral primitivity directly: the only eigenvalue of modulus one is 1. Together with the positive-definite fixed point ρ , these are exactly the three conditions in Definition 8.1.1.6. The complementary-gap consequences are proved in Section D.5. $\square$

Theorem 8.7.4.2 (Primitive MPS with positive-definite fixed point is normal). Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ , then A is normal (has eventually full Kraus rank). No identity-in-the-one-step-span condition is needed.*

This connects the complementary-gap condition in Definition 8.6.1 directly to normality.

Proof. ✓ Proof By Lemma 8.7.4.1, A is strongly irreducible. Theorem 8.1.1.28 identifies strong irreducibility with eventual full Kraus rank, hence with normality. $\square$

8.8 One-step padding and exact word spans

Cumulative spanning, $T_N(A) = M_D(\mathbb{C})$, is weaker than exact-length spanning $S_L(A) = M_D(\mathbb{C})$. If $1 \in S_1(A)$, shorter words can be padded to a common length. Section D.6 proves this conversion and records a counterexample without the one-step padding condition.

Theorem 8.8.1 (Homogeneous word-span propagation under normalization). [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) Suppose

$$\sum_i A^i (A^i)^\dagger = 1, \quad (8.3)$$

and $S_L(A) = M_D(\mathbb{C})$. Then $S_m(A) = M_D(\mathbb{C})$ for every $m \geq L$. Equivalently, L -block injectivity propagates to every larger homogeneous length.

Proof. [✓ Proof](#) If $S_n(A) = M_D(\mathbb{C})$, then for every $X \in M_D(\mathbb{C})$, substituting (8.3) gives

$$X = \left(\sum_i A^i (A^i)^\dagger \right) X = \sum_i A^i ((A^i)^\dagger X) \in S_1(A) S_n(A) \subseteq S_{n+1}(A).$$

Indeed, $(A^i)^\dagger X \in S_n(A)$ because $S_n(A) = M_D(\mathbb{C})$. Thus $S_{n+1}(A) = M_D(\mathbb{C})$. Iterating this implication gives the result for every $m \geq L$. $\square$

Theorem 8.8.2 (Algebra span plus one-step padding implies normality). [Lean](#) [✓ Stmt](#) If $\text{alg}(A) = M_D(\mathbb{C})$ and $1 \in S_1(A)$, then A is normal.

Proof. [✓ Proof](#) Since $\text{alg}(A) = M_D(\mathbb{C})$, the ascending chain condition for finite-dimensional subspaces gives an N such that $T_N(A) = M_D(\mathbb{C})$. Apply Theorem D.6.5. $\square$

Remark 8.8.3 (Normal canonical form avoids one-step padding). The identity-in-the-one-step-span results in Section D.6 are needed only when cumulative spanning is upgraded to exact-length spanning. The normal-canonical theory used later in Chapters 9–10 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, places them in normal canonical form in the sense of [CPGSV16]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are not needed for the implication stated below.

8.9 Block injectivity from TP/primitive/irreducible blocks

Theorem 8.9.1 (TP-primitive irreducible tensor is injective after blocking). [Lean](#) [✓ Stmt](#) Let A be a left-canonical MPS tensor with $D > 0$. Suppose that A is tensor-irreducible and that its transfer map is primitive. Then there exists a positive blocking length L such that $A^{[L]}$ is one-site injective.

Proof. [✓ Proof](#) If $D = 1$, trace preservation gives a nonzero one-site Kraus matrix, and every nonzero scalar matrix spans $M_1(\mathbb{C})$. Hence A is one-site injective, so Lemma 2.4.3 gives the positive witness $L = 1$. Suppose now that $D \geq 2$. Theorem 9.11.1.1 gives a block-injectivity witness L . If $L = 0$, Corollary D.1.3 would force $D = 1$, a contradiction. Thus $L > 0$, and the blocked-chain equivalence identifies this L -block injectivity with one-site injectivity of the blocked tensor $A^{[L]}$. $\square$

Remark 8.9.2. Blocking preserves normality by Theorem D.4.1, proved in Section D.4.

Chapter 9

Canonical Form Reduction

This is the first of three chapters that together prove the Fundamental Theorem of matrix product states. Here an arbitrary translation-invariant MPS tensor is reduced to a weighted family of primitive blocks; Chapter 10 constructs the basis of normal tensors carried by those blocks and compares their coefficients, and Chapter 11 proves the theorem itself. The reduction follows [PGVWC07], [CPGSV16], [CPGSV21], and [Wol12, Chapter 6]. Its goal is to expose the primitive blocks inside the tensor. The transfer operator of an arbitrary tensor can be reducible and periodic. The reduction first splits invariant subspaces. For each irreducible trace-preserving block, it then powers the transfer map, extracts the adjoint-fixed cyclic projections, compresses to their corners, and proves that each corner is primitive and tensor-irreducible. The blocks are returned in a fixed gauge. The periodic decomposition theorem of [PGVWC07, Theorem 5] gives p translated p -periodic state summands when $p \mid N$ and a zero state otherwise. We state this fixed-length decomposition below before using its blocked cyclic-sector counterpart in Theorem 9.3.1.1.

Definition 9.1 (Periodic projector-component chain). Lean Lean Lean ✓ Stmt Let $A = (A^i)_{i=0}^{d-1}$ have bond dimension D , let $(Q_u)_{u \in \mathbb{Z}/p\mathbb{Z}}$ be cyclic projections, and fix a starting sector u . For each $0 \leq j < N$, the corresponding chain of length N has local matrices

$$B_{u,j}^i = Q_{u+j} A^i Q_{u+j+1}, \quad (9.1)$$

where sector indices are read modulo p . It has the same bond dimension D as A and repeats with period p .

Theorem 9.2 (Boundary traces and component-chain coefficients). Lean Lean ✓ Stmt Suppose that the orthogonal projections satisfy $\sum_u Q_u = \mathbb{1}$ and $Q_u A^i = A^i Q_{u+1}$. For a word $\sigma = (i_0, \dots, i_{N-1})$, define the boundary product

$$F_u(\sigma) = Q_u A^{i_0} Q_{u+1} A^{i_1} \dots Q_{u+N-1} A^{i_{N-1}} Q_{u+N}. \quad (9.2)$$

At every length $N \geq 0$,

$$V^{(N)}(A)_\sigma = \sum_{u \in \mathbb{Z}/p\mathbb{Z}} \text{tr} F_u(\sigma). \quad (9.3)$$

If $N > 0$, then $\text{tr} F_u(\sigma)$ is the coefficient of the explicit component chain from Definition 9.1. At $N = 0$, the boundary product is $F_u(\emptyset) = Q_u$, so

$$\text{tr}(\mathbb{1}) = \sum_{u \in \mathbb{Z}/p\mathbb{Z}} \text{tr}(Q_u). \quad (9.4)$$

This is not the sum of the ordinary empty component-chain coefficients: an empty chain evaluates to $\mathbb{1}$ and does not retain its starting projection.

Proof. ✓ Proof The cyclic relation and orthogonality give $F_u(\sigma) = Q_u A^{i_0} \dots A^{i_{N-1}}$. Hence

$$\mathrm{tr}(A^{i_0} \dots A^{i_{N-1}}) = \mathrm{tr} \left(\left(\sum_u Q_u \right) A^{i_0} \dots A^{i_{N-1}} \right) = \sum_u \mathrm{tr} F_u(\sigma). \quad (9.5)$$

For $N > 0$, multiplying the local matrices in (9.1) gives $F_u(\sigma)$, proving the coefficient identity. For $N = 0$, the same trace calculation reduces to (9.4). $\square$

Theorem 9.3 (Normalized periodic state-vector decomposition). Lean Lean Lean ✓ Stmt This is a normalized form of the state-vector conclusion in [PGVWC07, Theorem 5]. Let A be an irreducible left-canonical tensor whose peripheral transfer spectrum is the set of p -th roots of unity. If $p \mid N$, there are orthogonal cyclic projections $(Q_u)_{u \in \mathbb{Z}/p\mathbb{Z}}$ such that $\sum_u Q_u = \mathbb{1}$, $Q_u A^i = A^i Q_{u+1}$, and $Q_{u+N} = Q_u$. For every ring length $N \geq 0$, its coefficients satisfy the boundary-aware decomposition

$$V^{(N)}(A)_\sigma = \sum_{u \in \mathbb{Z}/p\mathbb{Z}} \mathrm{tr} F_u(\sigma). \quad (9.6)$$

For $N > 0$, this becomes the source's decomposition into the p explicit component chains:

$$|V^{(N)}(A)\rangle = \sum_{u \in \mathbb{Z}/p\mathbb{Z}} |V^{(N)}(B_u)\rangle. \quad (9.7)$$

Each B_u is a p -periodic MPS of bond dimension D with local matrices given by (9.1). At $N = 0$, (9.6) instead uses the boundary traces $\mathrm{tr}(Q_u)$ from (9.4); it does not identify them with ordinary empty-chain coefficients. If $p \nmid N$, then $|V^{(N)}(A)\rangle = 0$.

Proof. ✓ Proof Choose the cyclic projections supplied by periodicity. If $p \mid N$, then $Q_{u+N} = Q_u$, and Theorem 9.2 gives (9.6) at every length. For $N > 0$, the explicit component-chain decomposition follows by inserting $\sum_u Q_u = \mathbb{1}$ and multiplying the local corner matrices; this proves (9.7). The separate coefficient identity in Theorem 9.2 identifies its summands with the boundary traces. If $p \nmid N$, then $Q_{u+N} \neq Q_u$; cyclicity of the trace and orthogonality give $\mathrm{tr} F_u(\sigma) = 0$ for every u and σ . $\square$

Two normalizations appear. The unital orientation $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$ is used in [PGVWC07, Theorem 4]; the trace-preserving (left-canonical) orientation $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ is used in the TP-gauge and after-blocking reductions below. They exchange under the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$, whose transfer map is the Frobenius adjoint of $\mathcal{E}_A$.

9.1 Invariant-subspace decomposition and irreducible blocks

The block decomposition rests on one principle: a positive semidefinite fixed point of $\mathcal{E}_A$ that is not full rank has a proper invariant support, which block-diagonalizes the Kraus operators without changing the MPV family. Iterating the resulting splitting decomposes any tensor into irreducible blocks.

Theorem 9.1.1 (Invariant projection gives a two-block decomposition). Lean ✓ Stmt Suppose P is an orthogonal projection satisfying $(\mathbb{1} - P)A^i P = 0$ for every physical index i . Then there exist n, m with $n + m = D$ and MPS tensors A_1, A_2 of bond dimensions n and m such that A and the two-block tensor $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.3.3.

Proof. ✓ Proof Diagonalize P by a unitary U . Since $P^2 = P$, the diagonal form of P has only 0- and 1-eigenvalues, so the basis splits as $n + m = D$. Conjugating A by U preserves the MPV family by Theorem E.1.1. In this basis the relation $(\mathbb{1} - P)A^iP = 0$ makes each letter upper block triangular. Write the conjugated letters as

$$B^i = \begin{pmatrix} A_1^i & * \\ 0 & A_2^i \end{pmatrix}.$$

Since $B^{i_1} \dots B^{i_N}$ is again upper block triangular, its trace satisfies

$$\text{tr}(B^{i_1} \dots B^{i_N}) = \text{tr}(A_1^{i_1} \dots A_1^{i_N}) + \text{tr}(A_2^{i_1} \dots A_2^{i_N}). \quad (9.8)$$

Hence the block-diagonal tensor $A_1 \oplus A_2$ has the same MPV coefficients as the conjugated tensor. $\square$

Definition 9.1.2 (Invariant projection condition). Lean ✓ Stmt An MPS tensor A has an *invariant projection* if there exists an orthogonal projection P with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1} - P)A^iP = 0$ for all i . A tensor is *irreducible* if it has no nontrivial invariant projection (Definition 9.1.1.1).

Theorem 9.1.3 (PSD fixed point gives invariant projection). Lean ✓ Stmt Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let P be its support projection. Then $(\mathbb{1} - P)A^iP = 0$ for all i : the range of P is invariant under every A^i .

Proof. ✓ Proof From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i\rho(A^i)^\dagger$, one gets $(\mathbb{1} - P)A^i\rho(A^i)^\dagger(\mathbb{1} - P) = 0$ for each i . Since ρ is supported on the range of P , this forces $(\mathbb{1} - P)A^iP = 0$.

This is the finite-dimensional case of [Wol12, Proposition 6.10]: the support projection of any stationary state is sub-harmonic. The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Theorem 6.2]; see also [CPGSV21, Section IV.A.1]. $\square$

Theorem 9.1.4 (Two-block decomposition). Lean ✓ Stmt If $\mathcal{E}_A$ has a nonzero PSD fixed point ρ that is not positive definite, then there exist MPS tensors A_1, A_2 of smaller bond dimensions with $\mathcal{V}(A) = \mathcal{V}(A_1 \oplus A_2)$.

Proof. ✓ Proof The support projection P of ρ is neither 0 nor $\mathbb{1}$, since $\rho \neq 0$ and ρ is not positive definite. Theorem 9.1.3 gives $(\mathbb{1} - P)A^iP = 0$, so a unitary change of basis diagonalizing P puts A^i in upper-triangular block form with diagonal blocks B_{11}^i, B_{22}^i . Theorem E.1.1 preserves the MPV under the conjugation, and the trace identity (9.8) shows that deleting the off-diagonal blocks preserves all MPV coefficients. $\square$

9.1.1 Iterated reduction to irreducible blocks

Iterating the two-block splitting on bond dimension decomposes any tensor into irreducible blocks; the remaining lemmas record the spectral facts about irreducible unital blocks and the algebra-spanning input used later.

Definition 9.1.1.1 (Irreducible tensor). Lean ✓ Stmt An MPS tensor A is *irreducible* if it has no nontrivial invariant projection.

Theorem 9.1.1.2 (Irreducible block decomposition). Lean ✓ Stmt Every MPS tensor A admits a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ with each A_k irreducible, $\mathcal{V}(A) = \mathcal{V}(\bigoplus_k A_k)$, and $\sum_{k=1}^r D_k = D$.

Proof. ✓ Proof By strong induction on D . If A is irreducible, take $r = 1$. Otherwise a nontrivial invariant projection P gives the upper-triangular splitting of Theorem 9.1.4; deleting the off-diagonal part preserves the MPV family and yields two blocks of strictly smaller bond dimension. Apply the induction hypothesis to each block.

Both [PGVWC07, Theorem 4] and [CPGSV21, Section IV.A.1] obtain the block decomposition by the same invariant-projection splitting. $\square$

9.2 Nonzero blocks and the trace-preserving gauge

In an irreducible block decomposition some blocks may have all-zero Kraus operators. Such a block contributes nothing to a matrix product vector at any positive length and is therefore discarded. The structural direct-sum identity implies that the retained block dimensions satisfy $\sum_k D_k \leq D$ without using an empty-word coefficient. Each retained block is then placed in the trace-preserving gauge.

Theorem 9.2.1 (Blockwise TP-gauge normalization of irreducible blocks). Lean ✓ Stmt Suppose A generates the same MPV family as $\bigoplus_{k=1}^r B_k$, where each B_k is irreducible with at least one nonzero Kraus operator. Then there are nonzero weights $(\mu_k)_{k=1}^r$ and blocks $(C_k)_{k=1}^r$ such that, for every N and every σ ,

$$V^{(N)}(A)_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r \mu_k C_k\right)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(C_k)_\sigma, \quad (9.9)$$

each C_k is irreducible and left-canonical, $\sum_i (C_k^i)^\dagger C_k^i = \mathbb{1}$, with positive bond dimension. This is the blockwise form used in the arbitrary-tensor trace-preserving gauge below.

Proof. ✓ Proof The Perron–Frobenius TP-gauge construction is applied to each irreducible block using the gauge (6.29). Gauge equivalence preserves irreducibility and the MPV family, and the positive scaling factors are absorbed into the weights μ_k . $\square$

9.2.1 Trace-preserving gauge for arbitrary tensors

Theorem 9.2.1.1 (TP-gauge reduction for arbitrary tensors). Lean ✓ Stmt For any MPS tensor A , there exist nontrivial blocks $(B_k)_{k=1}^r$ with nonzero weights $(\mu_k)_{k=1}^r$, each block irreducible, left-canonical ($\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$), and of positive bond dimension, such that, for every $N \geq 1$ and every σ ,

$$V^{(N)}(A)_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r \mu_k B_k\right)_\sigma, \quad (9.10)$$

and $\sum_{k=1}^r D_k \leq D$.

Proof. ✓ Proof Theorem E.1.3 supplies the nonzero irreducible blocks $(C_k)_{k=1}^r$, with $V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r C_k)_\sigma$ for every $N \geq 1$ and every σ . Applying the Perron–Frobenius TP gauge (6.29) to each C_k gives a left-canonical block B_k with weight μ_k , and

$$V^{(N)}\left(\bigoplus_{k=1}^r C_k\right)_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r \mu_k B_k\right)_\sigma.$$

The gauge preserves every block dimension, so the structural bound $\sum_k D_k \leq D$ is unchanged. $\square$

9.3 Blocking, period removal, and primitive blocks

Blocking by an integer p carries MPV equalities, weights, transfer maps, and identifications of the blocked physical alphabet. For an irreducible trace-preserving block, powering first collapses the peripheral eigenvalue set to $\{1\}$; the powered map can nevertheless split into several cyclic corners. The reduction compresses to those corners and proves primitivity there. A later least-common-multiple step places a finite family of the resulting sectors over one blocked alphabet.

9.3.1 Period removal by cyclic sectors

The period-removing blocking of [CPGSV16, Section 2.3] has four distinct steps. Power the transfer map by its period, obtain the adjoint-fixed cyclic projections, compress to their corners, and prove that each corner restriction is irreducible. Only then does the peripheral set $\{1\}$ imply primitivity of the sector maps. The cyclic projections come from [Wol12, Theorem 6.6].

Theorem 9.3.1.1 (Period removal by blocking). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor. Then there is a period-removing length $m \geq 1$ such that $A^{[m]}$ admits:*

1. left-canonical sector tensors $(C_k)_{k=1}^m$ of nonzero bond dimension,
2. a unit-weight direct-sum MPV decomposition $A^{[m]} \sim \bigoplus_{k=1}^m C_k$,
3. orthogonal projections $(P_k)_{k=1}^m$ summing to $\mathbb{1}$ with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$,
4. commutation relations $P_k(A^{[m]})^i = (A^{[m]})^i P_k$ for every blocked letter i ,
5. the sector trace formula $f_{C_k}(\sigma) = \text{tr}(P_k(A^{[m]})^{\sigma_1} \dots (A^{[m]})^{\sigma_N})$,
6. linear isomorphisms onto the compression corners $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$,
7. transfer-intertwining identities $\varphi_k(\mathcal{E}_{C_k^\dagger}(X)) = \mathcal{E}_{(P_k A^{[m]})^\dagger}(\varphi_k(X))$,
8. multiplicativity and adjoint compatibility $\varphi_k(XY) = \varphi_k(X)\varphi_k(Y)$ and $\varphi_k(X^\dagger) = \varphi_k(X)^\dagger$.

This is the cyclic-sector part of the period-removal step of [CPGSV16, Section 2.3]; the projections P_k come from the peripheral spectrum of the adjoint transfer map ([Wol12, Theorem 6.6]).

Proof. ✓ Proof The peripheral cyclic-structure theorem identifies the period and its primitive root. The cyclic decomposition of the adjoint transfer map gives projections (P_k) with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$. Iterating the cyclic relation m times gives $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$, and the blocked-transfer identity $\mathcal{E}_{(A^{[m]})^\dagger} = (\mathcal{E}_{A^\dagger})^m$ identifies these with the adjoint transfer map of $A^{[m]}$. Apply the block decomposition from adjoint-fixed projections. $\square$

Theorem 9.3.1.2 (Unconditional primitive blocked sectors). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum $\{\gamma^j : 0 \leq j < m\}$ for a primitive m th root of unity γ , with blocked cyclic-sector decomposition $(C_u)_{u=1}^m, (P_u)_{u=1}^m$, and linear isomorphisms $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$. Assume explicitly that the P_u are orthogonal projections summing to $\mathbb{1}$ and satisfying $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$, that each sector has nonzero bond dimension, and that the compression maps satisfy*

$$\begin{aligned} \varphi_u(\mathcal{E}_{C_u^\dagger}(X)) &= \mathcal{E}_{(P_u A^{[m]})^\dagger}(\varphi_u(X)), \\ \varphi_u(XY) &= \varphi_u(X)\varphi_u(Y), \\ \varphi_u(X^\dagger) &= \varphi_u(X)^\dagger. \end{aligned} \tag{9.11}$$

Here $P_u A^{[m]}$ denotes the blocked tensor with letters $P_u(A^{[m]})^\alpha$. Then every sector tensor satisfies $\sigma_\partial(\mathcal{E}_{C_u}) = \{1\}$ and is tensor-irreducible.

Proof. ✓ Proof Theorem 9.4.1 gives irreducibility of $(\mathcal{E}_A^\dagger)^m$ on each corner P_u . The compression equivalences in (9.11) intertwine the adjoint transfer map of C_u with the corresponding corner restriction of $(\mathcal{E}_A^\dagger)^m$. The peripheral spectrum of the blocked adjoint map is

$$\sigma_\partial((\mathcal{E}_A^\dagger)^m) = \{(\gamma^j)^m : 0 \leq j < m\} = \{1\}, \quad (9.12)$$

because γ is a primitive m th root of unity. Hence each nonzero irreducible corner restriction is primitive. Primitivity and irreducibility are preserved across φ_u ; then use Theorem E.4.4 to pass from the adjoint blocked map to the primal transfer map of C_u . $\square$

Theorem 9.3.1.3 (Period removal with primitive irreducible sectors). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor. Then there is a period $m \geq 1$ and a unit-weight decomposition of $A^{[m]}$ into sector blocks $(C_k)_{k=1}^m$, each of which is left-canonical, has primitive transfer map, is tensor-irreducible, and has nonzero bond dimension. Equivalently, the cyclic-sector decomposition of $A^{[m]}$ already satisfies the trace-preservation, primitivity, and irreducibility hypotheses needed for the common-blocking step (Lemma E.5.5).*

Proof. ✓ Proof Left-canonicity makes $K_i := (A^i)^\dagger$ unital, irreducibility passes to the adjoint map, and the quantum Perron–Frobenius theorem provides a positive-definite fixed point; the cyclic peripheral-spectrum theorem and Theorem 9.3.1.1 then produce the cyclic-sector decomposition. Keep the projections and compression equivalences, and apply Theorem 9.3.1.2 to the sector blocks. $\square$

Theorem 9.3.1.4 (Common blocking length for cyclic sectors). Lean ✓ Stmt *If two tensors generate the same MPV family at every positive length, then the cyclic-sector decompositions on the two sides can be expressed at one common positive blocking length p . For every $N \geq 1$ and every blocked σ , each blocked tensor equals its powered nonzero part:*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}\left(\bigoplus_j (\mu_j^A)^p (A_j)^{[p]}\right)_\sigma, \quad (9.13)$$

$$V^{(N)}(B^{[p]})_\sigma = V^{(N)}\left(\bigoplus_k (\mu_k^B)^p (B_k)^{[p]}\right)_\sigma. \quad (9.14)$$

The two powered nonzero parts agree:

$$V^{(N)}\left(\bigoplus_j (\mu_j^A)^p (A_j)^{[p]}\right)_\sigma = V^{(N)}\left(\bigoplus_k (\mu_k^B)^p (B_k)^{[p]}\right)_\sigma. \quad (9.15)$$

Proof. ✓ Proof Start from Theorem E.5.4. With p_A, p_B the least common multiples of the period-removal lengths on the two sides, use $p = \text{lcm}(p_A, p_B)$. The prescribed-common-multiple lemma constructs both one-sided sector families at this length, and the positive-length blocking lemmas carry the one-sided tensor/nonzero-part equalities over to the common alphabet, giving (9.13) and (9.14). The two-sided blocking lemma applied to (E.16) gives (9.15). $\square$

9.4 Cyclic-sector irreducibility

The powered map is analyzed corner by corner before the resulting sector families are placed over a common blocked alphabet. The compression and common-alphabet identities are proved in Section E.5.

Lemma 9.4.1 (Sector irreducibility for the adjoint blocked map). Lean ✓ Stmt *Let A be a left-canonical irreducible MPS tensor with $D > 0$, let m be the period of its cyclic-sector decomposition, and let $(P_u)_{u=1}^m$ be the cyclic projections of the adjoint transfer map with $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$ and $\sum_u P_u = \mathbb{1}$. Then the restriction of $(\mathcal{E}_{A^\dagger})^m$ to each corner $P_u M_D(\mathbb{C}) P_u$ is irreducible.*

Definition 9.4.2 (Primitive irreducible cyclic-sector decomposition). Lean ✓ Stmt *A has primitive irreducible cyclic sectors of period $m \geq 1$ if there is a family $(C_k)_{k=1}^m$ with $A^{[m]} \sim \bigoplus_{k=1}^m C_k$ with unit weights, each C_k left-canonical, with primitive transfer map, tensor-irreducible, and of positive bond dimension.*

9.5 Translation-invariant canonical form

Definition 9.5.1 (Unnormalized positive-length PGVWC07 witness). Lean ✓ Stmt *An unnormalized positive-length PGVWC07 witness for a tensor A consists of finitely many blocks $(A_k)_{k=1}^r$ of positive bond dimensions and positive real weights $(a_k)_{k=1}^r$. Each block is unital, has only scalar transfer-map fixed points, and has a diagonal positive-definite fixed point Λ_k for the adjoint transfer map. Moreover,*

$$V^{(N)}(A)_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r a_k A_k\right)_\sigma \quad (N > 0),$$

and the retained bond dimensions satisfy $\sum_k \dim(A_k) \leq D$.

Theorem 9.5.2 (Construction of the positive-length PGVWC07 witness). Lean Lean Lean Lean ✓ Stmt *Every MPS tensor has an unnormalized positive-length PGVWC07 witness. More precisely, the construction has four stages. First, a nonzero irreducible block remains irreducible under nonzero scalar rescaling and can be put in a unital gauge with scalar fixed-point space and a diagonal positive-definite dual fixed point. Second, this construction is applied to each member of a finite nonzero irreducible block family. Third, the nonzero irreducible decomposition of an arbitrary tensor supplies that family and the total bond-dimension bound. Finally, these blocks, weights, and fixed points form a witness in the sense of Definition 9.5.1.*

Proof. ✓ Proof Follow [PGVWC07, Theorem 4]. The nonzero irreducible decomposition is obtained by strict dimension induction. On a nonzero current block, Perron–Frobenius gives a positive semidefinite eigenvector at a positive spectral radius. If it is singular, positivity makes its support invariant: $(\mathbb{1} - P)A^i P = 0$. After diagonalizing P , every letter is upper block triangular and the finite-ring trace splits as

$$\mathrm{tr}(A^{i_1} \dots A^{i_N}) = \mathrm{tr}(A_1^{i_1} \dots A_1^{i_N}) + \mathrm{tr}(A_2^{i_1} \dots A_2^{i_N}).$$

Discard all-zero blocks and repeat on the two strictly smaller diagonal blocks. This gives the finite nonzero irreducible family of Theorem E.1.3.

For each retained irreducible block, divide by the square root of its positive spectral radius and conjugate by the square root of its positive-definite Perron eigenvector. Theorem 6.8.6 gives

the unital identity

$$\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}.$$

Nonzero scalar rescaling preserves irreducibility. The proof passes from tensor irreducibility to irreducibility of the matrix action, carries invariant subspaces through the gauge similarity, and returns to tensor irreducibility using Theorem 8.7.3.4 and Lemma D.2.2. Hence Theorem 9.8.1 makes every fixed point scalar. Theorem 9.10.2 then diagonalizes a positive-definite dual fixed point by a unitary without changing unitality.

Apply this single-block construction to every member of the finite family. The square roots of the block spectral radii become positive weights, gauge invariance preserves every positive-length MPV coefficient, and the block dimensions are unchanged. Combining this blockwise result with the initial decomposition yields the arbitrary-input unnormalized form and its bound $\sum_k \dim(A_k) \leq D$. Together these conclusions give the structured witness. $\square$

Theorem 9.5.3 (Finite normalization of a nonempty PGVWC07 witness). [Lean](#) [Lean](#) [✓ Stmt](#) *Let W be an unnormalized positive-length witness with positive weights a_k . If some positive-length MPV coefficient of the original tensor is nonzero, then W has at least one block. For any nonempty witness, define*

$$s := \max_{1 \leq k \leq r} a_k, \\ \nu_k := \frac{a_k}{s}.$$

Then $s > 0$, every ν_k is positive, $|\nu_k| \leq 1$, and $|\nu_k| = 1$ for at least one k . For every $N > 0$,

$$V^{(N)}(A)_\sigma = s^N V^{(N)}\left(\bigoplus_k \nu_k A_k\right)_\sigma. \quad (9.16)$$

Proof. [✓ Proof](#) If the witness had no blocks, its weighted block tensor would have zero MPV coefficients at every positive length, contrary to the assumed nonzero coefficient. Thus $r > 0$. Since the finite family (a_k) is nonempty and every $a_k > 0$, its maximum s is positive. The definition $\nu_k = a_k/s$ gives $0 < \nu_k \leq 1$, and any maximizing index has $\nu_k = 1$. Finally, $a_k = s\nu_k$ for every k , so homogeneity of a length- N matrix product vector gives (9.16). $\square$

Theorem 9.5.4 (Normalized exact form for a nonzero MPV family). [Lean](#) [✓ Stmt](#) *Suppose that $V^{(N)}(A)_\sigma \neq 0$ for some $N > 0$ and some word σ . Then there are a scalar $s > 0$ and a nonempty finite family of positive-bond-dimension blocks $(A_k)_{k=1}^r$ with positive weights $(\mu_k)_{k=1}^r$ such that $s^{-1}A$ and $\bigoplus_k \mu_k A_k$ have identical MPV coefficients at every positive length. The weights satisfy $|\mu_k| \leq 1$, with equality for at least one block, and*

$$\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}, \\ \sum_i (A_k^i)^\dagger \Lambda_k A_k^i = \Lambda_k \quad (9.17)$$

for a diagonal positive-definite matrix Λ_k . Every fixed point of $\mathcal{E}_{A_k}$ is a scalar multiple of $\mathbb{1}$, and $\sum_k \dim(A_k) \leq D$.

Proof. ✓ Proof Choose the structured witness from Theorem 9.5.2. The assumed nonzero coefficient makes its block family nonempty, so Theorem 9.5.3 supplies $s > 0$ and the normalized positive weights. Equation (9.16) and homogeneity give

$$V^{(N)}(s^{-1}A)_\sigma = s^{-N}V^{(N)}(A)_\sigma = V^{(N)}\left(\bigoplus_k \mu_k A_k\right)_\sigma.$$

The witness satisfies all required block conditions and the bond-dimension bound. $\square$

Theorem 9.5.5 (Zero/nonzero alternative for the normalized exact form). Lean ✓ Stmt *For every MPS tensor A , one of the following alternatives holds:*

1. $V^{(N)}(A)_\sigma = 0$ for every $N > 0$ and every word σ ;
2. there are a scalar $s > 0$ and a nonempty normalized block family satisfying all conclusions of Theorem 9.5.4.

Proof. ✓ Proof Split according to whether there exist $N > 0$ and σ with $V^{(N)}(A)_\sigma \neq 0$. In that case apply Theorem 9.5.4. If no such pair exists, every positive-length coefficient vanishes. $\square$

Theorem 9.5.6 (Translation-invariant canonical form). Lean ✓ Stmt *Let A be a translation-invariant MPS representation of bond dimension D . There is a positive scalar s and a finite family of blocks A_k , possibly empty, such that the globally rescaled tensor $s^{-1}A$ has the same MPV coefficients on every positive-length ring as $\bigoplus_k \mu_k A_k^i$. The weights are positive and satisfy $1 \geq |\mu_k|$; if the family is nonempty, then some weight has norm 1. For each block,*

$$\begin{aligned} \sum_i A_k^i (A_k^i)^\dagger &= \mathbb{1}, \\ \sum_i (A_k^i)^\dagger \Lambda_k A_k^i &= \Lambda_k. \end{aligned} \tag{9.18}$$

Here Λ_k is diagonal, positive, and full-rank, and each block has positive bond dimension. Moreover, the only fixed points of $\mathcal{E}_{A_k}(X) = \sum_i A_k^i X (A_k^i)^\dagger$ are the scalar multiples of $\mathbb{1}$: for every X , the identity $\mathcal{E}_{A_k}(X) = X$ implies that there exists $c \in \mathbb{C}$ such that $X = c\mathbb{1}$. The total bond dimension of the decomposition is at most D .

This is the positive-length form of [PGVWC07, Theorem 4], with the global spectral-radius normalization step made explicit.

Proof. ✓ Proof Apply Theorem 9.5.5. If every positive-length coefficient vanishes, take $s = 1$ and the empty retained family; the block conditions and bond-dimension bound are vacuous, and the MPV identity is the zero identity. In the nonzero branch, use the nonempty normalized family supplied by the second alternative. $\square$

Remark 9.5.7 (CPSV blocked canonical form (source statement)). The blocked canonical-form reduction of [CPGSV16, Section 2.3] and [CPGSV21, Section IV.A] states that an arbitrary translation-invariant MPS tensor becomes, after blocking, a weighted family of normal tensors, up to a summand that vanishes on every positive-length ring. Four statements must be distinguished. Theorem 9.5.6 gives the unconditional PGVWC07 normalized exact form before period removal. Theorem 9.11.1.2 gives the prepared trace-preserving primitive-block reduction. For a family with a nonzero positive-length coefficient, Theorem F.3.8 normalizes the weights and Theorem 10.6.3.17 gives the normalized BNT form, including its per-site scalar. If every positive-length coefficient vanishes, there is no nonempty family from which to choose a unit-modulus weight; this exceptional zero family is represented by the empty retained family instead.

9.6 Normal tensors and canonical forms

This section records the CPSV definitions of *normal tensor* and *canonical form* from [CPGSV16]. The basis of normal tensors is stated in Chapter 10. Write

$$\sigma_{\partial}(\mathcal{E}) = \{\lambda \in \sigma(\mathcal{E}) : |\lambda| = r(\mathcal{E})\} \quad (9.19)$$

for the peripheral spectrum of a transfer map $\mathcal{E}$, where $r(\mathcal{E})$ is its spectral radius. The stronger conditions used later (Definition 9.9.2 and the canonical-form conditions of Chapter 10) add left-canonical normalization and spectral primitivity. The weight moduli are taken non-increasing, not strictly decreasing; equal-modulus and repeated-copy sectors are retained and compared by the coefficient argument of Chapter 10.

Definition 9.6.1 (CPSV normal tensor (NT)). Lean ✓ Stmt A tensor A of physical dimension d and bond dimension D is a *normal tensor* if, after the spectral-radius normalization of [CPGSV16], (i) A admits no nontrivial invariant orthogonal projection and (ii) the associated CPM $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ has spectral radius $r(\mathcal{E}_A) = 1$ and peripheral spectrum $\sigma_{\partial}(\mathcal{E}_A) = \{1\}$.

Remark 9.6.2. Condition (i) cannot be dropped. A reducible tensor may have peripheral spectrum $\{1\}$ while still admitting a nontrivial invariant projection, so condition (ii) alone does not characterize normality.

Theorem 9.6.3 (Algebraically normal left-canonical tensors are CPSV normal). Lean ✓ Stmt Let $D > 0$. If A is algebraically normal and left-canonical, then

$$A \text{ is a CPSV normal tensor.} \quad (9.20)$$

Proof. ✓ Proof Algebraic normality together with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ makes the transfer map strongly irreducible. It consequently has a positive-definite fixed point, no nontrivial invariant projection, and peripheral spectrum $\{1\}$. The fixed-point eigenvalue and left-canonical normalization give $r(\mathcal{E}_A) = 1$. Thus no scalar rescaling remains, and the two conditions in Definition 9.6.1 hold for A itself. $\square$

Theorem 9.6.4 (Normality and left-canonical normalization under entrywise conjugation). Lean Lean Lean Lean ✓ Stmt For a tensor or matrix, an overline denotes entrywise conjugation. If A and B are gauge equivalent through X , then $\overline{A}$ and $\overline{B}$ are gauge equivalent through $\overline{X}$. If A is left-canonical, then $\overline{A}$ is left-canonical. If A is algebraically normal, then $\overline{A}$ is algebraically normal. Independently, if A is a CPSV normal tensor, then $\overline{A}$ is a CPSV normal tensor.

Proof. ✓ Proof Entrywise conjugating $B^i = X A^i X^{-1}$ gives $\overline{B^i} = \overline{X} \overline{A^i} (\overline{X})^{-1}$. Conjugating the left-canonical identity gives

$$\sum_i (\overline{A^i})^\dagger \overline{A^i} = \overline{\sum_i (A^i)^\dagger A^i} = \overline{\mathbb{1}} = \mathbb{1}. \quad (9.21)$$

Entrywise conjugation maps the span of the length- N products of A onto the span of those of $\overline{A}$, so it preserves positive-length block injectivity and hence algebraic normality. For the normal-tensor assertion, the transfer map of $\overline{A}$ is the anti-linear conjugate $\sigma \circ \mathcal{E}_A \circ \sigma$ by (9.22). By Theorem 4.14.3 this conjugation preserves the spectral radius and primitivity, and by Theorem 9.6.6 entrywise conjugation preserves irreducibility. Hence both clauses of Definition 9.6.1 hold for $\overline{A}$. $\square$

Theorem 9.6.5 (Transfer maps and positive diagonal matrices under entrywise conjugation). Lean Lean Lean ✓ Stmt Let σ denote entrywise conjugation. For every tensor A , as operators on the matrix space,

$$\mathcal{E}_{\bar{A}} = \sigma \circ \mathcal{E}_A \circ \sigma; \quad (9.22)$$

evaluating at $\bar{X}$ gives, for every matrix X ,

$$\mathcal{E}_{\bar{A}}(\bar{X}) = \overline{\mathcal{E}_A(X)}. \quad (9.23)$$

If Λ is positive definite and diagonal, then

$$\bar{\Lambda} = \Lambda. \quad (9.24)$$

Proof. ✓ Proof Entrywise conjugation commutes with finite sums, matrix products, and conjugate transposition, which gives (9.22) on every matrix, and (9.23) follows since σ is an involution. The diagonal entries of a positive-definite complex matrix are real, while all off-diagonal entries of a diagonal matrix vanish. This proves (9.24). $\square$

Theorem 9.6.6 (Irreducibility under entrywise conjugation). Lean ✓ Stmt If A is irreducible, then $\bar{A}$ is irreducible.

Proof. ✓ Proof Suppose P is a nontrivial invariant orthogonal projection for $\bar{A}$, so $(1-P)\bar{A}^i P = 0$ for every i . Conjugating entrywise gives $(1-\bar{P})A^i \bar{P} = 0$, and $\bar{P}$ is again a Hermitian idempotent distinct from 0 and $\mathbb{1}$. Hence $\bar{P}$ is a nontrivial invariant orthogonal projection for A , contradicting the irreducibility of A . $\square$

Definition 9.6.7 (CPSV canonical form). Lean Lean Lean Lean Lean ✓ Stmt Let A have bond dimension D . It is in the canonical form of [CPGSV16, Section 2.3] if there are positive integers $D_1, \dots, D_r$, normal tensors A_k of bond dimension D_k , weights $\mu_k \in \mathbb{C}$, and, writing $S = \sum_k D_k$, a matrix $U \in \mathbb{C}^{S \times D}$ such that $S \leq D$ and, for every physical index i ,

$$UU^\dagger = \mathbb{1}_S, \quad A^i = U^\dagger \left(\bigoplus_{k=1}^r \mu_k A_k^i \right) U. \quad (\text{CF})$$

Thus the complement of the retained direct sum consists literally of zero bond-space coordinates. In particular, $r = 0$ is possible only when $A = 0$.

The convention in line 246 of [CPGSV16] is a separate normalization of these data, not part of canonical-form membership:

$$\begin{aligned} |\mu_k| &\leq 1 \quad \text{for every } k, \\ A \neq 0 &\implies \exists k, |\mu_k| = 1. \end{aligned} \quad (9.25)$$

No weight is required to be nonzero, and no ordering or separation condition is imposed.

The basis-of-normal-tensors refinement is Definition 10.1.1; it is not part of canonical form itself.

Theorem 9.6.8 (Removing zero-weight blocks at positive length). Lean ✓ Stmt Let $(A_k)_{k=1}^r$ be blocks with weights $(\mu_k)_{k=1}^r$, and put $K_{\text{act}} = \{k : \mu_k \neq 0\}$. Then the full weighted direct

sum and the direct sum over K_{act} have the same matrix product vectors at every positive length. Equivalently, for $N > 0$ and every configuration σ ,

$$\sum_{k=1}^r \mu_k^N V^{(N)}(A_k)_\sigma = \sum_{k \in K_{\text{act}}} \mu_k^N V^{(N)}(A_k)_\sigma. \quad (9.26)$$

The restriction to positive lengths is essential: at length zero, a zero-weight block contributes through $\mu_k^0 = 1$.

Proof. [✓ Proof](#) Expand both matrix product vectors using Theorem 2.5.9. If $\mu_k = 0$ and $N > 0$, then $\mu_k^N = 0$. The inactive part of the finite sum therefore vanishes, and reindexing the remaining terms by K_{act} gives (9.26). $\square$

Theorem 9.6.9 (Active part of CPSV canonical form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A have canonical-form data as in (CF), and put $K_{\text{act}} = \{k : \mu_k \neq 0\}$. Then

$$\mathcal{V}^+(A) = \mathcal{V}^+\left(\bigoplus_{k \in K_{\text{act}}} \mu_k A_k\right). \quad (9.27)$$

This conclusion uses no weight normalization. The bounds and unit-modulus condition in (9.25) remain separate assumptions on the canonical-form data.

Proof. [✓ Proof](#) The canonical-form reconstruction first identifies the positive-length matrix product vectors of A with those of the full weighted block sum. Theorem 9.6.8 then removes precisely the zero-weight summands. $\square$

Theorem 9.6.10 (Physical relabeling preserves CPSV normal and canonical form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $e : \{0, \dots, d' - 1\} \rightarrow \{0, \dots, d - 1\}$ be a bijection and set

$$(e^* A)^j = A^{e(j)}. \quad (9.28)$$

The transfer maps of A and $e^* A$ agree. In particular, physical relabeling preserves left-canonical normalization and CPSV normality. If A has canonical-form data

$$A^i = U^\dagger \left(\bigoplus_{k=1}^r \mu_k A_k^i \right) U,$$

then $e^* A$ has the same weights and coisometry and the relabeled normal blocks:

$$(e^* A)^j = U^\dagger \left(\bigoplus_{k=1}^r \mu_k (e^* A_k)^j \right) U. \quad (9.29)$$

Proof. [✓ Proof](#) A bijection of the physical alphabet only permutes the summands in $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$. Hence the transfer map, its spectral radius, its peripheral spectrum, and its invariant orthogonal projections are unchanged. Formula (9.29) is obtained from Formula (CF) by replacing i with $e(j)$. $\square$

Theorem 9.6.11 (Transfer maps under a bond-space gauge). [Lean](#) [✓ Stmt](#) If $B^i = X A^i X^{-1}$ for every physical index i , then there is an invertible matrix C such that

$$\mathcal{E}_B(Y) = C^{-1} \mathcal{E}_A(CY C^\dagger) (C^\dagger)^{-1}.$$

Proof. ✓ Proof Expand both transfer maps and take $C = X^{-1}$. □

Theorem 9.6.12 (Gauge invariance of transfer quasi-idempotence). Lean ✓ Stmt *Let A and B be gauge-equivalent MPS tensors and let $c \in \mathbb{C}$. Then*

$$\mathcal{E}_B^2 = c\mathcal{E}_B \iff \mathcal{E}_A^2 = c\mathcal{E}_A.$$

Proof. ✓ Proof Theorem 9.6.11 identifies the two transfer maps by an algebra automorphism. Such an automorphism preserves composition and scalar multiplication, and therefore preserves the displayed equation. □

Theorem 9.6.13 (Spectral radius under congruence similarity). Lean ✓ Stmt *For an invertible matrix C , the linear maps*

$$E \quad \text{and} \quad Y \mapsto C^{-1}E(CYC^\dagger)(C^\dagger)^{-1}$$

have the same spectral radius.

Proof. ✓ Proof The two maps are conjugate by the invertible linear transformation $Y \mapsto CYC^\dagger$, and therefore have the same spectrum. □

Theorem 9.6.14 (Primitivity under congruence similarity). Lean ✓ Stmt *For an invertible matrix C , a linear map E is primitive if and only if*

$$Y \mapsto C^{-1}E(CYC^\dagger)(C^\dagger)^{-1}$$

is primitive.

Proof. ✓ Proof Congruence similarity is an algebra isomorphism on the endomorphism algebra and hence preserves the peripheral spectrum. □

Theorem 9.6.15 (Gauge transport of CPSV normality). Lean ✓ Stmt *Let X be invertible and suppose*

$$B^i = XA^iX^{-1}$$

for every physical index i . If B is a CPSV normal tensor, then A is a CPSV normal tensor.

Proof. ✓ Proof With $S_X(Y) = XYX^\dagger$, the transfer maps satisfy

$$\mathcal{E}_B = S_X \circ \mathcal{E}_A \circ S_X^{-1}. \tag{9.30}$$

Similarity preserves the spectral radius and the peripheral spectrum. For irreducibility, start with an invariant orthogonal projection Q for one transfer map and form the support projection of the corresponding positive congruence $XQX^\dagger$ (using X^{-1} in the reverse direction). The intertwining in Formula (9.30) makes this support projection invariant for the other transfer map. Irreducibility forces its support to be zero or full, and invertibility of X then forces $Q = 0$ or $Q = 1$. Thus both conditions in Definition 9.6.1 are transported across the gauge. □

Theorem 9.6.16 (Positive blocking preserves CPSV normal tensors). Lean Lean ✓ Stmt *Let A be a CPSV normal tensor and let $L \geq 1$. Then the blocked tensor $A^{[L]}$ is a CPSV normal tensor.*

Proof. ✓ Proof Choose a trace-preserving gauge $B^i = XA^iX^{-1}$. Positive blocking gives

$$(B^{[L]})^I = X(A^{[L]})^I X^{-1}. \quad (9.31)$$

The trace-preserving primitive and irreducible structure of B is preserved by positive blocking, so Theorem 9.6.3 makes $B^{[L]}$ CPSV normal. Gauge transport (Theorem 9.6.15) then gives CPSV normality of $A^{[L]}$. $\square$

Theorem 9.6.17 (Literal CPSV canonical form is preserved by positive blocking). Lean Lean ✓ Stmt *Let A have canonical-form data*

$$A^i = U^\dagger \left(\bigoplus_{k=1}^r \mu_k A_k^i \right) U, \quad UU^\dagger = \mathbb{1}.$$

For every $L \geq 1$, $A^{[L]}$ has canonical-form data with the same coisometry U , the blocked normal tensors $A_k^{[L]}$, and the weights μ_k^L :

$$(A^{[L]})^I = U^\dagger \left(\bigoplus_{k=1}^r \mu_k^L (A_k^{[L]})^I \right) U. \quad (9.32)$$

Proof. ✓ Proof Write $I = (i_1, \dots, i_L)$. Since $L \geq 1$, the word I is nonempty. Repeatedly inserting $UU^\dagger = \mathbb{1}$ into $A^{i_1} \dots A^{i_L}$ gives

$$\begin{aligned} A^{i_1} \dots A^{i_L} &= U^\dagger \left(\bigoplus_{k=1}^r \mu_k^L A_k^{i_1} \dots A_k^{i_L} \right) U \\ &= U^\dagger \left(\bigoplus_{k=1}^r \mu_k^L (A_k^{[L]})^I \right) U. \end{aligned}$$

The preceding theorem gives CPSV normality of every blocked block $A_k^{[L]}$, and the displayed identity is exactly the retained direct-sum reconstruction. $\square$

Theorem 9.6.18 (CPSV canonical-form representative after blocking). Lean ✓ Stmt *For every tensor A , there are a positive integer p , a bond dimension D' , and a tensor B of physical dimension d^p and bond dimension D' such that $B \in \text{CF}$ and, for every $N \geq 1$,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(B)_\sigma. \quad (9.33)$$

Here B is another tensor generating the same positive-length MPV family. The assertion $B \in \text{CF}$ is literal in the bond coordinates of B ; no assertion is made that $A^{[p]}$ itself is literally in canonical form in its original bond coordinates.

Proof. ✓ Proof Apply Theorem 9.11.1.3 to A and itself. It gives $p \geq 1$, weights μ_k , and left-canonical, primitive, tensor-irreducible blocks C_k such that, for every $N \geq 1$,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)} \left(\bigoplus_k \mu_k C_k \right)_\sigma. \quad (9.34)$$

First, Theorem 9.11.1.1 makes each C_k algebraically normal. Second, Theorem 9.6.3 combines this conclusion with the left-canonical identity to make each C_k a CPSV normal tensor. Hence

$$B := \bigoplus_k \mu_k C_k \quad (9.35)$$

satisfies the literal block-diagonal equation (CF), so $B \in \text{CF}$. Equation (9.34) gives the second part of (9.33). $\square$

Theorem 9.6.19 (Closed-chain expansion of CPSV canonical form). [Lean](#) [✓ Stmt](#) For canonical-form data as in (CF), every positive length N and every configuration σ satisfy

$$V^{(N)}(A)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(A_k)_\sigma. \quad (9.36)$$

Equivalently, $|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(A_k)\rangle$. This is the identity immediately following the canonical-form definition in [CPGSV16, Section 2.3].

Proof. [✓ Proof](#) Induction on a nonempty physical word w of length N , using (CF) at each letter and $UU^\dagger = \mathbb{1}_S$ between adjacent letters, gives

$$A^w = U^\dagger \left(\bigoplus_{k=1}^r \mu_k^N A_k^w \right) U. \quad (9.37)$$

Applying cyclicity of the trace and $UU^\dagger = \mathbb{1}_S$ to (9.37) gives

$$\text{tr}(A^w) = \text{tr} \left(\left(\bigoplus_{k=1}^r \mu_k^N A_k^w \right) UU^\dagger \right) = \text{tr} \left(\bigoplus_{k=1}^r \mu_k^N A_k^w \right). \quad (9.38)$$

Applying Theorem 2.5.9 to the right-hand side of (9.38) yields $\text{tr}(A^w) = \sum_{k=1}^r \mu_k^N \text{tr}(A_k^w)$, which is (9.36). $\square$

Theorem 9.6.20 (BNT characterization for CPSV canonical form). [Lean](#) [✓ Stmt](#) Let A have canonical-form data as in (CF), and let $(B_j)_{j=1}^g$ have positive bond dimensions. With $K_{\text{act}} = \{k : \mu_k \neq 0\}$, the family $(B_j)_{j=1}^g$ is a basis of normal tensors for A if and only if

$$\begin{aligned} & (\forall j, B_j \text{ is normal}) \\ & \wedge (\forall k \in K_{\text{act}}, \exists j, X_k, \phi_k, \forall i, A_k^i = e^{i\phi_k} X_k B_j^i X_k^{-1}) \\ & \wedge (\forall j \neq j', \nexists X, \phi, \forall i, B_{j'}^i = e^{i\phi} X B_j^i X^{-1}). \end{aligned} \quad (9.39)$$

Each displayed conjugacy also requires the two bond dimensions to agree, and every scalar $e^{i\phi}$ has modulus one. This is [CPGSV16, Proposition 2.7 and Appendix A], specialized to the literal canonical-form data.

Local fix (active blocks): coverage in (9.39) applies only to listed blocks with $\mu_k \neq 0$. The display defining canonical form permits $\mu_k = 0$, and a zero-weight block contributes nothing to any positive-length matrix product vector. Thus the theorem does not claim coverage of every syntactically listed block. The comparison is recorded in `docs/paper-gaps/cpsv16_bnt_characterization_active_blocks.tex`.

Proof. [✓ Proof](#) The canonical-form data supply normality of every A_k and the positive-length identity $|V^{(N)}(A)\rangle = \sum_k \mu_k^N |V^{(N)}(A_k)\rangle$. Substitution of these two facts into Theorem 10.1.5 gives (9.39). $\square$

The projector-closure result in Theorem 9.7.4 gives normal blocks and orthogonal ambient embeddings under the source hypotheses. Combining these embeddings into one coisometry gives Definition 9.6.7.

This source definition is distinct from the stronger normal-canonical-form conditions below. The renormalization fixed-point analysis uses a separate block-injective canonical-form criterion; it is not the CPSV canonical-form definition.

Remark 9.6.21 (Relation to the stronger conditions). A tensor A which is irreducible (Definition 9.1.1.1), has transfer map of spectral radius one, and has primitive transfer map is normal in the sense of Definition 9.6.1. The reverse implications from the stronger conditions of Definition 9.9.2 and of Chapter 10 require the fundamental-theorem step or the algebraic-to-spectral normality equivalence, established later.

9.6.1 Normalization conventions

Remark 9.6.1.1 (Scalar normalization conventions). The reductions use four scalar facts: scaling preserves injectivity; the transfer map under a scalar ζ rescales by $|\zeta|^2$; unit-modulus phase rescaling preserves $\sum_i (A^i)^\dagger A^i = \mathbb{1}$; and the MPV of a weighted block-diagonal tensor factorizes as $\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma$, with $\eta_k := \mu_k / |\mu_k|$.

Left-canonical normalization is required before the blocks can be compared. Without it, the implication $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k) \Rightarrow \forall k, \mathcal{V}(A_k) = \mathcal{V}(B_k)$ already fails for two scalar blocks: for $\mu = (1, 2)$, $A = (1, 3/2)$, and $B = (3, 1/2)$, the two direct sums generate identical MPV families but the first blocks differ at $N = 1$. Left-canonical normalization removes this particular ambiguity; the sectors are then matched, up to a permutation, by the exact fixed-length linear independence of Chapter 10, which requires no ordering of the weight moduli.

9.7 Projector criterion for canonical decomposition

Definition 9.7.1 (Invariant-projector closure). Lean ✓ Stmt An MPS tensor A has *invariant-projector closure* if every orthogonal projection P satisfying $PA^i = PA^i P$ for all i also satisfies $A^i P = PA^i P$ for all i . This is the projector hypothesis in the sufficient condition for canonical form of [CPGSV16, Section 2].

Theorem 9.7.2 (Decomposition into irreducible corners under projector closure). Lean ✓ Stmt Suppose A has invariant-projector closure. Then there are isometries $V_1, \dots, V_r$ with $V_k^\dagger V_k = \mathbb{1}$, pairwise orthogonal ranges, and $\sum_k V_k V_k^\dagger = \mathbb{1}$, such that $A^i V_k = V_k A_k^i$ for the corner tensors $A_k^i = V_k^\dagger A^i V_k$, every corner tensor A_k is irreducible, and $\mathcal{V}(A) = \mathcal{V}(\bigoplus_{k=1}^r A_k)$.

Proof. ✓ Proof Proceed by strong induction on the bond dimension. If A is irreducible, the single corner $V = \mathbb{1}$ suffices. Otherwise, A has a nontrivial invariant projection P , which commutes with every A^i by Theorem E.2.1. Factoring $P = V_1 V_1^\dagger$ and $\mathbb{1} - P = V_2 V_2^\dagger$ through support isometries by Lemma E.2.3 splits A into two corner tensors of strictly smaller bond dimension. For $j = 1, 2$, commutation gives $A^i V_j = V_j (V_j^\dagger A^i V_j)$, and each corner inherits invariant-projector closure by Lemma E.2.4. The two recursively obtained families are combined. Ranges from different corners are orthogonal because

$$V_1^\dagger V_2 = V_1^\dagger P(\mathbb{1} - P)V_2 = 0.$$

Equality of matrix product vectors follows from Lemma E.2.6. □

Definition 9.7.3 (No nontrivial periodic vectors). Lean ✓ Stmt An MPS tensor A has *no nontrivial p -periodic vectors* if, for every isometry V with $V^\dagger V = \mathbb{1}$ and $A^i V = V B^i$ for all i , with B irreducible (equivalently, for every restriction B of A to a minimal invariant subspace), and for every $\rho > 0$ and $r > 0$ with $\mathcal{E}_B(\rho) = r\rho$, every eigenvalue of $\mathcal{E}_B$ of modulus r equals r . The eigenvalue r is the spectral radius of $\mathcal{E}_B$ (Theorem 6.9.1), so the condition states that no block of the invariant-subspace decomposition of [CPGSV16, Section 2], rescaled to spectral radius one, has a peripheral eigenvalue $e^{2\pi i q/p} \neq 1$. Eigenvectors with such eigenvalues are the p -periodic vectors of [CPGSV16, Section 2].

Theorem 9.7.4 (Sufficient condition for canonical form). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose A has invariant-projector closure and no nontrivial p -periodic vectors. Then there are normal tensors $A_1, \dots, A_r$ of positive bond dimensions $D_1, \dots, D_r$, with $\sum_k D_k \leq D$, and nonzero weights $\mu_1, \dots, \mu_r$ such that

$$\mathcal{V}^+(A) = \mathcal{V}^+\left(\bigoplus_{k=1}^r \mu_k A_k\right). \quad (9.40)$$

Moreover, there are isometries $V_k : \mathbb{C}^{D_k} \rightarrow \mathbb{C}^D$ with pairwise orthogonal ranges such that

$$\begin{aligned} A^i V_k &= V_k (\mu_k A_k^i), \\ A^i &= \sum_k V_k (\mu_k A_k^i) V_k^\dagger. \end{aligned} \quad (9.41)$$

This is the sufficient condition for canonical form of [CPGSV16, Section 2], with that paper's convention $\sum_k D_k \leq D$ allowing zero blocks: a corner of A carrying the zero tensor cannot be rescaled to spectral radius one and is omitted because it vanishes at every positive length.

Proof. [✓ Proof](#) Decompose A into irreducible corners B_k along isometries V_k by Theorem 9.7.2; every matrix product vector coefficient of A is the sum of those of the corners. A corner whose matrices all vanish contributes nothing at positive length and is dropped. Every remaining corner has a Perron eigenpair $\mathcal{E}_{B_k}(\rho_k) = r_k \rho_k$ by Lemma E.3.1. The absence of p -periodic vectors, applied to the corner B_k , gives that every eigenvalue of $\mathcal{E}_{B_k}$ of modulus r_k equals r_k , so $A_k := r_k^{-1/2} B_k$ is normal by Lemma E.3.2. Setting $\mu_k := r_k^{1/2}$ gives $\mu_k A_k = B_k$ at the level of matrices, so for every word w of positive length N the weight cancels:

$$\mu_k^N \text{tr}(A_k^w) = (r_k^{1/2})^N (r_k^{-1/2})^N \text{tr}(B_k^w) = \text{tr}(B_k^w).$$

Summing over k gives (9.40), while the corner reconstruction gives (9.41). The retained dimensions satisfy $\sum_k D_k \leq \sum_k \text{tr}(V_k V_k^\dagger) = \text{tr}(\mathbb{1}) = D$. $\square$

Theorem 9.7.5 (Non-scalar fixed point gives a further split). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose A is unital, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If X is a Hermitian fixed point of $\mathcal{E}_A$ which is not a scalar multiple of $\mathbb{1}$, then there is a nonzero PSD fixed point ρ which is not positive definite, and consequently A admits a two-block MPV-equivalent decomposition with both block dimensions strictly smaller than D . This is the fixed-point shift and further decomposition step in [PGVWC07, Theorem 4].

Proof. [✓ Proof](#) Let $\lambda_{\max}$ be the largest eigenvalue of X . Then $\rho := \lambda_{\max} \mathbb{1} - X$ is positive semidefinite, singular, and nonzero because X is not scalar. Unitality and the fixed-point equation for X give $\mathcal{E}_A(\rho) = \rho$. Apply Theorem 9.1.4. The non-scalar hypothesis is needed because scalar multiples of $\mathbb{1}$ are fixed by every unital map and yield no nontrivial support projection. $\square$

9.8 Fixed points and canonical gauges of irreducible blocks

Theorem 9.8.1 (Scalar fixed points in an irreducible unital block). [Lean](#) [✓ Stmt](#) If A is irreducible and unital, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and $\mathcal{E}_A(X) = X$, then X is a scalar multiple of $\mathbb{1}$. This is the final fixed-point assertion in the proof of [PGVWC07, Theorem 4].

Proof. [✓ Proof](#) Tensor irreducibility gives irreducibility of the completely positive transfer map. By Wolf's Theorem 6.6 for irreducible unital Kraus maps, every fixed point is scalar. $\square$

Theorem 9.8.2 (Unital gauge from a full-rank Perron–Frobenius eigenvector). [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Let ρ be positive definite with $\mathcal{E}_A(\rho) = r\rho$ for some $r > 0$. For $B^i := r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$,

$$\sum_i B^i (B^i)^\dagger = \mathbb{1}. \quad (9.42)$$

If $r = 1$, then the unscaled gauge $B^i := \rho^{-1/2}A^i\rho^{1/2}$ is unital and generates the same MPV family as A . This is the spectral-radius normalization and full-rank fixed-point gauge of [PGVWC07, Theorem 4].

Proof. [Proof](#) The positive square root of ρ is invertible. Substituting $B^i = r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$ and using $\mathcal{E}_A(\rho) = r\rho$ gives

$$\sum_i B^i (B^i)^\dagger = r^{-1}\rho^{-1/2}\mathcal{E}_A(\rho)\rho^{-1/2} = \mathbb{1},$$

which is (9.42). For $r = 1$, the change from A to B is a similarity, so cyclicity of the trace gives equality of all MPV coefficients. $\square$

Definition 9.8.3 (CPSV canonical form II). [Lean](#) [Lean](#) [Stmt](#) In canonical form II of [CPGSV16, Appendix A], there are positive retained dimensions D_k , normal tensors A_k , weights μ_k , and, for $S = \sum_k D_k \leq D$, a matrix $U \in \mathbb{C}^{S \times D}$ such that, for every physical index i ,

$$\begin{aligned} UU^\dagger &= \mathbb{1}_S, \\ A^i &= U^\dagger \left(\bigoplus_k \mu_k A_k^i \right) U. \end{aligned} \quad (9.43)$$

Thus omitted bond-space coordinates are literally zero. Every retained block additionally satisfies

$$\begin{aligned} \sum_i (A_k^i)^\dagger A_k^i &= \mathbb{1}_{D_k}, \\ \mathcal{E}_{A_k}(\Lambda_k) &= \Lambda_k, \end{aligned}$$

where Λ_k is diagonal and positive definite. The two displayed equations are the only additional blockwise conditions; no ordering, injectivity, gauges, or basis-of-normal-tensors assumptions are included. The line-246 convention (9.25) remains a separate condition on the underlying canonical-form data.

Theorem 9.8.4 (Canonical form admits a canonical-form-II gauge). [Lean](#) [Lean](#) [Stmt](#) Let A be a tensor in CPSV canonical form, with ambient bond dimension D . Then there is a tensor B of the same bond dimension and an $G \in \text{GL}(D, \mathbb{C})$ such that, for every physical index i ,

$$B^i = GA^iG^{-1}. \quad (9.44)$$

Moreover, B is in canonical form II. This is the nonsingular gauge assertion of [CPGSV16, Appendix A].

Proof. [Proof](#) Write the retained part of the canonical-form decomposition as $C^i = \bigoplus_k \mu_k A_k^i$, and let $U \in \mathbb{C}^{S \times D}$ satisfy $UU^\dagger = \mathbb{1}_S$ and $A^i = U^\dagger C^i U$. For each normal block, first choose its pure trace-preserving Perron gauge and then diagonalize its positive definite fixed point by a unitary. This gives $X_k \in \text{GL}(D_k, \mathbb{C})$ and blocks B_k satisfying the two canonical-form-II identities. Put $X = \bigoplus_k X_k$ and extend it to the omitted zero coordinates by

$$G = U^\dagger XU + (\mathbb{1}_D - U^\dagger U). \quad (9.45)$$

Its inverse is

$$G^{-1} = U^\dagger X^{-1} U + (\mathbb{1}_D - U^\dagger U). \quad (9.46)$$

The identity $UU^\dagger = \mathbb{1}_S$ gives $GG^{-1} = G^{-1}G = \mathbb{1}_D$. It also gives

$$GU^\dagger = U^\dagger X, \quad (9.47)$$

and

$$UG^{-1} = X^{-1}U. \quad (9.48)$$

Therefore

$$GA^i G^{-1} = U^\dagger X C^i X^{-1} U = U^\dagger \left(\bigoplus_k \mu_k B_k^i \right) U, \quad (9.49)$$

which has the same ambient reconstruction and satisfies canonical form II. $\square$

Theorem 9.8.5 (Gauge equivalence under coisometric reconstruction). [Lean](#) [✓ Stmt](#) *Let A and B have common bond dimension D , and let C and E have common retained bond dimension n . Suppose that $U_A, U_B \in \mathbb{C}^{n \times D}$ satisfy $U_A U_A^\dagger = U_B U_B^\dagger = \mathbb{1}_n$ and, for every physical index i ,*

$$A^i = U_A^\dagger C^i U_A. \quad (9.50)$$

$$B^i = U_B^\dagger E^i U_B. \quad (9.51)$$

If C and E are gauge equivalent, then A and B are gauge equivalent.

Proof. [✓ Proof](#) Choose $X \in \text{GL}(n, \mathbb{C})$ such that, for every physical index i , $E^i = X C^i X^{-1}$. Equality of the Gram matrices $U_A U_A^\dagger$ and $U_B U_B^\dagger$ gives a unitary $W \in \mathbb{C}^{D \times D}$ with

$$U_B^\dagger = W U_A^\dagger. \quad (9.52)$$

$$U_B = U_A W^\dagger. \quad (9.53)$$

Extend the retained gauge through U_A by

$$G = U_A^\dagger X U_A + (\mathbb{1}_D - U_A^\dagger U_A). \quad (9.54)$$

Its inverse is

$$G^{-1} = U_A^\dagger X^{-1} U_A + (\mathbb{1}_D - U_A^\dagger U_A). \quad (9.55)$$

The coisometry identity gives $GU_A^\dagger = U_A^\dagger X$ and $U_A G^{-1} = X^{-1} U_A$. Hence, with $Y = WG$,

$$B^i = W U_A^\dagger E^i U_A W^\dagger = W G A^i G^{-1} W^\dagger = Y A^i Y^{-1}. \quad (9.56)$$

$\square$

Theorem 9.8.6 (Diagonal Perron normalization for an irreducible TP tensor). [Lean](#) [✓ Stmt](#) Let A be an irreducible MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that, for $B^i := U^\dagger A^i U$,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1},$$

$$\mathcal{E}_B(\Lambda) = \Lambda.$$

This proves the diagonal Perron normalization required in CFII. It does not prove that A is in the CPSV canonical form or that its block is normal; those require the corresponding primitivity and canonical-form hypotheses.

Proof. [✓ Proof](#) The TP hypothesis makes $\mathcal{E}_A$ a channel with a nonzero PSD fixed point. Irreducibility makes $\mathcal{E}_A$ an irreducible CP map, and Theorem 6.3.12 upgrades the fixed point to positive definite. The spectral theorem diagonalizes it by a unitary conjugation, which preserves the TP normalization. $\square$

9.9 Normal canonical form

This section defines the spectral *normal* canonical form used in the reduction. A separate block-injective canonical-form criterion is used in the renormalization fixed-point analysis. The supporting overlap consequences are collected in Section E.6 and Section 7.8.

Recall from Definition 2.7.3 and Lemma 7.3.1 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \text{Tr}(F_{AB}^N). \quad (9.57)$$

This notation enters the self-overlap clause of the canonical-form conditions below.

Remark 9.9.1 (Normalization convention). [\[PGVWC07, Theorem 4\]](#) uses the unital gauge $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$; here the blocks are in the dual TP gauge $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$. A positive-definite fixed point of the adjoint transfer map provides the change of gauge between the two (Theorem 6.6.3).

Definition 9.9.2 (Normal canonical form conditions). [Lean](#) [✓ Stmt](#) Scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfy the *normal canonical form conditions* if:

1. each A_k is irreducible,
2. each A_k satisfies $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each $\mathcal{E}_{A_k}$ is primitive (equivalently, $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$),
4. $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Here “normal” is the spectral sense of [\[CPGSV16\]](#), which by [\[SPGWC10, Proposition 3\]](#) is equivalent to the eventual block-injectivity formulation of Definition 2.4.4.

9.9.1 Separated one-copy normal canonical forms

The following specialization attaches BNT separation to the one-copy normal canonical form. Repeated representatives and their power-sum coefficients are instead recorded by the sector decomposition of Chapter 10.

Definition 9.9.1.1 (No gauge-phase-equivalent distinct blocks). Lean ✓ Stmt A finite block family has no gauge-phase-equivalent distinct blocks when, for any two distinct indices with the same bond dimension, the corresponding blocks are not gauge-phase equivalent.

Definition 9.9.1.2 (Normal canonical form with BNT separation). Lean ✓ Stmt This is a one-copy canonical-form condition: each listed block is one representative and repeated copies are not encoded as separate blocks. A family is in *normal canonical form with BNT separation* if it satisfies Definition 9.9.2 and distinct blocks of the same bond dimension are not gauge-phase equivalent. The weight moduli are non-increasing but need not be strictly decreasing, so equal-modulus blocks are allowed; the grouping into a BNT is governed by gauge-phase equivalence, not by distinctness of moduli. Repeated equal-modulus copies of one basis tensor are not separate blocks here; they appear as the weights $\mu_{j,q}$ in $c_N^{(j)} = \sum_q \mu_{j,q}^N$ in the sector decomposition below.

The off-diagonal decay needed for the BNT criterion follows from the separation condition together with the overlap-decay theorems of the preceding chapter.

Theorem 9.9.1.3 (Cross-overlap decay from explicit BNT hypotheses). Lean ✓ Stmt Let $(A_j)_{j=1}^r$ be a family in which each A_j is injective and left-canonical, $\sum_i (A_j^i)^\dagger A_j^i = 1$, and distinct equal-dimension blocks are not gauge-phase equivalent. Then $O_{A_j A_k}(N) \rightarrow 0$ for every pair $j \neq k$.

Proof. ✓ Proof If the bond dimensions differ, Theorem 7.5.4 gives the decay. If the bond dimensions agree, the non-equivalence hypothesis excludes gauge-phase equivalence, so Theorem 7.6.5 gives the decay. □

Lemma 9.9.1.4 (BNT from explicit blockwise hypotheses). Lean ✓ Stmt Let $(\mu_k, A_k)_{k=1}^r$ be a weighted block family. Assume each A_k is injective and left-canonical, $\sum_i (A_k^i)^\dagger A_k^i = 1$, that $O_{A_k A_k}(N) \rightarrow 1$ for each k , and that distinct equal-dimension blocks are not gauge-phase equivalent. Then the blocks form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.

Proof. ✓ Proof Injective blocks are normal. The block-diagonal decomposition theorem gives the MPV expansion of $\bigoplus_k \mu_k A_k$ in terms of the block MPVs. The self-overlap limits are part of the hypotheses, and Theorem 9.9.1.3 supplies the off-diagonal decay. Therefore Lemma F.1.10 gives the eventual linear independence required in Definition 10.1.1. □

The normal-canonical formulation encodes normality spectrally (primitive transfer map), the BNT definition algebraically (eventual block injectivity).

Remark 9.9.1.5 (Comparing the two normality conditions). The spectral normality of Definition 9.9.2 and the algebraic normality of Definition 2.4.4 agree: the Wielandt theory of Chapter 8 supplies the implication from primitivity to eventual block injectivity used here.

Lemma 9.9.1.6 (Restricted normal-CF-BNT hypotheses give a BNT). Lean ✓ Stmt On this one-copy surface, if a family is in normal canonical form with BNT separation and each block is also normal in the algebraic sense, then the blocks form a basis of normal tensors for the associated block-diagonal tensor. The blocks here are distinct representatives.

Proof. ✓ Proof The normal-canonical hypotheses already give the block-diagonal MPV expansion, the self-overlap normalization, and the eventual linear independence coming from irreducible trace-preserving overlap decay. Supplying algebraic normality for each block adds the remaining clause in Definition 10.1.1. $\square$

Remark 9.9.1.7. See Theorem 7.7.1 for the modulus-one eigenvalue rigidity statement for irreducible trace-preserving blocks.

9.9.2 Canonical form from primitivity

A normal block needs a primitive transfer map; an irreducible TP block may still have a cyclic peripheral spectrum. Raising the map to its period sends every peripheral eigenvalue to 1, but eigenvalue 1 can then have several cyclic corners and need not be simple on the full powered space. Period removal therefore proceeds in this order: power the map, obtain the adjoint-fixed cyclic projections, compress to their corners, prove each corner irreducible, and only then infer primitivity of the sector maps. This is the blocking step of [CPGSV16, Section 2.3].

Theorem 9.9.2.1 (Blocking yields a peripheral-spectrum primitive transfer map). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that $\sigma_\partial(\mathcal{E}_{A^{[p]}}) = \{1\}$.*

Proof. ✓ Proof Pass to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$, which is unital with irreducible Kraus map. The TP fixed-point theorem for $\mathcal{E}_A$ provides a positive definite fixed point for the adjoint of K , so Theorem 4.12.2.2 shows that every peripheral eigenvalue is a root of unity. Since $\sigma_\partial(\mathcal{E}_A)$ is finite (Theorem 4.12.2), Lemma 4.12.1.2 gives a common exponent p with $\mu^p = 1$ for every $\mu \in \sigma_\partial(\mathcal{E}_A)$, and Lemma 4.12.1.3 gives $\sigma_\partial(\mathcal{E}_A^p) = \{1\}$. Blocking by p takes the p th power of the transfer map, so $\mathcal{E}_{A^{[p]}}$ is primitive in the peripheral-spectrum sense of Definition 4.10.5. The blocked map need not remain irreducible: eigenvalue 1 may have several cyclic corners. The adjoint-fixed projections are therefore compressed before irreducibility and primitivity are asserted sector by sector in Theorem 9.3.1.3. This is the periodicity-removal step of [CPGSV16, Section 2.3] for an irreducible block already in TP gauge. $\square$

9.10 Left-canonical and dual diagonalization of irreducible blocks

Each irreducible block in trace-preserving gauge admits a unitary normalization with diagonal positive Perron data. This is the normalization component of CFII (Definition 9.8.3); the full CFII assertion also requires the CPSV canonical-form and normal-block hypotheses. The full PGVWC07 canonical form is Theorem 9.5.6.

Theorem 9.10.1 (Diagonal Perron data for an irreducible TP block). Lean ✓ Stmt *Let A be an irreducible MPS tensor in TP gauge, $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, with $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that:*

1. $B^i := U^\dagger A^i U$ is TP,
2. $\mathcal{E}_B(\Lambda) = \Lambda$,
3. Λ is diagonal and positive definite,
4. A and B generate the same MPV family.

These are the two normalization identities in the CFII definition of [CPGSV16, Appendix A]. The theorem does not by itself prove that A is in canonical form or that its block is normal. The unitary conjugation is performed inside TP gauge after TP normalization.

Proof. ✓ **Proof** Theorem 9.8.6 provides the unitary and the diagonal positive-definite fixed point, and unitary conjugation preserves trace preservation and the MPV family (Theorem E.1.1). $\square$

Theorem 9.10.2 (Dual diagonalization for irreducible unital blocks). Lean ✓ **Stmt** Let A be an irreducible MPS tensor in unital gauge $\sum_i A^i (A^i)^\dagger = \mathbb{1}$, with $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that, for $B^i := U^\dagger A^i U$,

$$\begin{aligned} \sum_i B^i (B^i)^\dagger &= \mathbb{1}, \\ \sum_i (B^i)^\dagger \Lambda B^i &= \Lambda. \end{aligned} \tag{9.58}$$

Moreover, A and B generate the same MPV family. This is the dual-map fixed-point and unitary-diagonalization step of [PGVWC07, Theorem 4].

Proof. ✓ **Proof** Apply Theorem 9.10.1 to the conjugate-transposed family $i \mapsto (A^i)^\dagger$: unitality of A is trace preservation for this family, whose transfer map is the dual of $\mathcal{E}_A$. Translating back gives (9.58). $\square$

9.11 Cyclic-sector isometries and primitive reduction

Theorem 9.11.1 (Spectral cyclic sectors with support isometries). Lean ✓ **Stmt** Let A be a left-canonical irreducible tensor, and suppose the adjoint transfer map $\mathcal{E}_{A^\dagger}$ has a positive-definite fixed point, is irreducible as a positive map, and has peripheral spectrum

$$\sigma_\partial(\mathcal{E}_{A^\dagger}) = \{\gamma^j : j \in \mathbb{Z}/m\mathbb{Z}\}, \tag{9.59}$$

where γ is a primitive m th root of unity and $m \geq 1$. Then the spectral cyclic-sector decomposition of $A^{[m]}$ has compression maps which may be chosen with rectangular support isometries V_k satisfying

$$\begin{aligned} V_k^\dagger V_k &= \mathbb{1}, \\ V_k V_k^\dagger &= P_k, \\ \varphi_k(X) &= V_k X V_k^\dagger. \end{aligned} \tag{9.60}$$

These isometries also realize the blocked corner letters $\varphi_k(C_k^\alpha) = P_k (A^{[m]})^\alpha P_k$.

Proof. ✓ **Proof** Theorem 7.9.1.1, applied to $\mathcal{E}_{A^\dagger}$, provides projections P_k satisfying $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$. Iterating m times gives $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$, and $(\mathcal{E}_{A^\dagger})^m = \mathcal{E}_{(A^{[m]})^\dagger}$ shows that each P_k is fixed by the adjoint transfer map of $A^{[m]}$. Applying Theorem E.5.3 to that blocked tensor gives the compressed sector tensors and the support isometries in (9.60). $\square$

9.11.1 Reduction to primitive blocks

This subsection completes the reduction after period removal. A TP-primitive-irreducible block is normal, and an arbitrary tensor becomes, after a positive blocking length, a weighted family of left-canonical primitive blocks whose total bond dimension is bounded by the original bond dimension. For two tensors with the same MPV family, the cyclic sectors can be expressed over one common blocked alphabet, which is the form used by the Fundamental Theorem.

Theorem 9.11.1.1 (TP-primitive irreducible tensor is normal). [Lean](#) [✓ Stmt](#) *Let A be a left-canonical MPS tensor with $D > 0$, tensor-irreducible, and with peripheral spectrum $\{1\}$. Tensor irreducibility implies irreducibility of the completely positive transfer map, so A is normal.*

Proof. [✓ Proof](#) The primitive peripheral spectrum, tensor irreducibility, and left-canonical normalization give primitive fixed-point data. The fixed point is positive definite by Lemma [E.4.8](#). Theorem [8.7.4.2](#) then gives normality. $\square$

An arbitrary translation-invariant tensor reduces, after blocking, to a weighted direct sum of peripheral-spectrum primitive blocks. These globally blocked pieces need not remain irreducible; the cyclic-sector decomposition below refines them into primitive irreducible blocks.

Theorem 9.11.1.2 (Arbitrary tensor to primitive block decomposition). [Lean](#) [✓ Stmt](#) *For any MPS tensor A , there exist $p \geq 1$, nonzero weights $(\mu_k)_{k=1}^r$, and blocks $(B_k)_{k=1}^r$ such that, for every blocked σ of positive length N ,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r \mu_k B_k\right)_\sigma. \quad (9.61)$$

The retained block dimensions satisfy $\sum_{k=1}^r D_k \leq D$. Each block is left-canonical and primitive:

$$\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1},$$

$$\sigma_\partial(\mathcal{E}_{B_k}) = \{1\}.$$

Each bond dimension is positive.

Proof. [✓ Proof](#) Theorem [9.2.1.1](#) supplies the left-canonical nontrivial blocks, the positive-length equality, and the bound $\sum_k D_k \leq D$. Apply Theorem [E.4.2](#) directly to this finite irreducible family. Its proof first applies Theorem [9.9.2.1](#) to each block and then takes the least common multiple of the resulting periods through Theorem [E.4.1](#). This gives one positive length p at which every blocked transfer map is primitive. Lemma [E.4.16](#) keeps the blocked weights nonzero, and Lemma [E.4.14](#) carries the positive-length equality through blocking, giving (9.61). $\square$

Theorem 9.11.1.3 (Unconditional common primitive block decompositions). [Lean](#) [✓ Stmt](#) *Let A and B have the same MPV family at every positive length. Then there is a positive blocking length at which both blocked tensors have the same positive-length MPV family as weighted direct sums of trace-preserving, primitive, tensor-irreducible blocks with positive bond dimensions and nonzero weights. The two weighted nonzero-sector families have the same positive-length MPV family.*

Proof. [✓ Proof](#) Apply Theorem [9.3.1.4](#) to choose the common blocking length and the two common cyclic-sector families. For either family, write G_k for its cyclic-sector blocks and (ν_x, C_x)

for the weights and blocks of the flattened common-sector family from Lemma [E.5.10](#). The common-alphabet nonzero-part identity gives, for every length N and blocked word σ ,

$$V^{(N)}\left(\bigoplus_k \mu_k^p G_k^{[p]}\right)_\sigma = V^{(N)}\left(\bigoplus_x \nu_x C_x\right)_\sigma.$$

Applying this identity on the two sides of the comparison yields a positive length p and weighted primitive block families such that, for every $N \geq 1$ and blocked word σ ,

$$\begin{aligned} V^{(N)}(A^{[p]})_\sigma &= V^{(N)}\left(\bigoplus_x \mu_x^A A_x\right)_\sigma, \\ V^{(N)}(B^{[p]})_\sigma &= V^{(N)}\left(\bigoplus_y \mu_y^B B_y\right)_\sigma. \end{aligned}$$

The two weighted nonzero parts are equal at every positive length. The structural properties and the nonvanishing of the weights come from the common-sector families. $\square$

Chapter 10

Basis of Normal Tensors

A *basis of normal tensors* (BNT) for an MPS tensor A is a finite family of normal blocks $\{A_j\}$ whose MPV states $\{|V^{(N)}(A_j)\rangle\}$ span $|V^{(N)}(A)\rangle$ at every positive length and are eventually linearly independent. The literal definition permits a dormant normal tensor with coefficient identically zero and therefore does not imply uniqueness. For active sector presentations, equal MPV families force a bijective gauge-phase matching $B_k = e^{i\phi_k} X_k A_j X_k^{-1}$, and the sector coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$ recover the per-copy weight multisets by Newton–Girard. The references are [PGVWC07], [CPGSV16], and [CPGSV21].

10.1 Bases of normal tensors

Definition 10.1.1 (Basis of Normal Tensors (BNT)). Lean ✓ Stmt A *basis of normal tensors* (BNT) for a total tensor A_{tot} consists of g block tensors $(A_1, \dots, A_g)$ with bond dimensions $(D_1, \dots, D_g)$ such that each A_j is normal (Definition 2.4.4); at each system size $N \geq 1$ the MPV of A_{tot} lies in the span of the block MPVs $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$; and for sufficiently large N those block MPVs are linearly independent. Here normality means eventual block injectivity, as in Definition 2.4.4. Thus this is the algebraic form of [CPGSV21, Definition IV.2], rather than its primitive-channel formulation. In the canonical-form characterization of [CPGSV16, Section 2.3], a BNT is a minimal family of representatives for $\tilde{A}_k^i = e^{i\phi_k} X_k A_j X_k^{-1}$ among the normal blocks.

Definition 10.1.2 (Basis of normal tensors, coefficient form). Lean ✓ Stmt The coefficient form of a basis of normal tensors: each block A_j is a CPSV normal tensor (Definition 9.6.1), the MPV of A is given at every positive length by explicit coefficients $c_j^{(N)} \in \mathbb{C}$ with $|V^{(N)}(A)\rangle = \sum_{j=1}^g c_j^{(N)} |V^{(N)}(A_j)\rangle$, and the block MPVs $\{|V^{(N)}(A_j)\rangle\}$ are eventually linearly independent. This is the literal coefficient-form definition of [CPGSV16, Section 2.3]. It differs from Definition 10.1.1 only in the chosen normality predicate. Theorem 10.2.5 proves the implication between the two forms.

Theorem 10.1.3 (Distinct BNT members are not gauge-phase equivalent). Lean ✓ Stmt *Distinct members of a basis of normal tensors are not gauge-phase equivalent after identifying equal bond dimensions.*

Proof. ✓ Proof At every sufficiently large length the matrix product vectors of the basis members are linearly independent. A gauge-phase equivalence between two distinct members would give

$|V^{(N)}(A_k)\rangle = \zeta^N |V^{(N)}(A_j)\rangle$ for a nonzero scalar ζ . Thus one vector would be a scalar multiple of the other, contradicting linear independence. $\square$

Theorem 10.1.4 (Printed BNT characterization). Not ready Let

$$A^i = \bigoplus_{k=1}^r \mu_k \tilde{A}_k^i$$

be a tensor in canonical form. According to [CPGSV16, Proposition 2.7], a family $(A_j)_j$ is a basis of normal tensors for A if and only if every listed normal block $\tilde{A}_k$ is gauge-phase equivalent to some A_j and no two distinct A_j are gauge-phase equivalent.

This is false when a coefficient μ_k vanishes. Such a block is invisible in every positive-length matrix product vector and need not occur in a basis recovered from those vectors. If only the active indices k with $\mu_k \neq 0$ are required to be represented, the characterization holds by Theorem 10.1.5.

Theorem 10.1.5 (Characterization by canonical-form representatives). Lean Lean ✓ Stmt Let A have a canonical-form MPV expansion

$$|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(\tilde{A}_k)\rangle$$

at every positive length, where the tensors $\tilde{A}_k$ are normal tensors of positive bond dimension. Set $K_{\text{act}} = \{k : \mu_k \neq 0\}$. Let $(A_j)_{j=1}^g$ be tensors of positive bond dimension. Then $(A_j)_{j=1}^g$ is a basis of normal tensors for A if and only if:

1. every A_j is normal;
2. for every $k \in K_{\text{act}}$ there are an index j , an invertible matrix X_k , and a phase $e^{i\phi_k}$ such that $\tilde{A}_k^i = e^{i\phi_k} X_k A_j^i X_k^{-1}$ for every physical index i ;
3. no two distinct tensors A_j and $A_{j'}$ are related in this way.

This is [CPGSV16, Proposition 2.7], with the following necessary local correction.

Local fix (active blocks): the coverage clause is restricted to K_{act} . The literal source canonical-form display permits $\mu_k = 0$, but such a summand contributes nothing to any positive-length MPV and therefore cannot be recovered from the BNT spanning property. Removing the zero-weight summands leaves every positive-length MPV unchanged and restores the nonzero-weight hypothesis of the source phase-class argument. The comparison is recorded in `docs/paper-gaps/cpsv16_bnt_characterization_active_blocks.tex`.

Proof. ✓ Proof First discard the zero-weight summands; they contribute zero at every positive length. Reindex the remaining summands, group them into the MPV phase classes of Definition 10.6.1.7, and choose one representative from each class. By Theorem 10.1.3, eventual linear independence of a BNT excludes gauge-phase equivalence between two distinct A_j .

Suppose that some phase-class representative is not gauge-phase equivalent to any A_j . Partition all phase classes into the unmatched classes T and the remaining classes. Every remaining representative is a nonzero scalar multiple of some $|V^{(N)}(A_j)\rangle$ at each positive length. Distinct unmatched representatives have vanishing mutual overlaps, and their overlaps with every A_j also vanish. Explicitly, for $t \neq t'$ in T and every j ,

$$\begin{aligned} O_{\tilde{A}_t \tilde{A}_{t'}}(N) &\longrightarrow 0, \\ O_{\tilde{A}_t A_j}(N) &\longrightarrow 0. \end{aligned}$$

By Lemma F.1.7, the combined family

$$\{|V^{(N)}(\tilde{A}_t)\rangle : t \in T\} \cup \{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g\}$$

is linearly independent for all sufficiently large N . Comparing the canonical-form expansion with the BNT expansion gives a relation

$$\sum_{t \in T} c_{N,t} |V^{(N)}(\tilde{A}_t)\rangle + \sum_{j=1}^g \alpha_{N,j} |V^{(N)}(A_j)\rangle = 0.$$

Linear independence forces every $c_{N,t}$ to vanish. Hence the coefficient of every unmatched phase class vanishes for all sufficiently large N . For an unmatched class t , let q range over its active canonical-form summands, write μ_q for the corresponding canonical-form weight, and write ζ_q for the phase relating that summand to the chosen representative. Then $c_{N,t} = \sum_{q \in t} (\zeta_q \mu_q)^N$. Every base $\zeta_q \mu_q$ is nonzero because only active summands occur. Eventual vanishing of this finite geometric sum would imply vanishing at $N = 0$, where its value is the positive number of summands in the class. This contradiction proves coverage. The normal-tensor phase theorem turns the resulting MPV phase relation into the stated gauge-phase relation.

Conversely, coverage gives, for each canonical-form summand, $|V^{(N)}(\tilde{A}_k)\rangle = \zeta_k^N |V^{(N)}(A_{j(k)})\rangle$. Hence

$$|V^{(N)}(A)\rangle = \sum_j \left(\sum_{k: j(k)=j} (\mu_k \zeta_k)^N \right) |V^{(N)}(A_j)\rangle.$$

This is the required BNT span at every positive length. Pairwise gauge-phase distinction and normality give eventual linear independence, completing the converse. $\square$

Theorem 10.1.6 (Failure of literal BNT uniqueness). Lean Stmt *There is a normal tensor with a one-block canonical-form expansion which admits two bases of normal tensors of different cardinalities. Hence the unrestricted uniqueness sentence of [CPGSV16, Appendix A, line 1148] does not follow from the literal definition of a basis of normal tensors.*

Proof. Proof Take physical dimension two and bond dimension one. Let A_0 be supported on physical symbol zero and let A_1 be supported on physical symbol one:

$$\begin{aligned} A_0^0 &= 1, \\ A_0^1 &= 0, \\ A_1^0 &= 0, \\ A_1^1 &= 1. \end{aligned}$$

Both transfer maps are the identity on the one-dimensional matrix algebra: $\mathcal{E}_{A_j}(X) = X$ for $j = 0, 1$. Hence both tensors are normal. At every positive length,

$$\begin{aligned} |V^{(N)}(A_0)\rangle &= |0 \cdots 0\rangle, \\ |V^{(N)}(A_1)\rangle &= |1 \cdots 1\rangle. \end{aligned}$$

Thus (A_0) and (A_0, A_1) are both bases of normal tensors for A_0 . Indeed, for every $N \geq 1$, the second expansion is $|V^{(N)}(A_0)\rangle = 1 \cdot |V^{(N)}(A_0)\rangle + 0 \cdot |V^{(N)}(A_1)\rangle$. Their index sets have cardinalities one and two, so they cannot be related by a bijection. $\square$

Theorem 10.1.7 (Restricted BNT uniqueness under converse coverage). [Lean](#) [Stmt](#) *Let A have a canonical-form MPV expansion*

$$|V^{(N)}(A)\rangle = \sum_{k=0}^{r-1} \mu_k^N |V^{(N)}(\tilde{A}_k)\rangle$$

at every positive length, where the tensors $\tilde{A}_k$ are normal and have positive bond dimensions. Let $(A_j)_{j=0}^{g_1-1}$ and $(B_l)_{l=0}^{g_2-1}$ be two bases of normal tensors for A , all of whose members have positive bond dimensions. Assume that every A_j and every B_l is related by a phase and an invertible gauge to a block $\tilde{A}_k$ with $\mu_k \neq 0$. Then there is a bijection $e : \{0, \dots, g_1 - 1\} \rightarrow \{0, \dots, g_2 - 1\}$ such that the bond dimensions of A_j and $B_{e(j)}$ agree and $B_{e(j)}^i = \zeta_j X_j A_j^i X_j^{-1}$, with $|\zeta_j| = 1$, for every j and every physical index i .

Proof. [Proof](#) For each A_j , choose an active canonical-form block $\tilde{A}_{\ell(j)}$ representing it, and then choose a member $B_{f(j)}$ representing that block. If $f(j) = f(j')$, then

$$A_j \sim_{\text{gp}} \tilde{A}_{\ell(j)} \sim_{\text{gp}} B_{f(j)} = B_{f(j')} \sim_{\text{gp}} \tilde{A}_{\ell(j')} \sim_{\text{gp}} A_{j'}.$$

Minimality of the first basis gives $j = j'$, so f is injective. Reversing the roles of the two bases gives an injection in the other direction. Since both index sets are finite, f is bijective; set $e = f$. Write the two relations through the chosen active block as

$$\begin{aligned} \tilde{A}_{\ell(j)}^i &= \zeta_{1,j} X_{1,j} A_j^i X_{1,j}^{-1}, \\ \tilde{A}_{\ell(j)}^i &= \zeta_{2,j} X_{2,j} B_{e(j)}^i X_{2,j}^{-1}. \end{aligned}$$

Solving the second identity for $B_{e(j)}^i$ and using the first gives

$$B_{e(j)}^i = (\zeta_{2,j}^{-1} \zeta_{1,j}) (X_{2,j}^{-1} X_{1,j}) A_j^i (X_{1,j}^{-1} X_{2,j}).$$

Denote the scalar and gauge in this identity by ζ_j and X_j . The gauge-phase relation gives

$$O_{B_{e(j)} B_{e(j)}}(N) = (\zeta_j \bar{\zeta}_j)^N O_{A_j A_j}(N) = |\zeta_j|^{2N} O_{A_j A_j}(N).$$

Normality gives $O_{A_j A_j}(N) \rightarrow 1$ and $O_{B_{e(j)} B_{e(j)}}(N) \rightarrow 1$; hence the last identity forces $|\zeta_j| = 1$. $\square$

10.2 Normal blocks and overlap rigidity

This section relates spectral normality to eventual block injectivity and records the overlap dichotomy that detects bond-dimension and gauge-phase matching between normal blocks.

Theorem 10.2.1 (Normality forces positive bond dimension). [Lean](#) [Stmt](#) *Every normal tensor has positive bond dimension.*

Proof. [Proof](#) If the bond dimension were zero, the bond-matrix space and its endomorphism algebra would each contain only zero. The transfer operator would therefore have spectral radius zero, contradicting the spectral-radius-one clause in the definition of a normal tensor. $\square$

Theorem 10.2.2 (Pure Perron normalization at spectral radius one). [Lean](#) [✓ Stmt](#) *Let A be irreducible and suppose that the spectral radius of $\mathcal{E}_A$ is one. There is a positive definite matrix σ satisfying*

$$\sum_i (A^i)^\dagger \sigma A^i = \sigma. \quad (10.1)$$

The tensor

$$B^i = \sigma^{1/2} A^i \sigma^{-1/2} \quad (10.2)$$

is left-canonical and irreducible, and A and B are related by a pure invertible similarity.

Proof. [✓ Proof](#) Positive-definite right and left Perron eigenvectors have the same positive eigenvalue by the trace pairing. The spectral-radius identity makes this eigenvalue equal to one. Thus the left Perron eigenvector is an adjoint fixed point, so its square-root gauge is trace-preserving. Similarity preserves irreducibility. $\square$

Theorem 10.2.3 (Spectral normality gives a pure trace-preserving gauge). [Lean](#) [✓ Stmt](#) *Let A be a normal tensor in the sense of [CPGSV16, Section 2.2]. Its bond dimension is necessarily positive, and there is a positive definite matrix σ such that the gauge obtained from σ is trace-preserving, primitive, and irreducible. No scalar rescaling is needed.*

Proof. [✓ Proof](#) Pure Perron normalization gives a trace-preserving gauge by a pure similarity. Irreducibility and primitivity are invariant under this similarity. $\square$

Theorem 10.2.4 (Spectral normality gives eventual block injectivity). [Lean](#) [✓ Stmt](#) *Every normal tensor in the sense of [CPGSV16, Section 2.2] is injective after blocking by some finite length; its positive bond dimension follows from normality.*

Proof. [✓ Proof](#) Apply Theorem 10.2.3. The trace-preserving, primitive, irreducible representative is eventually block-injective, and gauge invariance transports this conclusion back to the original tensor. $\square$

Theorem 10.2.5 (A coefficient-form BNT is an algebraic BNT). [Lean](#) [✓ Stmt](#) *If $(A_j)_{j=1}^g$ is a coefficient-form basis of normal tensors for A , then $(A_j)_{j=1}^g$ is an algebraic basis of normal tensors for A . Only this implication from spectral normality to algebraic normality is asserted.*

Proof. [✓ Proof](#) Every block has positive bond dimension, so spectral normality implies eventual block injectivity for each block. The spanning identities and eventual linear independence are unchanged. $\square$

Theorem 10.2.6 (Non-decaying rectangular overlap determines the bond dimension). [Lean](#) [✓ Stmt](#) *Let A and B be irreducible trace-preserving tensors of positive, possibly different, bond dimensions. If $|O_{AB}(N)| \rightarrow 1$, then their bond dimensions are equal.*

Proof. [✓ Proof](#) If the bond dimensions differed, the rectangular overlap-decay theorem would give $O_{AB}(N) \rightarrow 0$, and hence $|O_{AB}(N)| \rightarrow 0$, contradicting the assumed limit. $\square$

Theorem 10.2.7 (A normal tensor has a nonzero letter). [Lean](#) [✓ Stmt](#) *Let A be a CPSV normal tensor of positive bond dimension. Then $A^i \neq 0$ for some physical letter i .*

Proof. [✓ Proof](#) If every letter vanished, then the transfer map would vanish and its spectral radius would be zero, contradicting the spectral-radius-one normalization of a CPSV normal tensor. $\square$

Theorem 10.2.8 (Phase-equivalent normal tensors are gauge-phase equivalent). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B be normal tensors of positive, possibly different, bond dimensions. Suppose that, for some nonzero $\zeta \in \mathbb{C}$, $|V^{(N)}(B)\rangle = \zeta^N |V^{(N)}(A)\rangle$ for $N \geq 0$. Then the bond dimensions are equal and there is an invertible matrix X and a scalar η of unit modulus such that $B^i = \eta X A^i X^{-1}$ for every physical letter i . This is the gauge recovery used in [CPGSV16, Appendix A].

Proof. [✓ Proof](#) Apply Theorem 10.2.3 to both tensors. Gauge invariance of matrix product vectors preserves the phase relation, whose self-overlap identities imply unit modulus for the phase. The resulting rectangular cross-overlap has norm tending to one. If the bond dimensions differed, Theorem 10.2.6 would give a contradiction. Thus the bond dimensions agree, and the unit-overlap rigidity theorem gives a gauge-phase equivalence between the trace-preserving representatives. Conjugating by the two Perron gauges gives the asserted relation for A and B . $\square$

Theorem 10.2.9 (Normal-tensor overlap dichotomy). [Lean](#) [✓ Stmt](#) Let A and B be normal tensors of positive, possibly different, bond dimensions. Their self-overlaps satisfy

$$\begin{aligned} O_{AA}(N) &\rightarrow 1, \\ O_{BB}(N) &\rightarrow 1. \end{aligned}$$

Moreover, either $O_{AB}(N) \rightarrow 0$, or their bond dimensions are equal; call the common value D . Then there are $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C}$, with $|\zeta| = 1$, such that $B^i = \zeta X A^i X^{-1}$ for every physical letter i . In the latter case, $|O_{AB}(N)| \rightarrow 1$. This is [CPGSV16, Lemma A.2].

Proof. [✓ Proof](#) Put both tensors in their pure trace-preserving gauges. If their bond dimensions differ, rectangular overlap decay gives $O_{AB}(N) \rightarrow 0$. If their dimensions agree but they are not gauge-phase equivalent, the equal-dimension overlap-decay theorem gives the same conclusion. In the remaining case, $B^i = \zeta X A^i X^{-1}$. By Theorem 10.2.8, $|\zeta| = 1$. Hence

$$\begin{aligned} O_{AB}(N) &= \bar{\zeta}^N O_{AA}(N), \\ |O_{AB}(N)| &= |O_{AA}(N)| \rightarrow 1. \end{aligned}$$

$\square$

Theorem 10.2.10 (Exact phase alternative for normal tensors). [Lean](#) [✓ Stmt](#) Let A and B be normal tensors of positive, possibly different, bond dimensions. Then either $O_{AB}(N) \rightarrow 0$, or there is a phase $\zeta \in \mathbb{C}$, with $|\zeta| = 1$, such that $|V^{(N)}(B)\rangle = \zeta^N |V^{(N)}(A)\rangle$ for every chain length N . This is [CPGSV16, Corollary A.3].

Proof. [✓ Proof](#) Apply Theorem 10.2.9. In its second alternative, gauge-phase covariance gives $V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma$ for every configuration σ , and hence the asserted equality of vectors. $\square$

Theorem 10.2.11 (Unit overlap from eventual proportionality). [Lean](#) [✓ Stmt](#) Let A and B be tensors, possibly with different bond dimensions, whose self-overlaps tend to one. Suppose that, for all sufficiently large N , there is $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$. Then $|O_{AB}(N)| \rightarrow 1$.

Proof. [✓ Proof](#) The proportionality relation gives

$$\begin{aligned} O_{AB}(N) &= c_N O_{BB}(N), \\ O_{AA}(N) &= |c_N|^2 O_{BB}(N). \end{aligned}$$

Hence $|c_N|^2 \rightarrow 1$ and therefore $|c_N| \rightarrow 1$. The first identity now gives $|O_{AB}(N)| \rightarrow 1$. $\square$

Theorem 10.2.12 (Geometric proportionality phase for normal tensors). [Lean](#) [✓ Stmt](#) Let A and B be normal tensors of positive, possibly different, bond dimensions. Suppose that, for all sufficiently large N , their matrix product vectors are proportional by a nonzero scalar. Then there is $a \in \mathbb{C}$, with $|a| = 1$, such that $|V^{(N)}(A)\rangle = a^N |V^{(N)}(B)\rangle$ for every positive N . This is the one-normal-block consequence of [CPGSV16, Corollary A.3 and the proof of Theorem 2.10].

Proof. [✓ Proof](#) Normality gives $O_{AA}(N) \rightarrow 1$ and $O_{BB}(N) \rightarrow 1$, so Theorem 10.2.11 gives $|O_{AB}(N)| \rightarrow 1$. The first alternative in Theorem 10.2.10 gives $O_{AB}(N) \rightarrow 0$, hence $|O_{AB}(N)| \rightarrow 0$, a contradiction. In the remaining alternative, $|V^{(N)}(B)\rangle = \zeta^N |V^{(N)}(A)\rangle$ with $|\zeta| = 1$; take $a = \zeta^{-1}$. $\square$

Theorem 10.2.13 (Separated normal tensors are eventually independent). [Lean](#) [✓ Stmt](#) Let $(A_j)_{j=1}^g$ be a finite family of positive-bond-dimension normal tensors, no two of which are gauge-phase equivalent after identifying equal bond dimensions. Then the matrix product vectors $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$ are linearly independent for all sufficiently large N , as in [CPGSV16, Appendix A].

Proof. [✓ Proof](#) Put every tensor in the pure trace-preserving gauge of Theorem 10.2.3. Distinct gauge-phase classes have overlaps tending to zero, while each self-overlap tends to one. The finite Gram matrix therefore tends to the identity and is invertible for all sufficiently large lengths. Gauge invariance of matrix product vectors gives the conclusion for the original tensors. $\square$

10.3 Block-injective consequences for BNT families

The overlap separation of normal blocks yields a simultaneous product-algebra span after blocking. We first state the fixed-length span condition and then apply it to coefficient-form and sector BNTs.

Definition 10.3.1 (Fixed-length block-injective canonical-form condition). [Lean](#) [Lean](#) [✓ Stmt](#) For a fixed blocked length m , the fixed-length form of the block-injective canonical-form condition is

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = m\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Equivalently,

$$\forall X = (X_j)_j \in \bigoplus_{j=1}^g M_{D_j}(\mathbb{C}), \quad \exists (c_\omega(X))_{|\omega|=m}, \quad (X_j)_{j=1}^g = \sum_{|\omega|=m} c_\omega(X) (A_1^\omega, \dots, A_g^\omega).$$

This is the fixed-length form of the block-injective canonical-form condition of [CPGSV21, lines 2113–2117].

Theorem 10.3.2 (Gauge invariance of simultaneous fixed-length product spans). [Lean](#) [Lean](#) [✓ Stmt](#) Let $(A_j)_{j=1}^g$ and $(B_j)_{j=1}^g$ be tensor families with the same physical dimension and the same bond dimension D_j in each labelled block. Suppose that for every j there is an arbitrary invertible matrix $X_j \in \text{GL}_{D_j}(\mathbb{C})$ such that, for every physical index i ,

$$B_j^i = X_j A_j^i X_j^{-1}.$$

Then, at every fixed length L ,

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = L\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}) \iff \text{span}\{(B_1^\omega, \dots, B_g^\omega) : |\omega| = L\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Both directions hold for general invertible gauges; no unitarity condition is imposed. In particular, a separation certificate proved after passing to left-canonical representatives can be transported back to a source unital gauge carrying a positive dual fixed point. This supplies only the simultaneous-span transport; the subsequent source-gauge boundary and parent-Hamiltonian arguments are separate.

Proof. ✓ Proof Given a target tuple $(M_j)_j$, conjugate it backwards to $(X_j^{-1}M_jX_j)_j$. If the word tuples of $(A_j)_j$ span the product algebra, one common coefficient family $(c_\omega)_\omega$ expresses this conjugated tuple. Gauge covariance of word evaluation then gives

$$\sum_{|\omega|=L} c_\omega B_j^\omega = X_j \left(\sum_{|\omega|=L} c_\omega A_j^\omega \right) X_j^{-1} = M_j \quad (10.3)$$

for every j . Thus the $(B_j)_j$ word tuples span the product algebra. Replacing every X_j by X_j^{-1} proves the reverse implication. $\square$

Lemma 10.3.3 (Direct-sum block injectivity at the source length). Lean Lean Lean ✓ Stmt Let $(A_j)_{j=1}^g$ be irreducible left-canonical tensors with normalized self-overlaps and with no gauge-phase-equivalent distinct blocks. Suppose $g \geq 2$, $L_0 > 0$, and every A_j is block-injective at lengths L_0 , $L_0 + 1$, and $3(L_0 + 1)$. Then

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = 3(g-1)(L_0 + 1)\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

This is the direct-sum induction length in [PGVWC07, lines 1346–1408].

Proof. ✓ Proof Lemma F.3.1 gives homogeneous pair-trace separation at $S = 3(L_0 + 1)$. For a chosen anchor block, pair separation prescribes an arbitrary matrix there and zero on one other block. Multiplying $g - 1$ such homogeneous combinations removes the other blocks one at a time. Induction on the number of removed blocks therefore realizes every tuple supported on the anchor at length $(g - 1)S$. Summing over the anchors gives the full product algebra at $3(g - 1)(L_0 + 1)$. $\square$

Theorem 10.3.4 (Simultaneous one-letter span after blocking a BNT). Lean Lean Lean Lean ✓ Stmt Let $(A_j)_{j=1}^g$ be a basis of normal tensors, with bond dimensions D_j . There is a positive integer L such that the one-letter evaluations of the blocked tensors span the full product algebra:

$$\text{span}_{\mathbb{C}} \left\{ ((A_1^{[L]})^i, \dots, (A_g^{[L]})^i) : 0 \leq i < d^L \right\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Moreover, there is L_0 such that, for every $L \geq L_0$, the unblocked word tuples have the simultaneous span

$$\text{span}_{\mathbb{C}} \{(A_1^\omega, \dots, A_g^\omega) : |\omega| = L\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

This is the existence form of simultaneous block injectivity associated with [CPGSV16, lines 317–345]; no numerical bound on L or L_0 is asserted here.

Proof. ✓ Proof Put each BNT representative in its trace-preserving Perron gauge. The gauged family is irreducible and left-canonical, has normalized self-overlaps, and remains gauge-phase separated. Choose a common positive length p at which every gauged block is block-injective. Propagation of homogeneous word spans gives block injectivity at $p + 1$ and $3(p + 1)$.

Lemma 10.3.3 gives pairwise separation at length $S = 3(p + 1)$. Write $g \dot{-} 1 = \max\{g - 1, 0\}$. Multiplying one separator for each of the other blocks gives a selector of length $(g \dot{-} 1)S$ for every block. Transport these selectors back from the Perron gauges. Gauge invariance also gives block injectivity of every original block at length p . Concatenating the injective prefix with the selectors yields the full product-algebra span at

$$L = p + (g \dot{-} 1)S = p + 3(g \dot{-} 1)(p + 1) > 0.$$

The truncated factor makes the same construction valid for empty and singleton families. The same span provides selectors and block injectivity at length L . For every sufficiently large n , concatenate an $(n - L)$ -letter injective prefix with these selectors to obtain the full product-algebra span at length n . Finally, length- L words of A_j are the one-letter evaluations of $A_j^{[L]}$, which proves the blocked statement. $\square$

Theorem 10.3.5 (Sharp simultaneous span under a total-dimension bound). Lean ✓ Stmt Let $(A_j)_{j=1}^g$ be a basis of normal tensors with bond dimensions D_j . If $K > 0$ and $\sum_j D_j \leq K$, then there is a positive integer $L \leq 3K^5$ such that

$$\text{span}_{\mathbb{C}}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = L\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Proof. ✓ Proof Put every representative in its trace-preserving Perron gauge. Since $D_j \leq \sum_k D_k \leq K$, quantum Wielandt and homogeneous-span propagation make every representative block-injective at K^4 , $K^4 + 1$, and $3(K^4 + 1)$. Conjugation preserves the simultaneous word span.

For $g = 0$, take $L = 1$; the product algebra has only its zero element. For $g = 1$, take $L = K^4$. If $g \geq 2$, the three-block separation theorem gives

$$L = 3(g - 1)(K^4 + 1).$$

Positivity of the D_j gives $g = \sum_j 1 \leq \sum_j D_j \leq K$, and hence $L \leq 3(K - 1)(K^4 + 1) \leq 3K^5$. $\square$

Theorem 10.3.6 (Total dimension of the active BNT representatives). Lean ✓ Stmt Let literal canonical-form data of ambient bond dimension D have active phase classes j and chosen representatives of dimensions D_j . Then

$$\sum_j D_j \leq D.$$

Proof. ✓ Proof In every active phase class, choose its first copy. The class-copy equivalence sends these choices to distinct displayed block indices, and the first copy is the chosen representative. Thus the sum of the representative dimensions is bounded by the sum over all displayed blocks, which is at most D by the canonical-form dimension bound. $\square$

Theorem 10.3.7 (Sharp block-injective form for literal CPSV canonical form). Lean ✓ Stmt Let A be in literal CPSV canonical form with positive ambient bond dimension D . There is a basis of normal tensors $(A_j)_{j=1}^g$ and a positive integer $L \leq 3D^5$ such that

$$\text{span}_{\mathbb{C}} \{((A_1^{[L]})^i, \dots, (A_g^{[L]})^i) : 0 \leq i < d^L\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Hence A is in block-injective canonical form after blocking at most $3D^5$ sites. This is the proposition at [CPGSV16, lines 342–345].

Proof. ✓ Proof Choose one representative from every active matrix-product-vector phase class. These representatives form a basis of normal tensors, and their total bond dimension is at most D by Theorem 10.3.6. Apply Theorem 10.3.5 with $K = D$. A word of length L in A_j is a one-letter evaluation of $A_j^{[L]}$, so the resulting length- L span is exactly the asserted one-letter span. $\square$

Let A be a tensor in canonical form, and let $A_1, \dots, A_g$ be a basis of normal tensors of A . For suitable coefficients $\mu_{j,q}$ and invertible matrices $X_{j,q}$, write $M_j = \text{diag}(\mu_{j,1}, \dots, \mu_{j,r_j})$ and $X = \bigoplus_{j=1}^g \bigoplus_{q=1}^{r_j} X_{j,q}$. Then

$$A^i = X \left[\bigoplus_{j=1}^g (M_j \otimes A_j^i) \right] X^{-1} = \bigoplus_{j=1}^g \bigoplus_{q=1}^{r_j} \mu_{j,q} X_{j,q} A_j^i X_{j,q}^{-1}. \quad (10.4)$$

This is [CPGSV16, Section 2.3, Equations (20a)–(21)]. The decomposition (10.4) is used below to separate the normal blocks.

10.4 Permutation rigidity for bases of normal tensors

When two bases of normal tensors generate the same MPV subspace for all sufficiently large positive lengths, the blocks must agree up to a permutation and per-block gauge phases. The argument turns the eventual equality of spans into an invertible change-of-basis matrix, then analyses its asymptotics through the overlaps.

The change-of-basis matrix is now used to match the blocks. We first treat equal block counts, then remove that restriction.

Theorem 10.4.1 (Permutation rigidity for equal block counts). Lean ✓ Stmt Let $(A_j)_{j=1}^g$ and $(B_j)_{j=1}^g$ be injective families of positive bond dimension satisfying trace-preserving normalization, with self-overlaps tending to 1 and off-diagonal overlaps tending to 0 within each family. If the spans of the MPV states agree for all sufficiently large system sizes N , then there is a permutation π of $\{1, \dots, g\}$ such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent.

Proof. ✓ Proof For large N , Lemma F.2.2 gives an invertible matrix U_N relating the two MPV families. Fix j and write $|V^{(N)}(B_j)\rangle = \sum_i u_{ij}^{(N)} |V^{(N)}(A_i)\rangle$. With $G_A(N) = (O_{A_i A_\ell}(N))_{i,\ell}$ and $b_j(N) = (O_{A_i B_j}(N))_i$, this gives

$$G_A(N) \overline{u_j(N)} = b_j(N).$$

Moreover,

$$O_{B_j B_j}(N) = u_j(N)^\top G_A(N) \overline{u_j(N)}.$$

If all mixed overlaps $O_{A_i B_j}(N)$ tended to 0, then $b_j(N) \rightarrow 0$; since $G_A(N) \rightarrow 1$, one obtains $\overline{u_j(N)} \rightarrow 0$ and hence $u_j(N) \rightarrow 0$. Thus $O_{B_j B_j}(N) \rightarrow 0$, contradicting $O_{B_j B_j}(N) \rightarrow 1$. Thus each B_j has a matched block $A_{f(j)}$ with non-decaying mixed overlap. Theorems 7.5.4 and 7.6.5 then force equal bond dimension and gauge-phase equivalence for each match. Off-diagonal decay within the B -family makes f injective, hence bijective, and its inverse is the required permutation. $\square$

Lemma 10.4.2 (Permutation rigidity with asymptotically orthonormal overlaps). [Lean](#) [✓ Stmt](#) Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective tensors of positive bond dimension satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If, for all sufficiently large system sizes N ,

$$\text{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g_A\} = \text{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \leq k \leq g_B\},$$

then $g_A = g_B$, and there is a permutation π of the common block index set such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent for each j .

Proof. [✓ Proof](#) By Lemma F.1.10, both families are linearly independent for all sufficiently large N . Choose N large enough that both families are linearly independent and the assumed span equality holds. The common span then has the same dimension when computed from the A -family or the B -family, so $g_A = g_B$. After identifying the two index sets, Theorem 10.4.1 yields the permutation conclusion. $\square$

The two preceding results obtain the block matching from the asymptotic orthonormality of the overlaps. The same matching is obtained by the exact sector comparison of Section 10.8, which pairs the sectors of two canonical forms through a bijection fixed by their bond dimensions and gauge-phase equivalences. The proof of the Fundamental Theorem in Chapter 11 proceeds through the sector comparison.

The same overlap dichotomy also shows that the MPV states of distinct normal blocks are non-proportional at arbitrarily large lengths; this is an input to the exact sector matching later in the chapter.

Theorem 10.4.3 (Non-equivalent irreducible TP blocks are eventually non-proportional). [Lean](#)

[Lean](#) [✓ Stmt](#) Let A and B be irreducible trace-preserving blocks whose self-overlaps converge to 1. If they are not gauge-phase equivalent, then for every lower bound N_0 there is $N \geq N_0$ such that $|V^{(N)}(A)\rangle$ is not proportional to $|V^{(N)}(B)\rangle$. Consequently, in an already separated same-dimensional BNT block family, any two distinct blocks have such non-proportional MPV states at arbitrarily large lengths.

Proof. [✓ Proof](#) Suppose that for all sufficiently large N one had $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$. Then

$$O_{AA}(N) = |c_N|^2 O_{BB}(N), \tag{10.5}$$

$$O_{AB}(N) = c_N O_{BB}(N). \tag{10.6}$$

Since $O_{AA}(N) \rightarrow 1$ and $O_{BB}(N) \rightarrow 1$, the identity (10.5) gives $|c_N| \rightarrow 1$, and hence (10.6) gives $|O_{AB}(N)| \rightarrow 1$. The irreducible trace-preserving overlap decay theorem gives $O_{AB}(N) \rightarrow 0$ when $A \not\sim_{\text{gp}} B$, a contradiction. $\square$

10.5 Newton–Girard identities and power-sum recovery

The coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$ that appear in the two-layer BNT expansion are power sums of the copy weights. Recovering the individual weights from these sums is a Newton–Girard problem: equal power sums determine equal characteristic polynomials, hence equal multisets of roots.

The general matrix statement is Theorem 13.8.7. It identifies a characteristic polynomial from the first n power-sum traces.

The matched-sector comparison below applies the power-sum identities to a single matched pair, after the BNT permutation and phase gauges have identified which basis tensors are compared: the coefficient $\sum_q \mu_{j,q}^N$ then equals $\sum_q (\nu_{j,q} e^{i\phi_j})^N$, and Newton–Girard recovers the weight multiset with multiplicity.

Lemma 10.5.1 (Printed finite power-sum comparison). Not ready Suppose two finite lists of complex numbers have equal positive power sums for every exponent up to the larger list length. According to [CPGSV16, Lemma A.5], the list lengths agree and the two multisets are equal.

This is false when zero entries are allowed: the lists (1) and (1, 0) have equal power sums for every positive exponent. The conclusion becomes valid if every entry is nonzero, if the exponent-zero power sum is also compared, or if zero entries are deleted before comparing the multisets. The nonzero version is Theorem 10.5.2.

Theorem 10.5.2 (Finite-range equal power sums imply equal multisets). Lean Lean Lean Lean ✓ Stmt Let $\alpha, \beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for $1 \leq k \leq n$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.

Two variants relax the equal-cardinality assumption. If $\alpha : \{0, \dots, m-1\} \rightarrow \mathbb{C}$ and $\beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$ have no zero entries and their power sums agree for $1 \leq k \leq \max(m, n)$, then $m = n$ and the multisets are equal. Without a nonzero hypothesis, the list form $\sum_{k=1}^{x_a} \lambda_{a,k}^N = \sum_{k=1}^{x_b} \lambda_{b,k}^N$ for $1 \leq N \leq \max(x_a, x_b)$ gives equality of the nonzero multisets; if it also holds at $N = 0$ then $x_a = x_b$ and the full lists agree.

Proof. ✓ Proof Apply the finite-range Newton–Girard theorem to $\text{diag}(\alpha)$ and $\text{diag}(\beta)$ and extract the roots of $\prod_i (X - \alpha_i)$ and $\prod_i (X - \beta_i)$. For unequal cardinalities, pad both lists with zeros to length $\max(m, n)$; the nonzero hypothesis makes the padded-zero counts $\max(m, n) - m$ and $\max(m, n) - n$, forcing $m = n$. Without that hypothesis, delete the zero entries first (unchanged for positive N); the equality at $N = 0$ gives $x_a = x_b$. □

Corollary 10.5.3 (A nonzero finite family has a nonvanishing power sum). Lean ✓ Stmt Let $m \geq 1$ and let $\lambda_0, \dots, \lambda_{m-1} \in \mathbb{C}$ be nonzero. Then there is an exponent N with $1 \leq N \leq m$ such that $\sum_{q=0}^{m-1} \lambda_q^N \neq 0$. This is the consequence of the Appendix power-sum lemma used in the proof of Lemma L [CPGSV16, Appendix C.3, lines 1851–1858]; the finite-range comparison is [CPGSV16, Appendix A, lines 1155–1163].

Proof. ✓ Proof Suppose that every power sum with $1 \leq N \leq m$ vanishes. Comparing $\lambda_0, \dots, \lambda_{m-1}$ with the empty family in Theorem 10.5.2 gives $m = 0$, contradicting $m \geq 1$. □

10.6 The BNT canonical form on a sector decomposition

This section introduces the BNT canonical-form conditions of [CPGSV16, Section 2], defined on a sector decomposition with raw sector weights $\mu_{j,q}$ and coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$; no equal-modulus factorization is assumed.

10.6.1 Sector decompositions and phase matching

The sector decomposition and the MPV phase-equivalence relations introduced here determine the BNT canonical-form conditions below.

Definition 10.6.1.1 (Sector weights and sector decomposition). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [✓ Stmt](#) A sector-weight tuple over g basis blocks consists of multiplicities $r_j > 0$, nonzero weights $\mu_{j,q}$ for $q \in \{1, \dots, r_j\}$, and the sector coefficients

$$c_{S,j}^{(N)} := \sum_{q=1}^{r_j} \mu_{j,q}^N. \quad (10.7)$$

A *sector decomposition* consists of the basis blocks $(A_j)_{j=1}^g$ together with these multiplicities and weights. If A_j has bond dimension D_j , then the flattened copy family has total bond dimension

$$D_{\text{tot}} = \sum_{j=1}^g r_j D_j.$$

Definition 10.6.1.2 (Physical blocking of a sector decomposition). [Lean] [Stm] Let P be a sector decomposition with representatives A_j , multiplicities r_j , and copy weights $\mu_{j,q}$. Its physical p -site blocking has the same representative and copy indices, with

$$\begin{aligned} A_j &\longmapsto A_j^{[p]}, \\ \mu_{j,q} &\longmapsto \mu_{j,q}^p. \end{aligned}$$

The two-layer decomposition is [CPGSV16, lines 283–301]. Blocking preserves the copy index sets and bond dimensions, replaces each flattened representative by its p -site blocking, and raises each flattened weight to the p -th power.

Theorem 10.6.1.3 (Indices, dimensions, weights, and representatives under blocking). [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Physical blocking preserves the copy sets and bond dimensions. On the flattened copy index, it raises the weight to the p -th power and replaces the representative by its p -site blocking.

Proof. ✓ Proof These identities follow directly from the definition.

Lemma 10.6.1.4 (Sector decomposition formula). Lean Stmnt *For a sector decomposition S with associated tensor A_{tot} , the MPV satisfies*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{i=1}^g c_{S,j}^{(N)} V^{(N)}(A_j)_\sigma. \quad (10.8)$$

Two blocks are identified for canonical-form purposes when their MPV families differ by a scalar power. The next three definitions formalize this relation, the common cover it generates between two families, and the phase classes of a single family.

Definition 10.6.1.5 (MPV phase equivalence for nonzero-weight blocks). [Lean](#) [Stm](#) Two blocks A and B are MPV-phase equivalent if there is a nonzero scalar ζ such that, for every positive system size N and every physical word, $V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma$. The two bond dimensions are allowed to differ.

Enumerate $\mathcal{G}$. The induced equivalence Φ between the dependent grouped block coordinates and the original flattened listed coordinates gives

$$\sum_{x \in \mathcal{G}} D_x = \sum_{k=1}^r D_k. \quad (10.13)$$

For $x = (j, q)$, let $\hat{\mu}_x$ be the original weight of that copy and let $\widehat{B}_x^i = \zeta_x C_j^i$, with the required dimension identification. For $x \in K_0$, let $\hat{\mu}_x = 0$ and retain the original block $\widehat{B}_x = A_x$. Define the grouped tensor by

$$T_{\text{grp}}^i = \text{reindex}_{\Phi} \left(\bigoplus_{x \in \mathcal{G}} \hat{\mu}_x \widehat{B}_x^i \right). \quad (10.14)$$

If $G = \bigoplus_k Y_k$ is the block-diagonal gauge, with $Y_k = X_k$ on active indices and $Y_k = \mathbb{1}$ on K_0 , then, letter by letter,

$$\bigoplus_{k=1}^r \mu_k A_k^i = G T_{\text{grp}}^i G^{-1}, \quad (10.15)$$

and

$$A^i = U^\dagger G T_{\text{grp}}^i G^{-1} U. \quad (10.16)$$

Thus the class-copy coordinates and the inactive complement are explicit, while every zero-weight listed block remains present. This refinement does not delete the inactive coordinates; the active coisometric restriction is stated separately below.

Proof. ✓ Proof Enumerate the active indices by their phase classes and choose one normal representative in each class. The normal-tensor phase theorem gives the dimension equality, unit phase, and invertible gauge in (10.12). The BNT characterization proves that the representatives form a basis of normal tensors. Adjoin the inactive complement to the class-copy order. Reindexing the dependent block coordinates gives Φ and (10.13). The exact heterogeneous block-diagonal reindexing identity identifies the grouped tensor with the representative copies in the original retained-coordinate space. Block-diagonal conjugation then gives (10.15); substitution into (10.9) gives (10.16) with the original coisometry U . In the notation of [CPGSV16, Proposition 4.13], these identities are

$$\bigoplus_{k=1}^r \mu_k A_k^i = G T_{\text{grp}}^i G^{-1}, \quad (10.17)$$

$$A^i = U^\dagger G T_{\text{grp}}^i G^{-1} U, \quad (10.18)$$

$$U U^\dagger = \mathbb{1}. \quad (10.19)$$

□

Theorem 10.6.1.10 (Exact coisometric restriction to nonzero-weight blocks). Lean Lean Lean
Lean Lean ✓ Stmt Suppose the literal CPSV canonical-form data are

$$A^i = V^\dagger \left(\bigoplus_{k=1}^r \mu_k A_k^i \right) V, \quad (10.20)$$

$$V V^\dagger = \mathbb{1}. \quad (10.21)$$

Put $K_{\text{act}} = \{k : \mu_k \neq 0\}$ and $D_{\text{act}} = \sum_{k \in K_{\text{act}}} D_k$. There is a matrix $U \in \mathbb{C}^{D_{\text{act}} \times D}$ such that the following identities hold, the second for every physical index i :

$$UU^\dagger = \mathbb{1}, \quad (10.22)$$

$$A^i = U^\dagger \left(\bigoplus_{k \in K_{\text{act}}} \mu_k A_k^i \right) U. \quad (10.23)$$

Proof. ✓ Proof Let S select the flattened coordinates belonging to K_{act} . Distinct selected rows are orthonormal, so $SS^\dagger = \mathbb{1}$. The active weighted sum is the principal submatrix selected by S . Every omitted diagonal block is zero because its weight is zero, and therefore

$$\bigoplus_{k=1}^r \mu_k A_k^i = S^\dagger \left(\bigoplus_{k \in K_{\text{act}}} \mu_k A_k^i \right) S. \quad (10.24)$$

Set $U = SV$. Equations (10.20) and (10.24) give (10.23), while $UU^\dagger = S(VV^\dagger)S^\dagger = \mathbb{1}$ gives (10.22). $\square$

Definition 10.6.1.11 (Active BNT sector presentation). Lean Lean Lean Lean Lean ✓ Stmt
 An *active BNT sector presentation* of a tensor A is a nonempty sector decomposition P such that every representative has a positive number of copies, every copy has nonzero weight, every representative C_j is normal, the representative matrix product vectors are eventually linearly independent, and, for every $N > 0$,

$$V^{(N)}(A) = V^{(N)}(P). \quad (10.25)$$

This records the nonzero canonical summands constructed at [CPGSV16, lines 217–246]: dormant candidates are absent, whereas repeated nonzero copies and their weights remain explicit. The condition is stronger than the literal BNT definition, which permits the zero-coefficient candidate of Theorem 10.1.6.

Theorem 10.6.1.12 (Uniqueness of active BNT presentations). Lean ✓ Stmt *Two active BNT presentations of the same positive-length matrix product vector family have equally many representatives. After a permutation, corresponding representatives have equal bond dimension and are related by an invertible gauge and a unit-modulus phase.*

Only active presentations are compared. The conclusion is false for arbitrary literal BNTs by Theorem 10.1.6.

Proof. ✓ Proof Apply Theorem 10.1.5 first to the nonzero-weight direct-sum tensor of the first presentation, using the representatives of the second as a BNT, and then with the presentations interchanged. This gives a matching map in each direction. Eventual linear independence excludes gauge-phase-equivalent distinct representatives, so both maps are injective. Since the index sets are finite, they have equal cardinality, and either matching map is a bijection. Normal-tensor gauge rigidity gives the asserted gauge-phase relation for each matched pair. $\square$

Theorem 10.6.1.13 (Active representative sector decomposition). Lean Lean Lean Lean Lean ✓ Stmt *In the notation of Theorem 10.6.1.9, let $k(j, q) \in K_{\text{act}}$ be the listed block corresponding to the class-copy pair (j, q) . The active representatives C_j define a sector decomposition P_{act} with copy weights*

$$\lambda_{j,q} = \nu_{k(j,q)} \zeta_{k(j,q)}. \quad (10.26)$$

Every copy weight in P_{act} is nonzero. The representatives have simultaneous word-tuple separation at every sufficiently large length.

The sector decomposition contains only the active class-copy pairs. The full grouped tensor T_{grp} of Theorem 10.6.1.9 still contains every inactive zero-weight listed coordinate. For every positive length N , the grouped and active-sector tensors generate the same matrix product vector. The original tensor has the same positive-length matrix product vectors as both:

$$V^{(N)}(T_{\text{grp}}) = V^{(N)}(P_{\text{act}}). \quad (10.27)$$

$$V^{(N)}(A) = V^{(N)}(P_{\text{act}}). \quad (10.28)$$

Hence P_{act} is an active BNT sector presentation of A whenever at least one active listed block occurs. No equality at length zero is asserted, since deleting inactive bond coordinates changes the empty-word coefficient.

Proof. ✓ Proof The active class-copy equivalence reindexes the nonzero listed blocks. (10.26) combines the original coefficient with the unit phase relating its block to C_j ; both factors are nonzero. The basis-of-normal-tensors condition and Theorem 10.3.4 give eventual simultaneous word-tuple separation for the representatives. At positive length, every inactive contribution contains a positive power of its zero coefficient and vanishes. Reindexing the remaining finite sum by (j, q) gives (10.27). The original coisometric reconstruction and simultaneous block gauge preserve every positive-length closed-chain coefficient, yielding (10.28). $\square$

Definition 10.6.1.14 (BNT canonical form). Lean ✓ Stmt Given a sector decomposition P , these conditions define the normalized, left-canonical, aperiodic BNT surface used below. This is stronger than the abstract BNT of Definition 10.1.1 and the literal coefficient-form BNT of Definition 10.1.2: besides spanning and eventual linear independence, it fixes irreducible left-canonical representatives, their self-overlap normalization, gauge-phase separation, and the canonical-form weight normalization. It imposes no equal-modulus factorization or strict ordering on the raw weights. Explicitly, each basis block has positive bond dimension, is irreducible, and is left-canonical; the normalized self-overlap of every basis block converges to 1; the basis MPV family is eventually linearly independent; distinct same-dimension basis blocks are not gauge-phase equivalent after identifying equal bond dimensions; and the [CPGSV16, Section 2.3] canonical-form normalization holds,

$$\begin{aligned} \forall (j, q), \quad |\mu_{j,q}| &\leq 1, \\ \exists (j_*, q_*), \quad |\mu_{j_*, q_*}| &= 1. \end{aligned}$$

The MPV expansion takes the raw two-layer form

$$V^{(N)}(P)_\sigma = \sum_{j=1}^g \left(\sum_{q=1}^{r_j} \mu_{j,q}^N \right) V^{(N)}(P_j)_\sigma, \quad (10.29)$$

matching [CPGSV16, Section 2.3] and [CPGSV21, Definition IV.2]. The canonical-form normalization admits every example of the source paper (in particular $C \oplus D$, $C \oplus (-C)$, $C \oplus (1/2)C$, and $C \oplus (-C) \oplus (1/2)C$) and is not derivable from the per-block normality hypotheses alone.

Definition 10.6.1.15 (Single-sector decomposition). Lean ✓ Stmt For a tensor A , its single-sector decomposition has A as its only basis tensor, with one copy of weight 1.

Theorem 10.6.1.16 (Single-sector BNT canonical form). Lean ✓ Stmt Let A have positive bond dimension, be irreducible and left-canonical, and satisfy $O_{AA}(N) \rightarrow 1$. Then its single-sector decomposition is a BNT canonical form.

Proof. [✓ Proof](#) The identity

$$O_{AA}(N) = \|V^{(N)}(A)\|^2 \rightarrow 1 \neq 0. \quad (10.30)$$

shows that $V^{(N)}(A)$ is nonzero for all sufficiently large N . A family consisting of one nonzero vector is linearly independent. All remaining conditions are immediate from the hypotheses and the unit weight. $\square$

Theorem 10.6.1.17 (Normal tensors have single-sector BNT presentations). [Lean](#) [✓ Stmt](#) *Every normal tensor is gauge equivalent to a tensor whose single-sector decomposition is a BNT canonical form. The original tensor and the total tensor of this decomposition generate the same matrix product vectors at every positive length.*

Proof. [✓ Proof](#) Choose the pure trace-preserving Perron gauge. Its tensor is irreducible and left-canonical, and primitivity gives self-overlap converging to one. The preceding theorem supplies the BNT canonical form. If P denotes the total tensor of its single-sector decomposition, then gauge invariance and the sector-decomposition formula give, for every $N > 0$ and word σ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(B)_\sigma = 1^N V^{(N)}(B)_\sigma = V^{(N)}(P)_\sigma. \quad (10.31)$$

$\square$

Lemma 10.6.1.18 (Unit-modulus weight witness from the canonical-form normalization). [Lean](#) [✓ Stmt](#) *Under the BNT canonical form, the [CPGSV16, Section 2.3] normalization provides indices (j_*, q_*) with $|\mu_{j_*, q_*}| = 1$.*

Proof. [✓ Proof](#) This is the unit-modulus existential in the canonical-form normalization. $\square$

Corollary 10.6.1.19 (Basis blocks of a BNT canonical form are normal). [Lean](#) [✓ Stmt](#) *If P is in BNT canonical form, then every basis block P_j is normal in the algebraic sense of Definition 2.4.4.*

Proof. [✓ Proof](#) Each basis block has positive bond dimension, is irreducible and left-canonical, and has normalized self-overlap converging to 1. Theorem 7.10.3 applies to each block. $\square$

Theorem 10.6.1.20 (A BNT canonical form determines an algebraic BNT). [Lean](#) [✓ Stmt](#) *If P is in BNT canonical form, then its basis blocks form an algebraic basis of normal tensors for the total tensor associated with P . Only this implication from canonical form to algebraic BNT is asserted.*

Proof. [✓ Proof](#) The preceding corollary gives normality of every basis block. The sector coefficient formula (10.8) expresses the total MPV as a linear combination of the basis MPVs at each positive length, and the BNT sector data give their eventual linear independence. $\square$

Theorem 10.6.1.21 (Positive blocking preserves BNT canonical form). [Lean](#) [✓ Stmt](#) *Let P be a BNT canonical form and let $p \geq 1$. Then the blocked sector decomposition $P^{[p]}$ is again a BNT canonical form.*

Proof. [✓ Proof](#) For each representative A_j , normalized self-overlap gives peripheral period one, hence a primitive transfer map. Positive blocking therefore preserves irreducibility and left-canonical normalization. Moreover, $O_{A_j^{[p]} A_j^{[p]}}(N) = O_{A_j A_j}(Np) \rightarrow 1$. Let F_N be the linear isomorphism induced by concatenating blocked words, from the blocked configuration space at length N to the original one at length Np . For every representative, $F_N |V^{(N)}(A_j^{[p]})\rangle = |V^{(Np)}(A_j)\rangle$. Hence eventual linear independence is preserved. If two blocked representatives were

gauge-phase equivalent, then for some nonzero ζ and every N , $|V^{(N)}(A_k^{[p]})\rangle = \zeta^N |V^{(N)}(A_j^{[p]})\rangle$. Writing these two vectors as f_k and f_j gives $1 \cdot f_k = \zeta^N \cdot f_j$ in an eventually linearly independent family, and therefore $k = j$. Finally, $|\mu_{j,q}^p| = |\mu_{j,q}|^p \leq 1$, and a unit-modulus weight remains of unit modulus after taking its p -th power. $\square$

Theorem 10.6.1.22 (Common exact quantum-Wielandt length for BNT representatives).

[Lean](#) [✓ Stmt](#) *Let P be a BNT canonical form of total bond dimension D , with representatives A_j of bond dimensions D_j . For every $0 \leq j < g$, the representative A_j is D^4 -block injective: $S_{D^4}(A_j) = M_{D_j}(\mathbb{C})$.*

Proof. [✓ Proof](#) The BNT hypotheses make A_j irreducible and left-canonical, with $O_{A_j A_j}(N) \rightarrow 1$. The preceding theorem therefore makes A_j normal, and quantum Wielandt gives $S_{D_j^4}(A_j) = M_{D_j}(\mathbb{C})$. By Theorem F.2.6, $D_j^4 \leq D^4$. Write $D^4 = D_j^4 + k$ and apply positive-length propagation k times. $\square$

Theorem 10.6.1.23 (CPSV simultaneous block-injectivity bound). [Lean](#) [Lean](#) [✓ Stmt](#) *Let P be a BNT canonical form with total bond dimension D . There is a positive integer $N \leq 3D^5$ such that its representatives have the simultaneous fixed-length span*

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = N\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

This is the block-injectivity estimate in [CPGSV16, lines 340–345].

Proof. [✓ Proof](#) Every representative is block-injective at D^4 , and homogeneous-span propagation gives the same conclusion at $D^4 + 1$ and $3(D^4 + 1)$. If $g = 1$, take $N = D^4$. If $g \geq 2$, apply Lemma 10.3.3 with $L_0 = D^4$ and take

$$N = 3(g-1)(D^4 + 1).$$

The positive bond dimension of every representative gives $g \leq D$. Consequently $N \leq 3D^5$; the same estimate also covers the one-block case. The unit-modulus weight required by the BNT normalization ensures $D > 0$. $\square$

The matching arguments below read off three facts from a BNT canonical form: the raw coefficient identity, the cross-overlap decay between distinct basis blocks, and the combined-family linear independence.

Lemma 10.6.1.24 (Raw two-layer sector coefficient identity). [Lean](#) [✓ Stmt](#) *For every sector decomposition P , length N , and basis index j , $c_N^{(j)} = \sum_{q=1}^{r_j} \mu_{j,q}^N$. This is [CPGSV16, Section 2.3].*

Proof. [✓ Proof](#) The claim follows from the definition of the sector coefficient. $\square$

Lemma 10.6.1.25 (Cross-overlap decay between distinct basis blocks). [Lean](#) [✓ Stmt](#) *If P is a BNT canonical form and $j \neq k$, then $O_{P_j P_k}(N) \rightarrow 0$.*

Proof. [✓ Proof](#) Case on the bond dimensions. If they differ, the dimension-mismatch overlap-decay theorem applies. If they agree, the basis-distinctness clause rules out gauge-phase equivalence, so the equal-dimension overlap-decay theorem applies. $\square$

Lemma 10.6.1.26 (Combined-family eventual linear independence). Lean ✓ Stmt Let P and Q each be a BNT canonical form, and suppose every mixed overlap $O_{P_j Q_k}(N)$ tends to 0. Then the union family

$$\{|V^{(N)}(P_j)\}\}_{j=1}^{g_P} \cup \{|V^{(N)}(Q_k)\}\}_{k=1}^{g_Q}$$

is linearly independent for all sufficiently large N . This is [CPGSV16, Appendix A].

Proof. ✓ Proof Feed the per-family normalized self-overlaps, the within-family cross-overlap decay (Lemma 10.6.1.25), and the inter-family decay hypothesis into the two-family overlap-orthonormality criterion (Lemma F.1.9). □

Example 10.6.1.27 ($C \oplus (1/2)C$, unequal-modulus admissibility). Lean ✓ Stmt For any tensor C , this construction forms a single-sector, two-copy decomposition with basis tensor C and raw weights $(1, 1/2)$. Thus the sector data themselves allow unequal-modulus copies. If C additionally has positive bond dimension, is irreducible and left-canonical, has normalized nonzero self-overlap, and satisfies the BNT independence clause, these data satisfy Definition 10.6.1.14; those hypotheses are not consequences of the construction alone.

10.6.2 Single-copy sector decompositions

A weighted block sum is the one-copy case of the sector decomposition used below. Equal or repeated sectors require the full coefficient formalism.

10.6.3 Prepared-block construction of BNT canonical forms

The construction below separates the after-blocking preparation of suitable blocks from the normalized-weight BNT construction: the prepared blocks are trace-preserving, primitive, irreducible, and injective; normalized weights then quotient gauge-phase-equivalent blocks into sectors to produce the canonical form of Definition 10.6.1.14.

Theorem 10.6.3.1 (Prepared BNT blocks after blocking). Lean ✓ Stmt Let A be an MPS tensor. Then there is a blocking length $p \geq 1$, nonzero weights μ_k , and blocks B_k over the p -blocked alphabet such that:

1. each block has positive bond dimension;
2. each block is trace-preserving, primitive, tensor-irreducible, and one-site injective;
3. $A^{[p]}$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family at every positive length.

No weight-modulus normalization is asserted at this stage.

Proof. ✓ Proof Apply the common primitive irreducible block decomposition to (A, A) , giving a positive blocking length and a nonzero weighted family of trace-preserving primitive irreducible blocks with the positive-length MPV identity. Theorem F.3.2 supplies one common further blocking making every block one-site injective while preserving the other hypotheses; finally identify the iterated blocked alphabet with the direct one. □

With the prepared blocks in hand, normalized weights produce a BNT canonical form by quotienting gauge-phase-equivalent blocks into sectors.

Theorem 10.6.3.2 (BNT canonical form of the phase-class quotient). Lean ✓ Stmt Let $(B_k)_{k=1}^r$ have positive bond dimensions and suppose that every B_k is trace-preserving, primitive, and irreducible. Assume $\mu_k \neq 0$ and $|\mu_k| \leq 1$ for every k , and $|\mu_{k_*}| = 1$ for some k_* . Partition the blocks into matrix-product-vector phase classes, choose one representative from each class, and retain every class member as a copy of its representative with the corresponding phase-adjusted weight. The resulting sector decomposition is in BNT canonical form.

Proof. ✓ Proof The chosen representatives inherit irreducibility and trace preservation, while primitivity gives normalized self-overlap. Distinct phase classes give gauge-phase-distinct representatives. The collapsed phase-class construction supplies the eventual linear independence and retains every original block as one copy. Its phase multipliers have unit modulus, so the adjusted weights preserve the assumed bounds and unit witness. These are precisely the fields of Definition 10.6.1.14. $\square$

Theorem 10.6.3.3 (BNT from normal blocks with normalized weights). Lean ✓ Stmt Let $r \in \mathbb{N}$, and for each $k = 1, \dots, r$ let $D_k \in \mathbb{N}$, $\mu_k \in \mathbb{C}$, and let B_k be an MPS tensor of physical dimension d and bond dimension D_k . Assume:

1. **positive bond dimensions:** $D_k > 0$ for all k ,
2. **TP normalization:** $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$,
3. **primitivity:** $\mathcal{E}_{B_k}$ is primitive for each k ,
4. **irreducibility:** each B_k is irreducible,
5. **nonzero weights:** $\mu_k \neq 0$ for all k ,
6. **bounded weights:** $|\mu_k| \leq 1$ for all k ,
7. **global unit witness:** $|\mu_k| = 1$ for some k .

Then there exists a sector decomposition P such that P and $\bigoplus_{k=1}^r \mu_k B_k$ generate the same MPV family at every positive length, and P satisfies the basis-of-normal-tensors conditions (Definition 10.1.2) together with the multiplicity, span, and weight-bound conditions of Definition 10.6.1.14.

Proof. ✓ Proof The phase-class quotient groups gauge-phase-equivalent blocks into sectors with unit-modulus per-copy scalars. Under the TP, primitive, and irreducible hypotheses the phase-class construction applies, and for distinct representatives $B_j, B_{j'}$ ($j \neq j'$) the cross-overlap decays,

$$\begin{aligned} \lim_{N \rightarrow \infty} O_{B_j B_{j'}}(N) &= 0, \\ \lim_{N \rightarrow \infty} O_{B_j B_j}(N) &= 1, \end{aligned}$$

which gives the eventual linear-independence condition with no injectivity required. The resulting sector decomposition inherits the span and multiplicity conditions, and the bounded-weight and global-unit hypotheses give the weight-norm conditions. $\square$

The BNT construction preserves total bond dimension because CF summands assigned to the same BNT representative have equal bond dimension. Indeed, suppose trace-preserving primitive irreducible NTs X and Y of positive bond dimension have MPV families related by

$|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$ with $\zeta \neq 0$. The normalized self-overlaps give $|O_{XX}(N)| \rightarrow 1$. The phase relation gives the cross-overlap identity

$$\begin{aligned} O_{YX}(N) &= \zeta^N O_{XX}(N), \\ |O_{YX}(N)| &= |\zeta|^N |O_{XX}(N)|. \end{aligned}$$

Since the same normalization forces $|\zeta| = 1$, this gives $|O_{YX}(N)| \rightarrow 1$. If their bond dimensions differed, the rectangular overlap-decay theorem would give $O_{YX}(N) \rightarrow 0$, a contradiction.

Theorem 10.6.3.4 (Gauge realization of prepared phase-equivalent NTs). [Lean](#) [✓ Stmt](#) *Let X and Y be trace-preserving, primitive, irreducible NTs of positive bond dimensions. If their matrix-product-vector families differ by a nonzero scalar power, then their bond dimensions are equal and, after identifying the bond spaces, X and Y are gauge-phase equivalent. Neither positivity of the scalar nor unitarity of the gauge is asserted.*

Proof. [✓ Proof](#) Lemma F.3.4 identifies the bond dimensions. If the resulting same-dimensional blocks were not gauge-phase equivalent, Theorem 10.4.3 would give a positive length at which their matrix product vectors are not proportional. This contradicts the assumed scalar-power relation. The normalized self-overlaps required there follow from Theorem D.5.2. $\square$

Lemma 10.6.3.5 (Gauge realization with the fixed phase scalar). [Lean](#) [✓ Stmt](#) *Let X and Y be trace-preserving, primitive, irreducible NTs of positive bond dimensions. Fix $\zeta \in \mathbb{C} \setminus \{0\}$ and suppose that, for every positive length N and every physical word σ ,*

$$V^{(N)}(Y)_\sigma = \zeta^N V^{(N)}(X)_\sigma. \quad (10.32)$$

Then, after identifying the bond spaces, there exists $G \in \text{GL}_{D_Y}(\mathbb{C})$ such that, for every physical index i ,

$$Y^i = \zeta G X^i G^{-1}. \quad (10.33)$$

Proof. [✓ Proof](#) Theorem 10.6.3.4 gives an invertible G and a nonzero scalar η such that $Y^i = \eta G X^i G^{-1}$. By Theorem 2.3.19, $V^{(N)}(Y)_\sigma = \eta^N V^{(N)}(X)_\sigma$. Theorem D.5.2 gives consecutive positive lengths M and $M+1$ with nonzero coefficients of $V^{(M)}(X)$ and $V^{(M+1)}(X)$. Comparing the two scalar-power identities gives $\zeta^M = \eta^M$ and $\zeta^{M+1} = \eta^{M+1}$. Hence $\zeta = \eta$, which proves (10.33). $\square$

Theorem 10.6.3.6 (Dimension-preserving BNT form of prepared blocks). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 10.6.3.2, there is a BNT canonical-form sector decomposition P such that*

$$\begin{aligned} \mathcal{V}^+(P) &= \mathcal{V}^+ \left(\bigoplus_{k=1}^r \mu_k B_k \right), \\ D_{\text{tot}}(P) &= \sum_{k=1}^r D_k. \end{aligned} \quad (10.34)$$

Proof. [✓ Proof](#) Use the phase-class sector decomposition from Theorem 10.6.3.2. Its flattened copies are in bijection with the original blocks, and phase-equivalent blocks have equal bond dimension by Theorem 10.6.3.4. Summing the copy dimensions over this bijection gives (10.34). $\square$

Theorem 10.6.3.7 (Exact reindexing of heterogeneous weighted block sums). [Lean](#) [Lean](#) [✓ Stmt](#)
 Let $e : \{1, \dots, r\} \simeq \{1, \dots, r'\}$ satisfy $D_k = D'_{e(k)}$. Let Q_k be the coordinate permutation whose rows are indexed by the primed block and whose columns are indexed by the unprimed block. Suppose that, for every block and physical index,

$$\mu_k A_k^i = Q_k^\dagger (\mu'_{e(k)} B_{e(k)}^i) Q_k. \quad (10.35)$$

The induced equivalence E of flattened dependent block coordinates has a permutation matrix Q_E , with rows indexed by the primed coordinates and columns indexed by the unprimed coordinates. For every physical index i ,

$$\bigoplus_{k=1}^r \mu_k A_k^i = Q_E^\dagger \left(\bigoplus_{\ell=1}^{r'} \mu'_\ell B_\ell^i \right) Q_E. \quad (10.36)$$

Proof. [✓ Proof](#) Evaluate both sides on two flattened block coordinates. When their block labels agree, (10.35) gives the matrix entry. When the labels differ, both block-diagonal entries vanish. This proves (10.36). $\square$

Theorem 10.6.3.8 (Exact BNT reconstruction of prepared normal blocks). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 10.6.3.2, there are a BNT canonical-form sector decomposition P , an equivalence E of flattened bond coordinates, and $X \in \text{GL}_{D_{\text{tot}}(P)}(\mathbb{C})$. Write Q_E for the permutation matrix whose rows are indexed by the coordinates of P and whose columns are indexed by the original block coordinates. The first identity below holds, and the second holds for every physical index i :

$$D_{\text{tot}}(P) = \sum_{k=1}^r D_k, \quad (10.37)$$

$$\bigoplus_{k=1}^r \mu_k B_k^i = Q_E^\dagger (X P^i X^{-1}) Q_E. \quad (10.38)$$

Proof. [✓ Proof](#) Partition the prepared blocks into matrix-product-vector phase classes. For the copy q in class j , retain the precise class scalar $\zeta_{j,q}$ chosen in the weight $\zeta_{j,q} \mu_{j,q}$. The normal-tensor phase theorem supplies an invertible $G_{j,q}$ with

$$B_{j,q}^i = \zeta_{j,q} G_{j,q} C_j^i G_{j,q}^{-1}. \quad (10.39)$$

Thus the original weighted block equals the corresponding sector copy after conjugation, since its sector weight is $\zeta_{j,q} \mu_{j,q}$. Assemble the $G_{j,q}$ into a block-diagonal matrix X and apply Theorem 10.6.3.7. This gives (10.38); summing the matched copy dimensions gives (10.37). $\square$

Lemma 10.6.3.9 (Gauge equivalence of direct sums from per-block gauge equivalence). [Lean](#) [✓ Stmt](#) If each pair (A_k, B_k) is gauge equivalent, then the weighted block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.

Proof. [✓ Proof](#) Choose per-block gauge matrices via the hypothesis; then apply Lemma G.4.3. $\square$

Theorem 10.6.3.10 (Positive-length MPVs under coisometric reconstruction). [Lean](#) [✓ Stmt](#) Let A and B be MPS tensors, possibly with different bond dimensions. Suppose that U satisfies $UU^\dagger = \mathbb{1}$ and $A^i = U^\dagger B^i U$ for every physical index i . Then A and B generate the same matrix product vectors at every positive length.

Proof. ✓ **Proof** Multiply the reconstruction identities around a positive-length cycle. Each adjacent pair $UU^\dagger$ cancels, and cyclicity of the trace removes the remaining outer factors. $\square$

Theorem 10.6.3.11 (Active-sector BNT form of normalized CPSV canonical data). Lean

✓ **Stmt** Let A have CPSV canonical-form data satisfying the separate normalization condition (9.25), and assume $A \neq 0$. Put $K_{\text{act}} = \{k : \mu_k \neq 0\}$. Then there is a BNT canonical-form sector decomposition P such that

$$\begin{aligned} \mathcal{V}^+(A) &= \mathcal{V}^+(P), \\ D_{\text{tot}}(P) &= \sum_{k \in K_{\text{act}}} D_k. \end{aligned} \tag{10.40}$$

Thus the dimension is the total dimension of the active blocks, not the ambient bond dimension D . This statement records positive-length MPV equality; the exact coisometric reconstruction is stated next.

Proof. ✓ **Proof** Apply Theorem 10.6.3.12. Its reconstruction writes $A^i = U^\dagger B^i U$, where $UU^\dagger = \mathbb{1}$ and $B^i = XP^i X^{-1}$. Theorem 10.6.3.10 shows that A and B generate the same matrix product vectors at every positive length. Theorem 2.3.18 gives the same conclusion for B and P . Transitivity yields $\mathcal{V}^+(A) = \mathcal{V}^+(P)$, while the dimension identity is already part of the exact reconstruction theorem. $\square$

Theorem 10.6.3.12 (Exact active coisometric BNT reconstruction). Lean ✓ **Stmt** Let A have CPSV canonical-form data satisfying the separate normalization condition (9.25), assume $A \neq 0$, and put $K_{\text{act}} = \{k : \mu_k \neq 0\}$. There are a BNT canonical-form sector decomposition P , a coisometry $U \in \mathbb{C}^{D_{\text{tot}}(P) \times D}$, and $X \in \text{GL}_{D_{\text{tot}}(P)}(\mathbb{C})$. The first two identities below hold, and the third holds for every physical index i :

$$D_{\text{tot}}(P) = \sum_{k \in K_{\text{act}}} D_k, \tag{10.41}$$

$$UU^\dagger = \mathbb{1}, \tag{10.42}$$

$$A^i = U^\dagger (XP^i X^{-1}) U. \tag{10.43}$$

Proof. ✓ **Proof** By Theorem 10.6.1.10, the original tensor has the exact active reconstruction

$$A^i = V^\dagger T_{\text{act}}^i V, \tag{10.44}$$

$$VV^\dagger = \mathbb{1}. \tag{10.45}$$

Put every active normal block into a trace-preserving primitive irreducible gauge Y , and apply Theorem 10.6.3.8. Let Q be the permutation matrix of its flattened coordinate equivalence, so $QQ^\dagger = \mathbb{1}$. Reindexing Y by the same equivalence gives an invertible matrix W on the bond space of P and the identities

$$QY = WQ, \tag{10.46}$$

$$Y^{-1}Q^\dagger = Q^\dagger W^{-1}. \tag{10.47}$$

If X_0 is the prepared-block gauge, set $X = W^{-1}X_0$ and $U = QV$. Substitution in (10.44) gives (10.43), while

$$UU^\dagger = Q(VV^\dagger)Q^\dagger = \mathbb{1}.$$

This proves (10.42). The active enumeration gives (10.41). $\square$

Theorem 10.6.3.13 (Equal fundamental theorem for active CPSV canonical forms). [Lean](#)

[✓ Stmt](#) Let A and B have CPSV canonical-form data with weights $(\mu_k^A)_k$, $(\mu_k^B)_k$ and block dimensions $(D_k^A)_k$, $(D_k^B)_k$. Assume that both data satisfy the separate normalization condition (9.25), that $A \neq 0$ and $B \neq 0$, and that $\mathcal{V}^+(A) = \mathcal{V}^+(B)$. Put $K_A = \{k : \mu_k^A \neq 0\}$ and $K_B = \{k : \mu_k^B \neq 0\}$. Then there are BNT canonical-form sector decompositions P and Q such that

$$\begin{aligned} \mathcal{V}^+(A) &= \mathcal{V}^+(P), \\ \mathcal{V}^+(B) &= \mathcal{V}^+(Q), \\ D_{\text{tot}}(P) &= \sum_{k \in K_A} D_k^A, \\ D_{\text{tot}}(Q) &= \sum_{k \in K_B} D_k^B, \\ D_{\text{tot}}(P) &= D_{\text{tot}}(Q), \\ \sum_{k \in K_A} D_k^A &= \sum_{k \in K_B} D_k^B. \end{aligned} \tag{10.48}$$

After identifying the two total bond spaces by their equality, there is an invertible matrix X such that, for every physical index i ,

$$Q^i = X P^i X^{-1}. \tag{10.49}$$

The conclusion concerns the extracted active tensors P and Q . It does not assert equality of the ambient bond dimensions D_A and D_B , and it does not assert that A and B are gauge equivalent.

Proof. [✓ Proof](#) Apply Theorem 10.6.3.14 and retain its extracted sector decompositions, dimension identities, and conjugating matrix. Forget that the conjugating matrix is unitary. $\square$

Theorem 10.6.3.14 (Unitary refinement for active CPSV canonical forms). [Lean](#) [✓ Stmt](#) Let A and B have CPSV canonical-form data with weights $(\mu_k^A)_k$, $(\mu_k^B)_k$ and block dimensions $(D_k^A)_k$, $(D_k^B)_k$. Assume that both data satisfy the separate normalization condition (9.25), that $A \neq 0$ and $B \neq 0$, and that $\mathcal{V}^+(A) = \mathcal{V}^+(B)$. Put $K_A = \{k : \mu_k^A \neq 0\}$ and $K_B = \{k : \mu_k^B \neq 0\}$. Then there are BNT canonical-form sector decompositions P and Q such that

$$\begin{aligned} \mathcal{V}^+(A) &= \mathcal{V}^+(P), \\ \mathcal{V}^+(B) &= \mathcal{V}^+(Q), \\ D_{\text{tot}}(P) &= \sum_{k \in K_A} D_k^A, \\ D_{\text{tot}}(Q) &= \sum_{k \in K_B} D_k^B, \\ D_{\text{tot}}(P) &= D_{\text{tot}}(Q), \\ \sum_{k \in K_A} D_k^A &= \sum_{k \in K_B} D_k^B. \end{aligned} \tag{10.50}$$

After identifying the two total bond spaces, there is a unitary matrix U such that, for every physical index i ,

$$Q^i = U P^i U^\dagger. \tag{10.51}$$

This is the unitary choice noted in [CPGSV21, lines 1905–1906] and the canonical-form-II refinement of [CPGSV16, Corollary A.6, lines 1197–1199]. Again, there is no claim that $D_A = D_B$, and no gauge equivalence between A and B is asserted.

Proof. ✓ Proof Apply Theorem 10.6.3.11 to A and B . This gives P and Q , their BNT canonical-form properties, the first four identities in (10.50), and

$$\mathcal{V}^+(P) = \mathcal{V}^+(Q).$$

Corollary 11.3.2.3 gives the fifth identity and supplies U . Substituting the two active dimension formulas gives the last equality in (10.50). □

Theorem 10.6.3.15 (Equal-ambient gauge equivalence for CPSV canonical forms). Lean ✓ Stmt
Let A and B have the same ambient bond dimension D . Suppose each tensor has CPSV canonical-form data, both data satisfy the separate weight normalization condition (9.25), neither tensor is zero, and $\mathcal{V}^+(A) = \mathcal{V}^+(B)$. Then A and B are gauge equivalent: there is $G \in \text{GL}_D(\mathbb{C})$ such that, for every physical index i ,

$$B^i = GA^iG^{-1}. \tag{10.52}$$

Zero-weight listed blocks and unused ambient coordinates are allowed. The conclusion requires the common ambient dimension D and makes no claim for canonical forms with different ambient bond dimensions.

The cited sources supply the active canonical-form comparison. The passage back from the active tensors to the ambient tensors is the coisometric lifting of Theorem 9.8.5.

Proof. ✓ Proof Apply Theorem 10.6.3.12 to obtain BNT canonical forms P, Q , coisometries U_A, U_B , and retained-space gauges X_A, X_B with

$$\begin{aligned} A^i &= U_A^\dagger X_A P^i X_A^{-1} U_A, \\ B^i &= U_B^\dagger X_B Q^i X_B^{-1} U_B. \end{aligned} \tag{10.53}$$

Theorem 10.6.3.10, gauge invariance, and the hypothesis give

$$\mathcal{V}^+(P) = \mathcal{V}^+(A) = \mathcal{V}^+(B) = \mathcal{V}^+(Q). \tag{10.54}$$

Theorem 11.3.2.2 therefore identifies the retained dimensions and gives Y such that $Q^i = Y P^i Y^{-1}$. After identifying the retained spaces, the two middle tensors in (10.53) are conjugate by $X_B Y X_A^{-1}$. Theorem 9.8.5 aligns the two coisometries inside the common ambient space and yields (10.52). □

Combining the after-blocking preparation with the normalized-weight construction gives the arbitrary-input statement, conditional on the weight normalization.

Remark 10.6.3.16 (Arbitrary-input reduction). The reduction is conditional on the two weight-normalization hypotheses of Theorem F.3.7. The weight rescaling is carried out in Theorem 10.6.3.17 below, which discharges both hypotheses at the price of one positive scalar per site.

Theorem 10.6.3.17 (Normalized BNT form after blocking for a nonzero MPV family). Lean ✓ Stmt
Let A be an MPS tensor whose matrix-product family has a nonzero coefficient at some positive length. Then there exist a blocking length $p \geq 1$, a positive real m , and a sector

decomposition P over the blocked alphabet in the BNT canonical form of Definition 10.6.1.14, such that for every positive length N ,

$$V^{(N)}(A^{[p]}) = m^N V^{(N)}(P). \quad (10.55)$$

The scalar m^N is the per-site normalization cost of the line-246 weight choice; the hypotheses of Theorem F.3.7 are discharged rather than assumed.

Scope restriction (nonzero MPV family). The source proposition says “for any tensor”, but the zero MPV family cannot supply the global unit-modulus weight imposed immediately before it. The hypothesis above is exactly the exclusion of this degenerate case. The normalization issue and the proportional-MPV comparison are documented in `docs/paper-gaps/cpsv16_cf_normalization_and_proportional_comparison.tex`.

Proof. ✓ Proof Choose a word w with $\text{tr}(A^w) \neq 0$. For every $p \geq 1$, there is a $q \geq 1$ such that $\text{tr}((A^w)^{pq}) \neq 0$. Otherwise Newton–Girard applied to $(A^w)^p$ and the zero matrix would show that $(A^w)^p$, and hence A^w , is nilpotent, contradicting $\text{tr}(A^w) \neq 0$. Repeating w exactly pq times gives a nonzero coefficient of the p -blocked tensor. Thus the prepared family at the blocking length supplied by Theorem 10.6.3.1 is nonempty.

Normalize its weights by their maximal modulus m (Theorem F.3.8). By the weight-scaling identity of Lemma F.3.9,

$$\begin{aligned} V^{(N)}(A^{[p]}) &= V^{(N)}\left(\bigoplus_k \mu_k B_k\right) = V^{(N)}\left(\bigoplus_k m \frac{\mu_k}{m} B_k\right), \\ &= m^N V^{(N)}\left(\bigoplus_k \frac{\mu_k}{m} B_k\right) = m^N V^{(N)}(P). \end{aligned}$$

where the last identity is the prepared supplier of Theorem 10.6.3.3 applied to the normalized weights, which gives the BNT canonical form P . □

10.7 Coefficient comparison and copy-weight recovery

The exact matching of Section 10.8 produces, on each matched pair, a coefficient identity. This section turns that identity into equality of the per-copy weight multisets [CPGSV16, Appendix A]. No analytic non-decay estimate enters: the fixed-length linear independence isolates each sector coefficient exactly.

10.7.1 Matched-sector weight multiset equality

Given a matched pair $(j_0, \tilde{k}_0)$ and a nonzero scalar ζ with $c_N^{(j_0)}(P) = \zeta^N c_N^{(\tilde{k}_0)}(Q)$ past a threshold N_0 , this section records two conclusions: the multiset equality of the per-copy weights from power-sum recovery (Theorem F.2.5), and its upgrade to an explicit per-copy permutation. Together they give the $\mu_{j,q} = \nu_{j,q} e^{i\varphi_j}$ recovery of [CPGSV16, Appendix A]. In the equal-MPV case, Chapter 11 obtains this coefficient identity by expanding both total MPVs in the matched BNT basis and applying eventual linear independence (Lemma 11.1.2). For merely proportional total MPVs, a common length-dependent scalar remains in every matched coefficient identity and the scalar-free comparison need not hold; see Remark 11.2.6.

Lemma 10.7.1.1 (Matched-sector weight multiset equality). Lean ✓ Stmt *Let P, Q be sector decompositions with weights $\mu_{j,q}$ and $\nu_{k,q'}$, and fix indices $j_0, \tilde{k}_0$, and a nonzero scalar $\zeta \in \mathbb{C} \setminus \{0\}$. Suppose that for every $N > N_0$,*

$$c_N^{(j_0)}(P) = \zeta^N \cdot c_N^{(\tilde{k}_0)}(Q). \quad (10.56)$$

Then the per-sector multiplicities agree, $r_{j_0}(P) = r_{\tilde{k}_0}(Q)$, and the rescaled per-copy weight multisets coincide. This is [CPGSV16, Appendix A] read on a single matched sector.

Proof. ✓ Proof Expand both sides of the coefficient identity (10.56) into per-copy power sums $\sum_q \mu_{j_0,q}^N$ and $\sum_{q'} \nu_{\tilde{k}_0,q'}^N$, and distribute ζ^N through the right-hand sum to obtain, for every $N > N_0$,

$$\sum_q \mu_{j_0,q}^N = \sum_{q'} (\zeta \cdot \nu_{\tilde{k}_0,q'})^N.$$

Extend to all positive exponents and apply the unequal-cardinality multiset-recovery lemma (Theorem 10.5.2), using $\zeta \neq 0$ together with the weight non-vanishing assumption to ensure the input weights are nonzero. This yields the multiplicity match and the multiset equality. $\square$

Lemma 10.7.1.2 (Matched-sector weight-permutation extractor). Lean ✓ Stmt *Under the hypotheses of Lemma 10.7.1.1, the per-sector multiplicities agree and there is a permutation τ of the per-copy index set identifying the individual weights up to the gauge phase: for every q , $\nu_{\tilde{k}_0, \tau(q)} = \zeta^{-1} \cdot \mu_{j_0,q}$. This is the per-copy realisation of [CPGSV16, Appendix A]: the multiset equality from Lemma 10.7.1.1 is upgraded to a concrete indexing between the matched P - and Q -copies.*

Proof. ✓ Proof Invoke Lemma 10.7.1.1 to obtain the cardinality match and the multiset equality. From this multiset-map equality extract a permutation matching the elements pointwise (proved by induction on the cardinality). Compose with the cardinality identification and cancel ζ via $\zeta \neq 0$ to convert the gauge factor on the right of the pointwise identity to ζ^{-1} on the left. $\square$

10.8 Strong existential matching by exact linear independence

The exact matching reads [CPGSV16, Appendix A] as a statement at fixed length: equal MPV families force each basis sector of one form to pair with a basis sector of the other. The non-vanishing of the sector coefficients isolates the matched sector exactly, with no limit of a single-sector projection.

Theorem 10.8.1 (Exact single-sector matching). Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family (at all positive lengths). For every sector j of P there exist a sector k of Q with matched bond dimension $\dim_P(j) = \dim_Q(k)$, a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap. No per-sector unit-modulus hypothesis is required.*

Proof. ✓ Proof Equality at every positive length gives eventual nonzero proportionality with scalar 1. Apply Theorem F.2.1.1 to the chosen sector. $\square$

Theorem 10.8.2 (Strong existential matching, full-basis form). [Lean](#) [✓ Stmt](#) *Let P, Q be two BNT canonical forms with the same MPV family at every positive length. Then for every sector k of Q there exist a sector j of P with matched bond dimension $\dim_P(j) = \dim_Q(k)$, a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap.*

Proof. [✓ Proof](#) Convert equality to eventual nonzero proportionality with scalar 1 and apply Theorem [F.2.1.2](#). $\square$

10.8.1 Bijective matching by symmetry

Applying the exact matching once more with P and Q swapped gives the reverse direction; the basis-distinctness clause forces injectivity both ways, so $g_a = g_b$ and the source relabelling $B_k = e^{i\phi_k} X_k A_{j_k} X_k^{-1}$ holds on the full basis [[CPGSV16](#), Appendix A].

Theorem 10.8.1.1 (Bijective matching by symmetry). [Lean](#) [✓ Stmt](#) *Let P, Q be two BNT canonical forms with the same MPV family at every positive length. Then there is a bijection $\beta : J(Q) \simeq J(P)$ such that for every $k \in J(Q)$ the bond dimensions match, the basis tensors $P_{\beta(k)}$ and Q_k are gauge-phase equivalent after identifying the matched bond dimensions, and the basis cross-overlap does not tend to zero.*

Proof. [✓ Proof](#) Apply the full-basis matcher in both directions, using symmetry of the same-MPV relation. Lemma [F.2.1.3](#) converts these two families of matches into the required bijection. $\square$

Chapter 11

Proof of the Fundamental Theorem

For BNT canonical forms P and Q , write the weighted direct sums over normal-tensor sectors as $P^i = \bigoplus_{j,q} \mu_{j,q} P_j^i$ and $Q^i = \bigoplus_{k,q} \nu_{k,q} Q_k^i$. The proportional theorem matches the normal sectors by a bijection β , unit phases ζ_k , and invertible matrices X_k satisfying

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}.$$

It does not determine the copy weights. In the equal-MPV case, eventual linear independence gives

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q).$$

Finite power-sum comparison then gives $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. Thus the sector phases do not become equal: each phase cancels against the inverse phase in its matched copy weight. The remaining block conjugacies assemble into $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$, and a sector-coordinate permutation yields the literal global gauge.

Throughout, “eventually” means “for all sufficiently large N ”. At a fixed length in that range, BNT linear independence detects exact coefficient equalities even when a weight of modulus less than 1 makes its powers decay.

11.1 Coefficient identities and copy weights

Let P and Q be BNT canonical forms. Their basis sectors are normalized, left-canonical, and aperiodic, and their raw copy weights satisfy the global normalization of Definition 10.6.1.14. This is the normalized surface used for the proof. The literal CPSV canonical form is brought to this surface by the reduction and normalization of Chapters 9 and 10.

The proportional statement below allows nonzero projective proportionality for all sufficiently large lengths. The source assumption of proportionality at every positive length is a special case. No copy-weight comparison is needed for its sector-level conclusion.

Theorem 11.1.1 (Proportional sector matching with unit phases). [Lean](#) [✓ Stmt](#) *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities $\dim P_{\beta(k)} = \dim Q_k$,*

scalars $\zeta_k \in \mathbb{C}$, and block gauges X_k such that $|\zeta_k| = 1$ and, for every sector k and physical index i ,

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}. \quad (11.1)$$

Diagrammatically, the gauge relation (11.1) reads

with every open virtual end a matrix index. Equivalently, for every positive length N and word σ ,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma.$$

This is [CPGSV16, Appendix MPV theorem statement, lines 1167–1170, and proof line 1182].

Proof. ✓ Proof Theorem G.1.1 gives a bijection $\beta : J(Q) \simeq J(P)$ together with equal bond dimensions, gauge-phase equivalence, and

$$O_{P_{\beta(k)}Q_k}(N) \not\rightarrow 0$$

for every matched pair. Its proof defines $T = \{j \in J(P) : O_{P_jQ_k}(N) \rightarrow 0 \text{ for every } k\}$, substitutes exact Q -sector relations for $j \notin T$, applies linear independence to $\{P_j : j \in T\} \cup \{Q_k : k \in J(Q)\}$, and reverses direction; basis separation and finiteness turn the resulting matches into β .

Unpack each matched gauge-phase equivalence to choose an invertible matrix X_k and a nonzero scalar ζ_k . This gives $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ and, by taking traces around a positive-length word,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma.$$

Lemma G.1.2 applies to this MPV identity. Equivalently, the normalized self-overlaps satisfy

$$O_{Q_kQ_k}(N) = |\zeta_k|^{2N} O_{P_{\beta(k)}P_{\beta(k)}}(N),$$

and both outer sequences tend to 1, which forces $|\zeta_k| = 1$. □

Lemma 11.1.2 (Full-basis coefficient identity from matched phases). Lean ✓ Stmt Let P be a BNT canonical form, let Q be a sector decomposition, and assume P_{tot} and Q_{tot} have the same MPV family at every positive length. Suppose that there is a bijection $\beta : J(Q) \simeq J(P)$ and scalars $\zeta_k \in \mathbb{C}$ such that, for every $k \in J(Q)$, every positive length N , and every spin configuration σ of length N ,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma. \quad (11.2)$$

Then for every $k \in J(Q)$ there is N_0 such that, for all $N > N_0$, $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$. This is the coefficient-comparison step of [CPGSV16, Appendix MPV proof, lines 1187–1188] in full-basis form.

Proof. ✓ Proof For every N , expand the equality of total MPVs as

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = \sum_{k \in J(Q)} c_N^{(k)}(Q) |V^{(N)}(Q_k)\rangle.$$

The phase relation (11.2) gives $|V^{(N)}(Q_k)\rangle = \zeta_k^N |V^{(N)}(P_{\beta(k)})\rangle$, so reindexing the Q -sum by β onto the P -basis yields

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = \sum_{j \in J(P)} \zeta_{\beta^{-1}(j)}^N c_N^{(\beta^{-1}(j))}(Q) |V^{(N)}(P_j)\rangle.$$

For all sufficiently large N the P -basis MPV family $\{|V^{(N)}(P_j)\rangle\}_{j \in J(P)}$ is linearly independent, by the BNT eventual linear independence of P . Lemma F.1.11 then extracts the coefficient equalities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ from this eventual linear independence, for all sufficiently large N . $\square$

Definition 11.1.3 (Copy-weight matching for matched BNT sectors). Lean ✓ Stmt Let the sectors of two decompositions P and Q be matched by $\beta : J(Q) \simeq J(P)$, with sector phases ζ_k . A copy-weight matching consists of copy permutations τ_k such that, for all sectors k and copies q , $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. This is the copy-level conclusion of [CPGSV16, Appendix MPV proof, line 1188].

Definition 11.1.4 (Copy-weight matching from matched-sector coefficient identities). Lean ✓ Stmt The construction takes as input a sector matching $\beta : J(Q) \simeq J(P)$, nonzero sector phases ζ_k , and the eventual matched-sector coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q). \quad (11.3)$$

It returns the copy-weight matching obtained by choosing, in every sector, a copy permutation τ_k satisfying $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. This is the sectorwise construction in [CPGSV16, Appendix MPV proof, line 1188].

Proof. ✓ Proof Eventually the coefficient identity (11.3) in sector k reads

$$\sum_r \mu_{\beta(k), r}^N = \zeta_k^N \sum_q \nu_{k,q}^N = \sum_q (\zeta_k \nu_{k,q})^N.$$

Newton–Girard finite power-sum comparison (Lemma 10.7.1.2) applies sector by sector, producing a copy permutation τ_k with $\mu_{\beta(k), \tau_k(q)} = \zeta_k \nu_{k,q}$; since $\zeta_k \neq 0$ this is $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. $\square$

Theorem 11.1.5 (Equal-case sector and copy matching with differing bond dimensions). Lean ✓ Stmt Let P and Q be BNT canonical forms, possibly with different sector counts and different bond dimensions before matching. If their total tensors generate the same MPV family at every positive length, then there is a bijection $\beta : J(Q) \simeq J(P)$ such that each matched basis pair has equal bond dimension and is gauge-phase equivalent. The matched copy multiplicities agree, and there are unit phases ζ_k and copy bijections τ_k satisfying

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1},$$

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}.$$

Only the global weight normalization in Definition 10.6.1.14 is assumed; no unit-modulus copy is required in each sector. This is the sector-data form of [CPGSV16, Corollary 2.11 and Appendix A, lines 1182–1188].

Proof. ✓ Proof First, Theorem 10.8.1.1 gives the sector bijection, the bond-dimension equalities, and the gauge-phase relations. Choose their phases ζ_k and use the normalized self-overlaps to obtain $|\zeta_k| = 1$. Second, Lemma 11.1.2 expands the equality of the two total MPVs in the matched P -basis. Eventual linear independence gives, for every k and all sufficiently large N ,

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q).$$

Third, expand both coefficients as finite power sums. Their eventual equality determines the two multisets of nonzero copy weights, so the multiplicities agree and a copy bijection τ_k realizes the identity $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. $\square$

11.2 The equal-MPV global gauge

For equal MPV families, the sector phases occur twice: in the matched tensor relation and with the inverse power in the matched copy weights. They therefore cancel before the block gauges are assembled.

Definition 11.2.1 (Flattened block-diagonal gauge). Lean Lean ✓ Stmt Given per-block gauges $X_k \in \text{GL}_{D_k}(\mathbb{C})$, form the block-diagonal element $\bigoplus_k X_k$ and then reindex the dependent block-coordinate space with the standard bond index $\{0, \dots, \sum_k D_k - 1\}$ of the assembled tensor. The result is an element of $\text{GL}_{\sum_k D_k}(\mathbb{C})$.

Theorem 11.2.2 (Global gauge from matched sectors and copy weights). Lean ✓ Stmt Suppose that the sectors of two sector decompositions P and Q are matched by a bijection $\beta : J(Q) \simeq J(P)$, with bond-dimension equalities, copy permutations τ_k , nonzero phases ζ_k , and block gauges X_k . If, for every sector k , copy q , and physical index i ,

$$\begin{aligned} Q_k^i &= \zeta_k X_k P_{\beta(k)}^i X_k^{-1}, \\ \nu_{k,q} &= \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}, \end{aligned} \tag{11.4}$$

then, in the flattened Q -coordinates,

$$\begin{aligned} Q_{\text{flat}}^i &= X P_{\text{matched}}^i X^{-1}, \\ X &= \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k). \end{aligned}$$

This is the direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192], separated from the coefficient-comparison step that supplies the weight identities.

Proof. ✓ Proof For each matched copy,

$$\nu_{k,q} Q_k^i = (\zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}) (\zeta_k X_k P_{\beta(k)}^i X_k^{-1}) = \mu_{\beta(k), \tau_k(q)} X_k P_{\beta(k)}^i X_k^{-1}.$$

Thus every flattened weighted block is conjugated by its sector gauge X_k . Lemma G.4.3 combines these block conjugacies into the direct-sum gauge $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. $\square$

Lemma 11.2.3 (Equal-MPV matched copy-weight witnesses). Lean ✓ Stmt Let P and Q be BNT canonical forms whose total tensors have the same MPV family at every positive length. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities, unit-modulus phases ζ_k , and block gauges X_k such that $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. For the same phases there is a copy-weight matching: after a permutation of the copies inside each matched sector, $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$.

Proof. [✓ Proof](#) The full-basis matching theorem gives the bijection and block gauges. The self-overlap comparison

$$\langle V_N(Q_k), V_N(Q_k) \rangle = |\zeta_k|^{2N} \langle V_N(P_{\beta(k)}), V_N(P_{\beta(k)}) \rangle$$

together with the BNT normalization limits on both sides forces $|\zeta_k| = 1$. The matched MPV phase identities also give the eventual coefficient identities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$. Definition 11.1.4 applies the finite power-sum comparison in each sector and gives the copy-weight identities. $\square$

Theorem 11.2.4 (Full-basis global gauge in matched coordinates). [Lean](#) [✓ Stmt](#) *Let P and Q be BNT canonical forms whose total tensors have the same MPV family at every positive length. Then there exist:*

- a bijection $\beta : J(Q) \simeq J(P)$,
- bond-dimension equalities $\dim P_{\beta(k)} = \dim Q_k$,
- copy-number equalities $r_{\beta(k)}(P) = r_k(Q)$ and copy permutations τ_k ,
- unit-modulus phases ζ_k with $|\zeta_k| = 1$ and block gauges X_k ,

such that, for every sector k , copy q , and physical index i ,

$$\begin{aligned} Q_k^i &= \zeta_k X_k P_{\beta(k)}^i X_k^{-1}, \\ \nu_{k,q} &= \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}. \end{aligned}$$

Moreover, if $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$ is the flattened block-diagonal gauge, then, for every physical index i , the flattened Q -coordinates satisfy

$$Q_{\text{flat}}^i = X P_{\text{matched}}^i X^{-1}.$$

This is the explicit direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192].

Proof. [✓ Proof](#) Lemma 11.2.3 gives β , the bond-dimension identifications, the phases ζ_k , the block gauges X_k , and copy permutations τ_k satisfying $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. Theorem G.4.4 combines the block conjugacies into the flattened direct-sum conjugation by $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. $\square$

Theorem 11.2.5 (Literal equal-MPV gauge after sector-coordinate permutation). [Lean](#) [✓ Stmt](#) *Let P and Q be BNT canonical forms whose total tensors have the same MPV family at every positive length. Then the total bond dimensions agree; write their common value as D . There is*

$$Y \in \text{GL}_D(\mathbb{C}) \tag{11.5}$$

such that, for every physical index i ,

$$Q_{\text{tot}}^i = Y P_{\text{tot}}^i Y^{-1},$$

where P_{tot}^i and Q_{tot}^i are both regarded as matrices in $M_D(\mathbb{C})$ using the total bond-dimension equality. This is the BNT form of [CPGSV16, Corollary 2.11 and Appendix A, lines 1189–1192].

Proof. ✓ Proof Apply Theorem 11.2.4 to obtain the flattened gauge after the sectors of P have been reindexed by the bijection between basis blocks of P and Q . The sector-coordinate permutation ρ , induced by this bijection, the copy permutations, and the bond-dimension equalities, identifies the reindexed P -side direct sum with the total tensor P_{tot} , transported along the total bond-dimension equality. Composing the permutation gauge for ρ with the sector-permuted gauge gives the single matrix Y of (11.5). $\square$

Remark 11.2.6 (Proportional sector matching does not determine copy weights). The proportional theorem determines the normal sectors, but it does not determine the copy weights or imply conjugacy of the total tensors. Let C be a normalized normal tensor and consider the one-sector BNT forms

$$P = C \oplus \frac{1}{2}C, \quad Q = C \oplus \frac{1}{3}C.$$

Their positive-length MPV families are

$$V^{(N)}(P) = (1 + 2^{-N})V^{(N)}(C), \quad V^{(N)}(Q) = (1 + 3^{-N})V^{(N)}(C).$$

They are nonzero and proportional with

$$c_N = \frac{1 + 2^{-N}}{1 + 3^{-N}},$$

but $(c_N)_N$ is not a geometric sequence ξ^N . For a one-dimensional choice of C , the total matrices have eigenvalue multisets $\{1, \frac{1}{2}\}$ and $\{1, \frac{1}{3}\}$, so they are not conjugate, even after multiplication by a common phase.

Thus the scalar-free coefficient identity used in copy-weight recovery is an additional hypothesis in the proportional setting, not a consequence awaiting an analytic estimate. CPSV16 Theorem 2.10 asserts the sector matching proved in Theorem 11.3.2. The copy-weight recovery and global block gauge belong to the equal-MPV Corollary 2.11, where the proportionality scalar is identically 1.

11.3 The proportional and equal cases

Theorem 11.3.1 (Printed proportional fundamental theorem). Not ready *Let A and B be tensors in canonical form whose matrix product vectors are proportional at every positive length. According to [CPGSV16, Theorem 2.10], arbitrary chosen bases of normal tensors for A and B have the same cardinality and are related by a bijective family of gauge-phase equivalences.*

The statement is false if inactive sectors are admitted. Adjoining a normal tensor with coefficient zero changes the basis cardinality without changing the tensor or any positive-length matrix product vector. For active nonzero sector data, the asserted matching follows from Theorem 11.3.2.

Theorem 11.3.2 (Multi-block fundamental theorem for normal tensors). Lean ✓ Stmt *Let P and Q be BNT canonical forms (Definition 10.6.1.14), with bases of normal tensors $(P_j)_{j=1}^{g_a}$ and $(Q_k)_{k=1}^{g_b}$. Every listed sector has a positive number of copies, and every copy weight is nonzero. Thus no normal tensor with identically zero coefficient is included in either basis. If the total tensors $P_{\text{tot}}, Q_{\text{tot}}$ generate eventually nonzero proportional MPV families — for all sufficiently large N there is a nonzero scalar c_N with $V^{(N)}(P_{\text{tot}}) = c_N V^{(N)}(Q_{\text{tot}})$ — then $g_a = g_b$. After*

relabelling the basis of normal tensors of P by a bijection $\beta : J(Q) \simeq J(P)$, there are scalars $\zeta_k \in \mathbb{C}$ with $|\zeta_k| = 1$ and invertible matrices Y_k such that, for every k and every physical index i ,

$$Q_k^i = \zeta_k Y_k P_{\beta(k)}^i Y_k^{-1}.$$

Thus the paired normal tensors have the same bond dimension ($\dim P_{\beta(k)} = \dim Q_k$) and the same gauge-phase class.

Proof. ✓ Proof Apply Theorem 11.1.1. Its partition-and-substitution argument gives a bijection $\beta : J(Q) \simeq J(P)$, equal bond dimensions, invertible block gauges X_k , and phases ζ_k of unit modulus. Taking $Y_k = X_k$ gives exactly the asserted relations. $\square$

11.3.1 Unitary gauges on normalized sectors

Lemma 11.3.1.1 (Phase normalization for an irreducible gauge). Lean ✓ Stmt Let A and B be left-canonical tensors of common positive bond dimension, with B irreducible. If X is invertible, $\zeta \neq 0$, and $B^i = \zeta X A^i X^{-1}$ for every i , then $|\zeta| = 1$.

Proof. ✓ Proof Let ρ be a nonzero positive fixed point of $\mathcal{E}_A$ and set $\sigma = X \rho X^\dagger$. Then σ is nonzero and positive, and $\mathcal{E}_B(\sigma) = |\zeta|^2 \sigma$. The map $\mathcal{E}_B$ also has a nonzero positive fixed point. Its irreducibility therefore forces $|\zeta|^2 = 1$. $\square$

Lemma 11.3.1.2 (Unitary replacement with the same phase). Lean ✓ Stmt Let A and B be left-canonical irreducible tensors of common positive bond dimension satisfying $B^i = \zeta X A^i X^{-1}$ for an invertible X and a nonzero scalar ζ . There is a unitary matrix U , with the same scalar ζ , such that $|\zeta| = 1$ and $B^i = \zeta U A^i U^\dagger$ for every i .

Proof. ✓ Proof Lemma 11.3.1.1 gives $|\zeta| = 1$. Substitution into the left-canonical identities gives

$$\sum_i A^{i\dagger} (X^\dagger X) A^i = X^\dagger X.$$

Thus $X^\dagger X$ is a positive-semidefinite fixed point of the adjoint transfer map of A . Irreducibility gives $X^\dagger X = c \mathbb{1}$ for some $c > 0$, so $X^{-1} = c^{-1} X^\dagger$ and $U = c^{-1/2} X$ is unitary. Moreover,

$$U A^i U^\dagger = c^{-1} X A^i X^\dagger = X A^i X^{-1}.$$

$\square$

Corollary 11.3.1.3 (Printed Canonical Form II unitary refinement). Not ready According to [CPGSV16, Corollary A.6], the gauge matrices in Theorem 2.10 and Corollary 2.11 may be chosen unitary when both tensors are in Canonical Form II.

Canonical Form II does not exclude inactive zero-weight blocks, so the literal equal-MPV conclusion can still fail at the bond-dimension and global-unitary level. After restricting to active nonzero sectors, the proportional sector gauges and the equal global gauge may be chosen unitary, as stated in the two corollaries below.

Corollary 11.3.1.4 (Canonical Form II unitary refinement of proportional matching). Lean ✓ Stmt Under the hypotheses of Theorem 11.3.2, all sectors are active and all copy weights are nonzero. There exist a bijection $\beta : J(Q) \simeq J(P)$, unitary matrices $U_k \in \mathbf{U}(\dim Q_k)$, and phases $\zeta_k \in \mathbb{C}$ with $|\zeta_k| = 1$, such that $\dim P_{\beta(k)} = \dim Q_k$ and, for every k and every physical index i ,

$$Q_k^i = \zeta_k U_k P_{\beta(k)}^i U_k^\dagger.$$

This is the proportional part of [CPGSV16, Corollary A.6, lines 1197–1199].

Proof. ✓ Proof The proportional sector-matching theorem gives a bijection β , invertible matrices X_k , and unit-modulus phases ζ_k with $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. The matched tensors are left-canonical and irreducible. Hence Lemma G.8.1 gives unitary matrices U_k satisfying $X_k P_{\beta(k)}^i X_k^{-1} = U_k P_{\beta(k)}^i U_k^\dagger$, and therefore $Q_k^i = \zeta_k U_k P_{\beta(k)}^i U_k^\dagger$. $\square$

11.3.2 The equal case

Corollary 11.3.2.1 (Printed equal-MPV fundamental theorem). Not ready According to [CPGSV16, Corollary 2.11], two canonical-form tensors generating the same matrix product vectors at every positive length have equal bond dimensions and are conjugate by one invertible matrix.

An inactive zero-weight block may be added to either tensor without changing any positive-length matrix product vector. It changes the ambient bond dimension and therefore invalidates the literal dimension and global conjugacy conclusion. For active nonzero sector data, the conclusion holds by Theorem 11.3.2.2.

Theorem 11.3.2.2 (Equal BNT canonical forms are globally conjugate). Lean ✓ Stmt Let P and Q be BNT canonical forms (Definition 10.6.1.14) whose total tensors $P_{\text{tot}}, Q_{\text{tot}}$ generate the same MPV family for every positive system size N . Every listed sector has a positive number of copies and every copy weight is nonzero, so inactive zero-weight summands are excluded. Then their total bond dimensions coincide; writing the common value as D , there is one invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that, for every physical index i ,

$$Q_{\text{tot}}^i = X P_{\text{tot}}^i X^{-1}.$$

Proof. ✓ Proof This is Theorem 11.2.5. The BNT comparison of Chapter 10 provides the basis relabelling, the per-block gauge-phase equivalences, and the copy-weight multiset matching, which combine into the block-diagonal global gauge. The exact linear-independence matching needs no per-sector unit-modulus hypothesis, and the BNT canonical form needs no one-site injectivity. $\square$

Corollary 11.3.2.3 (Canonical Form II unitary refinement in the equal case). Lean ✓ Stmt Let P and Q be BNT canonical forms whose total tensors generate the same MPV family at every positive length. Assume every listed sector has a positive number of copies and every copy weight is nonzero, as in Definition 10.6.1.14; the literal zero-weight CFII statement is not included. Their total bond dimensions agree; writing the common value as D , there is a unitary matrix $U \in \text{U}(D)$ such that, for every physical index i ,

$$Q_{\text{tot}}^i = U P_{\text{tot}}^i U^\dagger.$$

Proof. ✓ Proof For each matched sector, retain the phase in both identities

$$\begin{aligned} Q_k^i &= \zeta_k U_k P_{\beta(k)}^i U_k^\dagger, \\ \nu_{k,q} &= \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}. \end{aligned}$$

For every matched copy (k, q) ,

$$\nu_{k,q} \zeta_k = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)} \zeta_k = \mu_{\beta(k), \tau_k(q)}.$$

Hence $Q_{\text{flat}}^i = X P_{\text{match}}^i X^{-1}$. Repeating U_k on its matched copies gives

$$\begin{aligned} X &= \bigoplus_{k,q} U_k, \\ X^\dagger X &= \mathbb{1}. \end{aligned}$$

If Π is the permutation matrix restoring the original sector and copy order, then $\Pi^\dagger \Pi = \mathbb{1}$. Hence

$$\begin{aligned} U &= X\Pi, \\ U^\dagger U &= \mathbb{1}, \end{aligned}$$

and the permutation of matched coordinates gives

$$Q_{\text{tot}}^i = U P_{\text{tot}}^i U^\dagger.$$

□

Chapter 12

Symmetries, Physical String Order, and Virtual-Boundary Nondecay

This chapter collects the symmetry-theoretic consequences of the MPS Fundamental Theorem [PGVWC07], following the symmetry analysis of [PGWS⁺08]. For injective tensors, equality of periodic MPV families under a physical index action determines virtual gauges, and those gauges define projective representations on the bond space. We keep this virtual covariance distinct from symmetry of an infinite physical state, and distinguish physical endpoint string order from virtual-boundary nondecay.

12.1 Virtual Local Covariance and Projective Bond Actions

With a full group of on-site symmetries (rather than just a single unitary), the resulting gauges determine a projective representation on the bond space.

Definition 12.1.1 (Twisted tensor). [Lean](#) [✓ Stmt](#) Given a group representation $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$ on the physical index, the *g-twisted tensor* is defined, for each $i \in \{0, \dots, d-1\}$, by

$$\tilde{A}_g^i := \sum_{j=0}^{d-1} U(g)_{ij} A^j. \quad (12.1)$$

Definition 12.1.2 (MPV-family invariance under physical twists). [Lean](#) [✓ Stmt](#) The predicate called on-site symmetry requires $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$ for every $g \in G$, where $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$ acts on the physical index. This is equality of periodic MPV families and is the hypothesis used to construct virtual local gauges. It is not, by itself, the source notion that every reduced density operator of an infinite physical state is invariant under $U(g)^{\otimes N}$.

Lemma 12.1.3 (Functoriality of twisting). [Lean](#) [✓ Stmt](#) The twisted tensor satisfies the composition law

$$\tilde{A}_{gh} = \widetilde{(\tilde{A}_h)_g}, \quad (12.2)$$

i.e. twisting by gh is the same as first twisting by h and then by g .

Proof. ✓ Proof Expand both sides of (12.2) using (12.1) and $U(gh) = U(g)U(h)$, then swap the order of summation. □

Lemma 12.1.4 (Identity twist). Lean ✓ Stmt *Twisting by the identity group element is trivial: $\tilde{A}_1 = A$.*

Proof. ✓ Proof Since $U(1) = \mathbb{1}_d$, each component in (12.1) reduces to $\sum_j \delta_{ij} A^j = A^i$. □

Definition 12.1.5 (Blocked on-site action). Lean ✓ Stmt For a block of L physical sites, the induced action is the Kronecker-power representation

$$U^{[L]}(g) := U(g)^{\otimes L} \in \text{GL}_{d^L}(\mathbb{C}).$$

Lemma 12.1.6 (Twisting commutes with blocking). Lean ✓ Stmt For every $g \in G$ and $L \geq 0$,

$$\left(\widetilde{A^{[L]}}\right)_g^{U^{[L]}} = (\tilde{A}_g)^{[L]}. \quad (12.3)$$

Proof. ✓ Proof Decode a blocked physical index as a word of length L and use the word expansion from Lemma H.1.1.3. Reindexing the sum over words by the blocked index gives exactly the Kronecker-product matrix coefficients on the left of (12.3). □

Theorem 12.1.7 (MPV-family symmetry is preserved by blocking). Lean ✓ Stmt If A has MPV-family invariance under U , then $A^{[L]}$ has MPV-family invariance under $U^{[L]}$ for every L .

Proof. ✓ Proof Equation (12.3) identifies the twisted blocked tensor with the blocking of the twisted tensor. Equality of MPV families is preserved by blocking, so the defining equality for A yields the defining equality for $A^{[L]}$. □

Lemma 12.1.8 (Symmetric twists are gauge equivalent). Lean ✓ Stmt If A is injective and on-site symmetric under U , then for every $g \in G$ the twisted tensor $\tilde{A}_g$ is gauge equivalent to A .

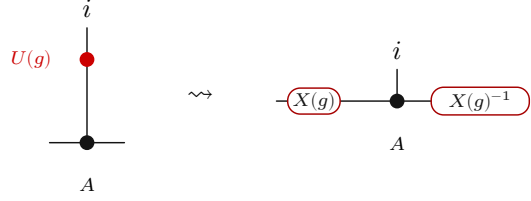
Proof. ✓ Proof On-site symmetry gives $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$, and the single-block Fundamental Theorem turns this equality of MPV families into gauge equivalence. □

Remark 12.1.9 (Fundamental-Theorem route for injective symmetry). The injective virtual-representation theorem is not proved from virtual-boundary nondecay. The on-site symmetry condition first identifies A and $\tilde{A}_g$ as the same MPV family; the single-block Fundamental Theorem gives the gauge in Lemma 12.1.8. Gauge uniqueness and the twist composition law then give Theorem 12.1.16. The virtual-boundary nondecay results later in this chapter use this virtual representation as an input.

Theorem 12.1.10 (Virtual symmetry equation). Lean ✓ Stmt If A is injective and on-site symmetric under U , then for every $g \in G$ there exist an invertible matrix $X(g) \in \text{GL}_D(\mathbb{C})$ and a nonzero scalar $\varphi(g) \in \mathbb{C}^\times$ (in fact $\varphi = 1$ in the single-block case) such that, for every physical index i ,

$$\tilde{A}_g^i = \varphi(g) X(g) A^i X(g)^{-1}. \quad (12.4)$$

Proof. ✓ Proof The physical action is transferred to the virtual level by the same local move that replaces the twisted tensor by a gauge-conjugated one:



Apply Lemma 12.1.8 to obtain $X(g)$, then set $\varphi(g) = 1$ in (12.4). $\square$

Lemma 12.1.11 (Gauge uniqueness up to scalar). [Lean](#) [✓ Stmt](#) *Let A be an injective tensor. If $X, Y \in \text{GL}_D(\mathbb{C})$ both satisfy $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i , then there exists a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

Proof. [✓ Proof](#) The matrix $Z = Y^{-1}X$ commutes with every A^i . Since A is injective, $\{A^i\}$ spans $M_D(\mathbb{C})$, so Z lies in the centre of $M_D(\mathbb{C})$. By Lemma 2.1.5, Z is a scalar matrix. Since Z is invertible, that scalar is nonzero, and $Y = u \cdot X$. $\square$

Theorem 12.1.12 (Gauge uniqueness from equality of MPV families). [Lean](#) [✓ Stmt](#) *Let A be an injective tensor and assume that A and B generate the same MPV family. If $X, Y \in \text{GL}_D(\mathbb{C})$ both satisfy $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i , then there is a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.*

Proof. [✓ Proof](#) This is exactly Lemma 12.1.11; the extra MPV hypothesis specifies the setting in which the two gauges are usually produced. $\square$

Definition 12.1.13 (Scalar 2-cochain). [Lean](#) [✓ Stmt](#) A scalar 2-cochain on a group G is a function $\omega : G \times G \rightarrow \mathbb{C}^\times$.

Definition 12.1.14 (Projective representation). [Lean](#) [✓ Stmt](#) Given a scalar 2-cochain ω , a projective representation of G on $\mathbb{C}^D$ is a map $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,

$$\rho(g)\rho(h) = \omega(g, h)\rho(gh). \quad (12.5)$$

Lemma 12.1.15 (Associativity forces the cocycle condition). [Lean](#) [✓ Stmt](#) *If ρ is a projective representation with factor system ω and $D > 0$, then ω satisfies the 2-cocycle identity*

$$\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k). \quad (12.6)$$

Proof. [✓ Proof](#) Expand $\rho(g)(\rho(h)\rho(k))$ and $(\rho(g)\rho(h))\rho(k)$ using the projective multiplication law (12.5) and equate the results via matrix associativity. The hypothesis $D > 0$ allows cancellation of the invertible matrix factor, giving (12.6). $\square$

Theorem 12.1.16 (Virtual representation theorem for injective MPS). [Lean](#) [✓ Stmt](#) *Let A be an injective MPS tensor that is on-site symmetric under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$. Then there exist:*

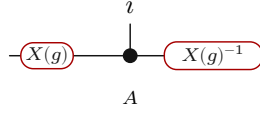
1. a scalar 2-cochain $\omega : G \times G \rightarrow \mathbb{C}^\times$, and
2. a map $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,

$$\rho(g)\rho(h) = \omega(g, h)\rho(gh).$$

such that, defining the gauge matrices $X(g) := \rho(g^{-1})$, one has, for every $g \in G$ and $i \in \{0, \dots, d-1\}$,

$$\tilde{A}_g^i = X(g) A^i X(g)^{-1}. \quad (12.7)$$

In tensor-network notation, for each fixed g this is the local gauge relation



with the black node denoting the tensor named in the surrounding formula and the two red side nodes acting on the virtual legs. In particular, ρ is a projective representation of G on the bond space $\mathbb{C}^D$.

Proof. ✓ Proof Since A is injective and on-site symmetric, each twisted tensor $\tilde{A}_g$ is gauge equivalent to A (Lemma 12.1.8). Choose $X(g) \in \text{GL}_D(\mathbb{C})$ satisfying (12.7) for all i .

By functoriality (Lemma 12.1.3), the product $X(h)X(g)$ also intertwines A with $\tilde{A}_{gh}$. Since $X(gh)$ does the same, gauge uniqueness (Lemma 12.1.11) gives a nonzero scalar $\omega'(h, g)$ such that

$$X(h)X(g) = \omega'(h, g)X(gh). \quad (12.8)$$

Defining $\rho(g) := X(g^{-1})$ converts (12.8) to the standard law (12.5), where $\omega(g, h) := \omega'(g^{-1}, h^{-1})$. □

Corollary 12.1.17 (2-cocycle identity for the factor system). Lean ✓ Stmt If $D > 0$, the factor system ω from Theorem 12.1.16 satisfies the 2-cocycle identity for all $g, h, k \in G$:

$$\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k).$$

Proof. ✓ Proof Theorem 12.1.16 gives ρ and ω . By Lemma 12.1.15, the cocycle identity (12.6) follows from associativity of matrix multiplication and invertibility of the representation matrices. □

12.2 Cohomology Classes of Virtual Symmetry Cocycles

Two cocycles may differ by a coboundary yet represent the same projective-representation class. The following definitions and results make this precise and establish the quotient $H^2(G, \mathbb{C}^\times)$.

Definition 12.2.1 (Coboundary). Lean ✓ Stmt A scalar 2-cocycle ω is a *coboundary* if there exists $\varphi: G \rightarrow \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}. \quad (12.9)$$

Definition 12.2.2 (Cohomologous cocycles). Lean ✓ Stmt Two cocycles ω_1, ω_2 are *cohomologous* (written $\omega_1 \sim \omega_2$) if there exists $\varphi: G \rightarrow \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h). \quad (12.10)$$

Theorem 12.2.3 (Gauge independence of the cocycle class). [Lean](#) [✓ Stmt](#) If the bond dimension satisfies $D \geq 1$ and two projective representations ρ_1, ρ_2 (with cocycles ω_1, ω_2) both arise as virtual representations of the same injective MPS tensor A under an on-site symmetry U , then $\omega_2 \sim \omega_1$, and hence also $\omega_1 \sim \omega_2$ by symmetry.

Proof. [✓ Proof](#) Gauge uniqueness gives nonzero scalars $f(k)$ with $X_2(k) = f(k) X_1(k)$ for every physical twist k . Since the projective representation is $\rho_a(g) = X_a(g^{-1})$, setting $k = g^{-1}$ gives $\rho_2(g) = f(g^{-1})\rho_1(g)$. Equivalently, for each fixed g the two local gauges

$$\begin{array}{c} i \\ | \\ \boxed{X_1(g^{-1})} \text{---} \bullet \text{---} \boxed{X_1(g^{-1})^{-1}} \\ A \end{array} \quad \text{and} \quad \begin{array}{c} i \\ | \\ \boxed{X_2(g^{-1})} \text{---} \bullet \text{---} \boxed{X_2(g^{-1})^{-1}} \\ A \end{array}$$

describe the same twisted tensor, so they differ by a scalar. Set $\alpha(g) := f(g^{-1})$. Substituting $\rho_2(g) = \alpha(g)\rho_1(g)$ into the projective law (12.5) and cancelling the invertible matrix $X_1((gh)^{-1}) = \rho_1(gh)$ yields

$$\omega_2(g, h) = \alpha(g)\alpha(h)\alpha(gh)^{-1}\omega_1(g, h). \quad (12.11)$$

Thus ω_2 is the coboundary multiple of ω_1 with cochain α . This is precisely the orientation required by (12.10) for the stated relation $\omega_2 \sim \omega_1$. Equivalently,

$$\omega_1(g, h) = \alpha(g)^{-1}\alpha(h)^{-1}\alpha(gh)\omega_2(g, h),$$

so the inverse cochain also gives the symmetric conclusion $\omega_1 \sim \omega_2$. $\square$

Definition 12.2.4 (Multiplicative 2-cocycle condition). [Lean](#) [✓ Stmt](#) A scalar 2-cochain $\omega : G \times G \rightarrow \mathbb{C}^\times$ is a 2-cocycle if it satisfies, for all $g, h, k \in G$,

$$\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k).$$

Theorem 12.2.5 (A projective factor system is a cocycle). [Lean](#) [✓ Stmt](#) If $D > 0$ and $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$ obeys $\rho(g)\rho(h) = \omega(g, h)\rho(gh)$, then ω is a $\mathbb{C}^\times$ -valued 2-cocycle.

Proof. [✓ Proof](#) Associativity gives the scalar identity of Lemma 12.1.15. Because the inclusion $\mathbb{C}^\times \hookrightarrow \mathbb{C}$ is injective, the same identity holds in $\mathbb{C}^\times$, which is precisely Definition 12.2.4. $\square$

Definition 12.2.6 (Second cohomology quotient). [Lean](#) [✓ Stmt](#) The second cohomology set is the quotient

$$H^2(G, \mathbb{C}^\times) := Z^2(G, \mathbb{C}^\times)/\sim, \quad (12.12)$$

where $\sim$ is the coboundary-equivalence relation.

Definition 12.2.7 (Agreement of factor-system cohomology classes). [Lean](#) [✓ Stmt](#) The relation called projective equivalence holds precisely when the two factor systems are cohomologous. It compares their classes in $H^2(G, \mathbb{C}^\times)$; it does not assert an intertwining equivalence between the underlying representation spaces.

Theorem 12.2.8 (Factor-system class agreement). [Lean](#) [✓ Stmt](#) Let ρ_1, ρ_2 be projective representations with factor systems ω_1, ω_2 . Then

$$\rho_1 \sim_{\text{proj}} \rho_2 \iff \omega_1 \sim \omega_2. \quad (12.13)$$

Proof. [✓ Proof](#) This follows from the definition of projective equivalence as cohomology-class equality for factor systems. $\square$

12.2.1 Non-triviality via the commutator phase

For an abelian symmetry group, the antisymmetric commutator phase is invariant under a change of cocycle representative. A value not equal to 1 therefore witnesses non-triviality; no converse is asserted here.

Definition 12.2.1.1 (Commutator phase). [Lean](#) [✓ Stmt](#) The *commutator phase* of a 2-cochain ω at a pair g, h is $\beta_\omega(g, h) := \omega(g, h)\omega(h, g)^{-1}$. It is defined for any $\mathbb{C}^\times$ -valued 2-cochain; the cocycle condition is not needed.

Theorem 12.2.1.2 (The commutator phase is a class invariant). [Lean](#) [✓ Stmt](#) If $\omega_1 \sim \omega_2$ and $gh = hg$, then $\beta_{\omega_1}(g, h) = \beta_{\omega_2}(g, h)$.

Proof. [✓ Proof](#) Write $\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h)$ for a coboundary witness φ . Substituting into the ratio and using $gh = hg$ and the commutativity of $\mathbb{C}^\times$ gives

$$\begin{aligned}\beta_{\omega_1}(g, h) &= \frac{\varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h)}{\varphi(h)\varphi(g)\varphi(hg)^{-1}\omega_2(h, g)} \\ &= \frac{\omega_2(g, h)}{\omega_2(h, g)} = \beta_{\omega_2}(g, h),\end{aligned}\tag{12.14}$$

where the φ -factors cancel since $\varphi(gh) = \varphi(hg)$. $\square$

Definition 12.2.1.3 (Non-trivial cohomology class). [Lean](#) [✓ Stmt](#) A factor system ω has a *non-trivial class* when it is not cohomologous to the constant cocycle 1.

Theorem 12.2.1.4 (Commutator-phase non-triviality test). [Lean](#) [✓ Stmt](#) If a commuting pair g, h has $\beta_\omega(g, h) \neq 1$, then ω has a non-trivial class.

Proof. [✓ Proof](#) The trivial cocycle has commutator phase 1 at every pair, so by Theorem 12.2.1.2 and (12.14), any cocycle cohomologous to it has $\beta = 1$ at every commuting pair. Contrapositively, $\beta_\omega(g, h) \neq 1$ for one commuting pair forces ω to be non-cohomologous to the trivial cocycle. $\square$

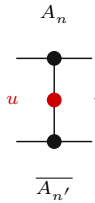
12.3 Physical Endpoint String Order and Virtual-Boundary Nondecay

The source notion of string order uses physical endpoint operators and a nonidentity unitary string [PGWS⁺08]. We also study a restricted virtual-boundary nondecay predicate and a virtual local-covariance relation. These notions are stated separately below; equivalence with physical state symmetry requires additional hypotheses.

Definition 12.3.1 (Twisted transfer map). [Lean](#) [✓ Stmt](#) Given an MPS tensor A and a physical-index matrix u , the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n, n'} \langle n' | u | n \rangle A_n X A_{n'}^\dagger.\tag{12.15}$$

Diagrammatically one inserts the physical operator u on the doubled local transfer cell:

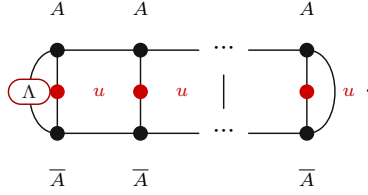


The upper and lower black nodes denote A_n and $A_n^\dagger$, and the red node labelled u couples the physical legs.

Definition 12.3.2 (Stationary-boundary block-twist functional). [Lean](#) [✓ Stmt](#) The *stationary-boundary block-twist functional* is

$$T_L(A, u, \Lambda) := \text{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})). \quad (12.16)$$

In tensor-network notation this is the length- L doubled chain with a copy of u inserted on each physical rung:



L sites

The boundary matrix Λ is contracted into the virtual boundary, the red nodes labelled u sit on the physical rungs, and the right boundary is traced.

Definition 12.3.3 (Periodic finite-chain symmetry overlap). [Not ready](#) For the periodic length- L matrix product vector $|\Psi_L(A)\rangle$, the source quantity is literally

$$R_L(u) := \langle \Psi_L(A) | u^{\otimes L} | \Psi_L(A) \rangle. \quad (12.17)$$

Identifying this periodic overlap with the normalized stationary-boundary functional $T_L(A, u, \Lambda)$ requires a thermodynamic-boundary and normalization hypothesis. Thus Definition 12.3.2 must not be read as the literal source definition of $R_L(u)$.

Definition 12.3.4 (Physical string correlator). [Lean](#) [✓ Stmt](#) For physical operators $x, y \in M_d(\mathbb{C})$, a twist $u \in M_d(\mathbb{C})$, and boundary state Λ , the transfer-form string correlator of [PGWS⁺08, display SOPMP, lines 176–181] is

$$S_N(x, y, u) := \text{tr}(\Lambda \mathcal{E}_x \mathcal{E}_u^N(\mathcal{E}_y(\mathbb{1}))). \quad (12.18)$$

This is the physical-endpoint expression corresponding to $\langle \Psi | x \otimes u^{\otimes N} \otimes y | \Psi \rangle$.

Definition 12.3.5 (Fixed-endpoint physical string order). [Lean](#) [✓ Stmt](#) For fixed physical operators x, y, u , the correlator has physical string order if there is $s > 0$ such that $|S_N(x, y, u)| \rightarrow s$.

Definition 12.3.6 (Physical endpoint string order). [Lean](#) [✓ Stmt](#) A tensor has physical string order in the sense of [PGWS⁺08, display SOP, lines 112–122] if there exist a unitary $u \neq \mathbb{1}$ and physical endpoint operators x, y for which Definition 12.3.5 holds. The comparison with arbitrary virtual boundary matrices is a separate endpoint- realizability statement.

Lemma 12.3.7 (Twisted spectral-radius bound and peripheral rigidity). [Not ready](#) This is Lemma 1 of [PGWS⁺08]. Let A represent a pure finitely correlated state in the canonical gauge $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, $\mathcal{E}_A^*(\Lambda) = \Lambda$, with $\Lambda > 0$ and $\text{tr}(\Lambda) = 1$. For every physical unitary u ,

$$\rho(\mathcal{E}_u) \leq 1.$$

Equality holds if and only if there are a unitary V and $\theta \in [0, 2\pi)$ satisfying the following identity for every j in an eigenbasis $u = \sum_j e^{i\theta_j} |\tilde{j}\rangle\langle\tilde{j}|$:

$$V^\dagger \tilde{A}_j = e^{i(\theta - \theta_j)} \tilde{A}_j V^\dagger. \quad (12.19)$$

Moreover, $\mathcal{E}_u$ has at most one eigenvalue of modulus one.

Proof. Let $\mathcal{E}_u(V) = \lambda V$. Multiplication by $\Lambda V^\dagger$, the trace, and Cauchy–Schwarz give

$$\begin{aligned} |\lambda| \operatorname{tr}(V \Lambda V^\dagger) &\leq \left(\sum_j \operatorname{tr}(V \tilde{A}_j^\dagger \Lambda \tilde{A}_j V^\dagger) \right)^{1/2} \left(\sum_j \operatorname{tr}(\tilde{A}_j^\dagger V \Lambda V^\dagger \tilde{A}_j) \right)^{1/2} \\ &= \operatorname{tr}(V \Lambda V^\dagger). \end{aligned} \quad (12.20)$$

Faithfulness of Λ yields $|\lambda| \leq 1$. Equality in Cauchy–Schwarz is equivalent to (12.19); conversely that relation makes $e^{i\theta}$ an eigenvalue. It also gives $\mathcal{E}_A(V^\dagger V) = V^\dagger V$. The fixed space of the pure canonical transfer map is one-dimensional, so $V^\dagger V = c\mathbb{1}$ for some $c > 0$. Replacing V by $c^{-1/2}V$ makes it unitary without changing the intertwining relation.

If V, V' are peripheral eigenmatrices with eigenvalues $e^{i\theta}, e^{i\theta'}$, respectively, the equality relations give the exact identity

$$\mathcal{E}_A(V^\dagger V') = e^{i(\theta' - \theta)} V^\dagger V'.$$

Purity leaves only the peripheral eigenvalue 1, hence $\theta' = \theta$ modulo 2π . The one-dimensional fixed space then gives $V^\dagger V' \in \mathbb{C}\mathbb{1}$, proving uniqueness of the peripheral eigenvalue and its eigendirection. $\square$

Theorem 12.3.8 (Physical endpoint criterion for string order). Not ready This is Theorem 1 of [PGWS⁺08]. A pure finitely correlated state has physical string order if and only if there exist a unitary $\tilde{u} \neq \mathbb{1}$, a unitary virtual matrix V , and physical indices n, m such that

$$\mathcal{E}_{\tilde{u}}(V) = V, \quad \operatorname{tr}(V \Lambda A_n A_m^\dagger) \neq 0. \quad (12.21)$$

The transfer criteria below assume one-site injectivity. The literal physical equivalence additionally requires that sufficiently long physical endpoint operators realize the needed virtual boundary matrices.

Proof. Lemma 12.3.7 shows that a nonzero limit requires a unique peripheral eigenvalue $e^{i\theta}$. If V and Y are the corresponding right and left eigenmatrices, then $Y = \Lambda V^\dagger$, and the leading term retains its length-dependent phase:

$$S_N(x, y, u) = e^{iN\theta} \operatorname{tr}(Y \mathcal{E}_y(\mathbb{1})) \operatorname{tr}(\Lambda \mathcal{E}_x(V)) + o(1).$$

In the eigenbasis $u = \sum_j e^{i\theta_j} |\tilde{j}\rangle\langle\tilde{j}|$, shift the string unitary by the peripheral phase,

$$\tilde{u} := e^{-i\theta} u = \sum_j e^{i(\theta_j - \theta)} |\tilde{j}\rangle\langle\tilde{j}|.$$

Then $\mathcal{E}_{\tilde{u}}(V) = V$. Choosing $y = x^\dagger \tilde{u}$ reduces nonvanishing to the second condition in (12.21); if it holds for n, m , take $x = |n\rangle\langle m|$. The reverse implication follows from the same phase-retaining factorization and the endpoint-realizability hypothesis. $\square$

Theorem 12.3.9 (Physical state symmetry and twisted spectral radius). Not ready This is Theorem 2 of [PGWS⁺08]. For a pure finitely correlated state and a unitary $u \neq \mathbb{1}$, the physical state is invariant under $u^{\otimes N}$ on every N -site reduced density operator if and only if

$$\rho(\mathcal{E}_u) = 1.$$

The theorem below gives the corresponding virtual local-covariance criterion for a one-site-injective tensor. Passing between that condition and invariance of every physical reduced state is an additional step.

Proof. If $\rho(\mathcal{E}_u) = 1$, Lemma 12.3.7 gives a unitary V satisfying the local intertwining relation. Substitution of that relation into the adjoint transfer map, together with $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$, gives $\mathcal{E}_A^\dagger(V\Lambda V^\dagger) = V\Lambda V^\dagger$. This positive fixed point has trace one, so uniqueness gives $V\Lambda V^\dagger = \Lambda$, equivalently $V^\dagger\Lambda V = \Lambda$, which yields the physical symmetry. Conversely, physical invariance gives, for length-independent boundary tensors L, R ,

$$D^{-2} \leq \text{tr}(\varrho_N^2) = \text{tr}[L(\mathcal{E}_u \otimes \mathcal{E}_{u^\dagger})^N(R)].$$

Exponential decay would contradict this bound, so $\rho(\mathcal{E}_u) = 1$. □

Definition 12.3.10 (Virtual local covariance with boundary invariance). Lean ✓ Stmt The virtual local-covariance predicate holds for A, u, Λ if there exist a unitary matrix V and a phase μ with $|\mu| = 1$ such that $V^\dagger\Lambda V = \Lambda$ and, for every physical index i ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger. \quad (12.22)$$

The unitary V intertwines the physical action of u with a virtual conjugation, and the boundary state Λ is invariant under V .

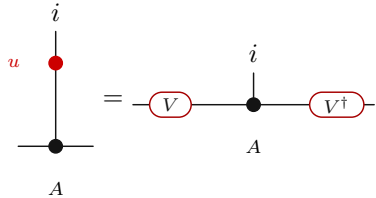
Definition 12.3.11 (Virtual-boundary nondecay). Lean ✓ Stmt For boundary matrices $X, Y \in M_D(\mathbb{C})$, set explicitly

$$R_L(u; X, Y) := \text{tr}(\Lambda X \mathcal{E}_u^L(Y)).$$

The virtual-boundary nondecay predicate holds for u and Λ if there exist X, Y and a positive constant $c > 0$ such that $c \leq \|R_L(u; X, Y)\|$ for every L . Thus some virtual-boundary sequence does not decay to zero.

Virtual-boundary nondecay is an existence statement over the virtual matrices X, Y , for a fixed twist u ; no claim is made for any particular boundary choice. Definition 12.3.6 records the physical-endpoint string-order notion from the source, which quantifies existentially over u, x, y and requires a positive limit rather than a lower bound uniform in the length. That paper also notes that the correlator for a particular endpoint choice can vanish even when the physical symmetry is present.

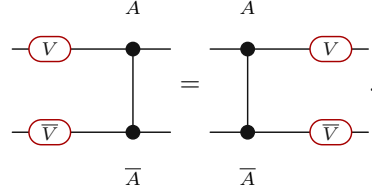
Definition 12.3.12 (Local physical intertwining). Lean ✓ Stmt The local physical intertwining relation is $\sum_j u_{ij} A^j = V A^i V^\dagger$. Locally, the physical action of u can therefore be pushed to the virtual legs:



Definition 12.3.13 (Transfer-map covariance). [Lean](#) [✓ Stmt](#) Transfer-map covariance is the identity that the transfer map commutes with virtual conjugation:

$$\mathcal{E}(VXV^\dagger) = V \mathcal{E}(X) V^\dagger. \quad (12.23)$$

In doubled-layer notation, the virtual operators slide through the local transfer cell:

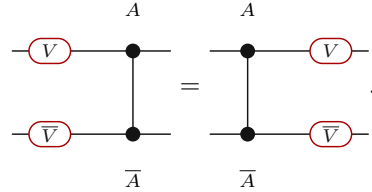


Definition 12.3.14 (Commutation in the doubled transfer picture). [Lean](#) [✓ Stmt](#) In the transfer-map language, the commutation of the doubled transfer matrix $E = \sum_i A_i \otimes \bar{A}_i$ with $V \otimes \bar{V}$ is

$$V \mathcal{E}_1(X) V^\dagger = \mathcal{E}_1(VXV^\dagger). \quad (12.24)$$

Theorem 12.3.15 (Transfer-map covariance and doubled commutation are equivalent). [Lean](#) [✓ Stmt](#) Transfer-map covariance and commutation in the doubled transfer picture are equivalent for any matrix V .

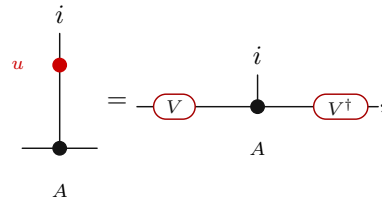
Proof. [✓ Proof](#) Both sides express the same local picture with opposite orientation:



The two formulas are the same identity read from opposite sides. □

Theorem 12.3.16 (Local physical intertwining implies transfer-map covariance). [Lean](#) [✓ Stmt](#) If V and u are unitary and the local physical intertwining relation holds, then the transfer map is covariant.

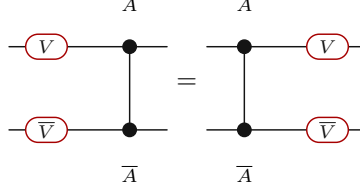
Proof. [✓ Proof](#) Starting from



one inserts $V^\dagger V = \mathbb{1}$ so that each local transfer cell is written in terms of conjugated Kraus operators, then replaces those operators using the intertwining relation. After that, the physical u -boxes disappear by the unitary Kraus-mixing identity. □

Theorem 12.3.17 (Transfer-map covariance implies local physical intertwining). [Lean](#) [✓ Stmt](#) Assume that A is one-site injective, $VV^\dagger = \mathbb{1}$, and the transfer map is covariant. Then there is u with $uu^\dagger = \mathbb{1}$ satisfying the local physical intertwining relation. In fact, covariance and $VV^\dagger = \mathbb{1}$ alone imply the conclusion; the injectivity hypothesis is unnecessary.

Proof. [✓ Proof](#) Set $B^i := V A^i V^\dagger$. The transfer-map covariance relation



implies that the two d -element Kraus families B and A define the same transfer map. Rectangular Kraus freedom (Theorem 13.10.12) converts $\mathcal{E}_B = \mathcal{E}_A$ into a unitary Kraus mixing matrix u with $B^i = \sum_j u_{ij} A^j$. $\square$

Theorem 12.3.18 (Twisted-transfer eigenvalues have modulus at most one). [Lean](#) [✓ Stmt](#) Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If $\mathcal{E}_u(X) = \lambda X$ for some nonzero $X \in M_D(\mathbb{C})$, then $|\lambda| \leq 1$.

Proof. [✓ Proof](#) By Lemma H.4.2, the twisted transfer map is the mixed transfer operator F_{A,B_u} , as in (H.6). Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. Theorem 7.4.1 then applies to the mixed transfer operator in that gauge, and similarity preserves the eigenvalue λ . $\square$

Theorem 12.3.19 (Modulus-one twisted eigenvalue implies gauge-phase equivalence). [Lean](#) [✓ Stmt](#) Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If

$$\mathcal{E}_u(X) = \lambda X \quad (12.25)$$

for some nonzero $X \in M_D(\mathbb{C})$ with $|\lambda| = 1$, then A is gauge-phase equivalent to the companion tensor B_u .

Proof. [✓ Proof](#) By Lemma H.4.2, the twisted transfer map is the mixed transfer operator F_{A,B_u} . Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. In that gauge, (12.25) becomes a modulus-one peripheral eigenvalue for an irreducible mixed transfer operator, so Theorem 7.7.1 yields gauge-phase equivalence. Undoing the common trace-preserving gauge gives the claim for A and B_u . $\square$

Theorem 12.3.20 (Virtual-boundary nondecay gives companion gauge-phase equivalence). [Lean](#) [✓ Stmt](#) Let A be injective, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If virtual-boundary nondecay holds for u with boundary parameter Λ , then A is gauge-phase equivalent to the companion tensor B_u .

Proof. [✓ Proof](#) If $\rho(\mathcal{E}_u) < 1$, Lemma H.6.1 forces every virtual-boundary sequence to decay to zero, contradicting Definition 12.3.11. Hence $\rho(\mathcal{E}_u) \geq 1$. Theorem 12.3.18 shows that every eigenvalue has modulus at most 1, so some peripheral eigenvalue has modulus exactly 1. Applying Theorem 12.3.19 then gives the gauge-phase equivalence. $\square$

Theorem 12.3.21 (Gauge-phase equivalence with the companion tensor yields a virtual unitary). [Lean](#) [✓ Stmt](#) Let A be injective, assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If A is gauge-phase equivalent to the companion tensor B_u , then there exist a unitary V and a phase μ with $|\mu| = 1$ such that, for every physical index i ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger. \quad (12.26)$$

Proof. [✓ Proof](#) Start from an invertible intertwiner X and a nonzero gauge scalar ζ between A and B_u . The positive matrix $Q := XX^\dagger$ satisfies $\mathcal{E}_A(Q) = |\zeta|^2 Q$. Since $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, irreducibility forces the two positive eigenvalues to agree, so $|\zeta| = 1$ and Q is a positive fixed point of $\mathcal{E}_A$. Uniqueness of positive fixed points for an injective normalized tensor then makes Q a positive scalar multiple of the identity. Rescaling X by the square root of that scalar produces a unitary V , and the original intertwining relation becomes (12.26), with $|\mu| = 1$. $\square$

Theorem 12.3.22 (Virtual unitary gives a twisted-transfer eigenmatrix). [Lean](#) [✓ Stmt](#) Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ and that, for every physical index i , $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$ with V unitary. Then V is an eigenmatrix of the twisted transfer map:

$$\mathcal{E}_u(V) = \mu V. \quad (12.27)$$

Proof. [✓ Proof](#) Expand $\mathcal{E}_u(V)$ using (12.15), insert the intertwining relation into each local term, and use $V^\dagger V = \mathbb{1}$ together with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. $\square$

Theorem 12.3.23 (Boundary-state invariance of the virtual unitary). [Lean](#) [✓ Stmt](#) Let A be injective, assume $uu^\dagger = \mathbb{1}$, and let Λ be a positive definite boundary state with $\text{tr}(\Lambda) = 1$ and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. If V is a unitary and μ a phase satisfying, for every physical index i ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger, \quad (12.28)$$

then

$$V^\dagger \Lambda V = \Lambda. \quad (12.29)$$

Proof. [✓ Proof](#) Relation (12.28) first shows that $V \Lambda V^\dagger$ is another positive fixed point of the adjoint transfer map. It has the same trace as Λ , so uniqueness of the normalized positive fixed point gives $V \Lambda V^\dagger = \Lambda$. Multiplying by $V^\dagger$ on the left and by V on the right yields (12.29). $\square$

Lemma 12.3.24 (Virtual local covariance implies virtual-boundary nondecay). [Lean](#) [✓ Stmt](#) Assume $\text{tr}(\Lambda) = 1$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If A, u, Λ satisfy virtual local covariance, then virtual-boundary nondecay holds for u .

Proof. [✓ Proof](#) Choose the boundary matrices $X = V^\dagger$ and $Y = V$, where V is the unitary from the virtual local-covariance datum. By Theorem 12.3.22, one has $\mathcal{E}_u^L(V) = \mu^L V$. Therefore

$$R_L(u; V^\dagger, V) = \mu^L \text{tr}(\Lambda) = \mu^L,$$

whose norm is constantly 1. $\square$

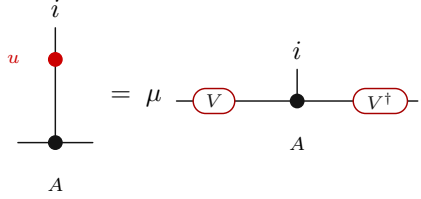
Theorem 12.3.25 (Virtual local covariance iff a unitary peripheral eigenmatrix exists). [Lean](#) [✓ Stmt](#) Let A be injective, and assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then virtual local covariance holds if and only if there exist a unitary $V \in M_D(\mathbb{C})$ and a phase μ with $|\mu| = 1$ such that $\mathcal{E}_u(V) = \mu V$. By Theorem 12.3.18, this witness form is equivalent to the peripheral-spectrum condition $\rho(\mathcal{E}_u) = 1$.

Proof. ✓ **Proof** The forward direction is Theorem 12.3.22. For the reverse direction, start from a unitary peripheral eigenmatrix. Theorem 12.3.19 gives gauge-phase equivalence with the companion tensor; then Theorem 12.3.21 yields a virtual unitary and phase satisfying the local covariance relation, and Theorem 12.3.23 gives the boundary-state invariance needed for virtual local covariance. $\square$

Theorem 12.3.26 (Virtual-boundary nondecay $\Leftrightarrow$ virtual local covariance). Lean ✓ **Stmt** For an injective MPS, assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then virtual-boundary nondecay for u holds if and only if A, u, Λ satisfy virtual local covariance.

Proof. ✓ **Proof** If virtual-boundary nondecay holds, Theorem 12.3.20 gives a gauge-phase equivalence with the companion tensor. Theorem 12.3.21 extracts a virtual unitary and phase from that equivalence, and Theorem 12.3.23 shows that the same unitary preserves the boundary state, so one obtains virtual local covariance. Conversely, Lemma 12.3.24 turns virtual local covariance into a nondecaying virtual-boundary sequence. $\square$

Theorem 12.3.27 (Virtual unitary from virtual-boundary nondecay). Lean ✓ **Stmt** For an injective MPS, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If virtual-boundary nondecay holds for u with boundary parameter Λ , then there exists a virtual unitary V and a unit-modulus scalar μ such that $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$ for all i . Diagrammatically this is the same local intertwining relation, but only up to an overall unit scalar on the virtual side:



Proof. ✓ **Proof** Virtual-boundary nondecay first gives gauge-phase equivalence with the companion tensor by Theorem 12.3.20. Theorem 12.3.21 then normalizes the resulting intertwiner to a unitary and produces the phase μ . $\square$

12.4 SPT Phase Labels and Virtual-Boundary Nondecay

The predicate in Definition 12.4.1 compares virtual cocycle labels. The source classification below concerns symmetric paths that remain within the MPS and corresponding parent-Hamiltonian framework, with the endpoint representations embedded into a common physical space as in [SPGC11].

Definition 12.4.1 (Equality of virtual cocycle labels). Lean ✓ **Stmt** For tensors A, B and a fixed physical representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$, the predicate called *same SPT phase* asserts the existence of factor systems $\omega_A, \omega_B : G \times G \rightarrow \mathbb{C}^\times$ and projective bond representations ρ_A, ρ_B such that

$$\begin{aligned}\tilde{A}_g^i &= \rho_A(g^{-1}) A^i \rho_A(g^{-1})^{-1}, \\ \tilde{B}_g^i &= \rho_B(g^{-1}) B^i \rho_B(g^{-1})^{-1},\end{aligned}\tag{12.30}$$

exactly, and $\omega_A \sim \omega_B$. Thus the predicate is equality of virtual factor-system classes. It neither quantifies over Hamiltonian paths nor states physical phase equivalence.

Remark 12.4.2 (Scalar and character conventions). These factor systems take values in $\mathbb{C}^\times$. For unitary bond representations one may choose unit-modulus representatives and obtain the physical classification label in $H^2(G, \mathbb{U}(1))$; this normalization is not an equality $\mathbb{C}^\times = \mathbb{U}(1)$. Moreover, Definition 12.4.1 requires exact tensor equations. A physical state symmetry may instead act on an MPS vector by a one-dimensional character $\chi(g)$, producing a scalar factor in the local tensor equation. Exact MPV-family equality and such a character are distinct hypotheses.

Theorem 12.4.3 (Classification along symmetric MPS parent-Hamiltonian paths). Not ready
For $p = 0, 1$, let A_p be an injective MPS tensor whose parent Hamiltonian H_p has a unique ground state and is symmetric under a unitary representation U^p of G on $\mathcal{H}_p$. Choose the endpoint phase gauges. After a common finite blocking, the embedded endpoints admit a continuous symmetric MPS path with corresponding translationally invariant, finite-range parent Hamiltonians that vary smoothly and remain uniformly gapped if and only if their virtual projective representations determine the same class in $H^2(G, \mathbb{U}(1))$.

More precisely, for the forward construction there exist a path-sector Hilbert space $\mathcal{H}_{\text{path}}$ and a linear representation U^{path} such that the endpoints embed into

$$\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \mathcal{H}_{\text{path}}, \quad U_g = U_g^0 \oplus U_g^1 \oplus U_g^{\text{path}}.$$

The converse quantifies over smooth paths whose unique ground states remain MPS and whose Hamiltonians are the corresponding parent Hamiltonians, as in [SPGC11, Section II.F, lines 929–953]. It does not assert invariance along an arbitrary gapped Hamiltonian path.

Proof. First deform each blocked injective MPS to its isometric form; the deformation preserves the symmetry relation

$$U_g \mathcal{P} = \mathcal{P}(V_g \otimes \overline{V}_g).$$

Along any continuous symmetric MPS path quantified in the theorem, the tensor and its induced virtual action V_g vary continuously. The cohomology class is discrete, so it cannot change along that path. This proves necessity within the MPS/parent-Hamiltonian framework.

Conversely, let the endpoint isometric forms have virtual actions V_g^0 and V_g^1 in the same cohomology class. Embed their bond spaces into the direct sum and interpolate between the maximally entangled bonds by

$$|\omega(\gamma)\rangle = (1 - \gamma) \sum_{i=1}^{D_0} |i, i\rangle + \gamma \sum_{i=D_0+1}^{D_0+D_1} |i, i\rangle.$$

The commuting parent Hamiltonians along this interpolation remain gapped. Their symmetry is

$$(V_g^0 \oplus V_g^1) \otimes (\overline{V}_g^0 \oplus \overline{V}_g^1) = \widehat{U}_g^0 \oplus \widehat{U}_g^1 \oplus \widehat{U}_g^{\text{path}}, \quad (12.31)$$

$$\widehat{U}_g^{\text{path}} = (V_g^0 \otimes \overline{V}_g^1) \oplus (V_g^1 \otimes \overline{V}_g^0).$$

Equality of the cohomology classes is exactly what makes the cross-term action $\widehat{U}^{\text{path}}$ a linear representation. Conjugating back by the endpoint isometries gives the common direct-sum physical representation and the required symmetric gapped path. $\square$

Theorem 12.4.4 (Virtual-boundary nondecay for injective MPV-symmetric tensors). Lean
✓ Stmt Let A be an injective tensor, symmetric under a unitary representation U . Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then the virtual-boundary nondecay predicate holds for every group element g .

Proof. ✓ Proof The virtual representation theorem produces $\rho(g^{-1}) \in \text{GL}_D(\mathbb{C})$ satisfying $\sum_j U(g)_{ij} A^j = \rho(g^{-1}) A^i \rho(g^{-1})^{-1}$. A direct computation using $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ shows that $\mathcal{E}_{U(g)}(\rho(g^{-1})) = \rho(g^{-1})$, so $\rho(g^{-1})$ is a nonzero eigenvector with eigenvalue 1. Theorem 12.3.19 gives gauge-phase equivalence with the companion tensor, and Theorem 12.3.21 extracts a virtual unitary and phase. By Theorem 12.3.23, this unitary preserves the boundary state, so Lemma 12.3.24 gives virtual-boundary nondecay. $\square$

Remark 12.4.5 (Virtual-boundary nondecay does not determine the SPT label). Theorem 12.4.4 applies to every injective symmetric tensor with canonical normalisations, irrespective of its cocycle class, so tensors in different SPT phases all satisfy virtual-boundary nondecay for every group element. This nondecay is equivalent to virtual local covariance (Theorem 12.3.26); the invariant that separates SPT phases is the cohomology class of the virtual representation cocycle (Theorem 12.4.3).

Chapter 13

Auxiliary Quantum-Channel Theory

This chapter collects quantum-channel and matrix-analysis material that is mathematically independent of the Fundamental Theorem proof chain of Chapters 2–11. The definitions of positive maps, quantum channels, irreducibility, and the Kadison–Schwarz inequality that are needed for the theorem appear in the earlier chapters; the results here extend that theory but are not required to state or prove the theorem.

Wolf’s finite-dimensional quantum-mechanics foundations include pure and mixed states, steering, operator-system extensions, spectrum preservation, and Wigner rigidity. The maximally entangled state and the Choi–Jamiołkowski isomorphism connect completely positive maps to positive semidefinite matrices. The abstract Schwarz inequality and multiplicative-domain theory express [Wol12, Chapter 5] in the operator-system language. The exponential-positivity, ergodicity, and spectral-characterization results complete the Perron–Frobenius programme of [Wol12, Chapter 6].

13.1 Bipartite systems and the Schmidt decomposition

Every pure state of a bipartite system can be brought into a normal form that exhibits its entanglement directly.

Definition 13.1.1 (Schmidt decomposition). [Lean](#) [✓ Stmt](#) For $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ with $d = \min\{d_A, d_B\}$, a Schmidt decomposition of ψ is a choice of orthonormal bases $\{e_j \in \mathbb{C}^{d_A}\}$, $\{f_j \in \mathbb{C}^{d_B}\}$ and nonnegative reals λ_j , $j = 1, \dots, d$, satisfying

$$\psi = \sum_{j=1}^d \sqrt{\lambda_j} |e_j\rangle \otimes |f_j\rangle, \quad \sum_j \lambda_j = \|\psi\|^2.$$

Lemma 13.1.2 (Schmidt decomposition, $d_A \leq d_B$ case). [Lean](#) [✓ Stmt](#) If $d_A \leq d_B$, every $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ admits a Schmidt decomposition.

Proof. [✓ Proof](#) Identify ψ with its coefficient matrix $C \in M_{d_A, d_B}(\mathbb{C})$. The spectral theorem applied to the positive semidefinite matrix $CC^\dagger$ supplies the eigenbasis $\{e_j\}$ of $\mathbb{C}^{d_A}$ and the eigenvalues λ_j ; the normalized images of the eigenvectors under $C^\dagger$, conjugated and completed to an orthonormal basis, supply $\{f_j\}$. $\square$

Proposition 13.1.3 (Schmidt decomposition). [Lean](#) [✓ Stmt](#) Every vector $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ admits a Schmidt decomposition. This is [Wol12, Proposition 1.1].

Proof. [✓ Proof](#) The case $d_A \leq d_B$ is Lemma 13.1.2. The case $d_B < d_A$ reduces to it by applying the lemma to ψ with its two factors exchanged, then exchanging the two resulting bases back. This matches Wolf’s proof sketch: a change of local bases in the coefficient matrix corresponds to left and right multiplication by two unitaries, so the Schmidt coefficients are its singular values. $\square$

Theorem 13.1.4 (Schmidt rank counts the nonzero Schmidt coefficients). [Lean](#) [✓ Stmt](#) For any Schmidt decomposition of ψ with coefficients $\{\lambda_j\}$, the number of nonzero λ_j equals $\text{SR}(\psi)$. This is [Wol12, Proposition 1.1].

Proof. [✓ Proof](#) The Schmidt decomposition factors the coefficient matrix of ψ through the isometries with columns $\{e_j\}$ and $\{f_j\}$, so its rank equals the rank of the diagonal matrix of Schmidt coefficients, which is the number of nonzero entries. $\square$

13.2 Quantum steering

A purification of a density operator lets an observer on the purifying system remotely prepare, or “steer,” any ensemble consistent with the reduced state.

Definition 13.2.1 (Purification). [Lean](#) [✓ Stmt](#) A vector $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ purifies a density operator $\rho \in M_{d_A}(\mathbb{C})$ if $\text{tr}_B[|\psi\rangle\langle\psi|] = \rho$.

Definition 13.2.2 (Convex decomposition). [Lean](#) [✓ Stmt](#) A convex decomposition of a density operator $\rho \in M_{d_A}(\mathbb{C})$ is a finite family of nonnegative weights λ_i summing to one together with density operators $\rho_i \in M_{d_A}(\mathbb{C})$ satisfying $\rho = \sum_i \lambda_i \rho_i$.

Theorem 13.2.3 (Partial-trace consequence of the transpose trick). [Lean](#) [✓ Stmt](#) For a bipartite vector $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ with coefficient matrix C and any linear map T with domain $M_{d_B}(\mathbb{C})$,

$$\text{tr}_B[(\text{id} \otimes T)(|\psi\rangle\langle\psi|)] = C (T^*(\mathbb{1}))^\top C^\dagger.$$

Proof. [✓ Proof](#) Tracing out the second factor of $(\text{id} \otimes T)(|\psi\rangle\langle\psi|)$ applies T to each rank-one block $|a\rangle\langle b| \mapsto C_{i,a} \bar{C}_{j,b}$ of the coefficient matrix and takes its trace; the trace-pairing identity $\text{tr}[T(M)] = \text{tr}[M T^*(\mathbb{1})]$ turns this into the displayed bilinear form in the rows of C . Wolf derives this identity for the canonical purification $\psi = (\sqrt{\rho} \otimes \mathbb{1})|\Omega\rangle$; the argument here uses only the coefficient matrix of ψ and linearity, so it holds for an arbitrary purification. $\square$

Theorem 13.2.4 (Quantum steering). [Lean](#) [✓ Stmt](#) Let $\rho \in M_{d_A}(\mathbb{C})$ be a density operator with purification $\psi \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$. For every convex decomposition $\rho = \sum_i \lambda_i \rho_i$ there is an instrument $\{T_i : M_{d_B}(\mathbb{C}) \rightarrow M_{d_A}(\mathbb{C})\}$ acting on Bob’s system such that

$$\lambda_i \rho_i = \text{tr}_B[(\text{id} \otimes T_i)(|\psi\rangle\langle\psi|)]$$

for every i . This is [Wol12, Proposition (Quantum steering)].

Proof. [✓ Proof](#) Write C for the coefficient matrix of ψ , so $C C^\dagger = \rho$, and let $C^+ := C^\dagger \rho^+$ using the generalized inverse of ρ on its support. Then $C C^+$ is the support projection of ρ , and $C^+ C$ is an orthogonal projection on $\mathbb{C}^{d_B}$ (the subspace of $\mathbb{C}^{d_B}$ needed to purify ρ). For each i , the support projection of ρ absorbs $\lambda_i \rho_i$ on both sides (its range lies in the range of

ρ), so $C(C^+(\lambda_i \rho_i)(C^+)^{\dagger})C^{\dagger} = \lambda_i \rho_i$. Transposing $C^+(\lambda_i \rho_i)(C^+)^{\dagger}$ gives an effect operator E_i with $C E_i^{\top} C^{\dagger} = \lambda_i \rho_i$; adding the transpose of the orthogonal complement of C^+C to one fixed E_{i_0} keeps this identity (the complement contributes zero under $C(\cdot)C^{\dagger}$) while making $\sum_i E_i = \mathbb{1}$. Taking T_i to be the Lüders instrument $\sigma \mapsto \sqrt{E_i} \sigma \sqrt{E_i}$ gives $T_i^*(\mathbb{1}) = E_i$, so $\{T_i\}$ is an instrument, and Theorem 13.2.3 recovers the displayed identity.

This route proves the proposition directly for an arbitrary purification and an arbitrary convex decomposition, in place of Wolf's route through the canonical purification $\psi = (\sqrt{\rho} \otimes \mathbb{1})|\Omega\rangle$ and the transposition operator θ with respect to its Schmidt basis. $\square$

13.3 Maximal weight in convex decomposition

Given a density operator ρ and a candidate term ρ_1 , how large a weight can ρ_1 carry in a convex decomposition of ρ ?

Definition 13.3.1 (Convex decomposition with a distinguished term). [Lean](#) [✓ Stmt](#) For density operators ρ, ρ_1 on the same space and $c \in \mathbb{R}$, ρ has a convex decomposition with ρ_1 carrying weight c when $\rho = \sum_i \lambda_i \rho_i$ for some finite index set, with every ρ_i a density operator, every $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, and some index i_1 with $\lambda_{i_1} = c$, $\rho_{i_1} = \rho_1$.

Lemma 13.3.2 (Congruence criterion for positive-semidefinite domination). [Lean](#) [✓ Stmt](#) Let A, B be positive semidefinite operators on the same space with $\ker(A) \subseteq \ker(B)$, and let $c > 0$. Then

$$A - cB \geq 0 \iff c\|A^{-1/2}BA^{-1/2}\|_{\infty} \leq 1,$$

where the inverse is taken on the range of A .

Proof. [✓ Proof](#) Write P for the support projection of A . The kernel inclusion compresses B onto the range of A , so $A^{-1/2}BA^{-1/2}$ lies entirely inside $\text{ran}(P)$, and conjugating by $A^{-1/2}$ and by $A^{1/2}$ give mutually inverse congruences

$$\begin{aligned} A^{-1/2}(A - cB)A^{-1/2} &= P - cA^{-1/2}BA^{-1/2}, \\ A^{1/2}(P - cA^{-1/2}BA^{-1/2})A^{1/2} &= A - cB, \end{aligned}$$

so $A - cB \geq 0$ iff $P - cA^{-1/2}BA^{-1/2} \geq 0$. Since $A^{-1/2}BA^{-1/2}$ is supported on $\text{ran}(P)$, this last inequality is in turn equivalent to $1 - cA^{-1/2}BA^{-1/2} \geq 0$, which by the C^* -identity $\|X\|_{\infty} \leq 1 \iff X \leq 1$ (for $X \geq 0$) holds iff $c\|A^{-1/2}BA^{-1/2}\|_{\infty} \leq 1$. $\square$

Lemma 13.3.3 (Necessity of the kernel inclusion). [Lean](#) [✓ Stmt](#) If ρ has a convex decomposition $\rho = \sum_i \lambda_i \rho_i$ giving ρ_1 a positive weight c , then $\ker(\rho) \subseteq \ker(\rho_1)$.

Proof. [✓ Proof](#) If $\rho v = 0$ for some vector v , every term of $\rho = \sum_i \lambda_i \rho_i$ is positive semidefinite, so each summand's quadratic form at v vanishes; in particular $c\langle v, \rho_1 v \rangle = 0$, and since $c > 0$ and ρ_1 is positive semidefinite, $\rho_1 v = 0$. $\square$

Theorem 13.3.4 (Maximal weight in convex decomposition). [Lean](#) [✓ Stmt](#) Let ρ and ρ_1 be two density operators acting on the same space and let $c > 0$. There is a convex decomposition of the form $\rho = \sum_i \lambda_i \rho_i$ with ρ_1 carrying weight c iff $\ker(\rho) \subseteq \ker(\rho_1)$ and

$$c\|\rho^{-1/2}\rho_1\rho^{-1/2}\|_{\infty} \leq 1,$$

where the inverse is taken on the range of ρ . Wolf's Proposition "Maximal weight in convex decomposition" [Wol12] states this with the inverse form $c \leq \|\rho^{-1/2}\rho_1\rho^{-1/2}\|_{\infty}^{-1}$; the two are equivalent

here because the kernel inclusion together with $\text{tr}(\rho_1) = 1$ (so $\rho_1 \neq 0$) forces $\|\rho^{-1/2}\rho_1\rho^{-1/2}\|_\infty > 0$, so dividing the displayed multiplicative inequality through by that norm is valid.

Proof. ✓ Proof Necessity of the kernel inclusion is Lemma 13.3.3.

The existence of a decomposition with weight c on ρ_1 is equivalent to $\rho - c\rho_1 \geq 0$: given the kernel inclusion, the remaining terms of the decomposition sum to $\rho - c\rho_1$, itself a positive-semidefinite combination; conversely, when $\rho - c\rho_1 \geq 0$ its trace is $1 - c \geq 0$, so either $c = 1$ and $\rho = \rho_1$ (a zero-trace positive-semidefinite matrix vanishes), or $c < 1$ and $\sigma := (1 - c)^{-1}(\rho - c\rho_1)$ is a density operator with $\rho = c\rho_1 + (1 - c)\sigma$.

Finally $\rho - c\rho_1 \geq 0$ is equivalent to $c\|\rho^{-1/2}\rho_1\rho^{-1/2}\|_\infty \leq 1$ by Lemma 13.3.2, applied with $A = \rho$, $B = \rho_1$ under the kernel inclusion. $\square$

13.4 Complete positivity from positivity

For a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, write T^* for its trace-pairing adjoint (Definition 4.5.1). Complete positivity is in general strictly stronger than positivity, but the two coincide once T 's range, or its adjoint's range, is commutative.

Definition 13.4.1 (Commutative range). Lean ✓ Stmt A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ has *commutative range* if $T(X)$ and $T(Y)$ commute for all $X, Y \in M_D(\mathbb{C})$. Since a family of pairwise-commuting elements generates a commutative subalgebra, this is the statement that T maps into a commutative subalgebra of $M_D(\mathbb{C})$.

Lemma 13.4.2 (Block quadratic form positivity for commuting images). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map and let $[a_{ij}]_{i,j=1}^n$ be a positive semidefinite block matrix with entries in $M_D(\mathbb{C})$ such that the images $T(a_{ij})$ pairwise commute. Then

$$\sum_{i,j=1}^n \langle \psi_i | T(a_{ij}) | \psi_j \rangle \geq 0$$

for every family of vectors $\{\psi_i\}_{i=1}^n \subset \mathbb{C}^D$.

Proof. ✓ Proof As the $T(a_{ij})$ mutually commute, choose a basis in which they are simultaneously diagonal, and write $\psi_{j,\beta}$ for the components of $|\psi_j\rangle$ in this basis. Then

$$\sum_{i,j=1}^n \langle \psi_i | T(a_{ij}) | \psi_j \rangle = \sum_{\beta} \langle \beta | T \left(\sum_{i,j} \bar{\psi}_{i,\beta} a_{ij} \psi_{j,\beta} \right) | \beta \rangle \geq 0,$$

by positivity of T and of $[a_{ij}]$: the bracketed matrix is $B_\beta^\dagger [a_{ij}] B_\beta$ for the column vector $B_\beta = (\psi_{j,\beta})_j$, hence positive semidefinite. $\square$

Lemma 13.4.3 (Complete positivity from positivity, commutative codomain). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with commutative range. Then T is completely positive.

Proof. ✓ Proof By Theorem 13.10.10 it suffices to show that the Choi matrix τ of T satisfies $\langle \phi | \tau | \phi \rangle \geq 0$ for every $\phi \in \mathbb{C}^D \otimes \mathbb{C}^D$. Let $a_{ij} \in M_D(\mathbb{C})$, for $i, j \in \{1, \dots, D\}$, be the blocks of $|\Omega\rangle\langle\Omega|$, the Choi matrix of the identity map (Theorem 13.10.6), and let $\psi_k := \phi(\cdot, k) \in \mathbb{C}^D$ be the reshaping of ϕ into a D -indexed family of vectors. Then

$$\langle \phi | \tau | \phi \rangle = \sum_{i,j=1}^D \langle \psi_i | T(a_{ij}) | \psi_j \rangle,$$

the block quadratic form of Lemma 13.4.2 at T , $[a_{ij}]$, and $\{\psi_i\}$. Since $[a_{ij}] \geq 0$ (it is the block family of the rank-one projector $|\Omega\rangle\langle\Omega|$) and, by commutative range, the images $T(a_{ij})$ pairwise commute, Lemma 13.4.2 gives $\langle\phi|\tau|\phi\rangle \geq 0$. $\square$

Lemma 13.4.4 (Positivity is trace-adjoint invariant). [Lean] [✓ Stmt] *If $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ is positive, then its trace-pairing adjoint $T^* : M_{D'}(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is positive.*

Proof. [✓ Proof] For $X \geq 0$, $T^*(X)$ is Hermitian since T preserves the adjoint. For every $B \geq 0$, the trace-pairing identity $\text{tr}[T^*(X)B] = \text{tr}[XT(B)]$ and positivity of X and of $T(B)$ (as T is positive) give $\text{tr}[T^*(X)B] \geq 0$; a Hermitian matrix with nonnegative trace pairing against every positive semidefinite matrix is itself positive semidefinite, so $T^*(X) \geq 0$. $\square$

Lemma 13.4.5 (Complete positivity is trace-adjoint invariant). [Lean] [✓ Stmt] *A linear map $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ is completely positive if and only if T^* is.*

Proof. [✓ Proof] Complete positivity in Kraus form is preserved by the trace-pairing adjoint, and the adjoint is an involution, so the two directions are the same statement applied to T and to T^* . $\square$

Lemma 13.4.6 (Complete positivity from positivity, commutative domain). [Lean] [✓ Stmt] *Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a positive linear map whose trace-pairing adjoint T^* has commutative range. Then T is completely positive.*

Proof. [✓ Proof] By Lemma 13.4.4, T^* is positive, and by hypothesis it has commutative range, so Lemma 13.4.3 gives that T^* is completely positive. Lemma 13.4.5 then gives that T is completely positive. $\square$

Theorem 13.4.7 (Complete positivity from positivity). [Lean] [✓ Stmt] *Let $T : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ be a positive linear map. If T has commutative range, or T^* has commutative range, then T is completely positive. This is [Wol12, Proposition 1.6]; the commutative-range hypothesis is the range-only reading of Wolf’s “ $\mathcal{A}$ or $\mathcal{B}$ commutative” hypothesis that his own proof establishes and that his remark immediately after the proof licenses for an arbitrary operator system mapping into a commutative algebra. See `docs/paper-gaps/wolf_prop16_cp_positivity_commutative_side.tex` for the precise relationship between the two phrasings.*

Proof. [✓ Proof] If T has commutative range, apply Lemma 13.4.3. If instead T^* has commutative range, apply Lemma 13.4.6. $\square$

13.5 Partial trace over subsystems

The following support-projector absorption theorems and the partial trace over the left factor of a bipartite space are generic bipartite-operator facts, with no tensor-network content; they are relocated here from the parent-Hamiltonian marginal-support chapter, whose own tripartite-lift content (Chapter ??) cites them across the chapter boundary.

Theorem 13.5.1 (Left absorption by the marginal support projector). [Lean] [✓ Stmt] *Let ρ be positive semidefinite on $H_L \otimes H_R$, and let P_L be the orthogonal projector onto the support of $\text{tr}_R \rho$. Then $(P_L \otimes \mathbb{1}_R)\rho = \rho$.*

Proof. [✓ Proof] Set $Q_L = \mathbb{1}_L - P_L$. Since $Q_L \text{tr}_R \rho = 0$, the adjoint identity gives

$$\text{tr}((Q_L \otimes \mathbb{1}_R)\rho) = \text{tr}(Q_L \text{tr}_R \rho) = 0.$$

Positivity of ρ then implies $(Q_L \otimes \mathbb{1}_R)\rho = 0$. Finally, $\mathbb{1}_{L \otimes R} - (Q_L \otimes \mathbb{1}_R) = P_L \otimes \mathbb{1}_R$ gives the claim. $\square$

Theorem 13.5.2 (Right absorption by the marginal support projector). [Lean](#) [✓ Stmt](#) Under the same assumptions, $\rho(P_L \otimes \mathbb{1}_R) = \rho$. For $H_L = H_A \otimes H_X$ and $H_R = H_B$, the two absorption identities are precisely the P_{AX} support-projector identities in [CPGSV16, Appendix D.2, lines 2228–2235].

Proof. [✓ Proof](#) Apply $(\cdot)^\dagger$ to the left absorption identity. Since $\rho^\dagger = \rho$ and $(P_L \otimes \mathbb{1}_R)^\dagger = P_L \otimes \mathbb{1}_R$,

$$\rho(P_L \otimes \mathbb{1}_R) = [(P_L \otimes \mathbb{1}_R)\rho]^\dagger = \rho^\dagger = \rho.$$

□

Definition 13.5.3 (Partial trace over the left factor). [Lean](#) [Lean](#) [✓ Stmt](#) For an operator ρ on $H_L \otimes H_R$, define $(\text{tr}_L \rho)_{r,s} := \sum_l \rho_{(l,r),(l,s)}$.

Theorem 13.5.4 (Left absorption by the right marginal support). [Lean](#) [✓ Stmt](#) Let ρ be positive semidefinite. If P_R is the support projector of $\text{tr}_L \rho$, then $(\mathbb{1}_L \otimes P_R)\rho = \rho$.

Proof. [✓ Proof](#) Set $Q = \mathbb{1}_R - P_R$. The preceding trace identity gives

$$\text{tr}((\mathbb{1}_L \otimes Q)\rho) = \text{tr}(Q \text{tr}_L \rho) = 0,$$

since P_R is the support projector of $\text{tr}_L \rho$. Positivity of ρ then gives $(\mathbb{1}_L \otimes Q)\rho = 0$, and hence $(\mathbb{1}_L \otimes P_R)\rho = \rho$. □

Theorem 13.5.5 (Right absorption by the right marginal support). [Lean](#) [✓ Stmt](#) Under the same assumptions, $\rho(\mathbb{1}_L \otimes P_R) = \rho$.

Proof. [✓ Proof](#) Take adjoints in the left absorption identity. □

Definition 13.5.6 (State-preparation map). [Lean](#) [✓ Stmt](#) Let ρ be a matrix on a finite-dimensional space B . The state-preparation map from matrices on A to matrices on $A \otimes B$ is

$$\mathcal{P}_\rho(X) = X \otimes \rho.$$

This is the elementary preparation operation used in the maps $\mathcal{T}_1$ and $\mathcal{S}_1$ of [CPGSV16, Appendix C.2, lines 1527–1533 and 1551–1555]. It is relocated here from the MPDO renormalization chapter (Chapter ??), whose Kraus-action and conditional-expectation consequences for the density-operator setting cite it across the chapter boundary.

13.6 Extending completely positive maps from operator systems

An *operator system* inside a finite-dimensional C^* -algebra is a subspace closed under the adjoint and containing the unit. Every positive map defined only on such a subspace, with values in a commutative algebra, is automatically completely positive by the remark following Theorem 13.4.7; extending it to a completely positive map on the whole algebra makes that observation useful for arbitrary target algebras as well.

The source states the extension theorem for an operator system inside any finite-dimensional C^* -algebra, that is, any direct sum of matrix algebras. What follows treats the one-summand case, where the ambient algebra is a single full matrix algebra $M_m(\mathbb{C})$; see `docs/paper-gaps/wolf_ch01_operator_system_extension_matrix_scope.tex`.

Theorem 13.6.1 (Left partial trace of a Kronecker product). [Lean](#) [✓ Stmt](#) For square matrices X on A and Y on B , $\text{tr}_A(X \otimes Y) = \text{tr}(X)Y$.

Proof. [✓ Proof](#) For all i, j , $[\text{tr}_A(X \otimes Y)]_{ij} = \sum_t X_{tt}Y_{ij} = \text{tr}(X)Y_{ij}$. □

Theorem 13.6.2 (Positivity of the maximally mixed matrix). [Lean](#) [✓ Stmt](#) For every natural number m , the matrix $m^{-1}\mathbb{1}_m$ is positive semidefinite.

Proof. [✓ Proof](#) The identity matrix is positive semidefinite, and m^{-1} is nonnegative. □

Theorem 13.6.3 (Trace of the maximally mixed matrix). [Lean](#) [✓ Stmt](#) If $m \geq 1$, then $\text{tr}(m^{-1}\mathbb{1}_m) = 1$.

Proof. [✓ Proof](#) Since $\text{tr}(\mathbb{1}_m) = m$, scalar linearity of the trace gives the result. □

Theorem 13.6.4 (Normalized partial-trace conditional expectation). [Lean](#) [Lean](#) [✓ Stmt](#) Let $m \geq 1$. The linear map $E : M_m(\mathbb{C}) \otimes M_d(\mathbb{C}) \rightarrow M_m(\mathbb{C}) \otimes M_d(\mathbb{C})$ defined by

$$E(A) = \mathbb{1}_m \otimes (m^{-1} \text{tr}_m(A)),$$

is trace-preserving and completely positive. It is a conditional expectation onto the unital $*$ -subalgebra $\mathbb{1}_m \otimes M_d(\mathbb{C})$: it is unital and idempotent, its range lies in that subalgebra, and it fixes the subalgebra pointwise. This is the one-block trace-preserving choice in the family described by [Wol12, Proposition 1.5 and Equation (1.40)]; it supplies the corresponding one-block codomain retraction for the extension theorem, not the general finite-dimensional codomain construction.

Proof. [✓ Proof](#) Interchange the two tensor factors, trace out the new right factor, prepare the maximally mixed state $m^{-1}\mathbb{1}_m$, and interchange the factors back. Theorem 13.6.2 and Theorem 13.6.3 show that the prepared matrix is positive semidefinite and has trace one. Theorem ??, Lemma ??, Lemma ??, and Lemma ?? then show that the resulting map is trace-preserving and completely positive. Expanding the composition gives

$$E(A) = \mathbb{1}_m \otimes (m^{-1} \text{tr}_m(A)).$$

The Kronecker-product partial-trace identity gives

$$\text{tr}_m(\mathbb{1}_m \otimes X) = mX,$$

so $E(\mathbb{1}_m \otimes X) = \mathbb{1}_m \otimes X$. Pointwise fixation gives both idempotence and unitality, while the evaluation identity gives the range inclusion. □

Theorem 13.6.5 (Coordinate direct-sum conditional expectation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let K be finite, let $m_k \geq 1$, and set

$$\mathcal{H} = \bigoplus_{k \in K} (\mathbb{C}^{m_k} \otimes \mathbb{C}^{d_k}).$$

For $A \in \text{End}(\mathcal{H})$, write A_{kk} for its k -th diagonal block. The map

$$E(A) = \text{diag}_{k \in K} (\mathbb{1}_{m_k} \otimes (m_k^{-1} \text{tr}_{m_k}(A_{kk})))$$

has zero off-diagonal output blocks. It is trace-preserving and completely positive, and it is a conditional expectation onto the coordinate subalgebra

$$\bigoplus_{k \in K} (\mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C})) \subseteq \text{End}(\mathcal{H}).$$

Thus its range lies in this subalgebra, it fixes the subalgebra pointwise, and it is unital and idempotent. This is the normalized trace-preserving coordinate choice obtained by specializing the block densities in [Wol12, Proposition 1.5 and Equation (1.40)] to maximally mixed densities. Relative to Wolf's $M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$ convention, the displayed $\mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C})$ order is obtained by interchanging the tensor factors. This theorem concerns only the displayed direct-sum coordinates; it does not conjugate the subalgebra by an arbitrary unitary or corestrict the codomain. This coordinate restriction is documented in `docs/paper-gaps/wolf_prop1_5_one_factor_scope.tex`.

Proof. ✓ Proof Compress A to the family $(A_{kk})_{k \in K}$ and apply the normalized partial-trace expectation from Theorem 13.6.4 in each coordinate. The coordinatewise Kraus theorem makes the resulting map on the finite sum completely positive, and its canonical full-matrix extension discards the off-diagonal blocks. Each coordinate preserves trace, so preservation of the total block trace implies preservation of the trace on $\text{End}(\mathcal{H})$. The displayed formula shows that the range lies in the coordinate subalgebra. The one-block fixation identity holds in every summand, hence E fixes that subalgebra pointwise; idempotence and unitality follow. $\square$

Theorem 13.6.6 (Positivity in unitary block coordinates). Lean Lean ✓ Stmt Let $e : H \simeq I$ be an equivalence of finite index sets and let U be a unitary matrix indexed by I . The mutually inverse coordinate changes

$$\Phi(A) = e^{-1}(U^\dagger AU)$$

and

$$\Phi^{-1}(X) = Ue(X)U^\dagger$$

both preserve positive semidefiniteness.

Proof. ✓ Proof Unitary conjugation preserves positive semidefiniteness, as does reindexing a matrix along an equivalence. Apply these facts in the two possible orders. $\square$

Theorem 13.6.7 (Unitary block-coordinate channels). Lean Lean ✓ Stmt Let $e : H \simeq I$ be an equivalence of finite index sets and let U be a unitary matrix indexed by I . Define

$$\Phi(A) = e^{-1}(U^\dagger AU)$$

and

$$\Phi^{-1}(X) = Ue(X)U^\dagger.$$

Both Φ and Φ^{-1} are trace-preserving and completely positive.

Proof. ✓ Proof The forward map is the composition of conjugation by $U^\dagger$ with reindexing by e^{-1} , while the inverse is reindexing by e followed by conjugation by U . Unitarity makes both single-Kraus conjugations trace preserving. Equivalence reindexing is trace preserving and completely positive, and composition preserves these properties. $\square$

Theorem 13.6.8 (Trace-preserving conditional expectation onto a matrix subalgebra). Lean ✓ Stmt For every unital $\ast$ -subalgebra $\mathcal{B} \subseteq M_n(\mathbb{C})$, there is a trace-preserving completely positive map $E : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ that is a conditional expectation onto $\mathcal{B}$: it is unital and idempotent, its range lies in $\mathcal{B}$, and it fixes every element of $\mathcal{B}$. This is the normalized trace-preserving choice in the family described by [Wol12, Proposition 1.5 and Equation (1.40)].

Proof. ✓ Proof By the finite-dimensional block form, choose unitary coordinates in which

$$\mathcal{B} = \bigoplus_k (\mathbb{1}_{m_k} \otimes M_{d_k}(\mathbb{C})).$$

Let E_0 be the coordinate direct-sum expectation and set

$$E = \Phi^{-1} \circ E_0 \circ \Phi.$$

Theorem 13.6.7 and closure under composition show that E is trace-preserving and completely positive. The identities $\Phi\Phi^{-1} = \text{id}$ and $E_0^2 = E_0$ give $E^2 = E$, and all three maps are unital. Finally, the block-membership equivalence identifies $\Phi(\mathcal{B})$ with the coordinate subalgebra. Thus the range of E lies in $\mathcal{B}$, and the pointwise fixation of that coordinate subalgebra by E_0 implies $E(B) = B$ for every $B \in \mathcal{B}$. $\square$

Definition 13.6.9 (Operator system). Lean ✓ Stmt A subspace $S \subseteq M_m(\mathbb{C})$ is an *operator system* if it contains the unit and is closed under the adjoint: $\mathbb{1} \in S$ and $X \in S \implies X^\dagger \in S$.

Definition 13.6.10 (Complete positivity on a subspace). Lean ✓ Stmt Let $S \subseteq M_m(\mathbb{C})$ be a subspace and $T : S \rightarrow M_n(\mathbb{C})$ a linear map. For a natural number k , write $T \otimes \text{id}_k$ for the map sending a $k \times k$ block matrix with entries in S to the block matrix obtained by applying T to every entry. Then T is *completely positive on S* if $T \otimes \text{id}_k$ sends every positive semidefinite such block matrix to a positive semidefinite matrix, for every k (the condition is vacuous at $k = 0$). This makes no reference to S being an operator system; it is applied below to an operator system S because that is the situation in which the extension theorem holds.

For the rest of this section fix an operator system $S \subseteq M_m(\mathbb{C})$ (Definition 13.6.9) and a linear map $T : S \rightarrow M_n(\mathbb{C})$ completely positive on S (Definition 13.6.10), and write

$$\tau(A) := \langle \Omega | (T \otimes \text{id}_n)(A) | \Omega \rangle \quad (A \in S \otimes M_n(\mathbb{C}))$$

for Wolf's functional, with Ω the maximally entangled vector of dimension n .

Lemma 13.6.11 (Norm domination for τ). Lean ✓ Stmt For every Hermitian $A \in S \otimes M_n(\mathbb{C})$,

$$\text{Re } \tau(A) \leq \|A\|_\infty \text{Re } \tau(\mathbb{1}).$$

Proof. ✓ Proof The matrix $\|A\|_\infty \mathbb{1} - A$ is positive semidefinite and lies in $S \otimes M_n(\mathbb{C})$, so complete positivity of T at level n gives that its image under $T \otimes \text{id}_n$ is positive semidefinite. Testing against Ω and using linearity of τ ,

$$\langle \Omega | (T \otimes \text{id}_n)(\|A\|_\infty \mathbb{1} - A) | \Omega \rangle = \|A\|_\infty \tau(\mathbb{1}) - \tau(A) \geq 0;$$

both sides are real since $\|A\|_\infty \mathbb{1} - A$ is Hermitian, and rearranging gives the bound. $\square$

Lemma 13.6.12 (Hahn–Banach extension of τ). Lean ✓ Stmt There is a $\mathbb{C}$ -linear functional τ' on $M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$ agreeing with τ on $S \otimes M_n(\mathbb{C})$ and satisfying $\text{Re } \tau'(A) \leq \|A\|_\infty \text{Re } \tau'(\mathbb{1})$ for every Hermitian $A \in M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$.

Proof. ✓ Proof The functional $f(A) := \|A\|_\infty \tau(\mathbb{1})$ is sublinear, so by Lemma 13.6.11 the Hahn–Banach theorem extends τ , dominated by f , from the Hermitian matrices of $S \otimes M_n(\mathbb{C})$ to those of $M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$ as a real-linear functional. Decomposing each matrix of $M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$ into its Hermitian and skew-Hermitian parts, both of which stay inside $S \otimes M_n(\mathbb{C})$ whenever the original matrix does (since S is closed under the adjoint), extends this further to a $\mathbb{C}$ -linear functional τ' agreeing with τ on all of $S \otimes M_n(\mathbb{C})$ and obeying the same bound throughout. $\square$

Lemma 13.6.13 (Positivity of a dominated functional). [Lean](#) [✓ Stmt](#) Let $\varphi : M_d(\mathbb{C}) \rightarrow \mathbb{C}$ be a $\mathbb{C}$ -linear functional satisfying $\operatorname{Re} \varphi(A) \leq \|A\|_\infty \operatorname{Re} \varphi(\mathbb{1})$ for every Hermitian $A \in M_d(\mathbb{C})$, with $\operatorname{Re} \varphi(\mathbb{1}) \geq 0$. Then $\operatorname{Re} \varphi(A) \geq 0$ for every positive semidefinite $A \in M_d(\mathbb{C})$.

Proof. [✓ Proof](#) The matrix $\|A\|_\infty \mathbb{1} - A$ is positive semidefinite (hence Hermitian) of norm at most $\|A\|_\infty$ whenever A is positive semidefinite, so the domination hypothesis applied to it gives

$$\operatorname{Re} \varphi(\|A\|_\infty \mathbb{1} - A) = \|A\|_\infty \operatorname{Re} \varphi(\mathbb{1}) - \operatorname{Re} \varphi(A) \leq \|A\|_\infty \operatorname{Re} \varphi(\mathbb{1});$$

since $\operatorname{Re} \varphi(\mathbb{1}) \geq 0$, rearranging gives $\operatorname{Re} \varphi(A) \geq 0$. $\square$

Lemma 13.6.14 (The Riesz matrix of the extension is positive semidefinite). [Lean](#) [✓ Stmt](#) Let τ' be a $\mathbb{C}$ -linear functional on $M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$ built from a $\mathbb{R}$ -linear functional by the Hermitian-decomposition construction of Lemma 13.6.12, and suppose $\operatorname{Re} \tau'(X) \geq 0$ for every positive semidefinite $X \in M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$. Let Y be the matrix representing τ' through the trace pairing, $\tau'(X) = \operatorname{tr}(YX)$. Then Y is positive semidefinite.

Proof. [✓ Proof](#) Because τ' is built by the Hermitian-decomposition construction, it sends the adjoint of a matrix to the conjugate of its value, so Y is Hermitian. For a vector v , the trace-pairing identity gives $v^\dagger Y v = \tau'(vv^\dagger)$, and $vv^\dagger$ is positive semidefinite, so the hypothesis gives $\operatorname{Re}(v^\dagger Y v) \geq 0$; since Y is Hermitian, $v^\dagger Y v$ is already real, hence nonnegative. $\square$

Lemma 13.6.15 (Reconstructing a Kraus map from τ'). [Lean](#) [✓ Stmt](#) Let τ' be as in Lemma 13.6.12 and let $Y = \sum_k v_k v_k^\dagger$ be any rank-one decomposition of its Riesz matrix. Then the map $T'(B)_{ij} := n \cdot \tau'(B \otimes |i\rangle\langle j|)$ equals $\sum_k K_k B K_k^\dagger$ for the reshaped operators $K_k : \mathbb{C}^m \rightarrow \mathbb{C}^n$, $K_k := \sqrt{n} v_k^\dagger$ (viewing v_k as an $m \times n$ matrix). Complete positivity of T' is assembled from this identity together with the positive semidefiniteness of Y at Theorem 13.6.17.

Proof. [✓ Proof](#) Both sides expand, entry by entry, to the same double sum $n \sum_k \sum_{p,q} \overline{v_k(p,i)} B_{pq} v_k(q,j)$. $\square$

Lemma 13.6.16 (Wolf's inversion formula holds for T on S). [Lean](#) [✓ Stmt](#) For every $B \in S$ and every i, j ,

$$\tau(B \otimes |i\rangle\langle j|) = \frac{1}{n} T(B)_{ij}.$$

Proof. [✓ Proof](#) Direct computation from the definition of τ , expanding the trace against Ω and collapsing the resulting sum using the structure of $B \otimes |i\rangle\langle j|$. $\square$

Theorem 13.6.17 (Extending cp maps from operator systems). [Lean](#) [✓ Stmt](#) Let $m, n \geq 1$, let $S \subseteq M_m(\mathbb{C})$ be an operator system, and let $T : S \rightarrow M_n(\mathbb{C})$ be a completely positive linear map on S . Then there is a completely positive map $T' : M_m(\mathbb{C}) \rightarrow M_n(\mathbb{C})$, in rectangular Kraus form, that agrees with T on S .

Proof. [✓ Proof](#) By Lemma 13.6.12, τ extends to a $\mathbb{C}$ -linear functional τ' on $M_m(\mathbb{C}) \otimes M_n(\mathbb{C})$ agreeing with τ on $S \otimes M_n(\mathbb{C})$ and obeying the same norm-domination bound throughout. By Lemma 13.6.13, τ' is nonnegative on positive semidefinite matrices, so by Lemma 13.6.14 the matrix Y representing τ' through the trace pairing is positive semidefinite, hence a sum of rank-one terms $Y = \sum_k v_k v_k^\dagger$. Lemma 13.6.15 reshapes these into a rectangular Kraus family for $T'(B)_{ij} := n \cdot \tau'(B \otimes |i\rangle\langle j|)$, so T' is completely positive. Lemma 13.6.16 gives $\tau(B \otimes |i\rangle\langle j|) = \frac{1}{n} T(B)_{ij}$ for $B \in S$, and τ' agrees with τ on $S \otimes M_n(\mathbb{C})$, so T' agrees with T on S . $\square$

13.7 Extending completely positive maps from operator systems in a direct sum of matrix algebras

The one-summand case above (Theorem 13.6.17) is generalized here to the source's full domain generality: the domain ranges over every finite-dimensional C^* -algebra, that is, every finite direct sum of matrix algebras $\bigoplus_{k < r} M_{d_k}(\mathbb{C})$. The generalization transports the one-summand theorem along the block-diagonal embedding of $\bigoplus_{k < r} M_{d_k}(\mathbb{C})$ into a single matrix algebra $M_M(\mathbb{C})$, $M = \sum_k d_k$.

Definition 13.7.1 (Operator system in a direct sum of matrix algebras). [Lean](#) [✓ Stmt](#) A subspace $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ is an *operator system* if it contains the unit and is closed under the entrywise adjoint: $\mathbb{1} \in S$ and $X \in S \implies X^\dagger \in S$, where $X^\dagger$ denotes the family $(X_k^\dagger)_{k < r}$.

Definition 13.7.2 (Complete positivity on a direct-sum operator system). [Lean](#) [✓ Stmt](#) Let $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ be a subspace and $T : S \rightarrow M_p(\mathbb{C})$ a linear map. The level- j ampliation of T is defined by bipartite slicing: for a family $X = (X_k)_{k < r}$ with $X_k \in M_{d_k}(\mathbb{C}) \otimes M_j(\mathbb{C})$ whose slice families lie in S ,

$$((T \otimes \text{id}_j)(X))_{ab} = T\left((X_k)_{ab}\right)_{k < r}, \quad a, b < j, \quad (13.1)$$

and T is *completely positive on S* if entrywise positivity ($\forall k < r, X_k \geq 0$) always implies $(T \otimes \text{id}_j)(X) \geq 0$, for every j . This is Definition 13.6.10 applied block by block, one summand at a time.

Definition 13.7.3 (Complete positivity on a direct sum of matrix algebras). [Lean](#) [✓ Stmt](#) A linear map $T : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ is *completely positive* if, for every natural number j and every family $X = (X_k)_{k < r}$ with $X_k \in M_{d_k}(\mathbb{C}) \otimes M_j(\mathbb{C})$, entrywise positivity implies

$$(\forall k < r, X_k \geq 0) \implies (T \otimes \text{id}_j)(X) \geq 0, \quad (13.2)$$

where the (a, b) slice of $(T \otimes \text{id}_j)(X)$ is T applied to the family of (a, b) slices of the matrices X_k . This is the direct-sum analogue of rectangular Kraus complete positivity (Definition 4.5.2).

Theorem 13.7.4 (Block-diagonal transport at every matrix level). [Lean](#) [Lean](#) [✓ Stmt](#) Let ι_j be the block-diagonal embedding

$$\iota_j : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \otimes M_j(\mathbb{C}) \longrightarrow M_{\sum_k d_k}(\mathbb{C}) \otimes M_j(\mathbb{C}). \quad (13.3)$$

Then $\iota_j(X)$ is positive semidefinite exactly when every X_k is positive semidefinite. Moreover, for all $a, b < j$,

$$(\iota_j(X))_{ab} = \iota((X_k)_{ab})_{k < r}. \quad (13.4)$$

Proof. [✓ Proof](#) The first assertion follows from the corresponding block-diagonal positivity equivalence and invariance of positivity under simultaneous row and column reindexing. The second follows by evaluating both sides of (13.4) at a pair of ambient matrix indices. $\square$

Theorem 13.7.5 (Complete positivity after tensoring with the identity). [Lean](#) [Lean](#) [✓ Stmt](#) If $\Phi(X) = \sum_i A_i X A_i^\dagger$ is a rectangular Kraus completely positive map, then $\Phi \otimes \text{id}_j$ is a Kraus completely positive map with Kraus operators $A_i \otimes \mathbb{1}_j$. In particular, $(\Phi \otimes \text{id}_j)(X)$ is positive semidefinite whenever X is.

Proof. [✓ Proof](#) Expanding matrix entries gives

$$(\Phi \otimes \text{id}_j)(X) = \sum_i (A_i \otimes \mathbb{1}_j) X (A_i \otimes \mathbb{1}_j)^\dagger. \quad (13.5)$$

Each summand preserves positive semidefiniteness, and so does their sum. $\square$

Theorem 13.7.6 (Transport through the block-diagonal embedding). [Lean](#) [Lean](#) [✓ Stmt](#) *Let ι be the block-diagonal embedding of $\bigoplus_{k < r} M_{d_k}(\mathbb{C})$ into $M_M(\mathbb{C})$, $M = \sum_k d_k$. If $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ is an operator system, then $\iota(S)$ is an operator system in $M_M(\mathbb{C})$; and if $T : S \rightarrow M_p(\mathbb{C})$ is completely positive on S , then the induced map $T'' : \iota(S) \rightarrow M_p(\mathbb{C})$, $T''(\iota(X)) = T(X)$, is completely positive on $\iota(S)$.*

Proof. [✓ Proof](#) The image of an operator system under an injective unital $*$ -homomorphism is an operator system. For complete positivity, a level- j matrix over the image $\iota(S)$ compresses blockwise to a family over S : the block-diagonal compression left-inverts the embedding on every slice,

$$(\iota_j(X))_{ab} = \iota((X_k)_{ab})_{k < r}, \quad \text{so} \quad (\text{compress}_j(\iota_j(X)))_k = X_k, \quad (13.6)$$

and each block is positive semidefinite by invariance of positivity under simultaneous row and column reindexing. Complete positivity of T on S applies to this family, and the linear equivalence of S and $\iota(S)$ converts the conclusion back to T'' . $\square$

Theorem 13.7.7 (Extending cp maps from operator systems, direct-sum domain). [Lean](#) [✓ Stmt](#) *Let $d_0, \dots, d_{r-1}$ be natural numbers, not all zero, let $p \geq 1$, let $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ be an operator system (Definition 13.7.1), and let $T : S \rightarrow M_p(\mathbb{C})$ be completely positive on S (Definition 13.7.2). Then there is a completely positive map $T' : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ (Definition 13.7.3) that agrees with T on S .*

Proof. [✓ Proof](#) Let $\iota = \iota_1$ be the block-diagonal embedding. By Theorem 13.7.6, $\iota(S)$ is an operator system in $M_M(\mathbb{C})$ and the induced map T'' is completely positive at every matrix level. Theorem 13.6.17 gives a rectangular Kraus extension $\tilde{T} : M_M(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ of T'' . Define $T' = \tilde{T} \circ \iota$.

To verify (13.2), fix j and an entrywise positive family X . The level embedding $\iota_j(X)$ is positive by Theorem 13.7.4. The Kraus ampliation theorem (Theorem 13.7.5) therefore gives $(\tilde{T} \otimes \text{id}_j)(\iota_j(X)) \geq 0$. Using (13.4) identifies this matrix with $(T' \otimes \text{id}_j)(X)$, proving complete positivity of T' . The defining agreement of $\tilde{T}$ with T'' shows that T' agrees with T on S . $\square$

Theorem 13.7.8 (Postcomposition with a Kraus completely positive map). [Lean](#) [✓ Stmt](#) *Let $F : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ be completely positive and let $E : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ have a Kraus representation. Then $E \circ F$ is completely positive.*

Proof. [✓ Proof](#) At each matrix level j , the ampliations satisfy

$$((E \circ F) \otimes \text{id}_j)(X) = (E \otimes \text{id}_j)((F \otimes \text{id}_j)(X)). \quad (13.7)$$

If every component of X is positive semidefinite, complete positivity of F makes the inner matrix positive semidefinite. The Kraus representation of $E \otimes \text{id}_j$ then preserves its positivity. $\square$

Theorem 13.7.9 (Extension with range in a matrix subalgebra). [Lean](#) [✓ Stmt](#) *Let $d_0, \dots, d_{r-1}$ be natural numbers, not all zero, let $p \geq 1$, let $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ be an operator system, and let $\mathcal{B} \subseteq M_p(\mathbb{C})$ be a unital $*$ -subalgebra. Suppose that $T : S \rightarrow M_p(\mathbb{C})$ is completely positive and $T(S) \subseteq \mathcal{B}$. Then there is a completely positive map $G : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ such that $G(\bigoplus_{k < r} M_{d_k}(\mathbb{C})) \subseteq \mathcal{B}$ and $G|_S = T$.*

Proof. [✓ Proof](#) Theorem 13.7.7 gives a completely positive extension F with values in $M_p(\mathbb{C})$. Choose the conditional expectation $E : M_p(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ from Theorem 13.6.8; its range lies in $\mathcal{B}$. Set $G = E \circ F$. Theorem 13.7.8 shows that G is completely positive, while $E(M_p(\mathbb{C})) \subseteq \mathcal{B}$. For $x \in S$, one has $F(x) = T(x) \in \mathcal{B}$, and therefore $G(x) = E(T(x)) = T(x)$. $\square$

Theorem 13.7.10 (Extending cp maps from operator systems to matrix subalgebras). [Lean](#)
[✓ Stmt](#) Let $d_0, \dots, d_{r-1}$ be natural numbers, not all zero, let $p \geq 1$, let $S \subseteq \bigoplus_{k < r} M_{d_k}(\mathbb{C})$ be an operator system, let $\mathcal{B} \subseteq M_p(\mathbb{C})$ be a unital $*$ -subalgebra, and let $T : S \rightarrow \mathcal{B}$ be completely positive, where positivity in $\mathcal{B}$ is read through its inclusion in $M_p(\mathbb{C})$. Then there is a linear map

$$\tilde{T} : \bigoplus_{k < r} M_{d_k}(\mathbb{C}) \longrightarrow \mathcal{B} \quad (13.8)$$

whose inclusion into $M_p(\mathbb{C})$ is completely positive and which satisfies $\tilde{T}|_S = T$.

Proof. [✓ Proof](#) Apply Theorem 13.7.9 to the composite of T with the inclusion $\mathcal{B} \hookrightarrow M_p(\mathbb{C})$. Since the resulting ambient map G takes every value in $\mathcal{B}$, it defines a map $\tilde{T}$ with codomain $\mathcal{B}$. Its composite with the inclusion is G , hence is completely positive, and equality in $M_p(\mathbb{C})$ gives $\tilde{T}|_S = T$. $\square$

13.8 Set spectra and spectral multiplicities

For a matrix $A \in M_d(\mathbb{C})$, the spectrum $\sigma(A)$ is a set: it does not record how often a root of the characteristic polynomial occurs. Wolf's spectrum-preserver hypothesis is precisely equality of these sets on Hermitian inputs. The next results justify the continuity argument in [Wol12, Chapter 1, Spectrum preserving maps] without replacing that hypothesis by multiplicity preservation.

Theorem 13.8.1 (Hermitian trace power sums). [Lean](#) [✓ Stmt](#) If A is Hermitian with eigenvalues $\lambda_0, \dots, \lambda_{d-1}$, listed with multiplicity, then for every $k \in \mathbb{N}$,

$$\mathrm{tr}(A^k) = \sum_{i=0}^{d-1} \lambda_i^k. \quad (13.9)$$

This includes $d = 0$, when both sides are empty sums.

Proof. [✓ Proof](#) Diagonalize $A = UDU^\dagger$, where $D = \mathrm{diag}(\lambda_0, \dots, \lambda_{d-1})$. Unitary conjugation commutes with powers and leaves the trace unchanged, while $D^k = \mathrm{diag}(\lambda_0^k, \dots, \lambda_{d-1}^k)$. $\square$

Definition 13.8.2 (Ordered simple-spectrum perturbation). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A = U \mathrm{diag}(\lambda_0, \dots, \lambda_{d-1}) U^\dagger$ be Hermitian with $\lambda_0 \geq \dots \geq \lambda_{d-1}$. For $\varepsilon \in \mathbb{R}$, define

$$\lambda_i(\varepsilon) = \lambda_i + \varepsilon(d - i). \quad (13.10)$$

Define the corresponding Hermitian matrix by

$$A_\varepsilon = U \mathrm{diag}(\lambda_0(\varepsilon), \dots, \lambda_{d-1}(\varepsilon)) U^\dagger. \quad (13.11)$$

Theorem 13.8.3 (Simple spectrum and convergence of the perturbations). [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $\varepsilon > 0$, then $\lambda_0(\varepsilon) > \dots > \lambda_{d-1}(\varepsilon)$, so the roots of χ_{A_ε} have no repetitions. Moreover, $A_\varepsilon \rightarrow A$ as $\varepsilon \rightarrow 0$. In particular, the positive sequence $\varepsilon_n = 1/(n+1)$ gives simple-spectrum matrices $A_{\varepsilon_n} \rightarrow A$. All assertions remain valid for $d = 0$.

Proof. ✓ Proof For $i < j$, monotonicity of the ordered eigenvalues and positivity of ε give

$$\lambda_j + \varepsilon(d - j) < \lambda_i + \varepsilon(d - i). \quad (13.12)$$

Thus the diagonal entries in (13.11) are distinct. The same formula is continuous in ε , and at $\varepsilon = 0$ it is the spectral decomposition of A . $\square$

Theorem 13.8.4 (Set spectrum determines multiplicities at simple spectrum). Lean ✓ Stmt Let $A, B \in M_d(\mathbb{C})$. Suppose $\sigma(B) = \sigma(A)$ and the roots of χ_A have no repetitions. Then the root multisets of χ_A and χ_B are equal.

Proof. ✓ Proof The common spectrum is the common finite set underlying the two root multisets. The roots of χ_A are distinct, so this set has d elements. Both characteristic polynomials split over $\mathbb{C}$ and have degree d , so the root multiset of χ_B also has cardinality d . Hence it cannot repeat any root, and the two root multisets agree. $\square$

Theorem 13.8.5 (Hermitian trace powers from simple set spectrum). Lean ✓ Stmt Let $A, B \in M_d(\mathbb{C})$ be Hermitian. Suppose $\sigma(B) = \sigma(A)$ and the roots of χ_A have no repetitions. Then $\text{tr}(B^k) = \text{tr}(A^k)$ for every $k \in \mathbb{N}$.

Proof. ✓ Proof Theorem 13.8.4 gives equality of the root multisets. Formula (13.9) then gives equality of trace powers. $\square$

Theorem 13.8.6 (Continuity of trace powers). Lean Lean ✓ Stmt For each $k \in \mathbb{N}$, the function $A \mapsto \text{tr}(A^k)$ on $M_d(\mathbb{C})$ is continuous. Thus $A_n \rightarrow A$ implies $\text{tr}(A_n^k) \rightarrow \text{tr}(A^k)$.

Proof. ✓ Proof Finite matrix multiplication is continuous, hence so is $A \mapsto A^k$; the trace is a finite sum of matrix entries. $\square$

Theorem 13.8.7 (Newton–Girard trace recursion). Lean Lean ✓ Stmt If $A, B \in M_n(\mathbb{C})$ satisfy $\text{tr}(A^k) = \text{tr}(B^k)$ for all $k \geq 1$, then A and B have the same characteristic polynomial. It is enough to assume this equality for $1 \leq k \leq n$.

Proof. ✓ Proof Let e_m denote the m th elementary symmetric polynomial in the eigenvalues, with $e_0 = 1$. The Newton–Girard recursion

$$me_m = \sum_{i=1}^m (-1)^{i-1} e_{m-i} \text{tr}(A^i) \quad (13.13)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces $\text{tr}(A^i)$. Equal power sums for A and B therefore give equal coefficients, hence equal characteristic polynomials. $\square$

Theorem 13.8.8 (Characteristic-polynomial preservation from Hermitian set spectra). Lean ✓ Stmt Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be complex linear. Suppose that $T(A)$ is Hermitian whenever A is Hermitian and that

$$\sigma(T(A)) = \sigma(A) \quad (13.14)$$

for every Hermitian A . Then every Hermitian A satisfies

$$\chi_{T(A)} = \chi_A. \quad (13.15)$$

Proof. [✓ Proof](#) Fix Hermitian A and use the positive simple-spectrum perturbations A_{ε_n} from Theorem 13.8.3. The hypothesis (13.14) and Theorem 13.8.5 give

$$\mathrm{tr}(T(A_{\varepsilon_n})^k) = \mathrm{tr}(A_{\varepsilon_n}^k) \quad (13.16)$$

for every n and k . Linearity makes T continuous in finite dimension. Since $A_{\varepsilon_n} \rightarrow A$, continuity from Theorem 13.8.6 yields $\mathrm{tr}(T(A)^k) = \mathrm{tr}(A^k)$. For $1 \leq k \leq d$, Newton–Girard identities determine all coefficients of the degree- d characteristic polynomial, proving $\chi_{T(A)} = \chi_A$. When $d = 0$, the same argument consists of empty sums and degree-zero characteristic polynomials. $\square$

For a finite Hermitian matrix, the characteristic polynomial records the ordered eigenvalues with their multiplicities. This determines the matrix up to unitary conjugation.

Theorem 13.8.9 (Hermitian matrices with the same characteristic polynomial). [Lean](#) [✓ Stmt](#) *Let $A, B \in M_d(\mathbb{C})$ be Hermitian. If their characteristic polynomials agree, then there is a unitary matrix $U \in M_d(\mathbb{C})$ such that*

$$B = UAU^\dagger. \quad (13.17)$$

Proof. [✓ Proof](#) Equality of the characteristic polynomials gives equality of the ordered eigenvalue lists. Write the two spectral decompositions as

$$A = U_A D U_A^\dagger. \quad (13.18)$$

Likewise,

$$B = U_B D U_B^\dagger. \quad (13.19)$$

Set

$$U = U_B U_A^\dagger. \quad (13.20)$$

This matrix is unitary, and cancellation of $U_A^\dagger U_A$ gives

$$B = UAU^\dagger. \quad (13.21)$$

$\square$

13.9 Pure states, projective rays, and Wigner’s theorem

A nonzero vector $v \in \mathbb{C}^d$ determines a ray $[v]$ and the normalized rank-one matrix

$$P_{[v]} = \frac{|v\rangle\langle v|}{\langle v, v \rangle}. \quad (13.22)$$

The quotient in (13.22) is unchanged when v is multiplied by a nonzero scalar.

Definition 13.9.1 (Projective pure-state matrix). [Lean](#) [✓ Stmt](#) For a ray $p = [v]$ in $\mathbb{C}^d$, define

$$P_p = \frac{|v\rangle\langle v|}{\langle v, v \rangle}. \quad (13.23)$$

Definition 13.9.2 (Projective transition probability). [Lean](#) [✓ Stmt](#) For each ray p , fix a nonzero representative p^{rep} . Define

$$\text{tp}(p, q) = \frac{\langle p^{\text{rep}}, q^{\text{rep}} \rangle \langle q^{\text{rep}}, p^{\text{rep}} \rangle}{\langle p^{\text{rep}}, p^{\text{rep}} \rangle \langle q^{\text{rep}}, q^{\text{rep}} \rangle}. \quad (13.24)$$

Theorem 13.9.3 (Projective rays as normalized pure states). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The matrix P_p is an orthogonal projection with trace one, and

$$\text{tr}(P_p P_q) = \text{tp}(p, q). \quad (13.25)$$

The map $p \mapsto P_p$ is injective. Moreover, a unitary U and coordinatewise conjugation act by

$$P_{U \cdot p} = U P_p U^\dagger, \quad P_{\bar{p}} = P_p^\top. \quad (13.26)$$

These are the pure-state matrix identities used in [Wol12, Chapter 1, Wigner’s theorem].

Proof. [✓ Proof](#) The rank-one identity $|v\rangle\langle v|^2 = \langle v, v \rangle |v\rangle\langle v|$ gives idempotence, while taking the adjoint gives Hermiticity. The trace of $|v\rangle\langle v|$ is $\langle v, v \rangle$, and

$$\text{tr}(P_p P_q) = \frac{\langle p^{\text{rep}}, q^{\text{rep}} \rangle \langle q^{\text{rep}}, p^{\text{rep}} \rangle}{\langle p^{\text{rep}}, p^{\text{rep}} \rangle \langle q^{\text{rep}}, q^{\text{rep}} \rangle} = \text{tp}(p, q). \quad (13.27)$$

Equality $P_p = P_q$ implies that a representative of one ray is a nonzero scalar multiple of a representative of the other. Finally, matrix multiplication sends $|v\rangle\langle v|$ to $|Uv\rangle\langle Uv|$, and coordinatewise conjugation exchanges the two factors of the outer product. $\square$

Theorem 13.9.4 (Recognizing pure states by the characteristic polynomial). [Lean](#) [✓ Stmt](#) Let $A \in M_d(\mathbb{C})$ be Hermitian, and let p be a projective ray. If

$$\chi_A = \chi_{P_p}, \quad (13.28)$$

then there is a projective ray q such that

$$P_q = A. \quad (13.29)$$

Proof. [✓ Proof](#) The matrix P_p is Hermitian. Equality of the characteristic polynomials therefore gives a unitary U such that

$$A = U P_p U^\dagger. \quad (13.30)$$

Taking $q = U \cdot p$ and using $P_{U \cdot p} = U P_p U^\dagger$ proves the claim. $\square$

Theorem 13.9.5 (Normalized and unnormalized pure states). [Lean](#) [✓ Stmt](#) For every nonzero $v \in \mathbb{C}^d$,

$$\langle v, v \rangle P_{[v]} = |v\rangle\langle v|. \quad (13.31)$$

Proof. [✓ Proof](#) Substitute the definition $P_{[v]} = \langle v, v \rangle^{-1} |v\rangle\langle v|$ and use $\langle v, v \rangle \neq 0$. $\square$

Theorem 13.9.6 (Pure-state extensionality for linear maps). [Lean](#) [✓ Stmt](#) Let $T, S : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be complex-linear maps. If

$$T(P_p) = S(P_p) \quad (13.32)$$

for every ray p in $\mathbb{C}^d$, then $T = S$.

Proof. ✓ Proof For $v \neq 0$, multiply the equality at $p = [v]$ by $\langle v, v \rangle$ to obtain $T(|v\rangle\langle v|) = S(|v\rangle\langle v|)$. The same equality is immediate for $v = 0$. Rank-one extensionality now gives $T = S$. $\square$

Theorem 13.9.7 (Transition probability on representatives). Lean ✓ Stmt For arbitrary nonzero representatives $v, w \in \mathbb{C}^d$,

$$\text{tp}([v], [w]) = \frac{\langle v, w \rangle \langle w, v \rangle}{\langle v, v \rangle \langle w, w \rangle}. \quad (13.33)$$

Proof. ✓ Proof By the trace-product identity above, $\text{tp}([v], [w]) = \text{tr}(P_{[v]}P_{[w]})$. Moreover,

$$\text{tr}(P_{[v]}P_{[w]}) = \frac{\text{tr}(|v\rangle\langle v||w\rangle\langle w|)}{\langle v, v \rangle \langle w, w \rangle} = \frac{\langle v, w \rangle \langle w, v \rangle}{\langle v, v \rangle \langle w, w \rangle}, \quad (13.34)$$

which gives the displayed quotient. $\square$

Theorem 13.9.8 (Characteristic polynomial of two pure states). Lean Lean Lean Lean ✓ Stmt There are no projective rays when $d = 0$. When $d = 1$, every pure-state matrix is the identity and

$$\chi_{P_p+P_q}(X) = X - 2. \quad (13.35)$$

If $d \geq 2$, then

$$\chi_{P_p+P_q}(X) = X^{d-2}((X-1)^2 - \text{tr}(P_pP_q)). \quad (13.36)$$

Proof. ✓ Proof For $d \geq 2$, choose nonzero representatives v, w of p, q , and define $A \in M_{d,2}(\mathbb{C})$ and $B \in M_{2,d}(\mathbb{C})$ by

$$A = \begin{pmatrix} v & w \end{pmatrix}. \quad (13.37)$$

Define the second factor by

$$B = \begin{pmatrix} \langle v, v \rangle^{-1} v^\dagger \\ \langle w, w \rangle^{-1} w^\dagger \end{pmatrix}. \quad (13.38)$$

Then $AB = P_p + P_q$. The characteristic-polynomial identity for rectangular products gives

$$\chi_{AB}(X) = X^{d-2}\chi_{BA}(X). \quad (13.39)$$

Direct calculation yields

$$\text{tr}(BA) = 2. \quad (13.40)$$

It also gives

$$\det(BA) = 1 - \text{tr}(P_pP_q). \quad (13.41)$$

Substitution into the quadratic characteristic polynomial of BA proves (13.36). In dimension one, trace one forces each pure-state matrix to equal the identity. The zero-dimensional claim follows because a projective ray would require a nonzero vector in $\mathbb{C}^0$. $\square$

Theorem 13.9.9 (Recovering transition probability from sum characteristic polynomials). [Lean](#)

[✓ Stmt](#) For projective rays p, q, r, s in $\mathbb{C}^d$, if

$$\chi_{P_p+P_q} = \chi_{P_r+P_s}, \quad (13.42)$$

then

$$\text{tr}(P_p P_q) = \text{tr}(P_r P_s). \quad (13.43)$$

Proof. [✓ Proof](#) For $d \geq 2$, apply (13.36) to both sums and cancel the nonzero polynomial X^{d-2} . Equality of the remaining constant terms gives the result. In dimension one both traces equal one, while in dimension zero there are no rays. $\square$

Theorem 13.9.10 (Projective Wigner rigidity). [Lean](#) [✓ Stmt](#) Let f be a map on the rays of $\mathbb{C}^d$ such that

$$\text{tp}(f(p), f(q)) = \text{tp}(p, q) \quad (13.44)$$

for all rays p, q . Then there is a unitary U such that either $f(p) = U \cdot p$ for every p , or $f(p) = U \cdot \bar{p}$ for every p .

Proof. [✓ Proof](#) The canonical linear isometry between $\mathbb{C}^d$ and its Euclidean-space realization induces inverse maps on rays. This identification preserves transition probabilities and commutes with both the unitary action and coordinatewise conjugation. Apply projective Wigner rigidity on the Euclidean-space realization and return the two alternatives through the inverse identification. $\square$

Theorem 13.9.11 (Wigner's theorem). [Lean](#) [✓ Stmt](#) Let F be a bijection of the normalized pure states P_p such that $\text{tr}(F(P)F(Q)) = \text{tr}(PQ)$ for all normalized pure states P, Q . Then there is a unitary U such that either $F(P) = UPU^\dagger$ for every P , or $F(P) = UP^\top U^\dagger$ for every P . This is a disjunction, not an exclusive alternative. For $d = 0$ the set of pure states is empty, while for $d = 1$ it is a singleton and the two alternatives coincide.

Proof. [✓ Proof](#) Use the bijection between rays and the range of $p \mapsto P_p$ to regard F as a map of projective rays. The trace identity $\text{tr}(P_p P_q) = \text{tp}(p, q)$ shows that this map preserves transition probabilities. Projective Wigner rigidity gives a unitary U and either $p \mapsto U \cdot p$ or $p \mapsto U \cdot \bar{p}$. The identities $P_{U \cdot p} = U P_p U^\dagger$ and $P_{\bar{p}} = P_p^\top$ give the two stated formulas. $\square$

Theorem 13.9.12 (Spectrum preserving maps). [Lean](#) [✓ Stmt](#) Let $T : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be complex linear. Suppose that

$$T(X^\dagger) = T(X)^\dagger \quad (13.45)$$

for every $X \in M_d(\mathbb{C})$, and that

$$\sigma(T(A)) = \sigma(A) \quad (13.46)$$

for every Hermitian $A \in M_d(\mathbb{C})$. Then there is a unitary U such that either

$$T(X) = UXU^\dagger \quad (13.47)$$

for every $X \in M_d(\mathbb{C})$, or

$$T(X) = UX^\top U^\dagger \quad (13.48)$$

for every $X \in M_d(\mathbb{C})$. This is the spectrum-preserver classification in [Wol12, Chapter 1, Spectrum preserving maps].

Proof. [✓ Proof](#) The adjoint identity shows that $T(A)$ is Hermitian whenever A is Hermitian. Hence Theorem 13.8.8 gives

$$\chi_{T(A)} = \chi_A \quad (13.49)$$

for every Hermitian A . For each projective ray p , recognize $T(P_p)$ by its characteristic polynomial and choose a ray $f(p)$ such that

$$P_{f(p)} = T(P_p). \quad (13.50)$$

For rays p, q , linearity and characteristic-polynomial preservation give

$$\chi_{P_{f(p)}+P_{f(q)}} = \chi_{T(P_p+P_q)} = \chi_{P_p+P_q}. \quad (13.51)$$

Theorem 13.9.9 therefore shows that f preserves transition probabilities. Projective Wigner rigidity gives a unitary U and either

$$f(p) = U \cdot p \quad (13.52)$$

for every p , or

$$f(p) = U \cdot \bar{p} \quad (13.53)$$

for every p . Thus T agrees on every P_p with, respectively, $X \mapsto UXU^\dagger$ or $X \mapsto UX^\top U^\dagger$. Pure-state extensionality gives the corresponding equality on all matrices. $\square$

13.10 Maximally entangled states and Choi matrices

The maximally entangled state and the flip operator give the basic matrices used by the Choi–Jamiołkowski correspondence.

Definition 13.10.1 (Maximally entangled state). [Lean](#) [✓ Stmt](#) For every $d \in \mathbb{N}$, define $|\Omega\rangle\langle\Omega|$ on $\mathbb{C}^d \otimes \mathbb{C}^d$ using

$$|\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle. \quad (13.54)$$

When $d \geq 1$, this is the *maximally entangled state*: a rank-one projector with $(|\Omega\rangle\langle\Omega|)_{(i_1, i_2), (j_1, j_2)} = \frac{1}{d} \delta_{i_1 i_2} \delta_{j_1 j_2}$ and $\text{tr}(|\Omega\rangle\langle\Omega|) = 1$.

Definition 13.10.2 (Flip operator). [Lean](#) [✓ Stmt](#) The *flip operator* F on $\mathbb{C}^d \otimes \mathbb{C}^d$ is

$$F = \sum_{i, j=0}^{d-1} |ij\rangle\langle ji|. \quad (13.55)$$

Its matrix entries are

$$F_{(i_1, i_2), (j_1, j_2)} = \begin{cases} 1, & i_1 = j_2 \text{ and } i_2 = j_1, \\ 0, & \text{otherwise.} \end{cases}$$

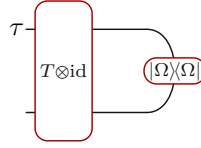
Definition 13.10.3 (Two-dimensional antisymmetric vector). [Lean](#) [✓ Stmt](#) In $\mathbb{C}^2 \otimes \mathbb{C}^2$, let $v = |01\rangle - |10\rangle$. Equivalently, $v_{(0,1)} = 1$, $v_{(1,0)} = -1$, and all other coefficients are zero.

Definition 13.10.4 (Tensor product of a map with identity). [Lean](#) [✓ Stmt](#) Given a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the *tensor extension* $T \otimes \text{id}$ acts on bipartite matrices $X \in M_{D \times D}(\mathbb{C})$ by applying T to each “slice”:

$$(T \otimes \text{id})(X)_{(i_1, i_2), (j_1, j_2)} = (T(X^{(i_2, j_2)}))_{i_1 j_1}, \quad (13.56)$$

where $X_{ab}^{(i_2, j_2)} = X_{(a, i_2), (b, j_2)}$ is the bipartite slice.

The Choi–Jamiołkowski isomorphism bends the input legs of the channel T around the maximally entangled pair $|\Omega\rangle$, presenting the map as the matrix $\tau = (T \otimes \text{id})|\Omega\rangle\langle\Omega|$ on the doubled space:



Definition 13.10.5 (Choi matrix). [Lean](#) [✓ Stmt](#) The *Choi matrix* of a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is

$$\tau = (T \otimes \text{id})(|\Omega\rangle\langle\Omega|) \in M_{D \times D}(\mathbb{C}). \quad (13.57)$$

Concretely, $\tau_{(i_1, i_2), (j_1, j_2)} = \frac{1}{D} (T(E_{i_2 j_2}))_{i_1 j_1}$ where $E_{i_2 j_2}$ is the matrix unit. This is the Choi–Jamiołkowski convention of [Wol12, Proposition 2.1].

Theorem 13.10.6 (Choi matrix of the identity map). [Lean](#) [✓ Stmt](#) The identity channel has Choi matrix $|\Omega\rangle\langle\Omega|$.

Proof. [✓ Proof](#) Apply the identity map in the definition of the Choi matrix. $\square$

Lemma 13.10.7 (The Choi correspondence is one-to-one). [Lean](#) [Lean](#) [✓ Stmt](#) Let $D \geq 1$. Two linear maps $T, S : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ with the same Choi matrix are equal.

Proof. [✓ Proof](#) The (i_2, j_2) slice of $|\Omega\rangle\langle\Omega|$ is $D^{-1}E_{i_2 j_2}$, so equality of the Choi matrices gives $D^{-1}T(E_{i_2 j_2}) = D^{-1}S(E_{i_2 j_2})$ entrywise for every matrix unit $E_{i_2 j_2}$. Cancel the nonzero factor D^{-1} and extend from the matrix units to all of $M_D(\mathbb{C})$ by linearity. $\square$

Theorem 13.10.8 (Choi matrix of transposition). [Lean](#) [✓ Stmt](#) Let $\theta(X) = X^T$ be matrix transposition on $M_D(\mathbb{C})$, with $D \geq 1$. Then

$$(\theta \otimes \text{id})(|\Omega\rangle\langle\Omega|) = \frac{1}{D}F. \quad (13.58)$$

This is [Wol12, Equation 3.1].

Proof. [✓ Proof](#) The (i_2, j_2) slice of $|\Omega\rangle\langle\Omega|$ is $D^{-1}E_{i_2 j_2}$. Transposition sends this matrix unit to $D^{-1}E_{j_2 i_2}$, whose (i_1, j_1) entry is D^{-1} exactly when $i_1 = j_2$ and $i_2 = j_1$. $\square$

Theorem 13.10.9 (Choi decomposition of a Kraus map). [Lean](#) [✓ Stmt](#) Let $D \geq 1$ and let $T(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger$. Define the vector v_i by $(v_i)_{(a, b)} = D^{-1/2} K_i(a, b)$, without transposing the matrix indices. Then

$$\tau_T = \sum_{i=0}^{r-1} |v_i\rangle\langle v_i|. \quad (13.59)$$

Proof. [✓ Proof](#) The (i_2, j_2) slice of $|\Omega\rangle\langle\Omega|$ is $D^{-1}E_{i_2 j_2}$. Therefore the $((i_1, i_2), (j_1, j_2))$ entry of the Choi matrix is

$$\frac{1}{D} \sum_x K_x(i_1, i_2) \overline{K_x(j_1, j_2)}, \quad (13.60)$$

which is the corresponding entry of the stated sum of outer products. $\square$

Theorem 13.10.10 (Complete positivity iff the Choi matrix is positive semidefinite). [Lean](#) [✓ Stmt](#) Let $D \geq 1$. A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is completely positive (in the Kraus sense) if and only if its Choi matrix $\tau \geq 0$. This is the $d = d'$ specialization of [Wol12, Proposition 2.1].

Proof. [✓ Proof](#) ($\Rightarrow$) If $T(X) = \sum_i K_i X K_i^\dagger$, then $\tau = \sum_i |v_i\rangle\langle v_i|$ where $(v_i)_{(a,b)} = c K_i(a, b)$ and $c = 1/\sqrt{D}$. This is a sum of rank-one PSD matrices.

($\Leftarrow$) If $\tau \geq 0$, write $\tau = \sum_m |v_m\rangle\langle v_m|$ by spectral decomposition, define $K_m(a, b) = v_m(a, b)/c$, and recover $T(X) = \sum_m K_m X K_m^\dagger$ by linearity on matrix units. $\square$

Unitary changes of Kraus operators preserve the corresponding completely positive map.

Theorem 13.10.11 (Unitary freedom in Kraus operators). [Lean](#) [✓ Stmt](#) Let U be a unitary $r \times r$ matrix ($U^\dagger U = \mathbb{1}$) and suppose $K_j = \sum_\ell U_{j\ell} \tilde{K}_\ell$. Then $\{K_j\}$ and $\{\tilde{K}_\ell\}$ define the same Kraus map: for every $X \in M_D(\mathbb{C})$,

$$\sum_j K_j X K_j^\dagger = \sum_\ell \tilde{K}_\ell X \tilde{K}_\ell^\dagger.$$

Proof. [✓ Proof](#) Substitute the combination formula, expand the triple sum over j, ℓ, ℓ' , swap summation order, and use the entry-wise form of $U^\dagger U = \mathbb{1}$ to collapse to $\ell = \ell'$ diagonal terms. $\square$

Theorem 13.10.12 (Rectangular Kraus freedom). [Lean](#) [✓ Stmt](#) If $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$ for all X and $r_2 \leq r_1$ (where $\{B_\alpha\}_{\alpha=0}^{r_1-1}$ is the larger family and $\{A_j\}_{j=0}^{r_2-1}$ the smaller), then there exists a rectangular isometry V ($r_1 \times r_2$, $V^\dagger V = \mathbb{1}_{r_2}$) such that $B_\alpha = \sum_j V_{\alpha j} A_j$.

Proof. [✓ Proof](#) Pad the smaller family by zero operators to a family A' of size r_1 . Evaluating equality of the two maps on matrix units and taking matrix entries gives equality of the Gram matrices of the Stinespring vectors, $M_B^\dagger M_B = M_{A'}^\dagger M_{A'}$. Hence the assignment from each column of $M_{A'}$ to the corresponding column of M_B is well-defined and isometric on their common span. Extend this isometry to the ambient space $\mathbb{C}^{r_1}$ and let W be its unitary matrix. Restricting W to the original r_2 coordinates gives a rectangular matrix V with $V^\dagger V = \mathbb{1}_{r_2}$. Reading the column identity entrywise yields $B_\alpha = \sum_j V_{\alpha j} A_j$. $\square$

13.11 No information without disturbance

An *instrument* on $M_d(\mathbb{C})$ (Definition ??) is a finite family $\{T_\alpha : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})\}$ of completely positive maps whose sum is trace-preserving; the outcome α occurs with probability $\text{tr}[T_\alpha(\rho)]$ on the input ρ . The instrument leaves the input undisturbed on average when $\sum_\alpha T_\alpha = \text{id}$. In that case the outcome distribution is the same for every input, so the measurement returns no information about ρ . This is the proposition “No information without disturbance” of [Wol12, Chapter 2].

Lemma 13.11.1 (Domination by the maximally entangled projector). [Lean](#) [✓ Stmt](#) Let A on $\mathbb{C}^d \otimes \mathbb{C}^d$ be positive semidefinite with $A \leq |\Omega\rangle\langle\Omega|$. Then $A = a|\Omega\rangle\langle\Omega|$ for a non-negative scalar a .

Proof. [✓ Proof](#) The projector $|\Omega\rangle\langle\Omega|$ is the rank-one matrix built from the vector $|\Omega\rangle$, so this is Lemma A.2.2 with $c = 1$ and $\psi = \Omega$, read after transporting along a bijection between the pair index set of the bipartite space $\mathbb{C}^d \otimes \mathbb{C}^d$ and a single index set of the same cardinality d^2 . $\square$

Theorem 13.11.2 (No information without disturbance). [Lean](#) [✓ Stmt](#) Let $d \geq 1$ and let $\{T_\alpha : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})\}$ be a finite family of completely positive maps with $\sum_\alpha T_\alpha = \text{id}$. Then for every α there is a constant $c_\alpha \geq 0$ with $T_\alpha = c_\alpha \text{id}$.

Proof. [✓ Proof](#) Pass to Choi matrices. The Choi matrix of id is $|\Omega\rangle\langle\Omega|$ and the Choi correspondence is linear, so the hypothesis reads $|\Omega\rangle\langle\Omega| = \sum_\alpha \tau_\alpha$, where τ_α is the Choi matrix of T_α . Each τ_α is positive semidefinite because T_α is completely positive, so every summand is dominated by the whole sum: $\tau_\alpha \leq |\Omega\rangle\langle\Omega|$. Then Lemma 13.11.1 gives $\tau_\alpha = c_\alpha |\Omega\rangle\langle\Omega|$ with $c_\alpha \geq 0$. The right-hand side is the Choi matrix of $c_\alpha \text{id}$, so Lemma 13.10.7 gives $T_\alpha = c_\alpha \text{id}$. $\square$

Theorem 13.11.3 (Normalization of the outcome weights). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 13.11.2 the constants satisfy $c_\alpha \geq 0$ and $\sum_\alpha c_\alpha = 1$.

Proof. [✓ Proof](#) Evaluate $\sum_\alpha T_\alpha = \text{id}$ at the identity matrix and take the trace: the left-hand side gives $d \sum_\alpha c_\alpha$ and the right-hand side gives d . Divide by $d \geq 1$. $\square$

Theorem 13.11.4 (The outcome probability is independent of the input). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 13.11.2 there is, for every α , a constant $c_\alpha \geq 0$ with $\text{tr}[T_\alpha(\rho)] = c_\alpha$ for every ρ of unit trace. The probability of the outcome α therefore does not depend on the input, so no information is gained.

Proof. [✓ Proof](#) By Theorem 13.11.2, $T_\alpha(\rho) = c_\alpha \rho$, whose trace is $c_\alpha \text{tr}(\rho) = c_\alpha$. $\square$

Corollary 13.11.5 (No information without disturbance for an instrument). [Lean](#) [✓ Stmt](#) Let $\{\Phi_i\}$ be a quantum instrument on $M_d(\mathbb{C})$ with $d \geq 1$ whose total channel is the identity. Then for every outcome i the probability $p_i(\rho)$ is a non-negative constant, the same for every ρ of unit trace.

Proof. [✓ Proof](#) The component maps of an instrument are completely positive and their sum is the identity by hypothesis, so Theorem 13.11.4 applies. $\square$

13.12 Additional results on multiplicative domains and order

13.12.1 Abstract Schwarz maps and their multiplicative domains

Definition 13.12.1.1 (Schwarz inequality for a linear map). [Lean](#) [✓ Stmt](#) A complex-linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ satisfies the Schwarz inequality if, for every $A \in M_D(\mathbb{C})$,

$$E(A^\dagger A) - E(A^\dagger)E(A) \succeq 0. \quad (13.61)$$

This is [Wol12, Equation (5.2)].

Definition 13.12.1.2 (Abstract one-sided multiplicative domains). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a complex-linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, set

$$\begin{aligned}\mathcal{A}_R(E) &= \{A : E(XA) = E(X)E(A) \text{ for every } X\}, \\ \mathcal{A}_L(E) &= \{A : E(AX) = E(A)E(X) \text{ for every } X\}, \\ \mathcal{A}(E) &= \mathcal{A}_R(E) \cap \mathcal{A}_L(E).\end{aligned}$$

These are [Wol12, Equations (5.11)–(5.12)], with the source’s convention that the subscript records the side on which A multiplies the varying matrix.

Theorem 13.12.1.3 (Abstract multiplicative-domain characterization). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If E satisfies the Schwarz inequality, then

$$\begin{aligned}A \in \mathcal{A}_R(E) &\iff E(A^\dagger A) = E(A^\dagger)E(A), \\ A \in \mathcal{A}_L(E) &\iff E(AA^\dagger) = E(A)E(A^\dagger).\end{aligned}$$

Consequently, $A \in \mathcal{A}(E)$ exactly when both equalities hold. This is [Wol12, Equations (5.13)–(5.14)].

Proof. [✓ Proof](#) Suppose first that equality holds at $A^\dagger A$. Apply (13.61) to $A + tY$ and $A + itY$ for real t . Set

$$\begin{aligned}L &= E(A^\dagger Y) - E(A^\dagger)E(Y), \\ R &= E(Y^\dagger A) - E(Y^\dagger)E(A), \\ Q &= E(Y^\dagger Y) - E(Y^\dagger)E(Y) \succeq 0.\end{aligned}$$

Expansion and the equality at $A^\dagger A$ give, for every real t ,

$$\begin{aligned}t(L + R) + t^2 Q &\succeq 0, \\ it(L - R) + t^2 Q &\succeq 0.\end{aligned}$$

Nonnegativity for both signs of every real parameter forces $L + R = 0$ and $L - R = 0$. Hence $L = R = 0$, which is precisely

$$\begin{aligned}E(A^\dagger Y) &= E(A^\dagger)E(Y), \\ E(Y^\dagger A) &= E(Y^\dagger)E(A).\end{aligned}$$

Replacing Y by $X^\dagger$ yields $E(XA) = E(X)E(A)$. The second characterization follows by applying the same argument to $A^\dagger$. The converse implications follow by evaluating the defining product identities at $A^\dagger$. $\square$

Theorem 13.12.1.4 (Kraus maps satisfy the abstract Schwarz inequality). [Lean](#) [✓ Stmt](#) Every unital Kraus map satisfies the abstract Schwarz inequality of Definition 13.12.1.1.

Proof. [✓ Proof](#) A Kraus map preserves adjoints. The Kadison–Schwarz inequality (5.3) therefore gives, for every $A \in M_D(\mathbb{C})$,

$$E(A^\dagger A) - E(A^\dagger)E(A) = E(A^\dagger A) - E(A)^\dagger E(A) \succeq 0.$$

$\square$

13.12.2 Kraus specialization and full algebraic structure

Definition 13.12.2.1 (Right and left multiplicative domains). [Lean](#) [Lean](#) [✓ Stmt](#) For a Kraus map E , define

$$\begin{aligned}\mathcal{A}_R(E) &:= \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}), E(XY) = E(X)E(Y)\}, \\ \mathcal{A}_L(E) &:= \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}), E(YX) = E(Y)E(X)\}.\end{aligned}$$

We follow the convention that $\mathcal{A}_R(E)$ controls right multiplication and $\mathcal{A}_L(E)$ controls left multiplication. These are Kraus-map specializations of the preceding abstract domains, with the names interchanged: here the subscript records the side on which the varying factor is appended, whereas the preceding convention records the side occupied by the fixed element.

Definition 13.12.2.2 (Full multiplicative domain). [Lean](#) [✓ Stmt](#) The *multiplicative domain* of E is $\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$.

Remark 13.12.2.3 (Comparison of the two naming conventions). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Write $\mathcal{A}_R^W, \mathcal{A}_L^W$ for Wolf's convention and $\mathcal{A}_R^K, \mathcal{A}_L^K$ for the Kraus convention of this subsection. For the linear map associated with a Kraus family,

$$\begin{aligned}\mathcal{A}_R^W &= \mathcal{A}_L^K, \\ \mathcal{A}_L^W &= \mathcal{A}_R^K, \\ \mathcal{A}^W &= \mathcal{A}^K.\end{aligned}$$

Theorem 13.12.2.4 (Multiplicative-domain characterization). [Lean](#) [Lean](#) [✓ Stmt](#) Let E be a unital Kraus map. Then

$$\begin{aligned}X \in \mathcal{A}_R(E) &\iff E(XX^\dagger) = E(X)E(X)^\dagger, \\ X \in \mathcal{A}_L(E) &\iff E(X^\dagger X) = E(X)^\dagger E(X).\end{aligned}$$

Together these are the Kraus-map specialization of Theorem 13.12.1.3.

Proof. [✓ Proof](#) If X lies in a one-sided multiplicative domain, evaluate the defining identity at $Y = X^\dagger$. Conversely, the left-domain implication is exactly Theorem 5.2.4, and the right-domain implication follows by applying Theorem 5.2.4 to $X^\dagger$. $\square$

Theorem 13.12.2.5 (One-sided multiplicative domains are subalgebras). [Lean](#) [Lean](#) [✓ Stmt](#) For a unital Kraus map E , both $\mathcal{A}_R(E)$ and $\mathcal{A}_L(E)$ are unital subalgebras of $M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Closure under addition follows from linearity of E . For closure under multiplication, if $X, Y \in \mathcal{A}_R(E)$, then $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$ for all Z , using Theorem 13.12.2.4 iteratively; the $\mathcal{A}_L(E)$ case is analogous. Containment of $\mathbb{1}$ follows from unitality of E . $\square$

Theorem 13.12.2.6 (Multiplicative domain is a *-subalgebra). [Lean](#) [✓ Stmt](#) For a unital Kraus map E , the full multiplicative domain $\mathcal{A}(E)$ is a *-subalgebra of $M_D(\mathbb{C})$. This is the algebraic content of [Wol12, Theorem 5.7].

Proof. [✓ Proof](#) If $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$, then

$$\begin{aligned}E(X^\dagger X) &= E(X)^\dagger E(X), \\ E(XX^\dagger) &= E(X)E(X)^\dagger.\end{aligned}$$

The characterization of Theorem 13.12.2.4 shows that $X^\dagger \in \mathcal{A}(E)$. Combined with Theorem 13.12.2.5, this gives a *-subalgebra. $\square$

Theorem 13.12.2.7 (Restriction to multiplicative domain is a $*$ -homomorphism). [Lean](#) [Stmnt](#) If E is unital, then the restricted map $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \rightarrow M_D(\mathbb{C})$ is a $*$ -algebra homomorphism. This is the homomorphism conclusion of [Wol12, Theorem 5.7].

Proof. [Proof](#) Multiplicativity on $\mathcal{A}(E)$ follows from the definition. For $X \in \mathcal{A}(E)$, the Kraus commutation relation (5.9), its adjoint, and unitality give $E(X^\dagger) = E(X)^\dagger$, so the restriction preserves adjoints. $\square$

13.12.3 Positive maps preserve order and spectral intervals

The elementary order-preservation results used below are proved in Section A.1.

Theorem 13.12.3.1 (Order intervals are preserved). [Lean](#) [Stmnt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$. If $a \leq 0 \leq b$ and $a\mathbb{1} \leq A \leq b\mathbb{1}$, then

$$a\mathbb{1} \leq T(A) \leq b\mathbb{1}. \quad (13.62)$$

This is the matrix form of [Wol12, Equation (5.21)].

Proof. [Proof](#) By Theorem A.1.1, $T(a\mathbb{1}) \leq T(A) \leq T(b\mathbb{1})$. For the lower bound, $T(a\mathbb{1}) = aT(\mathbb{1})$. Since $a \leq 0$ and $T(\mathbb{1}) \leq \mathbb{1}$, one has $aT(\mathbb{1}) \geq a\mathbb{1}$. For the upper bound, $T(b\mathbb{1}) = bT(\mathbb{1}) \leq b\mathbb{1}$ since $b \geq 0$. These bounds give (13.62). $\square$

Theorem 13.12.3.2 (Spectral interval contractivity). [Lean](#) [Stmnt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$, and let $A \in M_D(\mathbb{C})$ be Hermitian. If $\text{spec}(A) \subseteq [a, b]$ with $a \leq 0 \leq b$, then

$$\text{spec}(T(A)) \subseteq [a, b].$$

This is [Wol12, Equation (5.21)].

Proof. [Proof](#) For Hermitian matrices, the inclusion $\text{spec}(A) \subseteq [a, b]$ is equivalent to the order bounds $a\mathbb{1} \leq A \leq b\mathbb{1}$. Apply Theorem 13.12.3.1 and translate (13.62) back into spectral bounds. $\square$

13.13 Positive Schwarz maps outside complete positivity

The following example shows that the Schwarz inequality does not force complete positivity.

13.13.1 A positive Schwarz map that is not completely positive

Example 13.13.1.1 (Transpose-trace map on $M_2(\mathbb{C})$). [Lean](#) [Stmnt](#) Consider the linear map $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ defined by

$$T(A) = \frac{1}{2}A^T + \frac{1}{4}\text{tr}(A)\mathbb{1}.$$

This is the map from [Wol12, Example 5.3].

Theorem 13.13.1.2 (The transpose-trace map is positive). [Lean](#) [Stmnt](#) The map from Example 13.13.1.1 is positive.

Proof. ✓ Proof It is a non-negative linear combination of the transpose map and the map $A \mapsto \text{tr}(A)\mathbb{1}$, and both of these maps send positive semidefinite matrices to positive semidefinite matrices. $\square$

Theorem 13.13.1.3 (The transpose-trace map satisfies the Schwarz inequality). Lean ✓ Stmt For every $A \in M_2(\mathbb{C})$, $T(A^\dagger A) - T(A^\dagger)T(A) \geq 0$, where T is the map from Example 13.13.1.1.

Proof. ✓ Proof Write the Schwarz gap as $F(A)$. One checks that $F(A + \lambda\mathbb{1}) = F(A)$ for every scalar λ , so it is enough to treat the case $\text{tr}(A) = 0$. In that case a direct 2×2 computation shows that $F(A)$ dominates $\frac{1}{2}(A^\dagger A)^T$, hence is positive semidefinite. $\square$

Theorem 13.13.1.4 (The transpose-trace map is not completely positive). Lean ✓ Stmt The map from Example 13.13.1.1 is not completely positive.

Proof. ✓ Proof Compute the Choi matrix of T and test it on the antisymmetric vector $|01\rangle - |10\rangle$. The resulting expectation value is negative, so the Choi matrix is not positive semidefinite. $\square$

13.13.2 Two-variable operator Schwarz inequality

Lemma 13.13.2.1 (Two-variable operator Schwarz inequality, conditional form). Lean ✓ Stmt Let $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be a 2-positive complex-linear map. For every $A, B \in M_{D_{\text{in}}}(\mathbb{C})$ and all vectors $v, w \in \mathbb{C}^{D_{\text{out}}}$ such that $E(B^\dagger B)w = E(B^\dagger A)v$, we have

$$v^\dagger E(A^\dagger A)v \geq v^\dagger E(A^\dagger B)w.$$

This is the pseudoinverse-free form of [Wol12, Eq. (5.4)]: the usual statement $E(A^\dagger B)E(B^\dagger B)^{-1}E(B^\dagger A) \leq E(A^\dagger A)$ (with inverse taken on the range) follows by taking $w = E(B^\dagger B)^+ E(B^\dagger A)v$.

Proof. ✓ Proof

Write C for the horizontal concatenation $[A \ B]$. The block matrix $C^\dagger C$ is positive semidefinite. Applying the 2-positive ampliation entrywise yields the PSD block matrix $\begin{bmatrix} E(A^\dagger A) & E(A^\dagger B) \\ E(B^\dagger A) & E(B^\dagger B) \end{bmatrix}$. Evaluating the PSD quadratic form on the vector $(v, -w)$ and using the side condition $E(B^\dagger B)w = E(B^\dagger A)v$ to cancel the off-diagonal and diagonal terms yields $v^\dagger E(A^\dagger A)v - v^\dagger E(A^\dagger B)w \geq 0$. $\square$

Theorem 13.13.2.2 (Conjugate-transpose identity). Lean ✓ Stmt Let $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be a 2-positive complex-linear map. For every $A, B \in M_{D_{\text{in}}}(\mathbb{C})$,

$$E(B^\dagger A) = E(A^\dagger B)^\dagger.$$

Proof. ✓ Proof A 2-positive map is positive, and positive maps preserve adjoints. Applying this to $A^\dagger B$ and using $(A^\dagger B)^\dagger = B^\dagger A$ gives the identity. $\square$

Theorem 13.13.2.3 (Kernel inclusion for the two-variable Schwarz inequality). Lean ✓ Stmt Let $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be a 2-positive complex-linear map. For every $A, B \in M_{D_{\text{in}}}(\mathbb{C})$, $\ker(E(B^\dagger B)) \subseteq \ker(E(A^\dagger B))$.

Proof. ✓ Proof Rewrite $E(B^\dagger A)$ as $E(A^\dagger B)^\dagger$ (Lemma 13.13.2.2) in the amplified PSD block matrix $\begin{bmatrix} E(A^\dagger A) & E(A^\dagger B) \\ E(B^\dagger A) & E(B^\dagger B) \end{bmatrix}$ from the proof of Lemma 13.13.2.1: if $E(B^\dagger B)y = 0$ then testing the PSD quadratic form on $(0, y)$ forces $E(A^\dagger B)y = 0$. $\square$

Theorem 13.13.2.4 (Pseudoinverse range-inclusion witness). [Lean](#) [✓ Stmt](#) Let $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be a 2-positive complex-linear map. For every $A, B \in M_{D_{\text{in}}}(\mathbb{C})$,

$$E(B^\dagger B)(E(B^\dagger B)^+ E(B^\dagger A)) = E(B^\dagger A).$$

Proof. [✓ Proof](#) The kernel inclusion $\ker(E(B^\dagger B)) \subseteq \ker(E(A^\dagger B))$ (Lemma 13.13.2.3) gives, after taking adjoints, that the support projection of $E(B^\dagger B)$ absorbs $E(B^\dagger A)$ on the left. Since $E(B^\dagger B)E(B^\dagger B)^+$ is exactly that support projection, the displayed identity follows. $\square$

Theorem 13.13.2.5 (Two-variable Schwarz inequality for a two-positive map). [Lean](#) [✓ Stmt](#) Let $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be two-positive. Then, for every $A, B \in M_{D_{\text{in}}}(\mathbb{C})$,

$$E(A^\dagger B) E(B^\dagger B)^+ E(B^\dagger A) \leq E(A^\dagger A).$$

Proof. [✓ Proof](#) The range inclusion gives $E(B^\dagger B)E(B^\dagger B)^+ E(B^\dagger A) = E(B^\dagger A)$. Thus, for every vector v , the choice $w = E(B^\dagger B)^+ E(B^\dagger A)v$ satisfies the hypothesis of the conditional inequality. Substitution gives the result. $\square$

Theorem 13.13.2.6 (Two-variable operator Schwarz inequality). [Lean](#) [✓ Stmt](#) Let $E : M_{D_{\text{in}}}(\mathbb{C}) \rightarrow M_{D_{\text{out}}}(\mathbb{C})$ be positive and suppose that $E = T^*$ for a two-positive map $T : M_{D_{\text{out}}}(\mathbb{C}) \rightarrow M_{D_{\text{in}}}(\mathbb{C})$. For every $A, B \in M_{D_{\text{in}}}(\mathbb{C})$,

$$E(A^\dagger B) E(B^\dagger B)^+ E(B^\dagger A) \leq E(A^\dagger A),$$

where $(\cdot)^+$ is the Moore–Penrose pseudoinverse, i.e. the inverse taken on the range. This is [Wol12, Eq. (5.4)], with its independent input and output dimensions and without an assumed witness vector.

Proof. [✓ Proof](#) By trace-adjoint invariance of two-positivity (Theorem ??), $E = T^*$ is two-positive. Hence Theorem 13.13.2.5 applies to E with its rectangular input and output dimensions. The resulting inequality is precisely the displayed conclusion. $\square$

Definition 13.13.2.7 (Per- B Schwarz hypothesis). [Lean](#) [✓ Stmt](#) A complex-linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a fixed $B \in M_D(\mathbb{C})$ satisfy the per- B Schwarz hypothesis when

$$E(A^\dagger B) E(B^\dagger B)^+ E(B^\dagger A) \leq E(A^\dagger A)$$

for every $A \in M_D(\mathbb{C})$. A 2-positive map satisfies this for every B (Theorem 13.13.2.5).

Definition 13.13.2.8 (Equality-attaining set). [Lean](#) [✓ Stmt](#) For a fixed B , the set $\mathcal{A}_B \subseteq M_D(\mathbb{C})$ of A 's for which equality is attained in the two-variable operator Schwarz inequality:

$$E(A^\dagger B) E(B^\dagger B)^+ E(B^\dagger A) = E(A^\dagger A).$$

Theorem 13.13.2.9 (Vanishing linear coefficient of a nonnegative matrix quadratic). [Lean](#) [✓ Stmt](#) Let m be finite and $B, D \in M_m(\mathbb{C})$. If $tB + t^2D$ is positive semidefinite for every $t \in \mathbb{R}$, then $B = 0$.

Proof. [✓ Proof](#) Testing the quadratic form on a vector x reduces to the scalar statement: if $tb + t^2d \geq 0$ for every real t , where $b = x^\dagger Bx$ and $d = x^\dagger Dx \geq 0$, then $b = 0$. Evaluating at $t = \pm 1$ forces $d \geq 0$; evaluating at $t = -\text{Re}(b)/(d + 1)$ and completing the square forces $\text{Re}(b) = 0$, and comparing the imaginary parts at $t = \pm 1$ forces $\text{Im}(b) = 0$. Since this holds for every x , $B = 0$ by polarization. $\square$

Theorem 13.13.2.10 (Equality in the operator Schwarz inequality). [Lean](#) [✓ Stmt](#) Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map such that, for a given $B \in M_D(\mathbb{C})$, the per- B Schwarz hypothesis holds (Definition 13.13.2.7). Then for every $A \in \mathcal{A}_B$ (Definition 13.13.2.8) and every $X \in M_D(\mathbb{C})$,

$$E(A^\dagger B) E(B^\dagger B)^+ E(B^\dagger X) = E(A^\dagger X).$$

This is the source's equality theorem, local source `Notes/WolfNoteTexSource/ch05_schwarz_inequalities.tex`, line 198.

Proof. [✓ Proof](#) Write $Q(U, V) = E(U^\dagger V) - E(U^\dagger B)E(B^\dagger B)^+ E(B^\dagger V)$; the per- B hypothesis is $Q(U, U) \geq 0$ for every U , and $A \in \mathcal{A}_B$ means $Q(A, A) = 0$. Applying the hypothesis to $A + cX$ for $c \in \mathbb{C}$ and expanding bilinearly gives

$$Q(A + cX, A + cX) = cQ(A, X) + \bar{c}Q(X, A) + \bar{c}cQ(X, X) \geq 0.$$

Taking $c = t \in \mathbb{R}$ and applying Lemma 13.13.2.9 to the linear coefficient $Q(A, X) + Q(X, A)$ and the fixed nonnegative quadratic coefficient $Q(X, X)$ shows $Q(A, X) + Q(X, A)$ vanishes; taking $c = tI$ and applying the same lemma to $IQ(A, X) - IQ(X, A)$ shows this also vanishes. Combining the two gives $Q(A, X) = 0$, which is the displayed identity. $\square$

Corollary 13.13.2.11 (Converse of the equality criterion). [Lean](#) [✓ Stmt](#) If the identity of Theorem 13.13.2.10 holds for every $X \in M_D(\mathbb{C})$, then $A \in \mathcal{A}_B$. This is immediate from the theorem by specializing at $X = A$; the source states only the forward direction.

13.14 Exponential positivity for irreducible CP maps

Theorem 13.14.1 (Exponential positivity for irreducible CP maps). [Lean](#) [✓ Stmt](#) Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A) > 0. \quad (13.63)$$

This is the completely positive specialization of the forward implication in [Wol12, Theorem 6.2(3)].

Proof. [✓ Proof](#) By Theorem B.5.1, the first D terms in (13.63) already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail. $\square$

Theorem 13.14.2 (Exponential characterization of irreducibility). [Lean](#) [✓ Stmt](#) Let E be a completely positive map on $M_D(\mathbb{C})$. Then E is irreducible if and only if, for every $t > 0$ and every nonzero $A \geq 0$, one has $\exp(tE)(A) > 0$. This is the completely positive specialization of the equivalence in [Wol12, Theorem 6.2(3)]; Theorem 13.14.1 is its forward implication.

Proof. [✓ Proof](#) The forward implication is Theorem 13.14.1. Conversely, let P be an invariant orthogonal projection. Invariance gives $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, so induction yields $E^k(P) \in PM_D(\mathbb{C})P$ for every $k \geq 0$. Therefore every term, and hence the convergent series

$$\exp(tE)(P) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(P),$$

lies in the same corner. If $P \neq 0$, then at $t = 1$ the assumed positive definiteness is impossible unless $P = \mathbb{1}$, because a positive definite matrix cannot be supported on a proper corner. Thus every invariant projection is 0 or $\mathbb{1}$, so E is irreducible. $\square$

13.15 Ergodicity of irreducible channels

Theorem 13.15.1 (Unique density-matrix fixed point for an irreducible channel). [Lean](#) [✓ Stmt](#) *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$. Then there exists a unique density matrix σ such that $E(\sigma) = \sigma$. The fixed point σ is positive definite. This is the fixed-point conclusion in the forward direction of [Wol12, Corollary 6.3], specialized to completely positive trace-preserving maps.*

Proof. [✓ Proof](#) Every Cesàro mean (6.1) of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges to a density matrix σ . The telescope identity (6.2) gives $E(\sigma) = \sigma$. Theorem 6.3.3 makes σ positive definite. If τ is another density-matrix fixed point, Theorem 6.4.7 gives $\tau = c\sigma$. Taking traces yields $1 = c$, hence $\tau = \sigma$. $\square$

Theorem 13.15.2 (Cesàro convergence for irreducible channels). [Lean](#) [✓ Stmt](#) *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$, and let ρ be any density matrix. Then the Cesàro means (6.1) converge to the unique positive definite density-matrix fixed point of E . This is the Cesàro-convergence conclusion in the forward direction of [Wol12, Corollary 6.3], specialized to quantum channels.*

Proof. [✓ Proof](#) Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 13.15.1, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit. $\square$

13.16 A CP spectral characterization of irreducibility

Definition 13.16.1 (Restricted CP spectral properties). [Lean](#) [✓ Stmt](#) A completely positive map E on $M_D(\mathbb{C})$ has the *restricted CP spectral properties* used here if there exist a positive real number r , a positive definite right eigenvector ρ , and a positive definite left eigenvector σ for the adjoint map such that

$$\begin{aligned} E(\rho) &= r\rho, \\ E^\dagger(\sigma) &= r\sigma, \end{aligned} \tag{13.64}$$

every positive semidefinite right eigenvector for eigenvalue r is a scalar multiple of ρ , and the spectral radius of E is equal to r . The uniqueness clause concerns only positive semidefinite Perron eigenvectors. Unlike the full nondegenerate-eigenspace statement in [Wol12, Theorem 6.4], it does not assert that every complex eigenvector at r is proportional to ρ .

Theorem 13.16.2 (Irreducibility gives the restricted CP spectral properties). [Lean](#) [✓ Stmt](#) *Every nonzero irreducible completely positive map on $M_D(\mathbb{C})$ has the restricted CP spectral properties of Definition 13.16.1.*

Proof. [✓ Proof](#) Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 6.9.1. $\square$

Theorem 13.16.3 (Restricted CP spectral properties imply irreducibility). [Lean](#) [✓ Stmt](#) *If a completely positive map on $M_D(\mathbb{C})$ has the restricted CP spectral properties of Definition 13.16.1, then it is irreducible.*

Proof.  **Proof** Gauge by the positive definite left eigenvector in (13.64) and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 13.16.1. $\square$

Theorem 13.16.4 (Irreducibility iff the restricted CP spectral properties hold).   **Stmt** *Let E be a nonzero completely positive map on $M_D(\mathbb{C})$. Then E is irreducible if and only if it has the restricted CP spectral properties of Definition 13.16.1. This is a CP-map variant of [Wol12, Theorem 6.4]: uniqueness is required only among positive semidefinite Perron eigenvectors, not on the full eigenspace. Those restricted properties are nevertheless sufficient for the irreducibility equivalence.*

Proof.  **Proof** Combine Theorems 13.16.2 and 13.16.3. $\square$

Part I

Appendices

Appendix A

Schwarz Inequalities and Multiplicative Domains: Supporting Results

This appendix supports Chapter 5 by collecting the elementary order, trace-duality, rank-one, and weighted-trace arguments used there.

A.1 Trace duality and elementary order preservation

Let T denote a positive linear map between matrix algebras. For a linear map $E : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$, write E^* for the trace-pairing adjoint of Definition 4.5.1. The results below supply the order step in Theorem 13.12.3.1 and the trace duality used by the positive-retraction and fixed-point arguments.

Theorem A.1.1 (Positive maps are monotone). Lean ✓ Stmt *If $T : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ is positive and $A, B \in M_n(\mathbb{C})$ satisfy $A \leq B$, then $T(A) \leq T(B)$ in $M_m(\mathbb{C})$.*

Proof. ✓ Proof Since $B - A \geq 0$, positivity gives $T(B - A) \geq 0$. By linearity, this is $T(B) - T(A) \geq 0$. □

Theorem A.1.2 (Positive maps preserve adjoints). Lean ✓ Stmt *If $T : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ is positive, then $T(A^\dagger) = T(A)^\dagger$ for all $A \in M_n(\mathbb{C})$.*

Proof. ✓ Proof Regard T as the associated positive linear map (Definition 4.4.1). For positive linear maps between matrix C^* -algebras, $T(A^*) = T(A)^*$. Since the star on each matrix algebra is the conjugate transpose, $T(A^\dagger) = T(A^*) = T(A)^* = T(A)^\dagger$. □

Theorem A.1.3 (Nondegeneracy of the trace pairing). Lean ✓ Stmt *For $M \in M_D(\mathbb{C})$, $\text{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.*

Proof. ✓ Proof Testing against the matrix unit E_{ji} gives $\text{tr}(ME_{ji}) = M_{ij}$, so $\text{tr}(MN) = 0$ for all N forces every entry of M to vanish. □

Theorem A.1.4 (Trace-pairing adjoint identity). Lean ✓ Stmt *For every linear map $E : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$, every $\rho \in M_{D'}(\mathbb{C})$, and every $X \in M_D(\mathbb{C})$, one has*

$$\text{tr}(E^*(\rho)X) = \text{tr}(\rho E(X)). \tag{A.1}$$

Proof. ✓ Proof Write $X = \sum_{i,j} X_{ij} e_{ij}$ in the standard matrix units and use linearity in X . For $X = e_{ij}$ one has $\text{tr}(E^*(\rho) e_{ij}) = (E^*(\rho))_{ji}$, while the definition of E^* gives $(E^*(\rho))_{ji} = \text{tr}(\rho E(e_{ij}))$. $\square$

Theorem A.1.5 (Trace-pairing adjoint is involutive). Lean ✓ Stmt *For every linear map $E : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$, the trace-pairing adjoint of E^* is E .*

Proof. ✓ Proof Fix $X \in M_D(\mathbb{C})$ and an arbitrary matrix $N \in M_{D'}(\mathbb{C})$. Applying (A.1) first to $E^* : M_{D'}(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and then to E gives

$$\text{tr}((E^*)^*(X)N) = \text{tr}(XE^*(N)) = \text{tr}(E^*(N)X) = \text{tr}(NE(X)) = \text{tr}(E(X)N).$$

Both $(E^*)^*(X)$ and $E(X)$ lie in $M_{D'}(\mathbb{C})$, so, since this holds for every $N \in M_{D'}(\mathbb{C})$, nondegeneracy of the trace pairing on $M_{D'}(\mathbb{C})$ gives $(E^*)^* = E$. $\square$

Theorem A.1.6 (Self-duality of the positive semidefinite cone). Lean ✓ Stmt *Let $A \in M_D(\mathbb{C})$ be Hermitian. If $\text{tr}(AB) \geq 0$ for every positive semidefinite matrix B , then $A \geq 0$.*

Proof. ✓ Proof Test the hypothesis on rank-one positive semidefinite matrices $B = xx^\dagger$. The identity $\text{tr}(Axx^\dagger) = x^\dagger Ax$ and the hypothesis give $x^\dagger Ax \geq 0$ for every vector x , so the Hermitian matrix A is positive semidefinite. $\square$

Theorem A.1.7 (Trace-pairing adjoint of a positive map). Lean ✓ Stmt *If $E : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ is positive, then its trace-pairing adjoint $E^* : M_{D'}(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is positive.*

Proof. ✓ Proof Let $\rho \geq 0$. First $E^*(\rho)$ is Hermitian. Indeed, Theorem A.1.2 gives $E(A^\dagger) = E(A)^\dagger$, and hence, for matrix units e_{ij} ,

$$\overline{\text{tr}(\rho E(e_{ij}))} = \text{tr}(E(e_{ij})^\dagger \rho) = \text{tr}(E(e_{ji}) \rho) = \text{tr}(\rho E(e_{ji})).$$

For every $B \geq 0$, (A.1) gives $\text{tr}(E^*(\rho)B) = \text{tr}(\rho E(B))$. Since $E(B) \geq 0$, the right-hand side is non-negative by positivity of the trace product of two positive semidefinite matrices. Self-duality of the positive semidefinite cone therefore gives $E^*(\rho) \geq 0$. $\square$

Theorem A.1.8 (Trace-pairing adjoints preserve positivity exactly). Lean ✓ Stmt *A linear map $E : M_D(\mathbb{C}) \rightarrow M_{D'}(\mathbb{C})$ is positive if and only if its trace-pairing adjoint $E^* : M_{D'}(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is positive.*

Proof. ✓ Proof One direction is Theorem A.1.7. The other follows by applying the same theorem to E^* and using Theorem A.1.5. $\square$

A.2 Positive functionals and rank-one retractions

The next three results provide, respectively, the density, support, and scalarity steps in Theorem 5.3.1 and hence in its finite-star-algebra extension, Theorem 5.3.2.

Theorem A.2.1 (Density representation of positive matrix functionals). Lean ✓ Stmt *Let $f : M_D(\mathbb{C}) \rightarrow \mathbb{C}$ be a complex-linear functional, where $D > 0$. If $f(X) \geq 0$ for every $X \geq 0$ and $f(\mathbb{1}) = 1$, then there is a density matrix $\rho \in M_D(\mathbb{C})$ such that $f(X) = \text{tr}(\rho X)$ for every $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Define $F : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ by $F(X) = f(X)\mathbb{1}$. For $X \geq 0$ one has $f(X) \geq 0$, so $F(X) = f(X)\mathbb{1} \geq 0$ and F is positive. Set $\sigma = D^{-1}\mathbb{1}$ and $\rho = F^*(\sigma)$. Positivity of the trace-pairing adjoint gives $\rho \geq 0$, while (A.1) gives

$$\mathrm{tr}(\rho X) = \mathrm{tr}(\sigma F(X)) = f(X).$$

Taking $X = \mathbb{1}$ yields $\mathrm{tr}(\rho) = 1$. □

Lemma A.2.2 (Rank-one domination). Lean ✓ Stmt *Let $A \in M_D(\mathbb{C})$ be positive semidefinite, let $c \in \mathbb{C}$, and let $\psi \in \mathbb{C}^D$. If $A \leq c|\psi\rangle\langle\psi|$, then there is a non-negative scalar a such that $A = a|\psi\rangle\langle\psi|$.*

Proof. ✓ Proof The assertion is immediate when $\psi = 0$. Otherwise set $s = \langle\psi, \psi\rangle$, $P = |\psi\rangle\langle\psi|$, and $Q = s^{-1}P$. Then Q is the orthogonal projection onto $\mathbb{C}\psi$. Write $B = cP - A \geq 0$. Since $A + B = cP$ vanishes on the kernel of Q , positivity implies $A(\mathbb{1} - Q) = 0$. Taking adjoints gives $(\mathbb{1} - Q)A = 0$, and therefore $A = QAQ$. Finally,

$$\begin{aligned} PAP &= \langle\psi, A\psi\rangle P, \\ A &= s^{-2}\langle\psi, A\psi\rangle P. \end{aligned}$$

The coefficient is non-negative because $A \geq 0$ and $s > 0$. □

Theorem A.2.3 (Rank-one ray scalar rigidity). Lean ✓ Stmt *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be complex-linear. Suppose that for every $v \in \mathbb{C}^D$ there is a scalar $c_v \in \mathbb{C}$ such that $T(|v\rangle\langle v|) = c_v|v\rangle\langle v|$. Then there is a scalar $c \in \mathbb{C}$ such that $T = c \text{ id}$.*

Proof. ✓ Proof Write $P_v = |v\rangle\langle v|$. If $D = 0$, then $M_D(\mathbb{C})$ is the zero vector space, so the conclusion holds with $c = 0$. If $D = 1$, the single matrix unit is P_{e_0} . The hypothesis gives $T(P_{e_0}) = c_{e_0}P_{e_0}$, and linearity gives $T = c_{e_0} \text{ id}$ on all of $M_1(\mathbb{C})$.

It remains to consider $D \geq 2$. Set $c = c_{e_0}$. For $i \neq j$, the identities

$$\begin{aligned} P_{e_i+e_j} + P_{e_i-e_j} &= 2P_{e_i} + 2P_{e_j}, \\ c_{e_i+e_j}P_{e_i+e_j} + c_{e_i-e_j}P_{e_i-e_j} &= 2c_{e_i}P_{e_i} + 2c_{e_j}P_{e_j} \end{aligned}$$

give, from their (i, i) , (j, j) , and (i, j) entries, $c_{e_i+e_j} = c_{e_i-e_j} = c_{e_i} = c_{e_j}$. The imaginary parallelogram identity and its image under T are

$$\begin{aligned} P_{e_i+ie_j} + P_{e_i-ie_j} &= 2P_{e_i} + 2P_{e_j}, \\ c_{e_i+ie_j}P_{e_i+ie_j} + c_{e_i-ie_j}P_{e_i-ie_j} &= 2c_{e_i}P_{e_i} + 2c_{e_j}P_{e_j}. \end{aligned}$$

Comparing the same three entries gives $c_{e_i+ie_j} = c_{e_i-ie_j} = c_{e_i} = c_{e_j}$. Taking the pair $(0, i)$ shows that $c_{e_i} = c$ for every i . The hypothesis therefore gives $T(E_{ii}) = cE_{ii}$ for the diagonal matrix units. For $i \neq j$, the polarization identity

$$4E_{ij} = P_{e_i+e_j} - P_{e_i-e_j} + iP_{e_i+ie_j} - iP_{e_i-ie_j}$$

gives $T(E_{ij}) = cE_{ij}$. Thus $T = c \text{ id}$ on every matrix unit. □

The rank-one domination and rigidity results turn the vector-dependent coefficients in the one-factor retraction into a single positive functional; the density representation identifies that functional with a density matrix.

A.3 Faithful weighted traces and peripheral Schwarz equality

These two trace facts provide the adjointness and faithfulness steps in Theorem 5.5.1. For the Kraus map E , write E^* for its Kraus adjoint and let $\rho > 0$ denote the faithful fixed point used there.

Lemma A.3.1 (Trace pairing for Kraus map and adjoint map). Lean ✓ Stmt For all $X, Y \in M_D(\mathbb{C})$, $\text{tr}(X E(Y)) = \text{tr}(E^*(X) Y)$.

Lemma A.3.2 (Positive-definite trace test). Lean ✓ Stmt If $A > 0$, $X \geq 0$, and $\text{tr}(AX) = 0$, then $X = 0$.

Appendix B

Perron–Frobenius Theory for Channels and Transfer Maps: Supporting Results

This appendix supplies the compactness, gauge, similarity, and finite-sum calculations used in Chapter 6.

B.1 Density matrices, Brouwer’s theorem, and Cesàro limits

Write $\mathcal{D}_D$ for the density matrices of Definition 4.4.6. The next results provide the compact convex domain and the limit argument used for channel fixed points.

Theorem B.1.1 (Density matrices are compact). [Lean](#) [✓ Stmt](#) $\mathcal{D}_D$ is compact in the entrywise topology on $M_D(\mathbb{C})$.

Proof. [✓ Proof](#) The positive semidefinite cone is closed, being the preimage of the closed non-negative cone under continuous quadratic forms. If $\rho \geq 0$ with $\text{tr}(\rho) = 1$, then each entry satisfies $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$: the diagonal entries are non-negative and sum to 1, and the off-diagonal bound is the positive semidefinite Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. $\square$

Theorem B.1.2 (Density matrices are convex). [Lean](#) [✓ Stmt](#) $\mathcal{D}_D$ is convex.

Proof. [✓ Proof](#) A convex combination of positive semidefinite matrices is positive semidefinite, and the trace is linear. $\square$

Theorem B.1.3 (Density matrices are nonempty). [Lean](#) [✓ Stmt](#) For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.

Proof. [✓ Proof](#) $\frac{1}{D}\mathbb{1}_D$ is a density matrix. $\square$

Theorem B.1.4 (Channels preserve density matrices). [Lean](#) [✓ Stmt](#) If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.

Proof. [✓ Proof](#) Complete positivity implies positivity by Theorem 4.2.3, so $E(\rho) \geq 0$; trace preservation gives $\text{tr}(E(\rho)) = \text{tr}(\rho) = 1$. $\square$

Theorem B.1.5 (Brouwer fixed point on density matrices). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
 Let $D > 0$. Every continuous map from the compact convex set of density matrices in $M_D(\mathbb{C})$ to itself has a fixed point.

Proof. [✓ Proof](#) The proof starts from Brouwer's theorem for products of simplices, transfers it to closed cubes, and then to compact retracts of finite-dimensional real normed spaces. The density-matrix set is such a compact retract, using the explicit density retraction. $\square$

Lemma B.1.6 (Cesàro means remain density matrices). [Lean](#) [✓ Stmt](#) Let E be a quantum channel and $\rho \in \mathcal{D}_D$. Then, for every $N \geq 0$,

$$\frac{1}{N+1} \sum_{n=0}^N E^n(\rho) \in \mathcal{D}_D. \quad (\text{B.1})$$

Proof. [✓ Proof](#) Every iterate $E^n(\rho)$ belongs to $\mathcal{D}_D$ by Theorem B.1.4. Equation (B.1) is their convex average: positivity is preserved by sums and multiplication by $(N+1)^{-1}$, while its trace is $(N+1)^{-1} \sum_{n=0}^N 1 = 1$. $\square$

Lemma B.1.7 (Subsequential Cesàro limits are fixed points). [Lean](#) [✓ Stmt](#) Let E be a quantum channel, $\rho \in \mathcal{D}_D$, and $\psi : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $\psi(k) \rightarrow \infty$. If

$$\frac{1}{\psi(k)+1} \sum_{n=0}^{\psi(k)} E^n(\rho) \rightarrow \sigma,$$

then $\sigma \in \mathcal{D}_D$ and $E(\sigma) = \sigma$.

Proof. [✓ Proof](#) Lemma B.1.6 and closedness of the compact set $\mathcal{D}_D$ give $\sigma \in \mathcal{D}_D$. The telescope identity yields

$$E(\sigma_{\psi(k)+1}) - \sigma_{\psi(k)+1} = \frac{E^{\psi(k)+1}(\rho) - \rho}{\psi(k)+1}.$$

Both $E^{\psi(k)+1}(\rho)$ and ρ lie in the compact set $\mathcal{D}_D$, so their difference is uniformly bounded. The right-hand side tends to zero. Continuity of E and convergence of the subsequence therefore give $E(\sigma) - \sigma = 0$. $\square$

B.2 Canonical-gauge algebra

For the right-canonical gauge in Theorem 6.6.1, each summand becomes

$$\begin{aligned} (S^{-1}A^iS)(S^{-1}A^iS)^\dagger &= S^{-1}A^iSS^\dagger(A^i)^\dagger(S^\dagger)^{-1} \\ &= S^{-1}A^i\rho(A^i)^\dagger(S^\dagger)^{-1}. \end{aligned}$$

Hence

$$\sum_i A'^i(A'^i)^\dagger = S^{-1}\mathcal{E}_A(\rho)(S^\dagger)^{-1} = S^{-1}\rho(S^\dagger)^{-1} = \mathbb{1}.$$

For the left-canonical gauge in Theorem 6.6.3,

$$\begin{aligned} (SA^iS^{-1})^\dagger(SA^iS^{-1}) &= (S^\dagger)^{-1}(A^i)^\dagger S^\dagger SA^iS^{-1} \\ &= (S^\dagger)^{-1}(A^i)^\dagger \sigma A^iS^{-1}, \end{aligned}$$

and therefore

$$\sum_i (A'^i)^\dagger A'^i = (S^\dagger)^{-1} \left(\sum_i (A^i)^\dagger \sigma A^i \right) S^{-1} = (S^\dagger)^{-1} \sigma S^{-1} = \mathbb{1}.$$

Finally, for the square-root gauge (6.26), self-adjointness of $\sigma^{1/2}$ gives

$$\begin{aligned} (B^i)^\dagger B^i &= \sigma^{-1/2} (A^i)^\dagger \sigma^{1/2} \sigma^{1/2} A^i \sigma^{-1/2} \\ &= \sigma^{-1/2} (A^i)^\dagger \sigma A^i \sigma^{-1/2}. \end{aligned}$$

Summing, substituting the adjoint fixed-point equation, and cancelling the outer square-root factors yields

$$\sum_i (B^i)^\dagger B^i = \sigma^{-1/2} \left(\sum_i (A^i)^\dagger \sigma A^i \right) \sigma^{-1/2} = \sigma^{-1/2} \sigma \sigma^{-1/2} = \mathbb{1}.$$

Theorem B.2.1 (Trace-preserving gauge from an adjoint fixed point, Kraus form). [Lean](#) [✓ Stmt](#) Let $\{K_i\}_{i=0}^{d-1}$ be a finite Kraus family and let ρ be positive definite with $\sum_i K_i^\dagger \rho K_i = \rho$. Define the gauged family $B_i = \rho^{1/2} K_i \rho^{-1/2}$. Then B is trace-preserving: $\sum_i B_i^\dagger B_i = \mathbb{1}$.

Proof. [✓ Proof](#) Self-adjointness of $\rho^{1/2}$ gives

$$B_i^\dagger B_i = \rho^{-1/2} K_i^\dagger \rho^{1/2} \rho^{1/2} K_i \rho^{-1/2} = \rho^{-1/2} K_i^\dagger \rho K_i \rho^{-1/2}.$$

Summing, substituting the adjoint fixed-point equation, and cancelling the outer square-root factors yields

$$\sum_i B_i^\dagger B_i = \rho^{-1/2} \left(\sum_i K_i^\dagger \rho K_i \right) \rho^{-1/2} = \rho^{-1/2} \rho \rho^{-1/2} = \mathbb{1}.$$

□

Lemma B.2.2 (Square-root trace-preserving gauge is a gauge equivalence). [Lean](#) [✓ Stmt](#) If σ is positive definite and B is defined by (6.26), then A and B are gauge equivalent.

Proof. [✓ Proof](#) The gauge matrix is $\sigma^{1/2}$, which is invertible because σ is positive definite. Thus (6.26) has the form of the gauge relation (2.5). □

B.3 Similarity bookkeeping

Lemma B.3.1 (Similarity is an involution). [Lean](#) [✓ Stmt](#) For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, write $S_C(E)(X) = C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}$ for the similarity transform by C . The transforms by C and C^{-1} compose to the identity:

$$\begin{aligned} S_{C^{-1}}(S_C(E))(X) &= C [C^{-1} E(C(C^{-1} X (C^\dagger)^{-1}) C^\dagger) (C^\dagger)^{-1}] C^\dagger \\ &= E(X). \end{aligned} \tag{B.2}$$

Proof. [✓ Proof](#) Direct computation: the inner conjugation $C(C^{-1} X (C^\dagger)^{-1}) C^\dagger$ collapses to X using $C C^{-1} = \mathbb{1}$ and $(C^\dagger)^{-1} C^\dagger = \mathbb{1}$, and the outer conjugation by $C(\cdot) C^\dagger$ then cancels the inner $C^{-1}(\cdot) (C^\dagger)^{-1}$ on $E(X)$ for the same reason. □

Lemma B.3.2 (Irreducibility is invariant under similarity). [Lean](#) [✓ Stmt](#) For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transform $X \mapsto C^{-1}E(CXC^\dagger)(C^\dagger)^{-1}$ is irreducible if and only if E is.

Proof. [✓ Proof](#) The forward direction follows from Lemma 6.7.1 applied with the inverse C^{-1} , whose determinant is also nonzero: if the similarity transform of E by C is irreducible, then so is its similarity transform by C^{-1} , which equals E by (B.2). The reverse direction is Lemma 6.7.1 itself with $c = 1$. $\square$

B.4 Auxiliary Perron reductions

Theorem B.4.1 (Nonzero Kraus maps). [Lean](#) [✓ Stmt](#) If some Kraus operator $K_i \neq 0$, then the Kraus map $X \mapsto \sum_i K_i X K_i^\dagger$ is nonzero.

Proof. [✓ Proof](#) If the map vanishes, evaluating at $\mathbb{1}$ gives $\sum_i K_i K_i^\dagger = 0$. Since each summand $K_i K_i^\dagger$ is positive semidefinite, each $K_i K_i^\dagger = 0$ and hence each $K_i = 0$. Applying this to $K_i = (A^i)^\dagger$ and using $A^i \neq 0 \iff (A^i)^\dagger \neq 0$ shows that the adjoint transfer map $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ of a tensor with some $A^i \neq 0$ is nonzero. $\square$

Theorem B.4.2 (Kraus maps are completely positive). [Lean](#) [✓ Stmt](#) The Kraus map of a finite matrix family $\{K_i\}_{i=0}^{d-1}$ is completely positive.

Proof. [✓ Proof](#) The family $\{K_i\}$ itself is a Kraus representation of the map. $\square$

Theorem B.4.3 (Irreducibility passes to the conjugate-transposed Kraus family). [Lean](#) [✓ Stmt](#) If the Kraus map of $\{K_i\}_{i=0}^{d-1}$ is irreducible, then so is the Kraus map of the conjugate-transposed family $\{K_i^\dagger\}$.

Proof. [✓ Proof](#) If P is invariant for the Kraus map of $\{K_i^\dagger\}$, so that $(\mathbb{1} - P)K_i^\dagger P = 0$ for all i , then taking adjoints gives $PK_i(\mathbb{1} - P) = 0$; by Theorem 6.3.6, $\mathbb{1} - P$ is invariant for $\mathcal{K}_K$. Irreducibility of $\mathcal{K}_K$ forces $\mathbb{1} - P \in \{0, \mathbb{1}\}$, hence $P \in \{0, \mathbb{1}\}$, so the Kraus map of $\{K_i^\dagger\}$ is irreducible. $\square$

Theorem B.4.4 (Positive-definite adjoint eigenvector for irreducible Kraus families). [Lean](#) [✓ Stmt](#) Let $\{K_i\}_{i=0}^{d-1}$ be a family with $D > 0$ and some $K_i \neq 0$, whose Kraus map $\mathcal{K}_K$ is irreducible. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that

$$\sum_i K_i^\dagger \sigma K_i = r\sigma.$$

This combines the eigenvector-existence part of [Wol12, Theorem 6.5] with the irreducible upgrade to positive definiteness ([Wol12, Theorem 6.3(2)]).

Proof. [✓ Proof](#) The conjugate-transposed family $\{K_i^\dagger\}$ has the same Kraus map as the adjoint of $\mathcal{K}_K$, so it is completely positive (Theorem B.4.2), and by Theorem B.4.1 it is nonzero. By Theorem B.4.3, its Kraus map is irreducible, so it does not annihilate any nonzero positive semidefinite matrix, and Theorem 6.8.2 applies, giving $\sigma \geq 0$, $\sigma \neq 0$, and $r > 0$ with $\sum_i K_i^\dagger \sigma K_i = r\sigma$. Theorem 6.3.3 then gives that σ is positive definite. $\square$

Theorem B.4.5 (Real spectral-radius identity). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 6.9.1, the real-valued spectral radius of E is also equal to r .

Proof. [✓ Proof](#) This is the real-valued reformulation of Theorem 6.9.1. $\square$

B.5 Exponential truncation and scalar reformulations

Theorem B.5.1 (Exponential truncation is positive definite). [Lean](#) [✓ Stmt](#) *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then the finite exponential truncation satisfies*

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A) > 0. \tag{B.3}$$

This is the finite-sum core of the completely positive specialization of [Wol12, Theorem 6.2(3)].

Proof. [✓ Proof](#) If a nonzero vector v annihilated the quadratic form in (B.3), positivity of every summand would make $E^k(A)v = 0$ for $0 \leq k \leq D-1$. Expanding $(\mathbb{1} + E)^{D-1}(A)$ by the binomial theorem would then make it annihilate v , contrary to Theorem 6.3.1. $\square$

Appendix C

Peripheral Channel Structure and Transfer-Operator Gaps: Supporting Results

This appendix supports Chapter 7. It proves the algebraic mixed-transfer identities, an auxiliary trace-positivity result, the peripheral intertwiners behind the gauge-rigidity theorems, the Banach-algebra convergence facts behind the overlap-decay theorems, the rank-one Perron projection behind the primitive overlap limit, and the cyclic-decomposition machinery that removes peripheral periodicity.

C.1 Mixed-transfer powers and overlap traces

This section proves the self-transfer identity, the word expansion of the mixed transfer operator, and the trace identities used in Theorem 7.3.1 and Theorem 7.3.2. Recall the mixed transfer operator F_{AB} of Definition 7.2.1 and its rectangular form F_{AB}^{rect} of Definition 7.2.2.

Theorem C.1.1 (Self-transfer). Lean ✓ Stmt *Setting $B = A$ recovers the ordinary transfer map: $F_{AA} = \mathcal{E}_A$.*

Proof. ✓ Proof Substituting $B = A$ into (7.2) gives $F_{AA}(X) = \sum_i A^i X (A^i)^\dagger = \mathcal{E}_A(X)$ by (2.4). □

Lemma C.1.2 (Iterated mixed transfer). Lean ✓ Stmt *For every $X \in M_D(\mathbb{C})$ and $N \geq 0$,*

$$F_{AB}^N(X) = \sum_{\sigma \in \{0, \dots, d-1\}^N} A^\sigma X (B^\sigma)^\dagger, \quad (\text{C.1})$$

where $A^\sigma = A^{\sigma_1} \dots A^{\sigma_N}$ denotes the word evaluation of Definition 2.1.2.

Proof. ✓ Proof Induct on N . The base case is $F_{AB}^0(X) = X$. For the inductive step, apply F_{AB} to each summand in (C.1) and reindex using $\{0, \dots, d-1\}^{N+1} \cong \{0, \dots, d-1\} \times \{0, \dots, d-1\}^N$. □

Lemma C.1.3 (Trace expansion over matrix units). [Lean](#) [✓ Stmt](#) For any linear endomorphism T on $M_{D_1 \times D_2}(\mathbb{C})$, the operator trace expands as

$$\mathrm{Tr}(T) = \sum_{p=0}^{D_1-1} \sum_{q=0}^{D_2-1} (T(E_{pq}))_{pq}, \quad (\text{C.2})$$

where $E_{pq} \in M_{D_1 \times D_2}(\mathbb{C})$ has entry 1 in position (p, q) and zero elsewhere.

Proof. [✓ Proof](#) The operator trace is $\mathrm{Tr}(T) = \sum_{p,q} \mathrm{tr}(T(E_{pq})E_{qp})$, and $\mathrm{tr}(ME_{qp}) = M_{pq}$. $\square$

Theorem 7.3.1 and Theorem 7.3.2 combine Lemma C.1.2 and Lemma C.1.3 the same way, over the square and rectangular matrix-unit bases respectively: expanding $\mathrm{Tr}(T)$ by (C.2), applying (C.1) to each $T(E_{pq})$ for $T = F_{AB}^N$ or $T = (F_{AB}^{\mathrm{rect}})^N$, and reassembling the sums over matrix-unit indices gives

$$\mathrm{Tr}(T) = \sum_{\sigma} \mathrm{tr}(A^{\sigma}) \overline{\mathrm{tr}(B^{\sigma})} = \sum_{\sigma} V^{(N)}(A)_{\sigma} \overline{V^{(N)}(B)_{\sigma}} = O_{AB}(N).$$

C.2 Auxiliary trace positivity

Lemma C.2.1 (Trace product of positive semidefinite matrices). [Lean](#) [✓ Stmt](#) If $A, B \geq 0$, then $\mathrm{tr}(AB) \geq 0$.

Proof. [✓ Proof](#) Write $B = U\Lambda U^{\dagger}$ with U unitary and Λ diagonal. Then

$$\mathrm{tr}(AB) = \mathrm{tr}((U^{\dagger}AU)\Lambda) = \sum_i (U^{\dagger}AU)_{ii} \Lambda_{ii} \geq 0,$$

because $U^{\dagger}AU \geq 0$ and $\Lambda_{ii} \geq 0$ for every i . $\square$

C.3 Rectangular spectral-radius formulation

Lemma C.3.1 (Rectangular spectral radius bound). [Lean](#) [✓ Stmt](#) For normalized tensors of bond dimensions D_1 and D_2 , $\rho_{\mathrm{spec}}(F_{AB}^{\mathrm{rect}}) \leq 1$.

Proof. [✓ Proof](#) This is the MPS formulation of Theorem 4.10.10: the rectangular mixed transfer is the mixed Kraus map, and the two tensor normalization identities are its trace-preserving hypotheses. $\square$

C.4 Peripheral intertwiners and gauge rigidity

This section proves the intertwining relations behind Theorem 7.7.1 and Theorem 7.7.4: transporting a modulus-one mixed-transfer eigenvector through separate unital gauges produces Kraus-level intertwiners, and these intertwiners force equal bond dimension once they are invertible.

Lemma C.4.1 (Gauged peripheral eigenvector yields Kraus intertwining). [Lean](#) [✓ Stmt](#) Let A be a normalized tensor of bond dimension D_1 and B a normalized tensor of bond dimension D_2 . Suppose ρ_A, ρ_B are fixed points of the respective transfer maps and S_A, S_B are invertible matrices with $S_A S_A^{\dagger} = \rho_A$ and $S_B S_B^{\dagger} = \rho_B$ (for instance the positive definite Perron–Frobenius fixed points of Theorem 6.5.2 and their square roots, as in the right-canonical gauge of Theorem 6.6.1).

Gauge to unital families $A'^i = S_A^{-1} A^i S_A$ and $B'^i = S_B^{-1} B^i S_B$. If $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ satisfies $F_{AB}^{\text{rect}}(X) = \mu X$ with $|\mu| = 1$, then $X' = S_A^{-1} X (S_B^\dagger)^{-1}$ is nonzero, the families A' and B' are unital, and, for every i ,

$$X'(B'^i)^\dagger = \mu(A'^i)^\dagger X', \quad A'^i X' = \mu X' B'^i. \quad (\text{C.3})$$

Proof. ✓ Proof Embed the gauged families and transported eigenvector into the block matrices

$$C^i := \begin{pmatrix} A'^i & 0 \\ 0 & B'^i \end{pmatrix}, \quad Y := \begin{pmatrix} 0 & X' \\ 0 & 0 \end{pmatrix}, \quad (\text{C.4})$$

on $M_{D_1+D_2}(\mathbb{C})$; the matrices $\{C^i\}$ form a unital Kraus family, and $E_C(Y) = \mu Y$. The positive definite fixed points of the adjoint maps give a faithful weighted trace for the block map. Applying the Kadison–Schwarz inequality to Y and pairing its positive semidefinite gap with this fixed point shows that the gap vanishes. The Kraus relation of Theorem 5.2.2 then gives $Y(C^i)^\dagger = \mu(C^i)^\dagger Y$; applying the same argument to the complementary product gives $C^i Y = \mu Y C^i$. Expanding these block relations gives (C.3). $\square$

Lemma C.4.2 (Intertwining yields gauge-phase equivalence). Lean ✓ Stmt *Let A and B be tensors of the same bond dimension. Suppose invertible gauges take them to A' and B' , and suppose there are an invertible matrix X' and a scalar μ with $|\mu| = 1$ such that $A'^i X' = \mu X' B'^i$ for every physical index i . Then A and B are gauge-phase equivalent.*

Proof. ✓ Proof Compose the two gauge matrices with the intertwiner to obtain one invertible matrix conjugating A to a scalar multiple of B . $\square$

Combining these two lemmas proves Theorem 7.7.1: Lemma C.4.1 supplies the intertwining relations (C.3) for the separately gauged, unital families. The Kadison–Schwarz equality used inside its own proof makes $X'X'^\dagger$ and $X'^\dagger X'$ nonzero positive semidefinite fixed points of the gauged transfer maps; under irreducibility, Theorem 6.3.12 upgrades these to positive definite matrices, so the transported intertwiner X' is invertible. Lemma C.4.2 then converts the invertible intertwiner, together with the two gauge matrices, into a single invertible matrix realizing the gauge-phase relation $B^i = \bar{\mu} X A^i X^{-1}$ via the Skolem–Noether theorem (Theorem 3.2.4).

Lemma 7.5.1 gives the companion kernel-invariance argument, independent of irreducibility: if a modulus-one rectangular intertwiner (7.6) exists between injective tensors of bond dimensions D_1 and D_2 , then the first relation shows that $\ker(X)$ is invariant under every $(B^i)^\dagger$; since the $\{B^i\}$ span the full matrix algebra, so do the $\{(B^i)^\dagger\}$, so $\ker(X)$ is trivial and $D_2 \leq D_1$. Taking adjoints of the second relation and applying the same argument to $X^\dagger$ shows that $\ker(X^\dagger)$ is invariant under every $(A^i)^\dagger$ and hence trivial, giving $D_1 \leq D_2$. Combined with Lemma C.4.1, this proves $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$ whenever A and B are injective tensors of unequal bond dimension, which is exactly Theorem 7.5.2.

C.5 Spectral-radius decay and overlap limits

This section collects the Banach-algebra convergence facts underlying Theorem 7.6.4, Theorem 7.6.5, and the block-separation consequences of the transfer-operator gap.

Lemma C.5.1 (Powers vanish below unit spectral radius). Lean ✓ Stmt *Let a be an element of a complex Banach algebra with $\rho_{\text{spec}}(a) < 1$. Then $a^n \rightarrow 0$ as $n \rightarrow \infty$.*

Proof. ✓ Proof Choose r with $\rho_{\text{spec}}(a) < r < 1$. The Gelfand formula $\|a^n\|^{1/n} \rightarrow \rho_{\text{spec}}(a)$ gives $\|a^n\| \leq r^n$ for all sufficiently large n , and $r^n \rightarrow 0$. $\square$

Theorem 7.5.3 applies Lemma C.5.1 to the rectangular mixed transfer operator and then evaluates its powers at a fixed matrix. Theorem 7.6.4 combines the gap $\rho_{\text{spec}}(F_{AB}) < 1$ of Theorem 7.4.6 with the equal-bond-dimension specialization of this rectangular decay theorem. Since Tr is continuous on the finite-dimensional matrix-endomorphism algebra, this gives $\text{Tr}(F_{AB}^N) \rightarrow 0$ term by term over the finitely many matrix-unit entries of (C.2); combined with the overlap–trace identity (7.4), this proves Theorem 7.6.5.

Theorem C.5.2 (Cross-correlation decay). Lean ✓ Stmt *If A and B are injective normalized tensors that are not gauge-phase equivalent, then $\text{tr}(F_{AB}^N(X)) \rightarrow 0$ as $N \rightarrow \infty$ for every $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Theorem 7.6.4 gives $F_{AB}^N(X) \rightarrow 0$ in operator norm, and continuity of the trace gives the conclusion. $\square$

Theorem C.5.3 (Self-correlation persists). Lean ✓ Stmt *If ρ is a fixed point of $\mathcal{E}_A$, then $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$ for every $N \geq 0$. In particular, this applies to the distinguished Perron–Frobenius fixed point from Theorem 6.5.2.*

Proof. ✓ Proof Since $\mathcal{E}_A(\rho) = \rho$, one has $\mathcal{E}_A^N(\rho) = \rho$ for every N . Taking the trace proves the claim. $\square$

C.6 Rank-one Perron projection

This section proves the rank-one Perron limit (Theorem 7.11.1) underlying the primitive overlap convergence (Theorem 7.11.2). Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be trace-preserving with a fixed point $\rho \neq 0$, $\text{tr}(\rho) \neq 0$. Write $P := P_\rho$ for the fixed-point projection of Definition 4.11.1 (Chapter 4),

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)}\rho,$$

and $N := E - P$ for the complementary part. The power decomposition (Theorem 4.11.2, Chapter 4) gives $E^n = P + N^n$ for $n \geq 1$.

Lemma C.6.1 (Trace of the fixed-point projection). Lean ✓ Stmt *The fixed-point projection P has operator trace $\text{Tr}(P) = 1$.*

Proof. ✓ Proof P has the rank-one form $P(X) = f(X)\rho$ for the linear functional $f(X) = \text{tr}(X)/\text{tr}(\rho)$. The operator trace of a rank-one endomorphism $X \mapsto f(X)\rho$ equals $f(\rho)$, and $f(\rho) = \text{tr}(\rho)/\text{tr}(\rho) = 1$. $\square$

Under the hypothesis $\rho_{\text{spec}}(N) < 1$ of Theorem 7.11.1, Lemma C.5.1 gives $N^n \rightarrow 0$, so $\text{Tr}(E^n) = \text{Tr}(P) + \text{Tr}(N^n) \rightarrow \text{Tr}(P) = 1$ by Lemma C.6.1. This is exactly the mechanism recorded in Theorem 7.11.1.

When $E = \mathcal{E}_A$ is the transfer map of a normalized MPS tensor and ρ is its Perron–Frobenius fixed point (Theorem 6.5.2, Chapter 6), the overlap–trace identity (7.4) with $A = B$ gives $O_{AA}(N) = \text{Tr}(\mathcal{E}_A^N)$, so Theorem 7.11.1 yields the primitive self-overlap limit of Theorem 7.11.2. The hypothesis $\rho_{\text{spec}}(N) < 1$ is exactly the complementary transfer-map gap that the complementary spectral-radius theorem for primitive maps (Theorem 4.13.3, Chapter 4) supplies: for a primitive E all eigenvalues bounded by 1 in modulus and no trace-zero fixed point, every eigenvalue of $N = E - P$ has modulus strictly less than 1, and since $M_D(\mathbb{C})$ is finite-dimensional this bounds the spectral radius $\rho_{\text{spec}}(N)$ itself.

Theorem C.6.2 (Complementary gap at a prescribed fixed point). [Lean](#) [✓ Stmt](#) Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be an irreducible primitive channel, and let $\rho \neq 0$ be a positive semidefinite fixed point. Then $\text{tr}(\rho) \neq 0$ and, for $P = P_\rho$, $\rho_{\text{spec}}(E - P) < 1$.

Proof. [✓ Proof](#) Positive semidefiniteness and $\rho \neq 0$ imply $\text{tr}(\rho) \neq 0$. Every eigenvalue of $E - P$ has modulus strictly less than one by irreducibility and primitivity. Since $M_D(\mathbb{C})$ is finite-dimensional, the maximum eigenvalue modulus equals the spectral radius. $\square$

Theorem C.6.3 (Complementary spectral-radius gap for irreducible primitive tensors). [Lean](#) [✓ Stmt](#) Let A be an irreducible normalized MPS tensor whose transfer map $\mathcal{E}_A$ is primitive. Then there is a nonzero positive semidefinite fixed point ρ of $\mathcal{E}_A$ with $\text{tr}(\rho) \neq 0$ such that, for $P := P_\rho$ and $N := \mathcal{E}_A - P$, $\rho_{\text{spec}}(N) < 1$.

Proof. [✓ Proof](#) The channel fixed-point existence theorem (Theorem 6.5.1) supplies a nonzero positive semidefinite fixed point ρ . Apply Theorem C.6.2 to $\mathcal{E}_A$ and ρ . $\square$

This concretely instantiates the hypothesis of Theorem 7.11.1 and, via Theorem 7.11.2, gives the primitive self-overlap limit for every irreducible peripherally primitive normalized MPS tensor.

C.7 Periodicity removal

This section proves the cyclic-decomposition machinery of Theorem 7.9.1.1 and the divisibility statements of Theorem 7.9.2 and Theorem 7.9.3: normalizing a peripheral eigenvector to a unitary, forming its discrete Fourier (cyclic) spectral projections, and tracking how the channel shifts the resulting corners.

Lemma C.7.1 (Peripheral eigenvectors are invertible). [Lean](#) [✓ Stmt](#) Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If $X \neq 0$ satisfies $E(X) = \mu X$ with $|\mu| = 1$, then X is invertible.

Proof. [✓ Proof](#) By the Kadison–Schwarz inequality (5.3), the gap $G = E(X^\dagger X) - E(X)^\dagger E(X)$ is positive semidefinite. Pairing G with the positive-definite adjoint fixed point and using $E(X) = \mu X$ and $|\mu| = 1$ gives weighted trace zero, hence $G = 0$. Therefore $E(X^\dagger X) = E(X)^\dagger E(X) = X^\dagger X$, so $X^\dagger X$ is a nonzero positive semidefinite fixed point. Irreducibility upgrades it to a positive-definite, hence invertible, matrix, so $\det(X) \neq 0$. $\square$

Lemma C.7.2 (Peripheral eigenvalues: product closure). [Lean](#) [✓ Stmt](#) Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If μ, ν are peripheral eigenvalues of E , then so is $\mu\nu$.

Proof. [✓ Proof](#) Take eigenvectors X and Y for μ and ν , respectively. The Kadison–Schwarz equality at X places X in the multiplicative domain. Thus the right multiplicative identity (5.12) gives $E(YX) = E(Y)E(X) = (\nu\mu)YX$. Since X and Y are invertible by Lemma C.7.1, $YX \neq 0$, and $|\mu\nu| = 1$. $\square$

Lemma C.7.3 (Scalar fixed points for irreducible unital maps). [Lean](#) [✓ Stmt](#) If E is irreducible and unital, then every fixed point of E is a scalar multiple of $\mathbb{1}$.

Proof. [✓ Proof](#) If $E(X) = X$, then the Hermitian matrices $X + X^\dagger$ and $i(X - X^\dagger)$ are also fixed by E . The positive-semidefinite fixed-point uniqueness theorem gives $X + X^\dagger = a\mathbb{1}$ and $i(X - X^\dagger) = b\mathbb{1}$, hence $X = \frac{1}{2}(a - ib)\mathbb{1}$. $\square$

Lemma C.7.4 (Peripheral unitary eigenvector). [Lean](#) [✓ Stmt](#) *For an irreducible unital Schwarz map with faithful adjoint fixed point, each peripheral eigenvalue admits a unitary eigenvector.*

Proof. [✓ Proof](#) For an eigenvector X at a peripheral eigenvalue γ , the Kadison–Schwarz equality gives $E(X^\dagger X) = E(X)^\dagger E(X) = X^\dagger X$, so $X^\dagger X$ is a nonzero positive semidefinite fixed point. Irreducible uniqueness of the positive semidefinite fixed point makes it a positive scalar $c\mathbb{1}$; rescaling X by $c^{-1/2}$ produces a unitary eigenvector. $\square$

Lemma C.7.5 (Powers on a peripheral-unitary orbit). [Lean](#) [✓ Stmt](#) *If $E(U) = \mu U$ with $|\mu| = 1$ and U unitary, then $E(U^n) = \mu^n U^n$ for every n .*

Proof. [✓ Proof](#) The Kadison–Schwarz equality at the unitary eigenvector places U in the multiplicative domain, so the right multiplicative identity (5.12) gives $E(U^n) = E(U)^n = \mu^n U^n$ by induction on n . $\square$

Lemma C.7.6 (Normalized peripheral unitary). [Lean](#) [✓ Stmt](#) *Peripheral unitary eigenvectors can be normalized compatibly with the faithful fixed-point state.*

Proof. [✓ Proof](#) Given a unitary eigenvector U for a primitive m -th root γ , Lemma C.7.5 gives $E(U^m) = U^m$, a fixed point that is scalar by Lemma C.7.3: $U^m = \alpha\mathbb{1}$. Unitarity of U^m forces $|\alpha| = 1$; multiplying U by an m -th root β of α^{-1} produces $(\beta U)^m = 1$ while preserving the eigenvector equation. $\square$

Lemma C.7.7 (Cyclic projections from a peripheral unitary). [Lean](#) [✓ Stmt](#) *A primitive m -th peripheral root yields m orthogonal projections summing to $\mathbb{1}$ and cyclically permuted by the channel.*

Proof. [✓ Proof](#) Given a unitary U of order m with $E(U) = \gamma U$ for a primitive m -th root γ , the Fourier projections $P_k := m^{-1} \sum_{j=0}^{m-1} \gamma^{kj} U^j$ are mutually orthogonal and sum to $\mathbb{1}$ by discrete Fourier orthogonality of the characters $j \mapsto \gamma^{kj}$; since $E(U^j) = \gamma^j U^j$ (Lemma C.7.5), each Fourier mode is shifted by one step, giving $E(P_{k+1}) = P_k$. $\square$

Combining these lemmas proves Theorem 7.9.1.1: Lemma C.7.6 supplies a unitary generator of exact order m for a primitive m -th peripheral root, and Lemma C.7.7 builds the m Fourier projections out of its powers, using Lemma C.7.5 to identify how the channel shifts them.

C.7.1 Cyclic corners and restricted primitivity

Definition C.7.1.1 (Corner preservation). [Lean](#) [✓ Stmt](#) For an orthogonal projection P and linear map E , the corner $PM_D(\mathbb{C})P$ is preserved if $E(PXP) = PE(PXP)P$ for every X .

Definition C.7.1.2 (Corner subspace). [Lean](#) [✓ Stmt](#) The corner subspace associated with P is $\{PXP : X \in M_D(\mathbb{C})\}$.

Definition C.7.1.3 (Restricted corner map). [Lean](#) [✓ Stmt](#) The restriction of E to an invariant corner $PM_D(\mathbb{C})P$.

Definition C.7.1.4 (Irreducible restriction on a corner). [Lean](#) [✓ Stmt](#) Irreducibility of the corner-restricted map.

Definition C.7.1.5 (Rank of an orthogonal projection). [Lean](#) [✓ Stmt](#) The rank of an orthogonal projection $P \in M_D(\mathbb{C})$, equivalently the dimension of its range.

Lemma C.7.1.6 (Compression isometry for an orthogonal projection). [Lean](#) [✓ Stmt](#) Let $P \in M_D(\mathbb{C})$ be an orthogonal projection of rank n . The corner algebra $PM_D(\mathbb{C})P$ is linearly isomorphic to the full matrix algebra $M_n(\mathbb{C})$. The isomorphism is built from the spectral diagonalisation of P by conjugating the top-left $n \times n$ block by the unitary diagonalising P .

Lemma C.7.1.7 (Rank equals trace for an orthogonal projection). [Lean](#) [✓ Stmt](#) The rank of an orthogonal projection P equals its trace: $n = \text{tr}(P)$.

Proof. [✓ Proof](#) The eigenvalues of an orthogonal (Hermitian idempotent) projection are 0 or 1: the eigenvalues λ satisfy $\lambda^2 = \lambda$ from $P^2 = P$. The trace is the sum of the eigenvalues, which counts the eigenvalue-1 multiplicities, i.e. the rank n . $\square$

Lemma C.7.1.8 (Power maps preserve each cyclic sector). [Lean](#) [✓ Stmt](#) Appropriate powers of the channel preserve each sector corner.

Proof. [✓ Proof](#) Induction on the exponent, using the left- and right-multiplicative domain identities on each sector projection P_k , shows $T^n(P_{k+n \bmod m} X P_{k+n \bmod m}) = P_k T^n(X) P_k$; specializing to $n = m$, where $P_{k+m} = P_k$, gives that T^m preserves the corner $P_k M_D(\mathbb{C}) P_k$. $\square$

Lemma C.7.1.9 (Primitivity passes to a fixed corner). [Lean](#) [✓ Stmt](#) Let $P \in M_D(\mathbb{C})$ be a nonzero idempotent and let E be a linear map on $M_D(\mathbb{C})$ that preserves the corner $PM_D(\mathbb{C})P$, is primitive, and fixes the corner projection, $E(P) = P$. Then the restriction of E to the corner $PM_D(\mathbb{C})P$ is again primitive.

Proof. [✓ Proof](#) The corner projection P lies in its own corner and is fixed by the restriction, so 1 is a peripheral eigenvalue of the restriction. Conversely, every eigenvalue of the restriction of unit modulus arises from a nonzero corner element X with $E(X) = \mu X$; viewing X as an element of $M_D(\mathbb{C})$ exhibits μ as a peripheral eigenvalue of E . Primitivity of E forces $\mu = 1$, so the peripheral spectrum of the restriction is exactly $\{1\}$ and the restriction is primitive. $\square$

Lemma C.7.1.10 (Irreducibility of restricted sector dynamics). [Lean](#) [✓ Stmt](#) Let T be irreducible, let $P_0, \dots, P_{m-1}$ be orthogonal projections summing to $\mathbb{1}$ and cyclically permuted by T , and suppose that every orthogonal projection Q with $QP_k = P_k Q = Q$ that is invariant under the corner restriction of T^m to $P_k M_D(\mathbb{C}) P_k$ admits an orthogonal projection R invariant under T on the full algebra with $Q = 0 \iff R = 0$ and $Q = P_k \iff R = \mathbb{1}$. Then, for every k , the restriction of T^m to the corner $P_k M_D(\mathbb{C}) P_k$ is irreducible.

Proof. [✓ Proof](#) Let Q be an invariant projection for the corner restriction of T^m on $P_k M_D(\mathbb{C}) P_k$. The orbit-sum hypothesis supplies an ambient invariant projection R for T with $Q = 0 \iff R = 0$ and $Q = P_k \iff R = \mathbb{1}$. Irreducibility of T forces $R = 0$ or $R = \mathbb{1}$, hence $Q = 0$ or $Q = P_k$ by the two biconditionals. $\square$

Lemma C.7.1.11 (Primitivity of restricted sector dynamics). [Lean](#) [✓ Stmt](#) Each sector restriction is primitive.

Proof. [✓ Proof](#) Every peripheral eigenvalue of T is an m -th root of unity, so the peripheral spectrum of T^m is exactly $\{1\}$; combined with $T^m(P_k) = P_k$, Lemma C.7.1.9 makes the corner restriction of T^m to $P_k M_D(\mathbb{C}) P_k$ primitive. $\square$

Lemma C.7.1.12 (Corner invariance at the period). [Lean](#) [✓ Stmt](#) The channel raised to the period preserves every cyclic corner.

Proof. [✓ Proof](#) The same induction as in Lemma C.7.1.8, with the cyclic shift $k \mapsto k+1$ replaced by a permutation σ of the sector index set, shows that $T^{\text{ord}(\sigma)}$ preserves each corner $P_k M_D(\mathbb{C}) P_k$. $\square$

C.7.2 Group structure and divisibility

Lemma C.7.2.1 (Peripheral product closure). [Lean](#) [✓ Stmt](#) *The product of two peripheral eigenvalues is peripheral.*

Proof. [✓ Proof](#) Peripheral eigenvalues admit unitary eigenvectors (Lemma C.7.4). If U_α, U_β are unitary eigenvectors for α, β , the multiplicative-domain identity gives $E(U_\alpha U_\beta) = E(U_\alpha)E(U_\beta) = \alpha\beta U_\alpha U_\beta$, and $U_\alpha U_\beta \neq 0$ since both factors are unitary. $\square$

Lemma C.7.2.2 (Peripheral inverse closure). [Lean](#) [✓ Stmt](#) *The inverse of a peripheral eigenvalue is peripheral.*

Proof. [✓ Proof](#) For a unitary eigenvector U_α with $|\alpha| = 1$, conjugating the eigenvector equation gives $E(U_\alpha^\dagger) = \bar{\alpha} U_\alpha^\dagger = \alpha^{-1} U_\alpha^\dagger$, and $U_\alpha^\dagger \neq 0$. $\square$

Theorem 7.9.1.4 follows the same pattern: given a peripheral unitary eigenvector U for γ (Lemma C.7.4), the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ -eigenvector X , so the scalar fixed-point lemma for irreducible unital maps (Lemma C.7.3) gives $XU^\dagger = c\mathbb{1}$, hence $X = cU$ and the eigenspace is one-dimensional.

The divisibility statements of Theorem 7.9.2 and Theorem 7.9.3 follow from Lemma C.7.7 and Theorem C.7.1.7: the m cyclic Fourier projections $P_0, \dots, P_{m-1}$ sum to $\mathbb{1}$ and are cyclically permuted by E , so trace preservation gives $\text{tr}(P_{k+1}) = \text{tr}(P_k)$ for every k , hence $D = \text{tr}(\mathbb{1}) = \sum_{k=0}^{m-1} \text{tr}(P_k) = m \text{tr}(P_0)$. Since the trace of an orthogonal projection is a natural number (Theorem C.7.1.7), $m \mid D$.

This periodicity-removal chain — unitary normalization, Fourier projections, cyclic Kraus shift (Lemma 7.9.1.2), the blocked cyclic corners of this section, and restricted primitivity — is exactly the machinery behind the period-removing blocking theorems of Chapter 9: [Wol12, Theorem 6.6] supplies the cyclic peripheral spectrum of the adjoint transfer map, and Theorem 9.3.1.1 (Section 9.3.1) and Theorem 9.11.1 apply it to realize the cyclic-sector decomposition of a blocked irreducible tensor with rectangular support isometries.

Appendix D

Wielandt Bound: Supporting Results

This appendix supports Chapter 8. It collects exact-word, cumulative-span, augmentation, blocking, complementary-gap, and one-step padding results used by the main quantitative argument.

D.1 Exact-word and block-injectivity support

This section proves the exact-word identities and low-dimensional facts used in Lemma 8.1.1.35, Theorem 8.1.1.36, Theorem 8.5.2, and Theorem 8.9.1.

Lemma D.1.1 (Word span products). Lean ✓ Stmt *One has $S_m(A)S_n(A) \subseteq S_{m+n}(A)$: the product of word spans at lengths m and n is contained in the word span at length $m + n$.*

Proof. ✓ Proof Every product $A^{w_1}A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1w_2}$, where $|w_1w_2| = m + n$. □

Lemma D.1.2 (Zero-length word span). Lean ✓ Stmt *If $D \geq 2$, then $S_0(A) \neq M_D(\mathbb{C})$.*

Proof. ✓ Proof The zero-length products consist only of the identity, so $S_0(A) = \text{span}_{\mathbb{C}}\{\mathbf{1}\}$ has dimension 1. Since $\dim M_D(\mathbb{C}) = D^2 \geq 4$, this span cannot equal $M_D(\mathbb{C})$. □

Corollary D.1.3 (0-block injectivity). Lean ✓ Stmt *If $D \geq 1$ and A is 0-block injective, then $D = 1$.*

Proof. ✓ Proof By Lemma 8.1.1.35, 0-block injectivity gives $S_0(A) = M_D(\mathbb{C})$. Lemma D.1.2 excludes $D \geq 2$, so the nonzero bond dimension must be 1. □

Lemma D.1.4 (Two-sided span from one nonzero matrix). Lean ✓ Stmt *If $C \in M_D(\mathbb{C})$ is nonzero, then*

$$\text{span}_{\mathbb{C}}\{RCS : R, S \in M_D(\mathbb{C})\} = M_D(\mathbb{C}). \tag{D.1}$$

This is the matrix-factorization input used in [PGVWC07, Lemma 3].

Proof. ✓ Proof Choose a nonzero entry C_{pq} . Then every matrix unit E_{ij} is a scalar multiple of $E_{ip}CE_{qj}$, so the span in (D.1) contains the standard matrix basis. □

Lemma D.1.5 (Exact words around a nonzero middle matrix). [Lean](#) [✓ Stmt](#) *If A is L -block injective and $X \neq 0$, then*

$$\text{span}_{\mathbb{C}}\{A^u X A^v : |u| = |v| = L\} = M_D(\mathbb{C}).$$

Proof. [✓ Proof](#) By L -block injectivity, the length- L word products span $M_D(\mathbb{C})$. Bilinearity of $(R, S) \mapsto RXS$ reduces the claim to Lemma D.1.4, with R and S chosen among those word products. $\square$

D.2 Cumulative-span and spectral linear algebra

This section supplies the finite-dimensional stabilization and spectral facts used in Theorem 8.2.2, Theorem 8.3.2, and Theorem 8.4.8.

The next two lemmas connect irreducible matrix actions with the cumulative word span used throughout Chapter 8.

Remark D.2.1 (Burnside input for cumulative spanning). The passage from irreducible tensors to full algebra spanning first passes through the irreducible action on $\mathbb{C}^D$. Burnside's theorem then says that the resulting irreducible subalgebra of $M_D(\mathbb{C})$ is all of $M_D(\mathbb{C})$. Once the algebra span is full, finite-dimensionality gives a single cumulative span that already equals $M_D(\mathbb{C})$.

Lemma D.2.2 (Irreducible action gives an irreducible tensor). [Lean](#) [✓ Stmt](#) *If the letters of A act irreducibly on $\mathbb{C}^D$, then A has no nontrivial invariant orthogonal projection.*

Proof. [✓ Proof](#) If P were a nontrivial invariant orthogonal projection, then its range would be invariant under every letter of A . Irreducibility of the action forces this range to be either 0 or all of $\mathbb{C}^D$, so P must be 0 or $\mathbb{1}$, a contradiction. $\square$

Lemma D.2.3 (The algebra span reaches a finite cumulative span). [Lean](#) [✓ Stmt](#) *If the generated unital algebra satisfies $\text{alg}(A) = M_D(\mathbb{C})$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.*

Proof. [✓ Proof](#) Every element of $\text{alg}(A)$ lies in some cumulative span $T_n(A)$: the generators lie in $T_1(A)$, scalar multiples of the identity lie in $T_0(A)$, and the family $\{T_n(A)\}_{n \geq 0}$ is closed under addition and under multiplication with lengths adding. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \subseteq T_1(A) \subseteq \dots$ stabilizes, so one can choose a single N with $T_N(A) = M_D(\mathbb{C})$. $\square$

Definition D.2.4 (Fitting decomposition). [Lean](#) [✓ Stmt](#) A *Fitting decomposition* of a linear endomorphism $f : V \rightarrow V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ -eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V ,
4. the generalized eigenspaces are linearly independent.

Theorem D.2.5 (Fitting decomposition exists). [Lean](#) [✓ Stmt](#) *Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.*

Proof. ✓ Proof Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead. $\square$

Theorem D.2.6 (Nilpotent power bound). Lean ✓ Stmt *If f is nilpotent on a space of dimension n , then $f^n = 0$.*

Proof. ✓ Proof A nilpotent endomorphism on an n -dimensional space has nilpotency index at most n . $\square$

Theorem D.2.7 (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound). Lean ✓ Stmt *Let $D > 0$. If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof By Theorem 8.1.1.34, if $T_n \neq T_{n+1}$, then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ by Lemma 8.1.1.32, after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality, which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0 and hence $T_{L_0} = M_D(\mathbb{C})$. $\square$

Theorem D.2.8 (Nonzero trace product). Lean ✓ Stmt *Let $D > 0$. If A is normal, then there exists a word w of length at most D^2 such that $\text{tr}(A^w) \neq 0$. This is the weaker D^2 corollary of [SPGWC10, Lemma 1]. The sharp source statement is Theorem 8.2.5.*

Proof. ✓ Proof By Theorem 8.2.1, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length at most D^2 . At least one has nonzero trace because $\text{tr}(\mathbb{1}) = D \neq 0$. $\square$

Theorem D.2.9 (Nonzero trace implies eigenvector). Lean ✓ Stmt *If $M \in M_D(\mathbb{C})$ has $\text{tr}(M) \neq 0$, then M has a nonzero eigenvalue $\mu \neq 0$ and a corresponding eigenvector $\varphi \neq 0$ satisfying $M\varphi = \mu\varphi$.*

Proof. ✓ Proof The trace is the sum of the eigenvalues of M , counted with multiplicity, so a nonzero trace gives a nonzero eigenvalue μ . Choosing a nonzero vector $\varphi \in \ker(M - \mu\mathbb{1})$ gives $M\varphi = \mu\varphi$. $\square$

Theorem D.2.10 (Eigenvalue extraction). Lean ✓ Stmt *Let $D > 0$. If A is normal, then there exist a word w_0 with $|w_0| \leq D^2$, a nonzero scalar $\mu \neq 0$, and a nonzero vector $\varphi \neq 0$ such that $A^{w_0}\varphi = \mu\varphi$.*

Proof. ✓ Proof By Theorem D.2.8, there is a word w_0 with $\text{tr}(A^{w_0}) \neq 0$ and $|w_0| \leq D^2$. By Theorem D.2.9, A^{w_0} has a nonzero eigenvector. $\square$

Theorem D.2.11 (Vector span stabilisation). Lean ✓ Stmt *If $K_n(A, \varphi) = K_{n+1}(A, \varphi)$, then $K_m(A, \varphi) = K_n(A, \varphi)$ for every $m \geq n$.*

Proof. ✓ Proof The argument is the same as for Theorem 8.1.1.33: left multiplication by A^i cannot escape the stabilised span. $\square$

Lemma D.2.12 (Vector span from matrix span). Lean ✓ Stmt *If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \neq 0$, then $K_N(A, \varphi) = \mathbb{C}^D$.*

Proof. ✓ Proof Every matrix $M \in M_D(\mathbb{C})$ lies in $T_N(A)$, so in particular every rank-one matrix sending φ to any target vector v lies in $T_N(A)$. Applying such matrices to φ recovers all of $\mathbb{C}^D$. $\square$

D.3 One-step augmentation

This section proves the one-step augmentation results used in Theorem 8.4.2 and Theorem 8.4.6.

Definition D.3.1 (One-step augmentation). Lean ✓ Stmt Let A be an MPS tensor with Kraus family $\{A^i\}_{i=0}^{d-1}$, and let $X \in M_D(\mathbb{C})$. The *one-step augmentation* of A by X is the tensor $\tilde{A}$ with physical dimension $d + 1$ whose first Kraus operator is X and whose remaining Kraus operators are those of A :

$$\begin{aligned}\tilde{A}^0 &= X, \\ \tilde{A}^{i+1} &= A^i.\end{aligned}$$

Lemma D.3.2 (Kraus operators lie in the one-step span). Lean ✓ Stmt Every Kraus operator A^i belongs to the one-step span $S_1(A)$.

Proof. ✓ Proof This is the length-one word with letter i . □

Lemma D.3.3 (Redundant one-step augmentation). Lean Lean Lean ✓ Stmt Let $X \in S_1(A)$, and let $\tilde{A}$ be the one-step augmentation of A by X . Then $S_n(\tilde{A}) = S_n(A)$ for every $n \geq 0$.

Proof. ✓ Proof The new generator X is already in $S_1(A)$, and each original generator remains in the augmented family. Hence the one-step spans agree. Multiplying word spans and arguing by induction on the word length gives equality at every length. □

Lemma D.3.4 (Normality and Kraus rank under one-step augmentation). Lean Lean ✓ Stmt Let $X \in S_1(A)$, and let $\tilde{A}$ be the one-step augmentation of A by X . If A is normal, then $\tilde{A}$ is normal, and $\text{kr}(\tilde{A}) = \text{kr}(A)$.

Proof. ✓ Proof Since all exact word spans are unchanged by Lemma D.3.3, eventual full word span for A gives eventual full word span for $\tilde{A}$. The equality of Kraus ranks is the equality of the dimensions of the common one-step span. □

Theorem D.3.5 (Wielandt case (2), invertible generator). Lean Lean ✓ Stmt Let A be a normal MPS tensor with bond dimension $D > 0$, and suppose that some Kraus operator A^{i_0} is invertible. Then

$$\begin{aligned}S_{D^2 - \text{kr}(A) + 1}(A) &= M_D(\mathbb{C}), \\ \iota(A) &\leq D^2 - \text{kr}(A) + 1.\end{aligned}$$

Proof. ✓ Proof The dimensions of the exact word spans cannot decrease, because left multiplication by the invertible Kraus operator is injective. If this dimension growth stopped before the full matrix algebra was reached, the same argument at every later length would contradict normality. Since $\dim S_1(A) = \text{kr}(A)$ and $\dim M_D(\mathbb{C}) = D^2$, the full algebra is reached by length $D^2 - \text{kr}(A) + 1$. The index bound follows from the definition of $\iota(A)$. □

Corollary D.3.6 (Wielandt case (2), invertible Kraus operator). Lean Lean ✓ Stmt Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive in the sense of Definition 8.1.1.3. If some Kraus operator A^{i_0} is invertible, then

$$\begin{aligned}S_{D^2 - \text{kr}(A) + 1}(A) &= M_D(\mathbb{C}), \\ \iota(A) &\leq D^2 - \text{kr}(A) + 1.\end{aligned}$$

Proof. ✓ Proof Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma D.3.2. $\square$

Corollary D.3.7 (Wielandt case (3), non-invertible Kraus operator). Lean Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive in the sense of Definition 8.1.1.3. If some Kraus operator A^{i_0} is non-invertible and has an eigenvector $\varphi \neq 0$ with nonzero eigenvalue μ , then*

$$S_{D^2}(A) = M_D(\mathbb{C}), \\ \iota(A) \leq D^2.$$

Proof. ✓ Proof Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma D.3.2. $\square$

D.4 Blocking and fixed-length spanning

This section proves the blocking and fixed-length constructions used in Theorem 8.4.8, Theorem 8.5.5, and Theorem 8.9.1.

Theorem D.4.1 (Blocking preserves normality). Lean ✓ Stmt *If A is normal and $L > 0$, then the blocked tensor $A^{[L]}$ is also normal.*

Proof. ✓ Proof Choose N with $S_N(A) = M_D(\mathbb{C})$. Then $\mathbb{1} \in S_N(A)$. For any $M \in S_N(A)$, write $\mathbb{1} = \sum_r c_r A^{w_r}$ with each $|w_r| = N$. Then

$$M = M\mathbb{1} = \sum_r c_r M A^{w_r} \in S_N(A) S_N(A) \subseteq S_{2N}(A). \quad (\text{D.2})$$

By Lemma D.1.1, iterating (D.2) gives $S_N(A) \subseteq S_{kN}(A)$ for every $k \geq 1$. Taking $k = L$ yields $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$. By Lemma D.4.2, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. $\square$

Lemma D.4.2 (Chunking: word span embeds into blocked word span). Lean ✓ Stmt *One has $S_{nL}(A) \subseteq S_n(A^{[L]})$ for all n, L .*

Proof. ✓ Proof Every length- nL word in the original alphabet can be split into n consecutive chunks of length L , each of which is a single blocked Kraus operator by (2.13). The claim follows by induction on n , using Lemma D.1.1. $\square$

Theorem D.4.3 (Word eigenvector gives blocked eigenvector). Lean ✓ Stmt *If $A^{w_0} \varphi = \mu \varphi$ for a word w_0 of length L , then $(A^{[L]})^{j_0} \varphi = \mu \varphi$, where j_0 is the encoded blocked index corresponding to w_0 .*

Proof. ✓ Proof By (2.13), $(A^{[L]})^{j_0} = A^{w_0}$. $\square$

Theorem D.4.4 (Blocked fixed-length matrix spanning). Lean ✓ Stmt *Suppose A has bond dimension $D > 0$, is normal, and $L > 0$, and we have:*

1. *a word w_0 of length L with eigenvector $A^{w_0} \varphi = \mu \varphi$ ($\mu \neq 0$, $\varphi \neq 0$),*
2. *a word τ_0 of length L with transpose eigenvector $(A^{\tau_0})^T \psi = \nu \psi$ ($\nu \neq 0$, $\psi \neq 0$),*
3. *a rank-one element $\varphi \psi^T \in S_m(A^{[L]})$ for some m .*

Then there exists N such that $S_N(A) = M_D(\mathbb{C})$.

Remark. The proof gives the explicit bound $N = ((D-1) + m + (D-1))L$; the theorem states only the existential conclusion.

Proof. ✓ Proof Set $B = A^{[L]}$, which is normal by Theorem D.4.1. The word eigenvectors transfer to single-index eigenvectors of B and B^T by Theorem D.4.3. Eigenvector spreading and padding then give $H_{D-1}(B, \varphi) = H_{D-1}(B^T, \psi) = \mathbb{C}^D$.

For each standard basis vector e_j , choose $R_j \in S_{D-1}(B^T)$ with $R_j \psi = e_j$. Reversing each word after transposition shows that $R_j^T \in S_{D-1}(B)$, and therefore $(\varphi \psi^T) R_j^T = \varphi e_j^T$ belongs to $S_{m+D-1}(B)$. Theorem 8.5.2, applied with $n = D - 1$ and $m' = m + D - 1$, now gives $S_{(D-1)+m+(D-1)}(B) = M_D(\mathbb{C})$. Finally, Lemma 8.5.4 transfers this identity to $S_{((D-1)+m+(D-1))L}(A) = M_D(\mathbb{C})$. $\square$

Theorem D.4.5 (Rank-one extraction for blocked tensor). Lean ✓ Stmt Suppose A has bond dimension $D > 0$, is normal, and $N_0 \geq 1$ with $S_{N_0}(A) = M_D(\mathbb{C})$. Then there exist words σ_0, τ_0 of length N_0 , nonzero vectors φ, ψ , nonzero scalars μ, ν , and $m \in \mathbb{N}$ such that:

1. $A^{\sigma_0} \varphi = \mu \varphi$,
2. $(A^{\tau_0})^T \psi = \nu \psi$,
3. $\varphi \psi^T \in S_m(A^{[N_0]})$.

Proof. ✓ Proof Set $B = A^{[N_0]}$. By (2.13) and $S_{N_0}(A) = M_D(\mathbb{C})$, one has $S_1(B) = M_D(\mathbb{C})$. Hence $\mathbb{1} \in S_1(B)$, so not all blocked generators can have trace zero. Choose j_0 with $\text{tr}(B^{j_0}) \neq 0$. By Theorem D.2.9, B^{j_0} has a nonzero eigenvector φ with eigenvalue $\mu \neq 0$. Writing j_0 as the blocked index corresponding to a word σ_0 of length N_0 gives $A^{\sigma_0} \varphi = \mu \varphi$. Applying the same argument to the transpose blocked tensor gives a word τ_0 of length N_0 , a nonzero vector ψ , and $\nu \neq 0$ with $(A^{\tau_0})^T \psi = \nu \psi$. Since A is normal and $N_0 \geq 1$, the blocked tensor B is normal by Theorem D.4.1. Therefore there exists M with $S_M(B) = M_D(\mathbb{C})$. In particular, $\varphi \psi^T \in S_M(B)$. Taking $m = M$ proves (3). $\square$

Theorem D.4.6 (Fixed-length word-span saturation via blocking). Lean ✓ Stmt If A is a normal MPS tensor with bond dimension $D > 0$, then there exists N such that $S_N(A) = M_D(\mathbb{C})$. This is a blocked fixed-length construction under the stated normality hypothesis. The quantitative result of [SPGWC10, Lemma 2(b)] is Theorem 8.4.5.

Proof. ✓ Proof By normality, some $N_0 > 0$ has $S_{N_0}(A) = M_D(\mathbb{C})$. Theorem D.4.5 produces eigenvectors and a rank-one element $\varphi \psi^T \in S_m(A^{[N_0]})$. Theorem D.4.4 then gives $S_{((D-1)+m+(D-1))N_0}(A) = M_D(\mathbb{C})$. $\square$

D.5 Complementary-gap consequences

This section proves the complementary-gap consequences used in Lemma 8.7.2.1, Lemma 8.7.4.1, and Theorem 8.7.4.2.

Theorem D.5.1 (Primitive overlap convergence). Lean ✓ Stmt If A is a primitive MPS tensor, then $O_{AA}(N) \rightarrow 1$.

Proof. ✓ Proof The claim follows from the self-overlap convergence under a complementary transfer-map gap (Theorem 7.11.2). $\square$

Theorem D.5.2 (Peripheral primitivity and irreducibility give normalized overlap). Lean ✓ Stmt Let A have positive bond dimension and satisfy $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. If A is tensor-irreducible and the transfer map is primitive, then $O_{AA}(N) \rightarrow 1$.

Proof. ✓ Proof Irreducible Perron–Frobenius theory supplies a nonzero positive fixed point. Peripheral primitivity gives a strict spectral-radius bound on the complementary transfer map. Thus A is a primitive MPS tensor, and Theorem D.5.1 applies. $\square$

Lemma D.5.3 (Primitive MPS tensors have nonzero word products). Lean ✓ Stmt *If A is a primitive MPS tensor, then for every $n \geq 0$ there exists a word w of length exactly n with $A^w \neq 0$.*

Proof. ✓ Proof The PSD fixed point ρ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. $\square$

Lemma D.5.4 (Iterated transfer does not vanish). Lean ✓ Stmt *If A is a primitive MPS tensor, then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.*

Proof. ✓ Proof By Lemma D.5.3, there exists a word w of length n with $A^w \neq 0$. The expansion $\mathcal{E}_A^n(\mathbb{1}) = \sum_{|u|=n} A^u (A^u)^\dagger$ contains the nonzero positive semidefinite summand $A^w (A^w)^\dagger$, so $\mathcal{E}_A^n(\mathbb{1}) \neq 0$. $\square$

Lemma D.5.5 (Fixed-point uniqueness from a complementary transfer-map gap). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ , then every fixed point σ of $\mathcal{E}_A$ is proportional to ρ : $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$.*

Proof. ✓ Proof Set $\sigma' = \sigma - \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$. Then $\text{tr}(\sigma') = 0$ and $(\mathcal{E}_A - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $\mathcal{E}_A - P_\rho$, contradicting the complementary transfer-map gap $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$. $\square$

Lemma D.5.6 (Complement powers tend to zero). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ , then $(\mathcal{E}_A - P_\rho)^n \rightarrow 0$ in operator norm.*

Proof. ✓ Proof The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the complementary transfer-map gap hypothesis. Apply the general result that powers of an operator with spectral radius < 1 tend to zero. $\square$

Lemma D.5.7 (Primitive tensors with positive-definite fixed point have irreducible transfer map). Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ , then the transfer map $\mathcal{E}_A$ is irreducible.*

The proof uses fixed-point uniqueness (Lemma D.5.5): every PSD fixed point is proportional to ρ , so Wolf’s criterion for irreducibility (a positive-definite fixed point together with uniqueness of positive fixed points implies irreducibility) applies directly.

Proof. ✓ Proof By Lemma D.5.5, if $\sigma \geq 0$ and $\mathcal{E}_A(\sigma) = \sigma$, then $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$. Since $\rho > 0$, Wolf’s uniqueness criterion for irreducibility applies. $\square$

D.6 Identity in the one-step span

This section proves the one-step padding conversions used in Theorem 8.8.2. It also records why cumulative spanning alone does not imply normality.

Remark D.6.1 (Identity-in-the-one-step-span condition). ✓ Stmt The *one-step padding condition* is $\mathbb{1} \in S_1(A)$: the identity matrix lies in the span of the Kraus operators $\{A^i\}_{i=0}^{d-1}$. This is an additional hypothesis. The normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ gives a quadratic identity and does not imply $\mathbb{1} \in S_1(A)$. Theorem 8.7.4.2 obtains normality from a positive-definite fixed point and a complementary gap without this padding condition.

Remark D.6.2 (Counterexample: cumulative spanning does not imply normality). ✓ Stmt In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ be the off-diagonal matrix units. Then $\text{alg}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd n and only diagonal matrices for even n . Hence $S_n(A) \neq M_2(\mathbb{C})$ for every n , and A is not normal. The one-step padding condition fails because $\mathbb{1} \notin S_1(A) = \text{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

Theorem D.6.3 (Word-span monotonicity from one-step padding). Lean ✓ Stmt If $\mathbb{1} \in S_1(A)$, then $S_n(A) \leq S_{n+1}(A)$ for every n .

Proof. ✓ Proof For any $M \in S_n(A)$, one has $M = M\mathbb{1} \in S_n(A)S_1(A) \subseteq S_{n+1}(A)$ by Lemma D.1.1. □

Theorem D.6.4 (Cumulative span equals word span under one-step padding). Lean ✓ Stmt If $\mathbb{1} \in S_1(A)$, then $T_n(A) = S_n(A)$ for every n .

Proof. ✓ Proof By Theorem D.6.3, the word spans form the chain $S_0(A) \leq S_1(A) \leq \dots \leq S_n(A)$. Therefore their sum $T_n(A) = S_0(A) + \dots + S_n(A)$ equals the largest term $S_n(A)$. □

Theorem D.6.5 (Cumulative spanning plus one-step padding implies normality). Lean ✓ Stmt If $T_N(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.

Proof. ✓ Proof Theorem D.6.4 gives $S_N(A) = T_N(A) = M_D(\mathbb{C})$. □

Appendix E

Canonical Form Reduction: Supporting Results

This appendix supports Chapter 9. It collects the projection, gauge, blocking, compression, and reindexing arguments used by the main reduction while keeping the source-named canonical-form statements and the principal reductions in the chapter.

Throughout, $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$, $A^{[p]}$ denotes blocking by p , and $V^{(N)}(A)_\sigma = \text{tr}(A^{\sigma_1} \dots A^{\sigma_N})$. Weighted direct sums are written $\bigoplus_k \mu_k A_k$.

E.1 Similarities, strict splitting, and zero blocks

Similarity invariance and strict dimension decrease justify the induction used in Theorem 9.1.1.2; the final result removes blocks that vanish at every positive length.

Theorem E.1.1 (Unitary conjugation preserves the MPV family). [Lean](#) [✓ Stmt](#) *If U is a unitary matrix, then A and $U^\dagger A U$ generate the same MPV family.*

Proof. [✓ Proof](#) By cyclicity of the trace, $\text{tr}((U^\dagger A^{i_1} U) \dots (U^\dagger A^{i_N} U)) = \text{tr}(A^{i_1} \dots A^{i_N})$. □

Theorem E.1.2 (Nontrivial invariant projection gives strict two-block reduction). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 9.1.1, assume in addition that $P \neq 0$ and $P \neq \mathbb{1}$. Then there exist n, m with $n + m = D$, $n < D$, and $m < D$, together with block tensors A_1, A_2 , such that A and $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.3.3.*

Proof. [✓ Proof](#) Apply Theorem 9.1.1. In the diagonal form of P , the 1-eigenspace is nonempty because $P \neq 0$, and the 0-eigenspace is nonempty because $P \neq \mathbb{1}$. Hence both block sizes are positive. Since $n + m = D$, this implies $n < D$ and $m < D$. □

Theorem E.1.3 (Nonzero irreducible block decomposition). [Lean](#) [✓ Stmt](#) *Every MPS tensor A admits a finite family of nonzero irreducible blocks $(B_k)_{k=1}^r$, each with at least one nonzero Kraus operator and positive bond dimension. For every $N \geq 1$ and every σ ,*

$$V^{(N)}(A)_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r B_k\right)_\sigma. \quad (\text{E.1})$$

Moreover, $\sum_{k=1}^r D_k \leq D$.

Proof. ✓ Proof Apply the irreducible block decomposition (Theorem 9.1.1.2) and retain the blocks having a nonzero Kraus operator. For every all-zero irreducible block C_j and every $N \geq 1$, $V^{(N)}(C_j)_\sigma = 0$, so (E.1) holds. The dimension inequality follows by restricting the structural identity $\sum_j D_j = D$ to the retained subset. $\square$

E.2 Projector closure and isometric corner decompositions

The results below turn reducing projections into support isometries, compress tensors to their corners, and reconstruct closed-chain traces from an orthogonal family of corners. They support Theorem 9.7.2 and Theorem 9.7.4.

Theorem E.2.1 (Invariant projections are reducing under projector closure). Lean ✓ Stmt Suppose that A has invariant-projector closure and that P is an orthogonal projection satisfying $(\mathbb{1} - P)A^iP = 0$ for all i . Then P reduces every tensor matrix: $PA^i = A^iP$ for all i .

Proof. ✓ Proof The lower-corner relation gives $(\mathbb{1} - P)A^i = (\mathbb{1} - P)A^i(\mathbb{1} - P)$. Applying projector closure to $\mathbb{1} - P$ gives $A^i(\mathbb{1} - P) = (\mathbb{1} - P)A^i(\mathbb{1} - P)$, hence the upper corner $PA^i(\mathbb{1} - P)$ also vanishes. Therefore $PA^i = PA^iP = A^iP$. $\square$

Theorem E.2.2 (Invariant-projector closure gives a reducing projection). Lean ✓ Stmt If A has invariant-projector closure and admits a nontrivial invariant projection, then it admits a nontrivial orthogonal projection commuting with every A^i .

Proof. ✓ Proof Apply Theorem E.2.1 to the projection supplied by Definition 9.1.2. $\square$

Lemma E.2.3 (Support isometry of an orthogonal projection). Lean ✓ Stmt Every orthogonal projection P on $\mathbb{C}^D$ factors through an isometry: there are $n \in \mathbb{N}$ and $V \in \mathbb{C}^{D \times n}$ such that

$$\begin{aligned} V^\dagger V &= \mathbb{1}, \\ VV^\dagger &= P, \\ n &= \text{tr}(P). \end{aligned} \tag{E.2}$$

Proof. ✓ Proof Since P is Hermitian and idempotent, its eigenvalues are 0 and 1. Writing $n = \text{tr}(P)$, unitary diagonalization gives

$$P = U \text{diag}(\underbrace{1, \dots, 1}_n, 0, \dots, 0)U^\dagger.$$

The matrix $V \in \mathbb{C}^{D \times n}$ formed by the first n columns of U satisfies (E.2). $\square$

The rectangular Gram-equality criterion that turns $B^\dagger B = A^\dagger A$ into a unitary U with $B = UA$ is a generic algebra fact, proved as Lemma ?? in Appendix ??.

Lemma E.2.4 (Corner tensors inherit projector closure). Lean ✓ Stmt Let A have invariant-projector closure, and let V be an isometry, $V^\dagger V = \mathbb{1}$, whose range projection $VV^\dagger$ commutes with every A^i . Then the corner tensor $B^i = V^\dagger A^i V$ again has invariant-projector closure.

Proof. ✓ Proof Let Q be an orthogonal projection with $QB^i = QB^iQ$ for all i . Then $VQV^\dagger$ is an orthogonal projection. Commutation of $VV^\dagger$ with each A^i gives $V^\dagger A^i = B^i V^\dagger$, so left-invariance of Q yields

$$(VQV^\dagger)A^i = VQB^iV^\dagger = VQB^iQV^\dagger = (VQV^\dagger)A^i(VQV^\dagger).$$

Projector closure for A yields $A^i(VQV^\dagger) = (VQV^\dagger)A^i(VQV^\dagger)$, and compressing both sides between $V^\dagger$ and V returns $B^iQ = QB^iQ$. $\square$

Lemma E.2.9 (Closed-chain equality from an exact isometric decomposition). [Lean](#) [✓ Stmt](#) Suppose that isometries V_k satisfy

$$\begin{aligned} V_k^\dagger A^i &= (\mu_k B_k^i) V_k^\dagger, \\ A^i &= \sum_k V_k (\mu_k B_k^i) V_k^\dagger. \end{aligned} \tag{E.5}$$

Then A and $\bigoplus_k \mu_k B_k$ have the same matrix product vectors at every positive length.

Proof. [✓ Proof](#) Expand the first letter by the reconstruction formula in (E.5). In each summand, move the remaining word through $V_k^\dagger$ using the intertwining relation. Cyclicity of the trace places $V_k^\dagger V_k = \mathbb{1}$ next to the weighted block word, leaving precisely its closed-chain trace. $\square$

E.3 Perron gauges and preservation under conjugation

This section records the positive Perron eigenpair, its spectral normalization, and the behavior of irreducibility and eigenvalues under the corresponding gauge.

Lemma E.3.1 (Perron eigenpair of a nonzero irreducible tensor). [Lean](#) [✓ Stmt](#) Let B be an irreducible MPS tensor of positive bond dimension, not all of whose matrices vanish. Then there are $\rho > 0$ and $r > 0$ with $\mathcal{E}_B(\rho) = r\rho$.

Proof. [✓ Proof](#) Irreducibility of the tensor makes the transfer map an irreducible completely positive map, and a nonzero matrix makes it nonzero, so it does not annihilate any nonzero PSD matrix. The Perron–Frobenius theorem for positive maps (Theorem 6.8.2) gives a nonzero PSD eigenvector with eigenvalue $r > 0$, and irreducibility upgrades it to a positive-definite one ([Wol12, Theorem 6.3(2)–(3)]). $\square$

Lemma E.3.2 (Spectral normalization of an irreducible block). [Lean](#) [✓ Stmt](#) Let B be an irreducible MPS tensor of positive bond dimension with $\mathcal{E}_B(\rho) = r\rho$ for some $\rho > 0$ and $r > 0$, and suppose every eigenvalue of $\mathcal{E}_B$ of modulus r equals r . Then $r^{-1/2}B$ is a normal tensor.

Proof. [✓ Proof](#) The rescaling identity $\mathcal{E}_{r^{-1/2}B}(X) = r^{-1}\mathcal{E}_B(X)$ gives $\mathcal{E}_{r^{-1/2}B}(\rho) = r^{-1}r\rho = \rho$, so ρ is a positive-definite fixed point of $\mathcal{E}_{r^{-1/2}B}$. The rescaled map remains irreducible and completely positive, so Theorem 6.9.1 identifies its spectral radius with the eigenvalue one. The eigenvalues of modulus one of $\mathcal{E}_{r^{-1/2}B}$ are the eigenvalues of $\mathcal{E}_B$ of modulus r , divided by r ; by hypothesis the only such eigenvalue is one, so the peripheral eigenvalue set is $\{1\}$. Irreducibility is unchanged by a nonzero rescaling. $\square$

Lemma E.3.3 (Range factorization of the support projection). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The support projection π of a positive semidefinite matrix ρ factors as $\pi = \rho W$ for a spectral pseudo-inverse W , witnessing that the range of π is the range of ρ . For a rectangular matrix Y , the support projection π of $YY^\dagger$ fixes Y , $\pi Y = Y$; and whenever $AY = YG$ for some matrix G —that is, A maps the range of Y into itself—the one-sided invariance $(\mathbb{1} - \pi)A\pi = 0$ holds.

Proof. [✓ Proof](#) In the spectral decomposition $\rho = U \text{diag}(\lambda) U^\dagger$, take $W = U \text{diag}(\lambda^+) U^\dagger$, where λ^+ inverts the positive eigenvalues and vanishes on the kernel. Then $\rho W = U \text{diag}(\mathbf{1}_{\lambda>0}) U^\dagger = \pi$. From $\pi(YY^\dagger) = YY^\dagger$, the matrix $((\mathbb{1} - \pi)Y)((\mathbb{1} - \pi)Y)^\dagger$ vanishes, hence $\pi Y = Y$. Finally,

$$(\mathbb{1} - \pi)A\pi = (\mathbb{1} - \pi)AY(Y^\dagger W) = (\mathbb{1} - \pi)Y G(Y^\dagger W) = 0.$$

$\square$

Theorem E.3.4 (Irreducibility under rescaled conjugation). [Lean](#) [✓ Stmt](#) *Let B be an irreducible tensor, let X be invertible, and let $c \neq 0$. Then the rescaled conjugated tensor with letters $cX^{-1}B^vX$ is irreducible. In particular, the rescaled unital gauge of an irreducible tensor is irreducible; this is the normalization of [CPGSV16, Section 2.3] applied to an irreducible corner.*

Proof. [✓ Proof](#) An invariant orthogonal projection $P \notin \{0, \mathbb{1}\}$ of the conjugated tensor makes the columns of $Y = XP$ span a subspace invariant under every letter of B : $B^vY = YG_v$, where $G_v = X^{-1}B^vXP$. The support projection of $YY^\dagger$ is then a nontrivial invariant orthogonal projection of B by Lemma E.3.3: it is nonzero because $Y \neq 0$, and it is not the identity because any nonzero vector in the kernel of P transports through $(X^\dagger)^{-1}$ to the kernel of $YY^\dagger$. This contradicts irreducibility of B . $\square$

Theorem E.3.5 (Unital Schwarz data for an irreducible corner). [Lean](#) [✓ Stmt](#) *Let B be an irreducible tensor with a positive definite transfer eigenvector, $\mathcal{E}_B(\rho) = r\rho$, where $\rho > 0$ and $r > 0$. Then the rescaled unital gauge $K^v = r^{-1/2}\rho^{-1/2}B^v\rho^{1/2}$ is a unital Kraus family, is an irreducible tensor with an irreducible transfer map, and its adjoint map has a positive definite fixed point. These are the hypotheses of the cyclic decomposition of the peripheral spectrum, [Wol12, Theorem 6.6].*

Proof. [✓ Proof](#) Unitality is Theorem 9.8.2; irreducibility is Theorem E.3.4. The adjoint of a unital family is trace-preserving, so its transfer map is a channel and has a positive semidefinite fixed point by the Cesàro argument; irreducibility of the adjoint map upgrades the fixed point to a positive definite one. $\square$

Lemma E.3.6 (Eigenvalue transport to the unital gauge). [Lean](#) [✓ Stmt](#) *In the setting of Theorem E.3.5, every transfer eigenvalue μ of B becomes the transfer eigenvalue μ/r of the rescaled unital gauge: the gauge conjugates the transfer map by the congruence with $\rho^{1/2}$ and divides it by r . In particular, the peripheral eigenvalues $e^{2\pi i q/p}$ described in [CPGSV16, Section 2.3] arise from the transfer eigenvalues of modulus r .*

Proof. [✓ Proof](#) If $\mathcal{E}_B(Z) = \mu Z$, then $Z' = \rho^{-1/2}Z\rho^{-1/2}$ is nonzero and satisfies $\mathcal{E}_K(Z') = (\mu/r)Z'$ by direct computation. $\square$

E.4 Blocking identities and primitive stability

These identities control transfer-map powers, weighted direct sums, and positive-length MPV equalities under repeated blocking. They are used in the common-length reduction of Theorem 9.3.1.4.

Blocking by a length p commutes with weighted direct sums up to the p th power of each weight: for nonzero weights (μ_k) and blocks (B_k) ,

$$\left(\bigoplus_k \mu_k B_k \right)^{[p]} \sim \bigoplus_k \mu_k^p B_k^{[p]}, \quad (\text{E.6})$$

where $\sim$ is equality of MPV families. When two block families are reblocked to a common length $p = m_k e_k$, the p -blocked alphabet is identified with the e_k -fold tensor power of the m_k -blocked alphabet, so each sector reblocked through its own period agrees with the directly p -blocked block.

Theorem E.4.1 (Common blocking period for a block family). [Lean](#) [✓ Stmt](#) Let $(B_k)_{k=1}^r$ be a finite family of positive-bond-dimension MPS tensors, each admitting a blocking period p_k making its transfer map primitive. Then $p = \text{lcm}(p_1, \dots, p_r)$ is a common period: $\mathcal{E}_{B_k^{[p]}}$ is primitive for every k .

Proof. [✓ Proof](#) Each p_k divides p , so primitivity of $\mathcal{E}_{B_k^{[p_k]}}$ carries over to $\mathcal{E}_{B_k^{[p]}}$ via the divisibility identity for transfer-map powers. $\square$

Theorem E.4.2 (Common primitive blocking for irreducible left-canonical blocks). [Lean](#) [✓ Stmt](#) Let $(B_k)_{k=1}^r$ be a finite family of positive-bond-dimension, left-canonical, tensor-irreducible MPS tensors. Then there is a single blocking length $p > 0$ such that $\mathcal{E}_{B_k^{[p]}}$ is primitive for every k .

Proof. [✓ Proof](#) Apply Theorem 9.9.2.1 to each block, then apply Theorem E.4.1 to the resulting finite family of positive periods. Their least common multiple is positive and makes every blocked transfer map primitive. $\square$

Theorem E.4.3 (Quasi-idempotence under physical blocking). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor and let $c \in \mathbb{C}$. If

$$\mathcal{E}_A^2 = c\mathcal{E}_A,$$

then, for every integer $L \geq 0$,

$$\mathcal{E}_{A^{[L]}}^2 = c^L \mathcal{E}_{A^{[L]}}.$$

Proof. [✓ Proof](#) The transfer map of $A^{[L]}$ is $\mathcal{E}_A^L$. Therefore

$$(\mathcal{E}_A^L)^2 = (\mathcal{E}_A^2)^L = (c\mathcal{E}_A)^L = c^L \mathcal{E}_A^L,$$

which is the required identity. $\square$

Lemma E.4.4 (Adjoint primitivity equivalence). [Lean](#) [Lean](#) [✓ Stmt](#) Let E be a linear endomorphism of a finite-dimensional complex inner product space. Then E is primitive if and only if its adjoint $E^\dagger$ is primitive. Equivalently, the eigenvalues λ of E with $|\lambda| = 1$ form the singleton $\{1\}$ if and only if the same holds for $E^\dagger$, because adjunction sends every eigenvalue to its complex conjugate.

Lemma E.4.5 (Primitivity under blocking by a multiple). [Lean](#) [✓ Stmt](#) Let A have positive bond dimension and let $p_0, p \geq 1$ with $p_0 \mid p$. If $\mathcal{E}_{A^{[p_0]}}$ is primitive, then $\mathcal{E}_{A^{[p]}}$ is primitive.

Theorem E.4.6 (Left-canonical normalization under blocking). [Lean](#) [✓ Stmt](#) If a finite matrix family satisfies

$$\sum_i (A^i)^\dagger A^i = \mathbb{1},$$

then every blocked family satisfies the same identity.

Proof. [✓ Proof](#) Expand the blocked matrices as products indexed by words and sum one letter at a time. $\square$

Theorem E.4.7 (Irreducibility from a faithful fixed point). [Lean](#) [✓ Stmt](#) Let E be a channel with a positive-definite fixed point. If all nonzero positive semidefinite fixed points of E are proportional, then E is irreducible.

Proof. ✓ Proof A nontrivial invariant projection would produce a positive semidefinite fixed point not proportional to the faithful one. $\square$

Lemma E.4.8 (Fixed-point criteria for irreducibility after primitive blocking). Lean Lean
Lean Lean Lean ✓ Stmt *Let A be left-canonical, tensor-irreducible, and have primitive transfer map. Then A has primitive fixed-point data (ρ, P_ρ, N) with $\rho > 0$ and $N^n \rightarrow 0$. For every blocking length P , the tensor $A^{[P]}$ is left-canonical, its transfer map is $\mathcal{E}_A^P$, and ρ is fixed by this blocked map. For every $n \geq 1$,*

$$\mathcal{E}_A^n = P_\rho + N^n. \quad (\text{E.7})$$

Finally, a channel with a positive-definite fixed point whose positive semidefinite fixed points are all proportional is irreducible as a map; irreducibility of a transfer map implies tensor irreducibility.

Proof. ✓ Proof Peripheral primitivity, tensor irreducibility, and left-canonical normalization first give primitive fixed-point data, and tensor irreducibility makes its positive semidefinite fixed point positive definite. Blocking preserves the left-canonical identity, replaces the transfer map by its positive power, and therefore preserves the fixed point. The defining complementary-power identity gives (E.7). The channel fixed-point criterion then gives map irreducibility from uniqueness of positive semidefinite fixed points, and the final implication converts map irreducibility to tensor irreducibility. $\square$

Theorem E.4.9 (Blocking preserves irreducibility for primitive irreducible blocks). Lean ✓ Stmt
Let A be left-canonical, irreducible, primitive, of positive bond dimension. For every $P > 0$, the blocked tensor $A^{[P]}$ is irreducible: there is no orthogonal projection Q with $Q \neq 0$, $Q \neq \mathbb{1}$ such that $(\mathbb{1} - Q)(A^{[P]})^\alpha Q = 0$ for every P -blocked index α .

Proof. ✓ Proof Lemma E.4.8 gives primitive fixed-point data with $\rho > 0$. In particular, if $P_\rho(X) = \text{tr}(X)\rho / \text{tr}(\rho)$ and $N := \mathcal{E}_A - P_\rho$, then

$$\begin{aligned} \mathcal{E}_A &= P_\rho + N, \\ N^n &\rightarrow 0. \end{aligned}$$

The primitive fixed-point theorem supplies this complementary-power convergence.

The blocked transfer map is the corresponding positive power:

$$\mathcal{E}_{A^{[P]}}(X) = \mathcal{E}_A^P(X). \quad (\text{E.8})$$

Hence ρ remains a positive-definite fixed point after blocking. Let $\sigma \geq 0$ be any fixed point of $\mathcal{E}_A^P$, set

$$\begin{aligned} c &= \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}, \\ \sigma' &= \sigma - c\rho. \end{aligned}$$

Then $\text{tr}(\sigma') = 0$, so $P_\rho(\sigma') = 0$, while $\mathcal{E}_A^P(\sigma') = \sigma'$. Since the fixed-point projection and the complement annihilate one another, iteration gives, for every $k \geq 1$,

$$\sigma' = \mathcal{E}_A^{Pk}(\sigma') = N^{Pk}(\sigma').$$

But $N^{Pk}(\sigma') \rightarrow 0$ as $k \rightarrow \infty$, hence $\sigma' = 0$. Thus every positive semidefinite fixed point of the blocked map is proportional to ρ .

Blocking preserves the left-canonical identity, so Theorem 4.9.6 makes the blocked transfer map a channel. Its positive-definite fixed point and the uniqueness just proved imply map irreducibility by Lemma E.4.8. The final implication in that lemma then shows that $A^{[P]}$ is tensor-irreducible. $\square$

Theorem E.4.10 (Extra blocking of primitive irreducible sector blocks). [Lean](#) [✓ Stmt](#) *Let A have positive bond dimension and be trace-preserving, primitive, and tensor-irreducible. For every $k > 0$,*

$$\sum_{\alpha} ((A^{[k]})^{\alpha})^{\dagger} (A^{[k]})^{\alpha} = \mathbb{1},$$

$$\sigma_{\partial}(\mathcal{E}_{A^{[k]}}) = \{1\}, \quad (\text{E.9})$$

and $A^{[k]}$ is tensor-irreducible. This k is a later common-refinement or injectivity length, distinct from the period-removal length that produced the sector block.

Proof. [✓ Proof](#) Trace preservation follows from blocking a trace-preserving tensor, primitivity from the transfer-power identity (E.8), and irreducibility from Theorem E.4.9. $\square$

Lemma E.4.11 (Blocking the weighted block sum). [Lean](#) [✓ Stmt](#) *For nonzero weights $(\mu_k)_{k=1}^r$, blocks $(B_k)_{k=1}^r$, and any $p \geq 1$, the blocking identity (E.6) holds, where $\sim$ denotes generation of the same MPV family at every length.*

Lemma E.4.12 (Positive-length equality survives blocking). [Lean](#) [✓ Stmt](#) *If two tensors have the same MPV coefficients at every positive length, then so do their blocked tensors after any positive blocking length.*

Theorem E.4.13 (Eventual nonzero proportionality survives blocking). [Lean](#) [✓ Stmt](#) *If two tensors have eventually nonzero proportional MPV coefficients, then so do their blocked tensors after any positive blocking length.*

Proof. [✓ Proof](#) If proportionality holds from length N_0 onward, a blocked chain of length $N \geq N_0$ expands to an original chain of length $Np \geq N_0$. Use the original proportionality scalar at length Np . $\square$

Lemma E.4.14 (Positive-length blocking of a weighted block sum). [Lean](#) [✓ Stmt](#) *If A and $\bigoplus_k \mu_k B_k$ have the same MPV coefficients at every positive length, then after any positive blocking length p , $A^{[p]}$ and $\bigoplus_k \mu_k^p B_k^{[p]}$ have the same MPV coefficients at every positive length.*

Lemma E.4.15 (Positive-length equality of powered block sums). [Lean](#) [✓ Stmt](#) *If $\bigoplus_j \mu_j^A A_j$ and $\bigoplus_k \mu_k^B B_k$ have the same MPV coefficients at every positive length, then after any positive common blocking length p , $\bigoplus_j (\mu_j^A)^p A_j^{[p]}$ and $\bigoplus_k (\mu_k^B)^p B_k^{[p]}$ have the same MPV coefficients at every positive length.*

Lemma E.4.16 (Nonzero weights survive blocking). [Lean](#) [✓ Stmt](#) *If $\mu \in \mathbb{C}$ is nonzero and $p \geq 1$, then $\mu^p \neq 0$; blocking preserves the nonzero-weight hypothesis of a weighted block decomposition.*

Lemma E.4.17 (Iterated blocking at the transfer-map level). [Lean](#) [✓ Stmt](#) *For any MPS tensor A and $m, n \geq 0$,*

$$\mathcal{E}_{(A^{[m]})^{[n]}} = \mathcal{E}_{A^{[mn]}}. \quad (\text{E.10})$$

E.5 Support compression and cyclic-sector families

Support isometries first realize the cyclic corners. The remaining results place finitely many cyclic-sector families over one blocked alphabet and identify their weighted nonzero parts.

Theorem E.5.1 (Supported corner compression with a support isometry). [Lean](#) [✓ Stmt](#) *Let A be a tensor normalized on the support of an orthogonal projection P , so that $PA^iP = A^i$ and $\sum_i (A^i)^\dagger A^i = P$. Then there are a corner tensor C , a linear isomorphism $\varphi : M_{\dim C}(\mathbb{C}) \xrightarrow{\sim} PM_D(\mathbb{C})P$, and a rectangular isometry $V : \mathbb{C}^{\dim C} \rightarrow \mathbb{C}^D$ such that*

$$\begin{aligned} V^\dagger V &= \mathbb{1}, \\ VV^\dagger &= P, \\ \varphi(X) &= VXV^\dagger. \end{aligned} \tag{E.11}$$

Moreover, $\varphi(C^i) = A^i$ for every physical letter, and for every word $\sigma = (\sigma_1, \dots, \sigma_N)$,

$$V^{(N)}(C)_\sigma = \text{tr}(PA^{\sigma_1} \dots A^{\sigma_N}). \tag{E.12}$$

Proof. [✓ Proof](#) Apply Theorem ?? to the family $(A^i)_i$. It gives C , φ , and V satisfying (E.11) and $\varphi(C^i) = A^i$. It remains to identify the coefficients of the associated words. For each word $\sigma = (\sigma_1, \dots, \sigma_N)$, cyclicity of the trace and $VV^\dagger = P$ give

$$V^{(N)}(C)_\sigma = \text{tr}(C^{\sigma_1} \dots C^{\sigma_N}) = \text{tr}(V^\dagger A^{\sigma_1} \dots A^{\sigma_N} V) = \text{tr}(PA^{\sigma_1} \dots A^{\sigma_N}),$$

which proves (E.12). $\square$

Theorem E.5.2 (Commuting projections give sector isometries). [Lean](#) [✓ Stmt](#) *Let a left-canonical tensor A commute with a family of orthogonal projections (P_k) summing to $\mathbb{1}$. Then A decomposes into compressed sector tensors (C_k) , with a unit-weight direct-sum MPV decomposition, corner isomorphisms $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$, and rectangular isometries V_k satisfying*

$$\begin{aligned} V_k^\dagger V_k &= \mathbb{1}, \\ V_k V_k^\dagger &= P_k, \\ \varphi_k(X) &= V_k X V_k^\dagger. \end{aligned} \tag{E.13}$$

The sector letters obey $\varphi_k(C_k^i) = P_k A^i P_k$.

Proof. [✓ Proof](#) Apply Theorem E.5.1 to the compression of A on each sector P_k . Orthogonality and $\sum_k P_k = \mathbb{1}$ give the direct-sum decomposition. For every word $\sigma = (\sigma_1, \dots, \sigma_N)$, the partition of unity and (E.12) give

$$V^{(N)}(A)_\sigma = \text{tr} \left(\left(\sum_k P_k \right) A^{\sigma_1} \dots A^{\sigma_N} \right) = \sum_k \text{tr}(P_k A^{\sigma_1} \dots A^{\sigma_N}) = \sum_k V^{(N)}(C_k)_\sigma.$$

This is the unit-weight matrix-product-vector identity. $\square$

Theorem E.5.3 (Adjoint-fixed projections give sector isometries). [Lean](#) [✓ Stmt](#) *Let A be left-canonical and let (P_k) be orthogonal projections summing to $\mathbb{1}$ and fixed by the adjoint transfer map. Then the same sector decomposition and support isometries of Theorem E.5.2 exist. In particular, the adjoint fixed-point condition $\mathcal{E}_A^\dagger(P_k) = P_k$ gives $P_k A^i = A^i P_k$, which gives the corner-letter identity $\varphi_k(C_k^i) = P_k A^i P_k$.*

Proof. ✓ Proof The fixed-point identity $\mathcal{E}_A^\dagger(P_k) = P_k$, together with left-canonicity $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, places P_k in the multiplicative domain of $\mathcal{E}_A^\dagger$, forcing $P_k A^i = A^i P_k$ for every letter A^i . The result then follows from Theorem E.5.2. $\square$

Theorem E.5.4 (Cyclic sectors after removing zero blocks). Lean ✓ Stmt *If A and B generate the same MPV family at every positive length, then after removing the all-zero blocks and applying the trace-preserving gauge there are weighted nonzero-block tensors $\bigoplus_j \mu_j^A A_j$ and $\bigoplus_k \mu_k^B B_k$ such that, for every $N \geq 1$ and σ ,*

$$V^{(N)}(A)_\sigma = V^{(N)}\left(\bigoplus_j \mu_j^A A_j\right)_\sigma, \quad (\text{E.14})$$

$$V^{(N)}(B)_\sigma = V^{(N)}\left(\bigoplus_k \mu_k^B B_k\right)_\sigma. \quad (\text{E.15})$$

The two nonzero parts agree:

$$V^{(N)}\left(\bigoplus_j \mu_j^A A_j\right)_\sigma = V^{(N)}\left(\bigoplus_k \mu_k^B B_k\right)_\sigma. \quad (\text{E.16})$$

Each nonzero block has primitive irreducible cyclic sectors: for each j , there is $m_j \geq 1$ with a unit-weight sector decomposition $A_j^{[m_j]} \sim \bigoplus_{u=1}^{m_j} C_{j,u}^A$, and similarly for B_k .

Proof. ✓ Proof Apply Theorem 9.2.1.1 separately to A and B . Its positive-length conclusion gives (E.14) and (E.15). Chaining these identities through $\mathcal{V}^+(A) = \mathcal{V}^+(B)$ gives (E.16). Finally apply Theorem 9.3.1.3 to every trace-preserving irreducible nonzero-weight block. $\square$

Lemma E.5.5 (Common reblocking of cyclic-sector families). Lean ✓ Stmt *Given a finite family $(B_k)_{k=1}^r$, each with primitive irreducible cyclic sectors, there is a common blocking length p such that each $B_k^{[p]}$ admits a unit-weight direct-sum decomposition over a single common blocked alphabet $d^{[p]}$ into left-canonical blocks with primitive transfer maps, tensor irreducibility, and positive bond dimensions.*

Lemma E.5.6 (Common reblocking at a prescribed multiple). Lean ✓ Stmt *With hypotheses as in Lemma E.5.5, and given any common multiple p of the per-block period-removal lengths (m_k) , the common reblocking can be performed at the prescribed length p .*

Lemma E.5.7 (Structural properties of the common reblocked sectors). Lean ✓ Stmt *Each sector tensor in the common reblocked cyclic-sector family of Lemma E.5.5 is trace-preserving, $\sum_i (B^i)^\dagger B^i = \mathbb{1}$, has primitive transfer map, $\sigma_\partial(\mathcal{E}_B) = \{1\}$, is tensor-irreducible, and has positive bond dimension.*

Lemma E.5.8 (Common-alphabet flattening). Lean ✓ Stmt *With $p = m_k e_k$ for each k , the iterated blocked tensor $(B_k^{[m_k]})^{[e_k]}$ agrees with $B_k^{[p]}$ in MPV family, after the canonical identification of the p -blocked alphabet $d^{[p]}$ with the e_k -fold tensor power of the m_k -blocked alphabet $d^{[m_k]}$.*

Lemma E.5.9 (Direct blocking in the common alphabet). Lean ✓ Stmt *For every original block B_k , direct blocking at the common length $p = m_k e_k$ agrees, as an MPV family, with the same block transported to the common blocked alphabet. Write $\widetilde{B}_k$ for this common-alphabet image. Then*

$$V^{(N)}(B_k^{[p]})_\sigma = V^{(N)}(\widetilde{B}_k)_\sigma.$$

Lemma E.5.10 (Reindexed nonzero-part identity). [Lean](#) [✓ Stmt](#) *The powered nonzero-part decomposition $\bigoplus_k \mu_k^p B_k^{[p]}$ carries over under the common-alphabet identification of Lemma E.5.8: the common-alphabet sectors have the same nonzero-part decomposition as the direct p -blocked tensors.*

E.6 Prepared normal-form consequences

The first theorem derives the overlap condition from a prepared primitive family. The second does not construct primitive blocks from an arbitrary tensor: it only orders a family that already satisfies all primitive, irreducible, trace-preserving, nonzero-weight, and positive-dimension hypotheses.

Theorem E.6.1 (Self-overlap is derived in normal canonical form). [Lean](#) [✓ Stmt](#) *If (μ_k, A_k) satisfies the normal canonical form conditions, then $O_{A_k A_k}(N) \rightarrow 1$ for every block k .*

Proof. [✓ Proof](#) Fix a block k . Irreducibility and the TP normalization give a nonzero positive semidefinite fixed point ρ of $\mathcal{E}_{A_k}$ with fixed-point projector P ; primitivity gives the complementary gap $\rho_{\text{spec}}(\mathcal{E}_{A_k} - P) < 1$ (Theorem 4.13.3). Theorem 7.11.2 then yields $O_{A_k A_k}(N) \rightarrow 1$. $\square$

Theorem E.6.2 (Reordering prepared primitive blocks into normal canonical form). [Lean](#) [✓ Stmt](#) *Suppose A generates the same MPV family as $\bigoplus_k \mu_k A_k$, and that each A_k is irreducible, satisfies $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$, and has primitive transfer map, all $\mu_k \neq 0$, and all bond dimensions are positive. Then there exist a blocking length $p > 0$, weights (ν_k) , and blocked tensors (B_k) such that $A^{[p]}$ generates the same MPV family as $\bigoplus_k \nu_k B_k$, and (ν_k, B_k) satisfies the normal canonical form conditions. The weight moduli need not be distinct; equal-modulus blocks are allowed.*

Proof. [✓ Proof](#) The given blocks are already primitive, so take $p = 1$. Reorder the blocks by non-increasing weight modulus, with ties allowed, and set $\nu_k := \mu_k$ and $B_k := A_k$ in that order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are preserved, so the reordered family satisfies Definition 9.9.2 and generates the same MPV family as A . $\square$

Remark E.6.3 (Scope of Theorem E.6.2). Theorem E.6.2 assumes all six block hypotheses in advance and produces the normal canonical form by reordering. It is not the arbitrary-input existence theorem of [CPGSV16, Section 2.3]; that primitive-block reduction, including its structural bond-dimension bound, is Theorem 9.11.1.2 in Section 9.11.1.

Appendix F

Basis of Normal Tensors: Supporting Results

This appendix supports Chapter 10. It proves the elementary normal-tensor consequences, Gram-matrix criteria, finite-dimensional change-of-basis facts, sector-blocking identities, direct-sum separation lemmas, and prepared-family bookkeeping used by the main BNT characterization, matching, and block-injectivity results.

F.1 Elementary normal-tensor and Gram-matrix support

Throughout this section, $V^{(N)}(A)$ denotes the length- N matrix product vector of a tensor A , and $O_{AB}(N)$ denotes the corresponding mixed overlap. We first record elementary consequences of the two notions of normality used in Chapter 10, and then prove the finite-family linear-independence criteria used in the BNT characterization and matching arguments.

Lemma F.1.1 (Positive bond dimensions in a coefficient-form BNT). [Lean](#) [✓ Stmt](#) *Every block in a coefficient-form basis of normal tensors has positive bond dimension.*

Proof. [✓ Proof](#) Eventual linear independence makes the matrix product vector of each block nonzero at some positive length. A tensor with zero bond dimension has zero matrix product vector at every length, which is a contradiction. $\square$

Lemma F.1.2 (Bond dimension one implies irreducibility). [Lean](#) [✓ Stmt](#) *Every tensor with a one-dimensional bond space is irreducible.*

Proof. [✓ Proof](#) The only orthogonal projections on a one-dimensional complex vector space are zero and the identity. $\square$

Lemma F.1.3 (Normality in bond dimension one). [Lean](#) [✓ Stmt](#) *A tensor with one-dimensional bond space and identity transfer map is a normal tensor.*

Proof. [✓ Proof](#) Irreducibility follows from the preceding lemma. The identity transfer map has spectral radius one, and its only eigenvalue is one. $\square$

Lemma F.1.4 (Perron gauges preserve primitivity). [Lean](#) [✓ Stmt](#) *Let σ be positive definite. The transfer map of the Perron gauge determined by σ is primitive if and only if the original transfer map is primitive.*

Proof. ✓ **Proof** The two transfer maps are similar through the invertible congruence $X \mapsto \sigma^{-1/2} X \sigma^{-1/2}$. Similar linear maps have the same spectrum, so the condition that 1 is the only eigenvalue on the unit circle is unchanged. $\square$

Lemma F.1.5 (Preservation of an algebraic BNT under positive-length MPV equality). Lean ✓ **Stmt** *Let A and A' have the same matrix product vectors at every positive length. If $(A_j)_{j=1}^g$ is a basis of normal tensors for A , then the same family is a basis of normal tensors for A' .*

Proof. ✓ **Proof** Substitute the positive-length MPV equality into the spanning identity for A . Normality and eventual linear independence depend only on the family $(A_j)_{j=1}^g$. $\square$

Corollary F.1.6 (Positive-length geometric proportionality phase). Lean ✓ **Stmt** *Let A and B be normal tensors of positive, possibly different, bond dimensions. Suppose that, for every positive N , their matrix product vectors are proportional by a nonzero scalar. Then there is $a \in \mathbb{C}$, with $|a| = 1$, such that $|V^{(N)}(A)\rangle = a^N |V^{(N)}(B)\rangle$ for every positive N . This is the positive-length one-normal-block consequence of [CPGSV16, Corollary A.3 and Theorem 2.10].*

Proof. ✓ **Proof** Positive-length nonzero proportionality implies eventual nonzero proportionality. Apply Theorem 10.2.12. $\square$

Lemma F.1.7 (Eventual linear independence from finite-index overlap orthonormality). Lean ✓ **Stmt** *Let I be a finite index set and let $(A_i)_{i \in I}$ be a finite family of tensors such that $O_{A_i A_i}(N) \rightarrow 1$ for each i , and $O_{A_i A_j}(N) \rightarrow 0$ whenever $i \neq j$. Then the MPV states $\{|V^{(N)}(A_i)\rangle\}_{i \in I}$ are linearly independent for all sufficiently large N .*

Proof. ✓ **Proof** The Gram matrix of the family $\{|V^{(N)}(A_i)\rangle\}_{i \in I}$ converges entrywise to the identity matrix. Hence for all sufficiently large N it is invertible, so the MPV states are linearly independent. $\square$

The same conclusion is often more convenient with an explicit threshold rather than the eventual quantifier.

Theorem F.1.8 (Existential BNT independence from overlap limits). Lean ✓ **Stmt** *If the self-overlaps of a finite block family tend to 1 and the cross-overlaps of distinct blocks tend to 0, then the MPV states are linearly independent for all sufficiently large system sizes.*

Proof. ✓ **Proof** This is Lemma F.1.10, restated with an explicit threshold N_0 . $\square$

Lemma F.1.9 (Eventual linear independence for two families). Lean ✓ **Stmt** *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be finite families of tensors. Suppose the overlaps inside each family are asymptotically orthonormal, and suppose every mixed overlap $O_{A_j B_k}(N)$ tends to 0. Then the combined MPV family*

$$\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g_A} \cup \{|V^{(N)}(B_k)\rangle\}_{k=1}^{g_B}$$

is linearly independent for all sufficiently large N .

Proof. ✓ **Proof** Apply the finite-index Gram-matrix criterion to the disjoint union of the two index sets. The within-family hypotheses give the diagonal and off-diagonal limits inside each block, and the mixed overlap hypothesis gives the off-diagonal limits between the two blocks. $\square$

The following two linear-algebra consequences package the eventual Gram-matrix criterion and its coefficient-extraction corollary.

Lemma F.1.10 (Eventual linear independence from overlap orthonormality). [Lean](#) [Lean](#) [✓ Stmt](#) Let $(A_j)_{j=1}^g$ be a finite family of tensors such that $O_{A_j A_j}(N) \rightarrow 1$ for each j and $O_{A_i A_j}(N) \rightarrow 0$ for $i \neq j$. Then the MPV states $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$ are linearly independent for all sufficiently large N .

Proof. [✓ Proof](#) Apply Lemma F.1.7 with $I = \{1, \dots, g\}$. □

Eventual linear independence is also what converts an equality of two finite linear combinations into equality of their coefficients; this is the mechanism behind the coefficient identities of Chapter 11.

Lemma F.1.11 (Eventual coefficient extraction from linear independence). [Lean](#) [✓ Stmt](#) If a finite family of vectors is linearly independent for all sufficiently large N , and two finite linear combinations of that family are equal for all sufficiently large N , then the corresponding coefficients are equal for all sufficiently large N .

Proof. [✓ Proof](#) Move the two linear combinations to one side. Linear independence forces each coefficient difference to vanish at every sufficiently large length. □

When two BNT families must be compared against each other, the same Gram-matrix criterion applies to their disjoint union, provided the mixed overlaps also decay. This is the form used in the exact sector matching of Section 10.8.

F.2 Change of basis, power sums, and sector bookkeeping

This section collects the finite-dimensional linear algebra behind alternative BNT comparisons, the extrapolation identities for finite sums of geometric sequences, and the bookkeeping identities for sector decompositions. A sector has copy weights $\mu_{j,q} \neq 0$, coefficient $c_{N,j} = \sum_q \mu_{j,q}^N$, representative bond dimension D_j , and total bond dimension $D_{\text{tot}} = \sum_j r_j D_j$.

Lemma F.2.1 (Invertible change of basis). [Lean](#) [✓ Stmt](#) Let $v_1, \dots, v_g$ and $w_1, \dots, w_g$ be two linearly independent families in a complex vector space with the same span. Then there exists an invertible matrix $U \in \text{GL}_g(\mathbb{C})$ such that $w_j = \sum_{i=1}^g U_{ij} v_i$ for every j .

Proof. [✓ Proof](#) Because the two families span the same subspace, each w_j has a unique expansion in the basis $\{v_i\}_{i=1}^g$; these coefficients form the columns of U . If $Ua = 0$, then the corresponding linear relation among the w_j is trivial because the w_j are linearly independent. Thus U is injective, hence invertible. □

Lemma F.2.2 (Eventual change of basis from equal spans). [Lean](#) [✓ Stmt](#) If two finite families of blocks have asymptotically orthonormal overlaps within each family, and if their MPV spans agree for all sufficiently large system sizes N , then for all sufficiently large N there exists an invertible matrix U_N expressing the MPV states of one family as linear combinations of the MPV states of the other.

Proof. [✓ Proof](#) By Lemma F.1.10, both families are linearly independent for all sufficiently large N . Hence, for all sufficiently large N , both families are linearly independent and the assumed span equality holds. Lemma F.2.1 applies at each such N . □

Lemma F.2.3 (Eventual vanishing of a finite geometric sum). [Lean](#) [✓ Stmt](#) Let $w_1, \dots, w_n$ be nonzero complex numbers. If $\sum_i c_i w_i^N = 0$ for every sufficiently large N , then the same equality holds for every non-negative exponent.

Proof. [✓ Proof](#) Induct on the number of terms. Subtracting w_1 times the equality at exponent N from the equality at exponent $N+1$ gives $\sum_{i=2}^n c_i(w_i - w_1)w_i^N = 0$. The induction hypothesis yields this equality at every exponent, so the original sum $S_N = \sum_i c_i w_i^N$ satisfies $S_{N+1} = w_1 S_N$ for every N . Since S_N eventually vanishes and $w_1 \neq 0$, it follows that $S_0 = 0$ and hence $S_N = 0$ for every N . $\square$

Lemma F.2.4 (Eventual power-sum equality holds at every exponent). [Lean](#) [✓ Stmt](#) Let $a_1, \dots, a_m$ and $b_1, \dots, b_n$ be nonzero complex numbers. If their power sums agree for every sufficiently large exponent, then $\sum_i a_i^k = \sum_j b_j^k$ for every non-negative exponent k .

Proof. [✓ Proof](#) Set $w_l = a_l$ for $l \leq m$ and $w_{m+l} = b_l$ for $l \leq n$, with coefficients $c_l = 1$ for $l \leq m$ and $c_l = -1$ for $l > m$. Then $\sum_{l=1}^{m+n} c_l w_l^k = \sum_i a_i^k - \sum_j b_j^k$. The hypothesis makes this expression zero for every sufficiently large k . Lemma F.2.3 makes it zero for every k , which is the required power-sum equality. $\square$

Theorem F.2.5 (Equal power sums imply equal multisets). [Lean](#) [✓ Stmt](#) Let $\alpha, \beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k \geq 1$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.

Proof. [✓ Proof](#) Construct diagonal matrices $\text{diag}(\alpha)$ and $\text{diag}(\beta)$. The power-sum hypothesis gives $\text{tr}(\text{diag}(\alpha)^k) = \text{tr}(\text{diag}(\beta)^k)$. By Theorem 13.8.7, the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. $\square$

Theorem F.2.6 (Representative dimensions are bounded by the total bond dimension). [Lean](#) [Lean](#) [✓ Stmt](#) Let the representative A_j have bond dimension D_j , and let D be the bond dimension of the flattened sector tensor. Then $D_j \leq D$. If every D_j is positive and there are g representatives, then also $g \leq D$.

Proof. [✓ Proof](#) Since $r_j > 0$, choose one copy of A_j . Its contribution D_j is one non-negative summand in the total dimension of the flattened copy family, so $D_j \leq D$.

For the second assertion, every representative has multiplicity $r_j \geq 1$, and each copy has positive dimension $D_j > 0$. Therefore

$$g = \sum_j 1 \leq \sum_j r_j \leq \sum_j r_j D_j = D.$$

$\square$

Theorem F.2.7 (Blocked sector coefficients and assembled tensor). [Lean](#) [Lean](#) [✓ Stmt](#) The coefficient of the blocked decomposition satisfies

$$c_{P^{[p]},j}^{(N)} = \sum_q (\mu_{j,q}^p)^N = c_{P,j}^{(Np)}. \quad (\text{F.1})$$

Moreover, blocking the tensor assembled from P gives the same MPV family as the tensor assembled from the blocked sector decomposition.

Proof. [✓ Proof](#) The coefficient identity in (F.1) is $(\mu^p)^N = \mu^{pN}$. The tensor identity is Theorem E.4.11 applied to the flattened copy family. $\square$

Lemma F.2.8 (Existence of an MPV phase class). [Lean](#) [✓ Stmt](#) A nonempty finite family of blocks has at least one MPV phase class.

Proof. [✓ Proof](#) Choose one original block index. Its occurrence in the phase-class enumeration supplies a phase class containing it. $\square$

Lemma F.2.9 (Regrouping a matrix sum by MPV phase classes). [Lean](#) [✓ Stmt](#) Let F_k be a finite family of square matrices indexed by the original blocks. If $k = (j, q)$ enumerates the copies in the MPV phase classes, then $\sum_j \sum_q F_{j,q} = \sum_k F_k$.

Proof. [✓ Proof](#) Evaluate both sides at a pair of matrix indices and apply the bijective enumeration of the finite phase-class quotient. $\square$

Definition F.2.10 (Single-copy sector decomposition). [Lean](#) [✓ Stmt](#) Given nonzero weights $(\mu_k)_{k=1}^r$ and blocks $(B_k)_{k=1}^r$, the *single-copy sector decomposition* has r sectors, one basis block B_k per sector, multiplicity 1, and sector weight μ_k . As a finite sum,

$$\sum_{k=1}^r \mu_k^N V^{(N)}(B_k)_\sigma = V^{(N)} \left(\bigoplus_k \mu_k B_k \right)_\sigma. \quad (\text{F.2})$$

Lemma F.2.11 (Single-copy sector decomposition preserves the represented family). [Lean](#) [✓ Stmt](#) The single-copy sector decomposition of $(\mu_k, B_k)_k$ represents the same MPS family as $\bigoplus_k \mu_k B_k$.

Proof. [✓ Proof](#) Expanding the sector decomposition gives the left-hand side of (F.2), which is the finite block-sum expression for $\bigoplus_k \mu_k B_k$. $\square$

Lemma F.2.12 (Sector-decomposition coefficient is not eventually zero). [Lean](#) [Lean](#) [✓ Stmt](#) For any sector decomposition P and any sector j , the coefficient $c_{P,j}^{(N)} = \sum_q \mu_{j,q}^N$ is not eventually zero in N . More generally, for every positive integer p , the subsequence $c_{P,j}^{(pN)}$ is not eventually zero.

Proof. [✓ Proof](#) If the coefficient vanished for every $N \geq M$, the finite geometric-sum extrapolation would make it vanish at every non-negative exponent. However, at exponent zero, $c_{P,j}^{(0)} = \sum_q \mu_{j,q}^0 = r_j > 0$, which is impossible. For the subsequence, apply the same argument to the nonzero bases $\mu_{j,q}^p$. $\square$

Lemma F.2.13 (Sector coefficient is not eventually zero). [Lean](#) [✓ Stmt](#) For a BNT canonical form P and any sector j , the sector coefficient $c_{P,j}^{(N)} = \sum_q \mu_{j,q}^N$ is not eventually zero in N . Only nonvanishing of the copy weights is used, not any modulus condition.

Proof. [✓ Proof](#) Apply Lemma F.2.12; its conclusion does not require the canonical-form assumptions. $\square$

F.2.1 Proportional matching and finite bijection extraction

The equal-MPV matching theorems in Chapter 10 are specializations of an eventual proportionality argument. We record that general mechanism and the finite-set construction used to package matches in both directions.

Theorem F.2.1.1 (Exact sector matching under eventual proportionality). [Lean](#) [✓ Stmt](#) Let P and Q be BNT canonical forms whose total MPV families are eventually nonzero and proportional. For every sector j of P , there is a sector k of Q with equal bond dimension, gauge-phase-equivalent basis tensor after the dimension identification, and non-decaying cross-overlap.

Proof. ✓ **Proof** Suppose the chosen sector decays against every sector of Q , and separate all such P -sectors from the remaining ones. For each remaining sector, choose a Q -sector with non-decaying overlap. Both blocks are normal by the canonical-form hypotheses. Theorem 10.2.9 rules out its decay alternative and supplies equal bond dimensions and a gauge-phase relation with a unit phase. Gauge-phase covariance therefore expresses that P -sector MPV as a geometric scalar multiple of the chosen Q -sector MPV.

Substitute these relations into the eventual proportionality identity. The combined-family Gram criterion makes the decaying residue together with the Q family eventually linearly independent. Exact coefficient extraction then makes the chosen sector coefficient eventually zero, contradicting its power-sum nonvanishing. Thus a non-decaying partner exists. A second application of Theorem 10.2.9 rules out decay and gives the required dimension equality and gauge-phase equivalence; no modulus-one overlap limit is assumed. $\square$

Theorem F.2.1.2 (Full-basis matching under eventual proportionality). Lean ✓ **Stmt** *Under the same hypotheses, every sector of Q has an equal-dimensional, gauge-phase-equivalent partner in P with non-decaying cross-overlap.*

Proof. ✓ **Proof** Reverse the eventual proportionality and apply Theorem F.2.1.1 to each sector of Q ; symmetry of the overlap converts the resulting witness to the stated orientation. $\square$

Lemma F.2.1.3 (Bijection from matches in both directions). Lean ✓ **Stmt** *Suppose every sector of Q has an equal-dimensional, gauge-phase-equivalent partner in P with non-decaying overlap, and the converse holds from P to Q . Then the forward partners may be chosen as a bijection between the two finite sector sets, preserving all three properties.*

Proof. ✓ **Proof** Basis separation makes each partner map injective: two sectors with the same partner would be gauge-phase equivalent to one another. The injections in both directions force equal finite cardinalities, so the forward map is bijective. $\square$

F.3 Direct-sum separation and prepared-family bookkeeping

We now prove the supporting separation and regrouping results used by the source-facing block-injectivity and supplier theorems in Chapter 10. The direct-sum argument follows the ground-space uniqueness mechanism of [PGVWC07]: disjoint local MPS spaces give pair-trace selectors, products of those selectors isolate one BNT block, and an injective prefix supplies an arbitrary matrix on that block. The remaining results identify dimensions while phase-equivalent prepared blocks are grouped into sectors and normalize the resulting finite weight family.

Lemma F.3.1 (Three-block separation of distinct BNT representatives). Lean Lean Lean ✓ **Stmt** *Let $A^1, \dots, A^r$ be irreducible left-canonical representatives with normalized self-overlaps, and suppose that distinct same-dimensional representatives are not gauge-phase equivalent. Fix $L_0 > 0$ such that every block is injective at lengths L_0 , $L_0 + 1$, and $3(L_0 + 1)$. Then each ordered pair $k \neq j$ is trace-separated at the common length $S = 3(L_0 + 1)$: the map*

$$(c_\omega)_{|\omega|=S} \mapsto \left(\sum_{|\omega|=S} c_\omega A_\omega^k, \sum_{|\omega|=S} c_\omega A_\omega^j \right)$$

is onto $M_{D_k}(\mathbb{C}) \oplus M_{D_j}(\mathbb{C})$. In particular, one may prescribe an arbitrary matrix on either block while making the same word combination vanish on the other block.

Proof. ✓ Proof Put $L = L_0 + 1$ and $S = 3L$. For a block A , write

$$\Gamma_n^A(X) = \sum_{|w|=n} \text{tr}(XA_w)|w\rangle, \quad \mathcal{G}_n(A) = \text{im } \Gamma_n^A.$$

Block injectivity at L and S makes both boundary maps injective; in particular,

$$\dim \mathcal{G}_L(A^q) = D_q^2 \quad (1 \leq q \leq r). \quad (\text{F.3})$$

Fix $k \neq j$ and first order the pair so that $D_j \leq D_k$. Suppose that a nonzero vector belongs to $\mathcal{G}_S(A^k) \cap \mathcal{G}_S(A^j)$. There are boundary matrices X and Y such that

$$0 \neq \psi = \Gamma_S^{A^k}(X) = \Gamma_S^{A^j}(Y). \quad (\text{F.4})$$

Injectivity at S gives $X \neq 0$ and $Y \neq 0$. Split every length- S word as uvw with $|u| = |v| = |w| = L$. Equality F.4 gives

$$\text{tr}(XA_u^k A_v^k A_w^k) = \text{tr}(YA_u^j A_v^j A_w^j). \quad (\text{F.5})$$

Because the length- L words of A^k span $M_{D_k}(\mathbb{C})$ and $X \neq 0$, the two-sided ideal generated by X is all of $M_{D_k}(\mathbb{C})$:

$$\text{span}\{A_u^k X A_v^k : |u| = |v| = L\} = M_{D_k}(\mathbb{C}). \quad (\text{F.6})$$

Indeed, expand arbitrary left and right matrix units in length- L words; a nonzero entry of X then produces every matrix unit. Thus, for each $Z \in M_{D_k}(\mathbb{C})$, choose coefficients a_{uv} with $Z = \sum_{u,v} a_{uv} A_u^k X A_v^k$ and set $R(Z) = \sum_{u,v} a_{uv} A_u^j Y A_v^j$. Cyclicity of the trace and F.5, applied to the word vwu , yield

$$\begin{aligned} \text{tr}(Z A_w^k) &= \sum_{u,v} a_{uv} \text{tr}(X A_v^k A_w^k A_u^k) \\ &= \sum_{u,v} a_{uv} \text{tr}(Y A_v^j A_w^j A_u^j) = \text{tr}(R(Z) A_w^j). \end{aligned}$$

Hence $\Gamma_L^{A^k}(Z) = \Gamma_L^{A^j}(R(Z))$ for every Z , and therefore

$$\mathcal{G}_L(A^k) \subseteq \mathcal{G}_L(A^j). \quad (\text{F.7})$$

Equations F.3 and F.7 give

$$D_k^2 = \dim \mathcal{G}_L(A^k) \leq \dim \mathcal{G}_L(A^j) = D_j^2.$$

Together with $D_j \leq D_k$, this forces $D_j = D_k$ and equality in F.7. This proves the unequal-dimension branch immediately: if $D_j < D_k$, a nonzero intersection is impossible. Reversing the initial ordering treats $D_k < D_j$ in the same way.

It remains to exclude the intersection when $D_j = D_k$. Gauge-phase separation and Theorem 10.4.3 give, beyond any prescribed threshold, a chain length $N > L$ for which the two periodic MPVs are not proportional. On the other hand, equality $\mathcal{G}_L(A^k) = \mathcal{G}_L(A^j)$ gives identical translated length- L constraints on the N -site ring. The parent-Hamiltonian uniqueness mechanism identifies the corresponding chain ground spaces with the one-dimensional MPV spans:

$$\mathcal{G}_{N,L}(A^k) = \mathbb{C} V^{(N)}(A^k), \quad \mathcal{G}_{N,L}(A^j) = \mathbb{C} V^{(N)}(A^j).$$

Equality of the local spaces therefore forces these two MPV lines to be equal, contradicting the choice of N . Consequently, for every distinct pair, including the unequal-dimensional pairs,

$$\mathcal{G}_S(A^k) \cap \mathcal{G}_S(A^j) = \{0\}. \quad (\text{F.8})$$

Finally, let Φ_{kj} be the displayed length- S word-evaluation map from coefficient families to $M_{D_k}(\mathbb{C}) \oplus M_{D_j}(\mathbb{C})$. Pair the target by

$$\langle (X, Y), (M, N) \rangle = \text{tr}(XM) + \text{tr}(YN).$$

A pair (X, Y) annihilates $\text{im}(\Phi_{kj})$ precisely when

$$\text{tr}(XA_w^k) + \text{tr}(YA_w^j) = 0 \quad (|w| = S),$$

or equivalently $\Gamma_S^{A^k}(X) = \Gamma_S^{A^j}(-Y)$. There is therefore a linear map

$$\Theta : \text{im}(\Phi_{kj})^\perp \longrightarrow \mathcal{G}_S(A^k) \cap \mathcal{G}_S(A^j), \quad \Theta(X, Y) = \Gamma_S^{A^k}(X) = \Gamma_S^{A^j}(-Y).$$

It is injective because both boundary maps are injective. Conversely, if ψ lies in the intersection, there are unique X and Z with $\Gamma_S^{A^k}(X) = \psi = \Gamma_S^{A^j}(Z)$; the pair $(X, -Z)$ annihilates $\text{im}(\Phi_{kj})$ and maps to ψ . This construction is the inverse of Θ , so the two spaces are isomorphic:

$$\text{im}(\Phi_{kj})^\perp \cong \mathcal{G}_S(A^k) \cap \mathcal{G}_S(A^j). \quad (\text{F.9})$$

The right-hand side vanishes by F.8, and hence so does the annihilator. The trace pairing is nondegenerate and all spaces are finite-dimensional, so $\text{im}(\Phi_{kj})^\perp = \{0\}$ implies that Φ_{kj} is surjective. Surjectivity first gives an arbitrary value on one block and zero on an anchor block; the selector construction then chooses $(1, 0)$ for the remaining pairs, and products of these homogeneous selectors give the simultaneous word-tuple span in Lemma 10.3.3. $\square$

Theorem F.3.2 (Common injective blocking for primitive irreducible families). [Lean](#) [✓ Stmt](#) *Let $(B_k)_{k=1}^r$ be a finite family of positive-bond-dimension MPS tensors, each trace-preserving, primitive, and tensor-irreducible. Then there is a single $L \geq 1$ such that each $B_k^{[L]}$ is one-site injective, and this blocking preserves trace preservation, primitivity, and irreducibility for every block.*

Proof. [✓ Proof](#) For each block, Theorem 8.9.1 gives a positive length at which it becomes one-site injective; the product of these lengths is a common length. Injectivity persists under positive further blocking, and Theorem E.4.10 preserves the other hypotheses. $\square$

Lemma F.3.3 (Total dimension after grouping by BNT representatives). [Lean](#) [✓ Stmt](#) *Suppose CF summands are grouped by BNT representative, every original copy weight is nonzero, and every summand assigned to a representative has that representative's bond dimension. Then the total bond dimension of the resulting BNT sector decomposition is the sum of the original block dimensions.*

Proof. [✓ Proof](#) The collapsed decomposition counts one representative dimension for each assigned copy. Let $e(j, q)$ denote the original CF summand assigned to the q th copy of representative P_j . Since $(j, q) \mapsto e(j, q)$ reindexes the original summands and each assigned summand has the representative's bond dimension,

$$\sum_{j=1}^g \sum_{q=1}^{r_j} \dim(P_j) = \sum_{j=1}^g \sum_{q=1}^{r_j} D_{e(j,q)} = \sum_{k=1}^r D_k.$$

This is the equality between the collapsed total dimension and the original direct-sum dimension. $\square$

Lemma F.3.4 (Prepared phase-equivalent NTs have the same bond dimension). [Lean](#) [✓ Stmt](#)
Let X and Y be trace-preserving, primitive, irreducible NTs of positive bond dimension. If their MPV families differ by a nonzero scalar power, then X and Y have the same bond dimension.

Proof. [✓ Proof](#) Choose $\zeta \neq 0$ with $|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$ for every positive length N . The normalized self-overlaps give $|O_{XX}(N)| \rightarrow 1$, and the phase relation gives

$$\begin{aligned} O_{YX}(N) &= \zeta^N O_{XX}(N), \\ |O_{YX}(N)| &= |\zeta|^N |O_{XX}(N)|. \end{aligned}$$

Since the same normalization forces $|\zeta| = 1$, this gives $|O_{YX}(N)| \rightarrow 1$. If the two bond dimensions differed, the rectangular overlap-decay theorem would force $O_{YX}(N) \rightarrow 0$, contradicting the preceding limit. $\square$

Lemma F.3.5 (Prepared BNT representative classes preserve bond dimension). [Lean](#) [✓ Stmt](#)
In a prepared family of positive-bond-dimension trace-preserving primitive irreducible NTs, every CF summand assigned to a BNT representative has that representative's bond dimension.

Proof. [✓ Proof](#) Apply Lemma F.3.4 to each representative and each assigned summand. $\square$

Lemma F.3.6 (Prepared collapsed BNT total dimension). [Lean](#) [✓ Stmt](#) *For prepared trace-preserving primitive irreducible NTs of positive bond dimension with all copy weights nonzero, grouping CF summands by BNT representative preserves the total bond dimension: the collapsed sector decomposition has total bond dimension equal to the sum of the original block dimensions.*

Proof. [✓ Proof](#) Every assigned summand shares the bond dimension of its representative, by the equal-dimension fact above. The displayed regrouping identity in Lemma F.3.3 then identifies the collapsed sum over pairs (j, q) with $\sum_{k=1}^r D_k$, the original direct-sum dimension. $\square$

Theorem F.3.7 (Conditional BNT form after blocking for an arbitrary tensor). [Lean](#) [✓ Stmt](#)
Let A be an MPS tensor. Then there exists a blocking length $p \geq 1$, nonzero weights μ_k , and blocks B_k over the blocked alphabet, such that:

1. every block has positive bond dimension;
2. every block is TP-normalized, primitive, irreducible, and injective;
3. $A^{[p]}$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family at every positive length.

Moreover, under $|\mu_k| \leq 1$ for every k and $|\mu_k| = 1$ for at least one k , there exists a sector decomposition P over the blocked alphabet in the BNT canonical form of Definition 10.6.1.14 such that $A^{[p]}$ and P generate the same MPV family at every positive length.

Proof. [✓ Proof](#) The after-blocking reduction supplies the TP-normalized primitive irreducible injective blocks and the positive-length MPV identity. The normalized-weight hypotheses are the choice that [CPGSV16, Section 2.3] makes; Theorem 10.6.3.3 then produces the sector decomposition, following the BNT selection of [CPGSV16, Appendix A]. The remaining rescaling step is recorded in Remark 10.6.3.16. $\square$

Theorem F.3.8 (Projective weight normalization). [Lean](#) [✓ Stmt](#) *For nonzero weights $\mu_1, \dots, \mu_r$ with $r \geq 1$ there is a positive real m such that every normalized weight satisfies*

$$\begin{aligned} \left| \frac{\mu_k}{m} \right| &\leq 1, \\ \frac{\mu_k}{m} &\neq 0, \end{aligned}$$

and at least one normalized weight has unit modulus, $|\mu_{k_0}/m| = 1$. This is the normalization that [CPGSV16, line 246] chooses for the canonical-form weights.

Proof. ✓ Proof Take $m = \max_k |\mu_k|$; the maximum over a nonempty finite family is attained, positive because every weight is nonzero, an upper bound for every modulus, and equal to the modulus of the attaining weight. □

Lemma F.3.9 (Common scalar rescaling of block weights). Lean ✓ Stmt *Multiplying every block weight in a block-diagonal MPS tensor by a scalar c multiplies every length- N MPV coefficient by c^N .*

Proof. ✓ Proof Expand the block-diagonal MPV coefficient as a finite sum over blocks and factor c^N out of the sum. □

Appendix G

Proof of the Fundamental Theorem: Supporting Results

This appendix supports Chapter 11. It proves phase normalization, coefficient extraction from matched gauges, direct-sum conjugation, the exact consequences of length-dependent proportionality, coordinate reindexing, and unitary refinements. The first part concerns coefficients and block gauges; the second identifies matched coordinates with the literal total bond space.

G.1 Matched phases and coefficient extraction

This section isolates two elementary steps used by the main proof. First, the sector partners obtained in both directions determine a finite bijection. Second, normalization of the matched normal tensors forces every gauge scalar to be a phase.

Theorem G.1.1 (Bijective matching under eventual proportionality). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total MPV families are eventually nonzero and proportional. There is a bijection $\beta : J(Q) \simeq J(P)$ such that every Q_k has an equal-dimensional, gauge-phase-equivalent partner $P_{\beta(k)}$ satisfying*

$$O_{P_{\beta(k)} Q_k}(N) \not\rightarrow 0.$$

Proof. ✓ Proof Apply the full-basis matching theorem first from Q to P and then to the reverse proportionality from P to Q . If P_j and $P_{j'}$ had the same partner Q_k , then symmetry and transitivity would give

$$P_j \sim_{\text{gp}} Q_k \sim_{\text{gp}} P_{j'} \implies P_j \sim_{\text{gp}} P_{j'},$$

contradicting BNT basis separation. Both partner maps are therefore injective. The two finite sector sets have equal cardinality, so the forward map is bijective and retains the displayed overlap limit together with the dimension and gauge data. $\square$

Lemma G.1.2 (Unit phase from matched normalized BNT blocks). Lean ✓ Stmt *Let P and Q be BNT canonical forms. If, for every positive length N and word σ , a sector P_j and a sector Q_k have MPV families related by*

$$V^{(N)}(Q_k)_\sigma = \zeta^N V^{(N)}(P_j)_\sigma, \tag{G.1}$$

then $|\zeta| = 1$.

Proof. ✓ Proof The BNT normalization gives self-overlap limits of modulus 1 for both sectors. The MPV identity (G.1) gives, for every positive N ,

$$O_{Q_k Q_k}(N) = |\zeta|^{2N} O_{P_j P_j}(N).$$

Since $O_{Q_k Q_k}(N) \rightarrow 1$ and $O_{P_j P_j}(N) \rightarrow 1$, taking limits forces $|\zeta|^{2N} \rightarrow 1$ and hence $|\zeta| = 1$. $\square$

G.2 Coefficient identities from gauge-phase equalities

The main coefficient theorem takes the matched phases as input. The next result shows how those phases are selected from gauge-phase equivalences before the linear-independence argument is applied.

Lemma G.2.1 (Full-basis coefficient identity from matched sectors). Lean ✓ Stmt *Let P and Q be BNT canonical forms with the same MPV family at every positive length, matched by a bijection $\beta : J(Q) \simeq J(P)$ with bond-dimension equalities and a gauge-phase equivalence between Q_k and $P_{\beta(k)}$ for every k . Then each sector carries a unit-modulus phase ζ_k , and eventually*

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q). \quad (\text{G.2})$$

This is the equal-MPV coefficient comparison of [CPGSV16, Appendix MPV proof, lines 1187–1188].

Proof. ✓ Proof For each matched sector, the gauge-phase equivalence gives $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$. Lemma G.1.2 gives $|\zeta_k| = 1$. Applying Lemma 11.1.2 to the same phases gives $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ for all sufficiently large N . $\square$

G.3 The scalar-threaded proportional identity

For proportional total MPVs, coefficient extraction retains the one scalar that relates the two total vectors at each length. This identity is exact, but it does not imply the scalar-free identity needed to recover copy weights; the counterexample in Remark 11.2.6 shows that no such implication is possible.

Lemma G.3.1 (Proportional matched-phase coefficient identity, scalar-threaded). Lean ✓ Stmt *Let P be a BNT canonical form and Q a sector decomposition whose total tensors generate eventually nonzero proportional MPV families. Suppose a bijection $\beta : J(Q) \simeq J(P)$ and scalars ζ_k satisfy the matched MPV phase identities (11.2) for every sector k . Then there is a single eventually-nonzero scalar sequence c_N and a threshold N_0 such that, for all $N > N_0$ and every $k \in J(Q)$,*

$$c_N^{(\beta(k))}(P) = c_N \zeta_k^N c_N^{(k)}(Q). \quad (\text{G.3})$$

The scalar c_N is the same for every sector k : it is the single global proportionality scalar of [CPGSV16, Theorem thm1, lines 1167–1170] at length N .

Proof. ✓ Proof For each N on the proportionality tail, expand both total MPVs in their sector bases and use the global proportionality scalar c_N :

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = c_N \sum_{k \in J(Q)} c_N^{(k)}(Q) |V^{(N)}(Q_k)\rangle.$$

The matched phase identities (11.2) give $|V^{(N)}(Q_k)\rangle = \zeta_k^N |V^{(N)}(P_{\beta(k)})\rangle$; because every sector is matched, reindexing the Q -sum by β rewrites the right-hand side entirely in the P -basis. The BNT eventual linear independence of the P -basis and Lemma F.1.11 then extract the coefficient identities (G.3) for all sufficiently large N . $\square$

G.4 Direct-sum conjugation

We now prove the general block-matrix identities used by the global gauge in Section 11.2. These statements do not choose or match sectors; they only combine conjugacies that have already been obtained.

Definition G.4.1 (Invertible matrix associated to a unitary). Lean ✓ Stmt A unitary matrix U determines an invertible matrix whose inverse is $U^\dagger$.

Lemma G.4.2 (Unitary block-diagonal gauges). Lean ✓ Stmt If every block gauge U_k is unitary, then the flattened block-diagonal gauge $U = \bigoplus_k U_k$ is unitary.

Proof. ✓ Proof The products are computed blockwise:

$$UU^\dagger = \bigoplus_k U_k U_k^\dagger = \bigoplus_k \mathbb{1} = \mathbb{1}.$$

$\square$

Lemma G.4.3 (Direct-sum conjugation by the flattened gauge). Lean Lean ✓ Stmt If, for every block k and physical index i , $B_k^i = X_k A_k^i X_k^{-1}$, then the weighted direct sums satisfy

$$\bigoplus_k \mu_k B_k^i = X \left(\bigoplus_k \mu_k A_k^i \right) X^{-1},$$

where X is the flattened block-diagonal gauge from Definition 11.2.1. Hence the two assembled tensors are gauge equivalent.

Proof. ✓ Proof Multiplication of block-diagonal matrices is computed blockwise. The scalar weight μ_k commutes with the conjugation on the k th block, and the reindexing from dependent block coordinates to the flattened bond index preserves products and inverses. $\square$

Theorem G.4.4 (Global gauge from a copy-weight matching). Lean ✓ Stmt Suppose that the sectors of P and Q are matched, with bond-dimension equalities, nonzero phases, and block gauges satisfying $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. If the same phases also give a copy-weight matching, then the flattened Q -tensor is obtained from the matched-coordinate P -tensor by the direct-sum gauge $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. This is only a reformulation of Theorem 11.2.2.

Proof. ✓ Proof The copy-weight matching supplies $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. Together with $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$, these are exactly the two hypotheses (11.4) of Theorem 11.2.2. $\square$

G.5 Additional proportional consequences

The following implications are useful when copy weights satisfy extra relations. They are not conclusions of CPSV16 Theorem 2.10. In particular, their additional hypotheses fail for the proportional BNT forms in Remark 11.2.6.

Theorem G.5.1 (Additional proportional global gauge under copy-weight matching). [Lean](#)
[✓ Stmt](#) Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then the proportional sector-matching theorem gives β , the bond-dimension equalities, the unit phases ζ_k , and the block gauges X_k . If one additionally assumes a copy-weight matching for these same phases, then the flattened Q -tensor is obtained from the matched-coordinate P -tensor by the direct-sum gauge $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$.

Proof. [✓ Proof](#) The proportional sector-matching theorem supplies β , X_k , and unit phases ζ_k satisfying $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. Since $|\zeta_k| = 1$, each ζ_k is nonzero. A copy-weight matching gives $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$, so Theorem G.4.4 gives the direct-sum gauge. $\square$

Theorem G.5.2 (Additional proportional global gauge under coefficient identities). [Lean](#)
[✓ Stmt](#) Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities, unit-modulus phases ζ_k , and block gauges X_k such that $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ for every sector k and physical index i . If one additionally assumes that these same phases satisfy the matched-sector coefficient identities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ for all sufficiently large N , then they determine a copy-weight matching and hence the flattened Q -tensor is obtained from the matched-coordinate P -tensor by the direct-sum gauge $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$.

Proof. [✓ Proof](#) The sector-matching theorem gives β , X_k , and $|\zeta_k| = 1$ with $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. Since $\zeta_k \neq 0$, the coefficient identities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ produce $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$ by Definition 11.1.4. Theorem G.5.1 then applies to this copy-weight matching. $\square$

G.6 Matched flattened coordinates

Let $\beta : J(Q) \simeq J(P)$ match the basis sectors and let τ_k match the copies in sector k . Flatten the pair (k, q) to the copy index of Q . The matched P -data on that index are the weight and tensor

$$\tilde{\mu}_{k,q} = \mu_{\beta(k), \tau_k(q)}, \quad \tilde{P}_{k,q} = P_{\beta(k)},$$

where the second expression is read after the matched bond-dimension identification. The block gauge repeats X_k on every copy q .

Definition G.6.1 (Matched weights, tensors, and repeated gauges). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The matched flattened data are the functions $\tilde{\mu}$, $\tilde{P}$, and $(k, q) \mapsto X_k$ defined above on the flattened copy coordinates of Q .

Lemma G.6.2 (Equality of the matched total bond dimensions). [Lean](#) [✓ Stmt](#) If matched sectors have equal bond dimensions and matched copy sets are bijective, then P and Q have the same total bond dimension.

Proof. [✓ Proof](#) Reindex first by the sector bijection and then by each copy bijection. Every summand $\dim P_{\beta(k)}$ becomes $\dim Q_k$, so

$$\sum_{j \in J(P)} r_j(P) \dim P_j = \sum_{k \in J(Q)} r_k(Q) \dim Q_k.$$

$\square$

Definition G.6.3 (Sector and bond-coordinate equivalences). [Lean](#) [Lean](#) [✓ Stmt](#) The sector bijection, copy bijections, and bond-dimension equalities induce an equivalence of the dependent triples (j, q, a) and (k, q', a') . After flattening the dependent bond coordinate, this gives an equivalence of the standard total bond-index sets.

Proof. [✓ Proof](#) Send (k, q, a) to $(\beta(k), \tau_k(q), a)$, using the bond-dimension equality to identify the last coordinate. The inverse uses β^{-1} and τ_k^{-1} . Both composites are the identity componentwise. Conjugating by the standard equivalence between dependent triples and a finite interval gives the flattened bond-coordinate equivalence. $\square$

Theorem G.6.4 (Equal-MPV gauge with all matched witnesses). [Lean](#) [✓ Stmt](#) *Under the hypotheses of the equal-MPV theorem, one may choose simultaneously the sector bijection, bond-dimension equalities, copy bijections, phases, block gauges, total-dimension equality, and the flattened global gauge. They satisfy the matched block and weight identities of Theorem 11.2.4, together with the conjugation of Q by the matched-coordinate P -tensor.*

Proof. [✓ Proof](#) Theorem 11.2.4 supplies all witnesses except the displayed equality of total dimensions. Apply Lemma G.6.2 to its sector and copy bijections. The resulting equality identifies the total bond spaces, while the global conjugation equation is unchanged. $\square$

G.7 Permutation gauges and literal coordinates

The matched P -tensor has the correct blocks but lists them in Q -order. A permutation matrix restores the literal order of P . We record its invertible and unitary forms before proving the coordinate conversion used in Theorem 11.2.5.

Definition G.7.1 (Invertible permutation gauge). [Lean](#) [✓ Stmt](#) For a permutation ρ of a finite bond-index set, let Π_ρ be its permutation matrix. The corresponding element of the general linear group is denoted G_ρ .

Lemma G.7.2 (Matrix and inverse identities for a permutation gauge). [Lean](#) [Lean](#) [✓ Stmt](#) *The matrix of G_ρ is Π_ρ , and the matrix of its inverse is $\Pi_{\rho^{-1}}$.*

Proof. [✓ Proof](#) The first identity is the definition of the permutation gauge. Since the inverse permutation reverses the coordinate relabelling, $G_\rho^{-1} = G_{\rho^{-1}}$, which gives the second identity. $\square$

Every row and every column of Π_ρ contains exactly one nonzero entry, equal to 1. Hence

$$\Pi_\rho \Pi_{\rho^{-1}} = \mathbb{1}, \quad \Pi_\rho^\dagger = \Pi_{\rho^{-1}}.$$

Together with Lemma G.7.2, these identities show that $G_\rho^\dagger = G_\rho^{-1}$, so the permutation gauge is unitary.

Lemma G.7.3 (Permutation conjugation reindexes both coordinates). [Lean](#) [✓ Stmt](#) *For a permutation ρ and a square matrix M , conjugation by the permutation matrix reindexes both coordinates by ρ :*

$$\Pi_\rho M \Pi_{\rho^{-1}} = (M_{\rho(i), \rho(j)})_{i,j}. \quad (\text{G.4})$$

Equivalently, the right-hand side is the submatrix of M along $\rho \times \rho$.

Proof. [✓ Proof](#) Left multiplication by Π_ρ selects row $\rho(i)$ in row i , and right multiplication by $\Pi_{\rho^{-1}}$ selects column $\rho(j)$ in column j . The two selections give the displayed submatrix. $\square$

The coordinate equivalence of Definition G.6.3 induces the required permutation ρ after the total-dimension identification. A convention with inverse indices merely replaces the chosen permutation by ρ^{-1} ; we use the convention in (G.4) to match the formal coordinate identity. Applying that identity to P_{tot}^i gives the matched-coordinate tensor. Composing G_ρ with the block-diagonal gauge therefore gives the literal gauge of Theorem 11.2.5. This also explains why the total-dimension cast and the sector permutation must occur in that order.

G.8 Unitary witness refinements

The main text proves that a specified gauge between left-canonical irreducible sectors can be replaced by a unitary without changing its phase. We record the existential form and then repeat the construction over every matched copy.

Lemma G.8.1 (Unitary gauge for left-canonical irreducible tensors). Lean ✓ Stmt *Let A and B be left-canonical irreducible tensors of common positive bond dimension that are gauge-phase equivalent. Then the gauge can be taken unitary: there are a unitary matrix U and a scalar ζ with $|\zeta| = 1$ such that, for every physical index i ,*

$$B^i = \zeta U A^i U^\dagger.$$

Proof. ✓ Proof Unpack the gauge-phase equivalence as $B^i = \zeta X A^i X^{-1}$ with X invertible and $\zeta \neq 0$. Lemma 11.3.1.2 replaces X by a unitary U while retaining this same phase and proves $|\zeta| = 1$. $\square$

Lemma G.8.2 (Unitary gauge in matched sector coordinates). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors generate the same MPV family at every positive length. There are a sector bijection β , copy bijections τ_k , phases ζ_k , sector unitaries U_k , and a unitary block-diagonal matrix X such that*

$$\begin{aligned} Q_k^i &= \zeta_k U_k P_{\beta(k)}^i U_k^\dagger, \\ \nu_{k,q} &= \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}, \\ |\zeta_k| &= 1. \end{aligned}$$

The matched sectors and copies have equal bond dimensions and multiplicities. If P_{match} denotes the corresponding reordered block assembly, then $Q_{\text{flat}}^i = X P_{\text{match}}^i X^{-1}$.

Proof. ✓ Proof Apply Lemma 11.3.1.2 to each matched sector, retaining the phase in the copy-weight identity. Repeating U_k on every copy gives $X = \bigoplus_{k,q} U_k$, which is unitary by Lemma G.4.2. For every matched copy (k, q) ,

$$\nu_{k,q} Q_k^i = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)} \zeta_k U_k P_{\beta(k)}^i U_k^\dagger = \mu_{\beta(k), \tau_k(q)} U_k P_{\beta(k)}^i U_k^\dagger.$$

These are precisely the blocks of $Q_{\text{flat}}^i = X P_{\text{match}}^i X^{-1}$. $\square$

Appendix H

Symmetries and Virtual-Boundary Nondecay: Supporting Results

This appendix supports Chapter 12. It collects permutation and gauge bookkeeping, factor-system identities, stationary-boundary twist calculations, common trace-preserving gauges, and the finite-dimensional limit arguments used in the virtual-boundary nondecay results. The notation is introduced locally in each part, and every reference to the main development is given by its stable label.

The physical index set is $I = \{0, \dots, d-1\}$, the bond algebra is $M_D(\mathbb{C})$, and G denotes a group. All sums below are finite. The first section treats permutation actions on I ; the second isolates scalar uniqueness for two gauges of the same injective tensor; the third records the elementary factor-system bookkeeping used in the cohomology discussion of Chapter 12.

H.1 Permutation reindexing and identity bookkeeping

The linear-combination twist $\tilde{A}_g^i = \sum_j U(g)_{ij} A^j$ is the standard form used in the virtual representation theorem. The following alternative formulation replaces the matrix action by a permutation of the physical index; it is useful for symmetries that act by reordering basis states rather than by a general linear map.

Definition H.1.1 (Permutation symmetry datum). [Lean](#) [✓ Stmt](#) An *on-site symmetry datum* for a group G on a physical space of dimension d consists of a matrix-valued map $u : G \rightarrow M_d(\mathbb{C})$ and a physical-index action $\sigma : G \rightarrow \text{End}(\{0, \dots, d-1\})$.

Definition H.1.2 (Permutation-twisted tensor). [Lean](#) [✓ Stmt](#) Given an on-site symmetry datum (u, σ) , the *permutation-twisted tensor* at group element g is

$$(A^{\sigma_g})^i := A^{\sigma(g)(i)}. \quad (\text{H.1})$$

In tensor-network notation the local tensor is unchanged and only the physical leg is relabelled:

$$\begin{array}{ccc}
\begin{array}{c} i \\ | \\ \bullet \\ | \\ A \end{array} & \xrightarrow{\sigma_g} & \begin{array}{c} \sigma(g)(i) \\ | \\ \bullet \\ | \\ A \end{array} .
\end{array}$$

Lemma H.1.3 (Identity permutation twist). [Lean](#) [✓ Stmt](#) If $\sigma(1) = \text{id}$, then $(A^{\sigma_1})^i = A^i$ for all i .

Proof. [✓ Proof](#) The claim follows from $\sigma(1)(i) = i$ for all i . $\square$

Composition order. The linear-combination twist (Lemma 12.1.3) composes as $\widetilde{(\widetilde{A_h})_g} = \widetilde{A_{gh}}$, applying h first then g . The permutation twist below uses the same left-action convention $\sigma(gh) = \sigma(g) \circ \sigma(h)$, so $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$; the inner permutation $\sigma(h)$ acts first on the index.

Lemma H.1.4 (Composition of permutation twists). [Lean](#) [✓ Stmt](#) If $\sigma(gh) = \sigma(g) \circ \sigma(h)$ for all $g, h \in G$, then

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}. \quad (\text{H.2})$$

Proof. [✓ Proof](#) Expanding both sides gives $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$. Diagrammatically this is just two consecutive relabellings of the same local tensor:

$$\begin{array}{ccccc}
\begin{array}{c} i \\ | \\ \bullet \\ | \\ A \end{array} & \xrightarrow{\sigma_h} & \begin{array}{c} \sigma(h)(i) \\ | \\ \bullet \\ | \\ A \end{array} & \xrightarrow{\sigma_g} & \begin{array}{c} \sigma(g)(\sigma(h)(i)) \\ | \\ \bullet \\ | \\ A \end{array} .
\end{array}$$

$\square$

H.1.1 Linear physical-index rotations

The same bookkeeping applies to a general matrix acting on the physical index.

Definition H.1.1.1 (Physical-index rotation). [Lean](#) [✓ Stmt](#) Given a matrix $M \in M_d(\mathbb{C})$ and an MPS tensor A , the *physical-index rotation* is the tensor $(A_M)^i := \sum_j M_{ij} A^j$. When the matrix is a unitary u , the rotation is written A_u .

Lemma H.1.1.2 (Matrix product vector under physical rotation). [Lean](#) [✓ Stmt](#) For a length- N physical configuration $s = (s_0, \dots, s_{N-1})$, physical rotation satisfies

$$\text{tr}((A_M)^{s_0} \dots (A_M)^{s_{N-1}}) = \sum_{t \in \{0, \dots, d-1\}^N} \left(\prod_{n=0}^{N-1} M_{s_n, t_n} \right) \text{tr}(A^{t_0} \dots A^{t_{N-1}}). \quad (\text{H.3})$$

Proof. [✓ Proof](#) Substitute $(A_M)^{s_n} = \sum_{t_n} M_{s_n, t_n} A^{t_n}$ at each site, distribute the finite sums through the ordered matrix product, and use linearity of the trace to obtain (H.3). $\square$

Lemma H.1.1.3 (Word expansion for a twisted tensor). [Lean](#) [✓ Stmt](#) For a word $b : \{0, \dots, L-1\} \rightarrow \{0, \dots, d-1\}$,

$$A_g^{b_0} \dots A_g^{b_{L-1}} = \sum_{v : \{0, \dots, L-1\} \rightarrow \{0, \dots, d-1\}} \left(\prod_{k=0}^{L-1} U(g)_{b_k v_k} \right) A^{v_0} \dots A^{v_{L-1}}. \quad (\text{H.4})$$

Proof. [✓ Proof](#) Substitute $A_g^{b_k} = \sum_{v_k} U(g)_{b_k v_k} A^{v_k}$ into the ordered product. Repeated distributivity gives one summand for each function v , scalar coefficients commute past the matrices and multiply, and the remaining ordered product is $A^{v_0} \dots A^{v_{L-1}}$. $\square$

H.2 Gauge ratios and scalar uniqueness

The virtual representation theorem (Theorem 12.1.16) relies on the fact that composing two gauge transformations yields another gauge up to a scalar factor.

Lemma H.2.1 (Gauge ratio commutes with tensor matrices). [Lean](#) [✓ Stmt](#) *Diagrammatically, the hypothesis is that the two local gauges*

$$\begin{array}{c}
 \begin{array}{ccc}
 \begin{array}{c} i \\ | \\ \bullet \\ B \end{array} & = & \begin{array}{c} i \\ | \\ \text{---} \text{---} X \text{---} \bullet \text{---} X^{-1} \text{---} \\ A \end{array}
 \end{array} \\
 \text{and} \\
 \begin{array}{ccc}
 \begin{array}{c} i \\ | \\ \bullet \\ B \end{array} & = & \begin{array}{c} i \\ | \\ \text{---} \text{---} Y \text{---} \bullet \text{---} Y^{-1} \text{---} \\ A \end{array}
 \end{array}
 \end{array}$$

produce the same target tensor B^i from the same source tensor A^i . If $X, Y \in \text{GL}_D(\mathbb{C})$ both conjugate an MPS tensor A to the same tensor B (i.e. $B^i = XA^iX^{-1} = YA^iY^{-1}$ for all i), then $Y^{-1}X$ commutes with every A^i .

Proof. [✓ Proof](#) Direct matrix calculation: conjugating both gauge relations and composing gives $(Y^{-1}X)A^i = A^i(Y^{-1}X)$. $\square$

Lemma H.2.2 (Gauge ratio is scalar for injective tensors). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, if A is additionally injective, then $Y^{-1}X = \lambda \mathbb{1}_D$ for some scalar $\lambda \in \mathbb{C}$.*

Proof. [✓ Proof](#) The same two-gauge picture shows that the ratio $Y^{-1}X$ acts invisibly on the local tensor. By Lemma H.2.1, $Y^{-1}X$ commutes with all A^i . Since $\{A^i\}$ spans $M_D(\mathbb{C})$, the matrix $Y^{-1}X$ commutes with all of $M_D(\mathbb{C})$. By Lemma 2.1.5, it is a scalar matrix. $\square$

Theorem H.2.3 (Projective multiplication law for gauge matrices). [Lean](#) [✓ Stmt](#) *Let A be an injective MPS tensor with on-site symmetry under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$. For each $g \in G$, let $X(g) \in \text{GL}_D(\mathbb{C})$ satisfy $\tilde{A}_g^i = X(g)A^iX(g)^{-1}$ for all i . Then for all $g, h \in G$, there exists $c(g, h) \in \mathbb{C}$ such that $X(h)X(g) = c(g, h)X(gh)$. In fact $c(g, h) \neq 0$ since both sides are invertible, so $c(g, h) \in \mathbb{C}^\times$. Diagrammatically, the two candidate gauges are competing local replacements for the same twisted tensor:*

$$\begin{array}{ccc}
 \begin{array}{c} i \\ | \\ \text{---} \text{---} X(h)X(g) \text{---} \bullet \text{---} (X(h)X(g))^{-1} \text{---} \\ A \end{array} & = & \begin{array}{c} i \\ | \\ \text{---} \text{---} X(gh) \text{---} \bullet \text{---} X(gh)^{-1} \text{---} \\ A \end{array}
 \end{array}$$

and both sides conjugate A^i to the same twisted tensor $\tilde{A}_{gh}^i$.

Proof. [✓ Proof](#) By the composition law $\tilde{A}_{gh} = \widetilde{(\tilde{A}_h)_g}$ (Lemma 12.1.3), both $X(h)X(g)$ and $X(gh)$ conjugate A to $\tilde{A}_{gh}$. Lemma H.2.2 then gives the scalar factor. $\square$

H.3 Cocycle equivalence and coboundaries

Let $\omega_1, \omega_2 : G \times G \rightarrow \mathbb{C}^\times$ be arbitrary scalar 2-cochains. The definitions of coboundary and cohomology are Definitions 12.2.1 and 12.2.2.

Theorem H.3.1 (Cohomology relation is an equivalence relation). [Lean](#) [✓ Stmt](#) *The relation “cohomologous to” is reflexive, symmetric, and transitive on the set of $\mathbb{C}^\times$ -valued scalar 2-cochains, and therefore also on the subset of cocycles.*

Proof. [✓ Proof](#) Reflexivity: take $\varphi = 1$. Symmetry: replace φ by φ^{-1} . Transitivity: compose the witnesses φ and ψ pointwise in (12.10). $\square$

Lemma H.3.2 (Coboundary iff cohomologous to the trivial cocycle). [Lean](#) [✓ Stmt](#) *A scalar 2-cochain ω is a coboundary if and only if $\omega \sim 1$, where $1(g, h) = 1$ for all g, h .*

Proof. [✓ Proof](#) Both directions follow by absorbing or introducing the factor $1(g, h) = 1$ via $x \cdot 1 = x$ in (12.10). $\square$

H.4 Stationary-boundary twist support

Fix an MPS tensor $A = (A^i)_{i=0}^{d-1}$, a physical matrix u , and a stationary boundary matrix $\Lambda \in M_D(\mathbb{C})$. The twisted transfer map $\mathcal{E}_u$ is Definition 12.3.1. The next constructions are virtual-boundary functionals; they should not be confused with the physical endpoint correlator of Definition 12.3.4.

Definition H.4.1 (Twisted companion tensor). [Lean](#) [✓ Stmt](#) For an MPS tensor A and a physical-index matrix u , define the companion tensor B_u by

$$B_u^n := \sum_{n'=0}^{d-1} \overline{u_{n'n}} A^{n'}. \quad (\text{H.5})$$

Lemma H.4.2 (Twisted transfer as a mixed transfer operator). [Lean](#) [✓ Stmt](#) *The twisted transfer map of A is the mixed transfer operator of the pair (A, B_u) :*

$$\mathcal{E}_u = F_{A, B_u}. \quad (\text{H.6})$$

Proof. [✓ Proof](#) Expand both definitions and rearrange the double sum. $\square$

Lemma H.4.3 (The companion tensor has the same transfer map). [Lean](#) [✓ Stmt](#) *If $uu^\dagger = \mathbb{1}_d$, then for every $X \in M_D(\mathbb{C})$ the companion family B_u satisfies*

$$\mathcal{E}_{B_u}(X) = \mathcal{E}_A(X).$$

Proof. [✓ Proof](#) Theorem 13.10.11 applies to the unitary combination defining B_u . Equivalently, expand $B_u^n = \sum_{n'} \overline{u_{n'n}} A^{n'}$. The coefficient of $A^j X (A^k)^\dagger$ in $\mathcal{E}_{B_u}(X)$ is

$$\sum_n \overline{u_{jn}} u_{kn} = (uu^\dagger)_{kj} = \delta_{kj}.$$

All off-diagonal terms vanish and the diagonal terms sum to $\mathcal{E}_A(X)$. $\square$

Definition H.4.4 (Virtual-boundary twist functional). [Lean](#) [✓ Stmt](#) For a boundary state Λ and boundary matrices $X, Y \in M_D(\mathbb{C})$, the *virtual-boundary twist functional* is

$$R_L(u; X, Y) := \text{tr}(\Lambda X \mathcal{E}_u^L(Y)). \quad (\text{H.7})$$

The special case $X = Y = \mathbb{1}$ is the stationary-boundary block-twist functional of Definition 12.3.2. The matrices X, Y act on the virtual level. In the physical string correlator of [PGWS⁺08] the string of twists is instead flanked by local operators x, y acting on physical sites; expanding such operators in the transfer picture produces particular virtual matrices, and [PGWS⁺08] argues that for injective tensors, physical operators on sufficiently many sites produce every virtual matrix. The results below are stated directly in this virtual-boundary form.

Lemma H.4.5 (Virtual-boundary nondecay prevents convergence to zero). [Lean](#) [✓ Stmt](#) If virtual-boundary nondecay holds for A, u, Λ , then for some boundary matrices X, Y the sequence $R_L(u; X, Y)$ does not tend to 0 as $L \rightarrow \infty$.

Proof. [✓ Proof](#) A uniform lower bound $c \leq \|R_L(u; X, Y)\|$ is incompatible with convergence to 0. $\square$

H.5 Kraus mixing, transfer expansions, and a common TP gauge

The spectral estimates in the main chapter compare A with B_u . They are placed in one trace-preserving gauge obtained from their common transfer map.

Lemma H.5.1 (Unitary Kraus mixing). [Lean](#) [✓ Stmt](#) If u is unitary, then replacing Kraus operators $\{A_i\}$ by their unitary mixtures $\{\sum_j u_{ij} A_j\}$ preserves the channel:

$$\sum_i \left(\sum_j u_{ij} A_j \right) Y \left(\sum_j u_{ij} A_j \right)^\dagger = \sum_i A_i Y A_i^\dagger. \quad (\text{H.8})$$

Proof. [✓ Proof](#) This is exactly the usual invariance of a Kraus decomposition under a unitary change of Kraus operators; see Theorem 13.10.11. $\square$

Definition H.5.2 (Common trace-preserving gauge for a twisted pair). [Lean](#) [✓ Stmt](#) A *common trace-preserving gauge setup* for (A, u) consists of a tensor B equal to the companion B_u and satisfying $\mathcal{E}_B(X) = \mathcal{E}_A(X)$ for every $X \in M_D(\mathbb{C})$. It also includes a positive definite matrix σ fixed by the adjoint transfer map of B , its positive square root S , and gauged tensors A', B' satisfying

$$\begin{aligned} B &= B_u, & \mathcal{E}_B(X) &= \mathcal{E}_A(X), & \mathcal{E}_{B^\dagger}(\sigma) &= \sigma, & \sigma &> 0, \\ S &= \sigma^{1/2}, & S^\dagger &= S, & A'^i &= S A^i S^{-1}, & B'^i &= S B^i S^{-1}. \end{aligned} \quad (\text{H.9})$$

In addition,

$$\begin{aligned} S S^{-1} &= S^{-1} S = \mathbb{1}, & S^\dagger (S^\dagger)^{-1} &= \mathbb{1}, & (S^{-1})^\dagger &= S^{-1}, \\ \sum_i (A'^i)^\dagger A'^i &= \mathbb{1}, & \sum_i (B'^i)^\dagger B'^i &= \mathbb{1}, \end{aligned} \quad (\text{H.10})$$

together with irreducibility of both A' and B' . These identities complete the common trace-preserving gauge data.

Definition H.5.3 (Construction of the common trace-preserving gauge). [Lean](#) [✓ Stmt](#) Let $D > 0$. If A is injective, $uu^\dagger = \mathbb{1}_d$, and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, define the common trace-preserving gauge setup for (A, u) by choosing the unique positive definite stationary state of the adjoint channel and using the data in Definition H.5.2.

Proof. [✓ Proof](#) Injectivity makes $\mathcal{E}_A$ irreducible. The adjoint channel therefore has a positive definite stationary state σ . By Lemma H.4.3, A and B_u have the same transfer map, hence the same adjoint fixed point. Put $S = \sigma^{1/2}$. Positivity makes S invertible, and direct substitution gives

$$\begin{aligned} \sum_i (SA^i S^{-1})^\dagger (SA^i S^{-1}) &= \mathbb{1}, \\ \sum_i (SB_u^i S^{-1})^\dagger (SB_u^i S^{-1}) &= \mathbb{1}. \end{aligned} \tag{H.11}$$

Similarity preserves irreducibility, so the two gauged families have all the asserted properties. $\square$

Theorem H.5.4 (Twisted eigenvalues survive the common gauge). [Lean](#) [✓ Stmt](#) If $V \neq 0$ and $\mathcal{E}_u(V) = \lambda V$, then λ is an eigenvalue of the mixed transfer map $F_{A', B'}$ in any common trace-preserving gauge setup.

Proof. [✓ Proof](#) By Lemma H.4.2, the original equation is $F_{A, B_u}(V) = \lambda V$. Multiplying on the left and right by S and using $S^\dagger = S$ gives

$$F_{A', B'}(SVS) = \lambda SVS.$$

Since S is invertible, $SVS \neq 0$. $\square$

H.6 Boundary and limit support

Lemma H.6.1 (Spectral radius < 1 forces virtual-boundary decay). [Lean](#) [✓ Stmt](#) If $\rho(\mathcal{E}_u) < 1$, then, as $L \rightarrow \infty$, for all boundary matrices $X, Y, \Lambda \in M_D(\mathbb{C})$ one has $R_L(u; X, Y) \rightarrow 0$.

Proof. [✓ Proof](#) In finite dimension, $\rho(\mathcal{E}_u) < 1$ implies $\mathcal{E}_u^L \rightarrow 0$ in operator norm. Composing with the fixed trace pairing $Z \mapsto \text{tr}(\Lambda X Z)$ gives the stated scalar convergence. $\square$

H.7 Routine universality corollary

The following consequence uses only the virtual MPV-family symmetry predicate and the canonical fixed-point hypotheses from the main chapter.

Theorem H.7.1 (Virtual-boundary nondecay agrees for two injective MPV-symmetric tensors). [Lean](#) [✓ Stmt](#) If A and B are injective, both on-site symmetric under a unitary representation U , and if $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, $\mathcal{E}_B(\mathbb{1}) = \mathbb{1}$, Λ_A and Λ_B are positive definite, $\text{tr}(\Lambda_A) = 1$, $\text{tr}(\Lambda_B) = 1$, $\mathcal{E}_A^\dagger(\Lambda_A) = \Lambda_A$, and $\mathcal{E}_B^\dagger(\Lambda_B) = \Lambda_B$, then for every $g \in G$, virtual-boundary nondecay holds for A with twist $U(g)$ if and only if it holds for B . In particular, since Definition 12.4.1 implies on-site symmetry for both tensors, this theorem applies to tensors in the same SPT phase; it applies equally to symmetric tensors in different SPT phases.

Proof. [✓ Proof](#) Both directions follow from Theorem 12.4.4, which shows that virtual-boundary nondecay holds universally for injective symmetric tensors with canonical normalisations. $\square$

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